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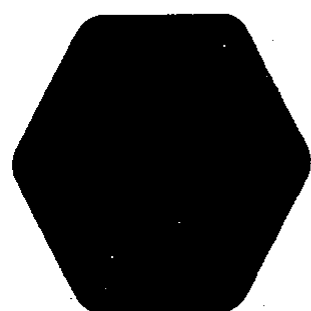
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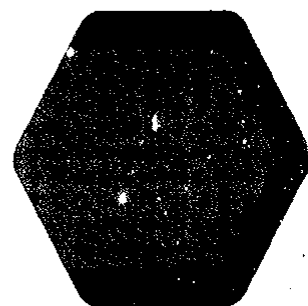
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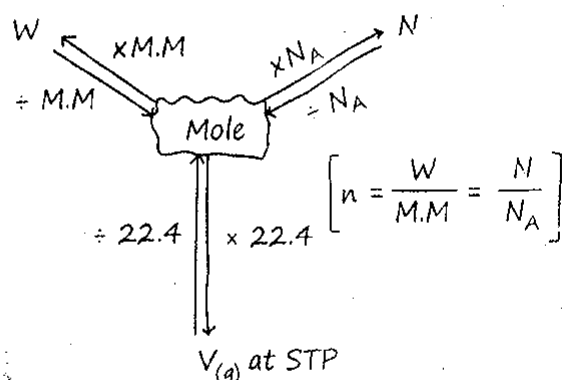
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Physical Chemistry



1. Mole calculation (Y MAP)



2. % Composition

A.W × no. of atoms
of given element
in one molecule of
compound

$$\% \text{ comp of element} = \frac{\text{A.W} \times \text{no. of atoms}}{\text{MW}_{\text{comp}}} \times 100$$

Q. Find % of Ca, C & O in CaCO_3 ?

$$\% \text{ Ca} = \frac{40 \times 1}{100} \times 100 = 40\%$$

$$\% \text{ C} = \frac{12 \times 1}{100} \times 100 = 12\%$$

$$\% \text{ O} = \frac{16 \times 3}{100} \times 100 = 48\%$$

3. Density

Density $\begin{cases} \rightarrow \text{Absolute density} \\ \rightarrow \text{Relative density} \end{cases}$

Case 1: For solids and liquid.

$$1. \text{ A.D(d)} = \frac{\text{Mass}}{\text{Volume}} \left(\frac{\text{g}}{\text{mL}} \right)$$

$$2. \text{ R.D} = \frac{d_{\text{sub}}}{d_{\text{H}_2\text{O at } 4^\circ\text{C}}} \quad [\text{R.D is unitless}]$$

P = pressure (atm)

Case 2: For gases

$$1. \text{ A.D(d)} = \frac{PM}{RT} \left(\frac{\text{g}}{\text{L}} \right)$$

M = MW
R = gas const
= 0.0821 atm mol⁻¹ K⁻¹

$$2. \text{ R.D} = \frac{M_1}{M_2} \quad (\text{at const P \& T}) \quad T = \text{Temp (K)}$$

= $\left(\frac{1}{12} \right)$ approx.

$$3. \text{ V.D} = \frac{M}{2}$$

Q. Find V.D. of SO_2 ?

$$\text{V.D} = M/2 = 64/2 = 32$$

4. Avg atomic wt (for element having isotopes)

$$\text{Avg at wt} = \frac{\% \text{ abundance of isotope}_1 \times \text{AW}_1 + \% \text{ abundance of isotope}_2 \times \text{AW}_2 + \dots}{100}$$

Q. Find avg. atomic wt?

$$\text{Isotopes \%abun} \cdot \text{Avg at wt} = \frac{75 \times 35 + 25 \times 37}{100}$$

$$\text{Cl}^{35} \quad 75\% \quad = \frac{3550}{100}$$

$$\text{Cl}^{37} \quad 25\% \quad = 35.5$$

5. Mean molar mass (for gaseous mix)

$$\text{Mean molar mass} = \frac{n_1 M_1 + n_2 M_2 + \dots}{n_1 + n_2 + \dots}$$

Q. Find mean molar mass

Gas Mole%

O_2 16 mean molar mass =

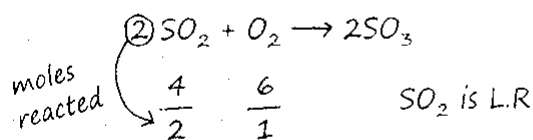
$$\text{N}_2 \quad 80 \quad \frac{16 \times 32 + 80 \times 28 + 3 \times 44 + 1 \times 64}{100}$$

$$\text{CO}_2 \quad 3 \quad = \frac{2948}{100} = 29.48$$

SO_2 1

6. Limiting reagent

n_i 4 6

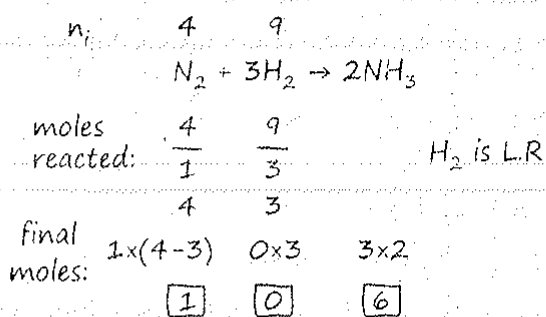


$$\frac{4}{2} \quad \frac{6}{1}$$

final moles

2×0	$(6-2) \times 1$	2×2
0	4	4

Q. 4 moles of N_2 is reacted with 9 moles of H_2 to form NH_3 find



1. Moles of NH_3 formed = 6
2. Wt of NH_3 formed = $6 \times 17 g$
3. Molecules of NH_3 formed = $6 \times N_A$
4. Volume of NH_3 formed = $6 \times 22.4 L$ at STP
5. Moles of excess reactant left = 1
6. Wt of excess reactant left = $1 \times 28 g$
7. Molecules of excess reactant left = $1 \times N_A$
8. Volume of excess reactant left = $1 \times 22.4 L$ at STP

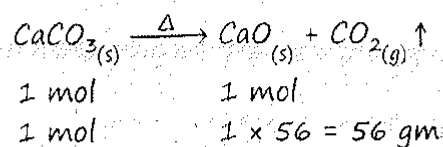
7. Stoichiometric calculation:

	Here	Balance	
	(1) C	(2) H	(3) O

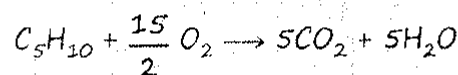
$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$$

Mole	1	2	1	2
Wt (g)	1×16	2×32	1×44	2×18
N	$1 \times N_A$	$2 \times N_A$	$1 \times N_A$	$2 \times N_A$
Vat STP	1×22.4	1×22.4	1×22.4	—

Q. Calculate wt of residue obtained when 1 mol of $CaCO_3$ is strongly heated?



Q. How many litre of O_2 (g) at STP is required for complete combustion of 2 moles of C_5H_{10} ?

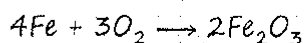


1 mol	$\frac{15}{2}$ mol
-------	--------------------

2 mol	15 mol
-------	--------

2 mol	$15 \times 22.4 L$ at STP
-------	---------------------------

Q. How many moles of O_2 are needed to produce 5 mol of Fe_2O_3 ?



3 mol	2 mol
-------	-------

$\frac{3}{2}$ mol	1 mol
-------------------	-------

$\frac{3}{2} \times 5$ mol	5 mol
----------------------------	-------

$\boxed{= 7.5 \text{ mol}}$

8. Empirical and molecular formula

Element	%comp.	A.W	$\frac{\%}{A.W}$	Simplest ratio	Simplest whole no. ratio
C	40.6	12	$\frac{40.6}{12} = 3.3$	$\frac{3.3}{3.3} = 1$	$1 \times 2 = 2$
H	5	1	$\frac{5}{1} = 5$	$\frac{5}{3.3} = 1.5$	$1.5 \times 2 = 3$
O	54.4	16	$\frac{54.4}{16} = 3.3$	$\frac{3.3}{3.3} = 1$	$1 \times 2 = 2$

$$E.F = C_2H_3O_2$$

$$E.F.W = 12 \times 2 + 1 \times 3 + 16 \times 2 = 59$$

$$M.F = (E.F)_n$$

$$\left(n = \frac{M.F.W}{E.F.W} \right)$$

$$n = \frac{118}{59} = 2$$

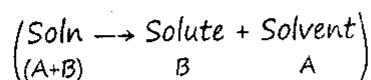
$$M.F = (C_2H_3O_2)_n$$

$$= (C_2H_3O_2)_2$$

$$[M.F = C_4H_6O_4]$$

Q. If M.W of comp = 118 then, find molecular formula.

9. Concentration terms



(1) $\% \frac{W}{W}$ (mass %)

$x \% \frac{W}{W} \Rightarrow x \text{ gm solute is present in } 100 \text{ gm sol}^n$

$$\left[\% \frac{W}{W} = \frac{W_{\text{Solute}}}{W_{\text{Sol}}^n} \times 100 \right]$$

(2) $\% \frac{W}{V}$

$x \% \frac{W}{V} \Rightarrow x \text{ gm solute in } 100 \text{ mL sol}^n$

$$\% \frac{W}{V} = \frac{W_{\text{Solute}} (g)}{V_{\text{Sol}}^n (\text{mL})} \times 100$$

(3) $\% \frac{V}{V}$

$x \% \frac{V}{V} \Rightarrow x \text{ mL solute in } 100 \text{ mL sol}^n$

$$\% \frac{V}{V} = \frac{V_{\text{Solute}} (\text{mL})}{V_{\text{Sol}}^n (\text{mL})} \times 100$$

(4) Molarity (M)

$$\left[M = \frac{n_B}{V_{\text{Sol}}^n (\text{L})} \right]$$

(5) Molality (m)

$$\left[m = \frac{n_B}{W_A (\text{kg})} \right]$$

(6) Normality (N)

$$[N = M \times n.f]$$

$$N = \frac{\text{No. of gm eq}}{V_{\text{Sol}}^n (\text{L})}$$

(7) Strength (S)

$$S = \frac{W_B}{V_{\text{Sol}}^n (\text{L})}$$

(8) PPM (Parts per million)

$$\text{PPM} = \frac{W_B}{W_{\text{Sol}}^n} \times 10^6$$

(9) PPb (Parts per billion)

$$\text{PPb} = \frac{W_B}{W_{\text{Sol}}^n} \times 10^9$$

$$\left. \begin{array}{l} \% \frac{W}{W} \\ \text{molality} \\ \text{PPm} \\ \text{PPb} \end{array} \right\} \text{Temp independent}$$

$$\left. \begin{array}{l} \% \frac{W}{V} \\ \% \frac{V}{V} \\ M \\ N \\ S \end{array} \right\} \text{Temp dependent}$$

Relation blw concⁿ terms

(1) $\% \frac{W}{W}$ & M (molarity)

$$\left[M = \frac{10 \times d}{MW_B} \right] \times \Rightarrow \% \frac{W}{W}$$

$$d \Rightarrow \text{density in } \frac{g}{\text{mL}}$$

$$MW_B = \text{molar mass of solute}$$

(2) Relation blw m and x

$$\left[m = \frac{x_B}{x_A} \times \frac{1000}{MW_A} \right]$$

$$x_B = \text{mole fraction of solute}$$

$$x_A = \text{mole fraction of solvent}$$

$$[x_A + x_B = 1] \quad x_A = \frac{n_A}{n_A + n_B}; \quad x_B = \frac{n_B}{n_A + n_B}$$

$$MW_A = \text{molar mass of solvent}$$

(3) Relation blw m and M

$$\left[M = \frac{md}{1 + mMW_B} \right]$$

(4) Relation blw S(g/L) and M

$$[S(\text{g/L}) = M \times MW_B]$$

Q. Density of 2M aq. CH_3COOH is 1.2 g/mL

find (1) $\frac{W}{W}$ (2) m (3) x_{solute}

$$(1) M = \frac{10 \times d}{MW_B} \quad \left(x = \% \frac{W}{W} \right)$$

$$2 = \frac{10 \times 1.2}{60}$$

$$x = 10$$

$$(10\% \frac{W}{W} \text{ CH}_3\text{COOH})$$

$$(2) M = \frac{md}{1 + m MW_B} \times \frac{1000}{1000}$$

$$2 = \frac{m \times 1.2}{1 + m \times 60} \times \frac{1000}{1000}$$

$$m = ?$$

$$(3) M = \frac{x_B}{1 - x_B} \times \frac{1000}{18}$$

Q. 6.022×10^{22} molecules of a compound 'x' has mass 10 gm. What is the molarity of solⁿ containing 5 gm of 'x' in 2 L?

$$\frac{6.022 \times 10^{22}}{6.022 \times 10^{23}} = \frac{10}{MW_x}$$

$$MW_x = 100$$

$$n_x = \frac{5}{100}$$

$$M_x = \frac{5}{100 \times 2} = 2.5 \times 10^{-2}$$

10. Molarity in diff cases

Case 1: On dilution

$$[M_1 V_1 = M_2 V_2]$$

M_1 = Molarity before dilution

V_1 = Volume before dilution

M_2 = Molarity after dilution

V_2 = Volume after dilution

Case 2: On mixing

$$M_{\text{mix}} = \frac{M_1 V_1 + M_2 V_2}{V_1 + V_2}$$

Case 3: Molarity of ion

$$M_{\text{ion}} = \frac{n_{\text{ion}}}{V_{\text{sol}} \times n(L)}$$

Q. Find molarity on mixing

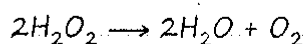
200 mL of 1M HCl + 100 mL of 0.5M HCl

$$M_{\text{mix}} = \frac{200 \times 1 + 100 \times 0.5}{300}$$

$$= \frac{200 + 50}{300} = \frac{250}{300} = \frac{5}{6}$$

11. Volume strength of H_2O_2

x 'V' H_2O_2



1L $\text{H}_2\text{O}_2(\text{aq})$ gives xL O_2 at S.T.P.

$$(1) [V.S = 11.2 \times M]$$

$$(N = M \times nf)$$

$$(2) [V.S = 5.6 \times N]$$

$$(3) [S(g/L) = M \times MW_B]$$

$$(4) \left[\% \frac{W}{V} = \frac{S}{10} \right]$$

12. % of free SO_3 (labelling of oleum)

% free SO_3 = y % strength oleum

$$\% \text{ free } \text{SO}_3 = \frac{80}{18} \times (y - 100)$$

13. % of available Cl_2

$$\% \text{ avail. } \text{Cl}_2 = \frac{3.5 \times M \times V}{W_{\text{gm}}}$$

14. Hardness of water in ppm

$$\text{Hardness} = \frac{W_{\text{CaCO}_3}}{\text{total mass of } \text{H}_2\text{O}} \times 10^6$$

(in terms of ppm of CaCO_3)

$$[d = 1 \text{ g/mL}]$$

Practice

Q. Find % $\frac{W}{W}$ of 5.6 V H_2O_2

$$(d = 1 \text{ g/mL}, MW_{H_2O_2} = 34)$$

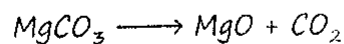
$$V.S. = 11.2 \times M \quad \left[M = \frac{1}{2} \right]$$

$$5.6 = 11.2 \times M$$

$$M = \frac{10 \times d}{MW_B} \quad x = 1.7$$

$$\frac{1}{2} = \frac{10 \times x \times 1}{34} \quad \left[1.7 \% \frac{W}{W} H_2O_2 \right]$$

Q. 20 g $MgCO_3$ sample decomposes on heating to give CO_2 and 8 gm MgO . Find % purity of $MgCO_3$ in sample?



$$84 \text{ gm} \longrightarrow 40 \text{ gm} \quad \frac{84}{40} \times 8 \text{ gm} \longrightarrow 8 \text{ gm}$$

$$\left[\% \text{ purity} = \frac{W_{\text{comp}}}{W_{\text{sample}}} \times 100 \right] = \frac{84}{5 \times 20} \times 100 = 84 \%$$

Q. Find x_B of 1 molal aq solⁿ?

$$m = \frac{x_B}{x_A} \times \frac{1000}{MW_A}$$

$$1 = \frac{x_B}{(1-x_B)} \times \frac{1000}{18}^{55.5} \quad \left[\frac{1}{55.5} = \frac{x_B}{1-x_B} \right]$$

Q. Concⁿ aq solⁿ of H_2SO_4 is 98% H_2SO_4 by mass & has density 1.8 gm/mL. Volume of acid required to make 1 Ltr of 0.1 M H_2SO_4 solⁿ is?

$$x = 98, d = 1.8 \quad M = \frac{10 \times d}{MW_B}$$

$$M = \frac{10 \times 98 \times 1.8}{98} = 18 \text{ M}$$

$$18 \times V_1 = 1 \times 0.1 \Rightarrow V_1 = \frac{1}{180} \text{ L}$$

Q. How many moles of $PbCl_2$ will be formed from a rxn b/w 6.5 gm PbO and 3.2 gm HCl ?

n_i	0.029	0.087	
	PbO	$+ 2HCl$	$\longrightarrow PbCl_2 + H_2O$
	0.029	0.087	
	1	2	$n_{PbO} = \frac{6.5}{224}$
moles reacted	0.029	0.0435	
	= 0.029		0.029

$$n_{HCl} = \frac{3.2}{36.5} = 0.087$$

PbO is limiting reagent

$$n_{PbCl_2} \text{ formed} = 0.029$$

1. Electromagnetic radiation

$$(1) \quad c = \nu \lambda$$

$\swarrow$ $\downarrow$ $\downarrow$
 $3 \times 10^8 \text{ m/s}$ frequency wave length

$$(2) \quad \nu = \frac{1}{\lambda}$$

$$1 \text{ fm} = 10^{-15} \text{ m}$$

$$1 \text{ pm} = 10^{-12} \text{ m}$$

$$1 \text{ \AA} = 10^{-10} \text{ m}$$

$$1 \text{ nm} = 10^{-9} \text{ m}$$

$$1 \mu\text{m} = 10^{-6} \text{ m}$$

Q. Which has highest frequency?

☒ (A) X-rays

(B) Yellow

(C) Infrared

(D) Radio wave

Electromagnetic Spectrum				Wave length ↑	freq ↓		
Cosmic rays	Gamma rays	X-Ray	U.V radiation	Visible	Infrared	Microwaves	Radio waves
ν_{max}				↓ VIBGYOR λ ↑, ν ↓			ν_{min}
λ_{min}							λ_{max}
Classical	gaadi	X	U	V	ln	my	range

2. Planck's quantum theory

$$(1) E = \frac{hc}{\lambda} = h\nu \text{ (for 1 quantum)}$$

$$(2) E = \frac{hc}{\lambda} n \text{ (for } n \text{ quantum)}$$

$$E = n \frac{12400}{\lambda(\text{\AA})} \text{ eV} [1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}]$$

Q. Find energy of a photon whose wave length is 310 nm.

$$\left(\begin{array}{l} \lambda = 310 \text{ nm} \\ = 3100 \text{ \AA} \end{array} \right) \quad E = 1 \times \frac{12400}{3100} = 4 \text{ eV}$$

$$E = 4 \times 1.6 \times 10^{-19} \text{ J}$$

$$= 6.4 \times 10^{-19} \text{ J}$$

Q. No. of photons emitted by a bulb of 40 watt in 1 min with 50% efficiency will be?

(Given: $\lambda = 620 \text{ nm}$)

$$\lambda = 6200 \text{ \AA} \quad E = P \times t$$

$$= 40 \times 60 \times \frac{1}{2}$$

$$= 1200 \text{ J}$$

$$\frac{600}{1200} = n \times \frac{12400}{6200}$$

$$n = \frac{600}{1.6 \times 10^{-19}}$$

$$= 375 \times 10^{19}$$

$$n = 3.75 \times 10^{21}$$

3. Bohr's theory (for $1e^-$ system)H, He^+ , Li^{2+} , Be^{3+} Na^{10+} etc.

(1) Orbit angular momentum

$$mvr = \frac{nh}{2\pi} = n\hbar \quad \left(\hbar = \frac{h}{2\pi} \right)$$

$$(2) \text{ Radius, } r = r_0 \times \frac{n^2}{Z} = a_0 \times \frac{n^2}{Z} = 0.53 \times \frac{n^2}{Z} \text{ \AA}$$

$$r_0 \text{ or } a_0 = \text{Bohr radius} = 0.53 \text{ \AA}$$

$$(3) \text{ Velocity } V = 2.18 \times 10^6 \times \frac{Z}{n} \text{ m/s}$$

$$(4) \text{ Total energy T.E} = -13.6 \times \frac{Z^2}{n^2} \frac{\text{ev}}{\text{atom}}$$

$$(5) \text{ T.E} = -\text{K.E} = \frac{\text{P.E}}{2}$$

$$(6) \text{ Time period } T \propto \frac{n^3}{Z^2}$$

$$(7) \text{ Frequency } f = \frac{1}{T} \quad f \propto \frac{Z^2}{n^3}$$

$$(8) \text{ Ionization energy I.E} = +13.6 \times Z^2 \frac{\text{ev}}{\text{atom}}$$

$$(9) \text{ Excitation energy E.E} = 13.6 \times Z^2$$

$$\left[1 - \frac{1}{n^2}\right] \frac{\text{ev}}{\text{atom}}$$

$$(10) \text{ Binding energy B.E} = +13.6 \times \frac{Z^2}{n^2} \frac{\text{ev}}{\text{atom}} = \frac{\text{I.E}}{n^2}$$

$$\begin{bmatrix} 3^{\text{rd}} \text{ E.S } n=4 \\ 4^{\text{th}} \text{ E.S } n=5 \end{bmatrix}$$

Q. Find r , V , T.E , K.E , P.E of He^+ in 1^{st} excited state?

$$n = 2 \quad ; \quad Z = 2$$

$$r = r_0 \times \frac{2^2}{2} = 2r_0 \text{ or } 2a_0 = 2 \times (0.53) \text{ \AA}$$

$$V = 2.18 \times 10^6 \times \frac{2}{2} \text{ m/s} = 2.18 \times 10^6 \text{ m/s}$$

$$\text{TE} = -13.6 \times \frac{2^2}{2^2} \text{ ev/atom} = -13.6 \text{ ev/atom}$$

$$\text{KE} = +13.6 \text{ ev/atom}$$

$$\text{PE} = -13.6 \times 2 = -27.2 \text{ ev/atom}$$

Q. Find orbit A.M of 5^{th} orbit?

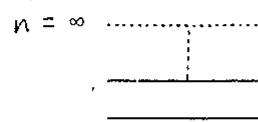
$$mvr = \frac{5h}{2\pi} = 2.5 \frac{h}{\pi}$$

Q. Find I.E of Li^{2+} ion?

$$\begin{aligned} \text{I.E} &= +13.6 \times 3^2 \\ &= +13.6 \times 9 \text{ ev/atom} \end{aligned}$$

$$\left[\frac{mv^2}{r} = \frac{KZe^2}{r^2} \right]$$

4. Hydrogen spectrum



$n = 6$ ————— Humphrey FIR

$n = 5$ ————— P fund IR

$n = 4$ ————— Brackett IR

$n = 3$ ————— Paschen IR

$n = 2$ ————— Balmer (Visible)

$n = 1$ ————— Lyman (U.V.)

Rydberg's formula

$$R_H = \text{Rydberg's const} \frac{1}{\lambda} = R_H Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$= 1.1 \times 10^7 \text{ m}^{-1} \quad (C = v\lambda) \left(\frac{1}{v} = \frac{1}{\lambda} \right)$$

Total no. of spectral lines during de. excitation from n_2 to n_1 orbit = $\frac{\Delta n(\Delta n + 1)}{2}$

$$\Delta n = n_2 - n_1$$

Total no. of spectral lines in particular series

$$= n_2 - n_1 \rightarrow \text{series no.}$$

$$\begin{cases} n_1 = \text{Lower energy level} \\ n_2 = \text{Higher energy level} \end{cases}$$

Q. Calculate λ of photon emitted when e^- makes transition from $n = 2$ to $n = 1$ in H atom?

$$\frac{1}{\lambda} = R_H (1)^2 \left[\frac{1}{1^2} - \frac{1}{2^2} \right]$$

$$\frac{1}{\lambda} = R_H \times \frac{3}{4}$$

$$\left[\lambda = \frac{4}{3R_H} \right]$$

Q. Calculate λ for 2nd line of Balmer series for He^+ ion?

$$\frac{1}{\lambda} = R_H (2)^2 \left[\frac{1}{2^2} - \frac{1}{4^2} \right]$$

$$\frac{1}{\lambda} = R_H \times \cancel{4} \times \frac{1}{\cancel{4}} \left[1 - \frac{1}{4} \right]$$

$$\lambda = \frac{4}{3R_H}$$

Q. Find series limit of Lyman series for Li^{2+} .

$Z = 3 \quad n_1 = 1 \quad n_2 = \infty$

$$\frac{1}{\lambda} = R_H (3)^2 \left[\frac{1}{1^2} - \frac{1}{\infty^2} \right]$$

$$\left[\lambda = \frac{1}{9R_H} \right]$$

Q. Find λ_{max} and λ_{min} for Lyman series for H atom?

$$\frac{1}{\lambda} = R_H (1)^2 \left[\frac{1}{1^2} - \frac{1}{\infty^2} \right]$$

$$\frac{1}{\lambda} = R_H (1)^2 \left[\frac{1}{1^2} - \frac{1}{2^2} \right] \quad \left[\lambda_{\text{min}} = \frac{1}{R_H} \right]$$

$$\left(\lambda_{\text{max}} = \frac{4}{3R_H} \right)$$

Q. Find $\bar{\nu}$ (wave no.) when $e^\ominus$ makes transition from 3rd E.S to ground state in Be^{3+} ion?

$$\frac{1}{\lambda} = \bar{\nu} = R_H (4)^2 \left[\frac{1}{1^2} - \frac{1}{4^2} \right]$$

$$Z = 4, n_1 = 1, n_2 = 4 \quad \bar{\nu} = \cancel{16} R_H \times \frac{15}{\cancel{16}}$$

$$[\bar{\nu} = 15 R_H]$$

Q. Find λ for H α line for Lyman series for He^+ ion? $n_1 = 1, n_2 = 2, Z = 2$

$\alpha \rightarrow 1^{\text{st}} \text{ line}$

$$\frac{1}{\lambda} = R_H (2)^2 \left[\frac{1}{1^2} - \frac{1}{2^2} \right]$$

$\beta \rightarrow 2^{\text{nd}} \text{ line}$

$\gamma \rightarrow 3^{\text{rd}} \text{ line}$

$$\frac{1}{\lambda} = \cancel{4} R_H \times \frac{3}{\cancel{4}} \left[\lambda = \frac{1}{3R_H} \right]$$

$\delta \rightarrow 4^{\text{th}} \text{ line}$

Q. Find total no. of spectral lines when $e^\ominus$ makes transition from 9th E.S to 1st E.S

$$\Delta n = n_2 - n_1 = 8 \quad n_2 = 10; n_1 = 2$$

$$\text{S.L} = \frac{8 \times (8 + 1)}{2}$$

$$= 4 \times 9$$

$$= 36$$

Q. Find spectral lines in Balmer series when $e^\ominus$ makes transition from 14th E.S to G.S.?

$$n_2 = 15 \quad n_1 = 2$$

$$\text{S.L in Balmer series} = 15 - 2 = 13$$

5. Debroglie hypothesis

$$(1) \lambda = \frac{h}{p}$$

$$(2) \lambda = \frac{h}{mv}$$

$$m = \text{mass (kg)}$$

$$v = \text{m/s}$$

$$(3) \lambda = \frac{h}{\sqrt{2m KE}}$$

$$V = \text{Volt potential}$$

$$KE = J$$

$$(4) \lambda = \frac{h}{\sqrt{2m qV}}$$

$$(5) \text{ for } e^\ominus \lambda = \frac{12.24}{\sqrt{V}} \text{ \AA} = \sqrt{\frac{150}{V}} \text{ \AA}$$

$$(6) 2\pi r = n\lambda \quad n = \text{no. of waves made in one revolution}$$

$$\left(r = r_0 \times \frac{n^2}{Z} \text{ \AA} = 0.53 \times \frac{n^2}{Z} \text{ \AA} \right)$$

Q. Find λ when an $e^\ominus$ is moving with velocity of 10^6 m/s

$$\lambda = \frac{h}{mv} = \frac{6.6 \times 10^{-34}}{9.1 \times 10^{-31} \times 10^6} \text{ m}$$

$$[\lambda \approx 7 \text{ \AA}]$$

Q. Find ratio of λ of α and proton ($\frac{\lambda_\alpha}{\lambda_p}$) accelerated through same potential diff.

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

$$\lambda \propto \frac{1}{\sqrt{mq}}$$

$$\frac{\lambda_\alpha}{\lambda_p} = \sqrt{\frac{m_p q_p}{m_\alpha q_\alpha}}$$

$$= \sqrt{\frac{1}{4} \times \frac{1}{2}}$$

$$= \frac{1}{\sqrt{8}} = \frac{1}{2\sqrt{2}}$$

$$\begin{array}{l} \alpha \quad p \\ m_\alpha = 4m_p \\ q_\alpha = 2q_p \end{array}$$

Q. Find λ when an e^- is accelerated by a potential diff of 15 V from rest?

$$\lambda = \sqrt{\frac{150}{V}} \text{ \AA}$$

$$\lambda = \sqrt{\frac{150}{15}}$$

$$[\lambda = \sqrt{10} \text{ \AA}]$$

6. Heisenberg's uncertainty principle

$$(1) \Delta x \cdot \Delta p \geq \frac{h}{4\pi} \quad \left(\frac{h}{4\pi} \right) = 0.52 \times 10^{-34}$$

$$(2) \Delta x \cdot \Delta v \geq \frac{h}{4\pi m}$$

$$(3) \Delta E \cdot \Delta t \geq \frac{h}{4\pi}$$

$$(4) \Delta x \cdot |\Delta \lambda| \geq \frac{\lambda^2}{4\pi} \quad \lambda = \text{wave length (m)}$$

Q. If $\Delta x = 2\Delta p$ then find Δv ?

$$\Delta x = 2m\Delta v \quad \Delta v = \sqrt{\frac{h}{8\pi m^2}}$$

$$\Delta x \cdot \Delta v = \frac{h}{4\pi m} \quad = \frac{1}{2m} \frac{h}{\sqrt{2\pi}}$$

$$2m \Delta v \times \Delta v = \frac{h}{4\pi m}$$

$$(\Delta v)^2 = \frac{h}{8\pi m^2}$$

Q. Δx is 10^{-5} m of a particle of mass 0.25 gm then find Δv ?

$$\Delta x \cdot \Delta v = \frac{0.52 \times 10^{-34}}{m}$$

$$10^{-5} \times \Delta v = \frac{0.52 \times 10^{-34}}{0.25 \times 10^{-3}}$$

$$[\Delta v = 2.1 \times 10^{-26} \text{ m/s}]$$

Q. A golf ball has mass $\overset{0.1 \text{ kg}}{100 \text{ gm}}$ and speed of 100 m/s. If the speed can be measured within accuracy of 2% find Δx ?

$$\Delta v = v \times \text{accuracy}$$

$$= 100 \times \frac{2}{100}$$

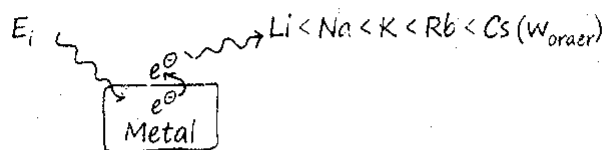
$$= 2 \text{ m/s}$$

$$\Delta x \cdot \Delta v = \frac{0.52 \times 10^{-34}}{m}$$

$$\Delta x \times 2 = \frac{0.52 \times 10^{-34}}{0.1}$$

$$[\Delta x = 0.26 \times 10^{-33} \text{ m}]$$

7. Photo electric effect (P.E.E)



$$[E_i = \Phi_{\min} + K.E_{\max}]$$

$$W_{\min}$$

$$[h\nu = h\nu_0 + KE_{\max}] \quad \left[KE_{\max} = \frac{1}{2} mV_{\max}^2 \right]$$

$$\left[\frac{hc}{\lambda} = \frac{hc}{\lambda_0} + KE_{\max} \right]$$

V = freq. of incident light (λ = W.L of incident light)

V_0 = Threshold freq (min) (λ_0 = Threshold W.L (max))

Q. Calculate KE(J) of an e^- emitted when radiation of frequency $\nu = 1.1 \times 10^{15} \text{ s}^{-1}$ hits the metal ($\nu_0 = 6 \times 10^{14} \text{ s}^{-1}$)

$$h\nu = h\nu_0 + KE$$

$$h \times 1.1 \times 10^{15} = h \times 6 \times 10^{14} + KE$$

$$KE = 5 \times h \times 10^{14}$$

$$= 5 \times 6.6 \times 10^{-34} \times 10^{14}$$

$$= 5 \times 6.6 \times 10^{-20} \text{ J}$$

Q. Light of $\lambda = 310 \text{ nm}$ is used in an exp of P.E.E with Li ($\phi = 2.5 \text{ eV}$)

Find (a) max KE (b) Stopping potential

$$E_i = \phi + KE_{\max}$$

$$\frac{12400}{\lambda(\text{\AA})} = \phi + KE_{\max}$$

$$\frac{12400}{3100} = 2.5 + KE_{\max}$$

$$4 - 2.5 = KE_{\max}$$

$$(a) KE_{\max} = 1.5 \text{ eV} = 1.5 \times 1.6 \times 10^{-19} \text{ J}$$

$$(b) [V_0 = 1.5 \text{ Volt}]$$

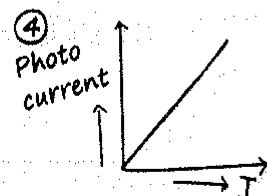
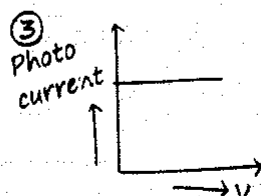
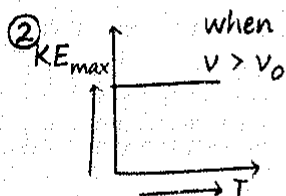
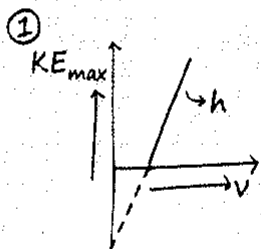
Imp. Points

1. Intensity $\uparrow$, no. of ejected e^- s $\uparrow$

2. Frequency $\nu \uparrow$, $KE_{\max} \uparrow$

3. P.E.E is obs only when $[\nu > \nu_0]$
 $E_i > \phi$

Graphs $h\nu = h\nu_0 + KE_{\max}$
 $[KE_{\max} = h\nu - h\nu_0]$



Q. The no of metals which will show P.E.E when light of 310 nm wavelength falls on following metals $3100 \text{ \AA}$

metal	K	Na	K	Mg	Cu	Ag	Fe	pt
$\phi(\text{eV})$	2.4	2.3	2.2	3.7	4.8	4.3	4.7	6

$$E_i = \frac{12400}{3100} = 4 \text{ eV}$$

8. Quantum no.

(1) Principal Q.N (n)

(a) It describes the shell or orbit or energy level

(b) $n = 1, 2, 3, 4, 5 \dots$

K, L, M, N, O $\dots$

(c) Orbit angular momentum

$$mvr = \frac{nh}{2\pi}$$

(d) Total orbitals = n^2
total e^- s = $2n^2$ } in n^{th} orbit

(2) Azimuthal Q.N (ℓ)

(a) It describes shape of subshell or sub orbit (sub energy level)

(b) $\ell = 0$ to $n - 1$

$$\left[\begin{array}{cccccc} \ell = 0 & 1 & 2 & 3 & 4 \\ & s & p & d & f & g \end{array} \right]$$

(c) Orbital angular momentum

$$= \sqrt{\ell(\ell+1)} \frac{h}{2\pi}$$

$$= \sqrt{\ell(\ell+1)} \hbar$$

(d) Total orbitals = $(2\ell + 1)$
total e^- s = $2(2\ell + 1)$ } in subshell ℓ .

(3) Magnetic Q.N (m or m_ℓ)

(a) Orientation of subshell.

(b) $m = -\ell$ to $+\ell$

$$\ell = 0 \quad \boxed{} \quad m = 0$$

$$\ell = 1 \quad \boxed{} \boxed{} \boxed{} \quad m = \pm 1, 0$$

$$\ell = 2 \quad \boxed{} \boxed{} \boxed{} \boxed{} \boxed{} \quad m = \pm 2, \pm 1, 0$$

(4) Spin Q.N (S or m_s)

(a) Spin either C.W or A.C.W

(b) $S = \pm \frac{1}{2}$

(c) Spin angular momentum

$$= \sqrt{S(S+1)} \frac{h}{2\pi}$$

$$= \sqrt{S(S+1)} \hbar$$

$$S = \frac{n}{2} \quad (n = \text{no. of unpaired } e^{\ominus} \text{ s})$$

(d) Total spin $= \pm \frac{n}{2}$

(e) Spin multiplicity $= (2S + 1)$

$$S = \frac{n}{2}$$

Q. Total no. of orbitals in

(a) $n = 3$

(b) $\ell = 2$

Orbitals $= n^2$

Orbitals $= (2\ell + 1)$

$= 3^2$

$= (2 \times 2 + 1)$

$= 9$

$= 5$

Q. Describe the subshell

n

ℓ

Subshell

3

0

3 s

4

1

4 p

5

2

5 d

2

2

~~2~~ d (not possible)

Q. Find orbital A.M

(a) 2 s $\sqrt{0(0+1)} \hbar = 0$

(b) 3 p $\sqrt{1(1+1)} \hbar = \sqrt{2} \hbar$

(c) 4 d $\sqrt{2(2+1)} \hbar = \sqrt{6} \hbar$

(d) 5 f $\sqrt{3(3+1)} \hbar = \sqrt{12} \hbar$

Q. Find orbit A.M

(a) 4 p 4 $\hbar$

(b) 5 d 5 $\hbar$

(c) 1 s 1 $\hbar$

(d) 6 f 6 $\hbar$

$$\left[\hbar = \frac{h}{2\pi} \right]$$

Q. Incorrect set of Quantum no.

	n	$0 \text{ to } n-1$ ℓ	$-\ell \text{ to } +\ell$ m	s
(A)	4	0	0	$+\frac{1}{2}$
(B)	3	1	1	$-\frac{1}{2}$
(C)	2	0	0	$-\frac{1}{2}$
(D)	3	1	-2	$+\frac{1}{2}$

Q. Find spin A.M when no. of U.P $e^{\ominus}$ s is 2

$n = 2 \quad S = \frac{n}{2} = \frac{2}{2} = 1$

S.A.M $= \sqrt{1(1+1)} \hbar = \sqrt{2} \hbar$

Q. Find spin multiplicity and total spin if no. of U.P $e^{\ominus}$ s is 2

$n = 2 \quad T.S = \pm \frac{n}{2} = \pm \frac{2}{2} = \pm 1$

S.M $= 2S + 1 = 2 \times 1 + 1 = 3$

$S = \frac{n}{2} = \frac{2}{2} = 1$

Q. No. of $e^{\ominus}$ s present in d-orbital?

(A) 2

(B) 6

(C) 10

(D) 14

Imp. Points

- (a) Total nodes $= n - 1$

(b) Radial or spherical nodes $= n - \ell - 1$

(c) Nodal plane $= \ell$

or

Angular node $= \ell$

for given subshell

2. Energy of subshell [Bohr burry rule or $(n + \ell)$ rule]

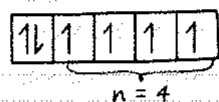
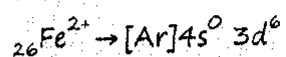
$E \propto (n + \ell)$ [if $(n + \ell)$ is same, $E \propto n$]

3. Magnetic moment $= \sqrt{n(n+2)} \text{ B.M}$

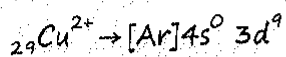
$n = \text{no. of U.P } e^{\ominus} \text{ s}$

Q. Find magnetic moment

(A) Fe^{2+}



(B) Cu^{2+}



$M.M = \sqrt{4(4+2)} = \sqrt{24}$ $M.M = \sqrt{1(1+2)} = \sqrt{3}$

Q. Find no. of total nodes, Nodal planes, angular nodes and Radial nodes

Subshell T.N R.N N.P A.N

$n = 3, l = 0$ 2 2 0 0
3s

$n = 4, l = 1$ 3 2 1 1
4p
 $n = 5, l = 2$ 4 2 2 2
5d

Q. Compare the energy of subshell

(A) 3p (B) 3s (C) 5d (D) 4p

$n = 3$ $n = 3$ $n = 5$ $n = 4$

$l = 1$ $l = 0$ $l = 2$ $l = 1$

$n+l = 4$ $n+l = 3$ $n+l = 7$ $n+l = 5$

$[C > D > A > B]$

Gas Laws #

1. Boyle's Law

$$P \propto \frac{1}{V} \text{ [at const } n, T \text{]}$$

$$[P_1 V_1 = P_2 V_2]$$

2. Charle's Law

$$V \propto T \text{ [at const } n \text{ and } P \text{]}$$

$$\left[\frac{V_1}{T_1} = \frac{V_2}{T_2} \right]$$

$$T = K$$

3. Gay Lussac's Law

$$P \propto T \text{ [at const } n \text{ and } V \text{]}$$

$$\left[\frac{P_1}{T_1} = \frac{P_2}{T_2} \right]$$

4. Avogadro's Law

$$V \propto n \text{ [at const } P \text{ and } T \text{]}$$

$$\left[\frac{V_1}{n_1} = \frac{V_2}{n_2} \right]$$

Combined Gas Law

$$\left[\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \right]$$

Ideal Gas Eqⁿ

$$PV = nRT$$

$$P \Rightarrow \text{atm}$$

$$V \Rightarrow L$$

$$n \Rightarrow \text{moles}$$

$$R \Rightarrow 0.0821 \text{ or } \frac{1}{12} \frac{L \text{ atm}}{\text{mol-K}}$$

$$T \Rightarrow K$$

Dalton's Law of partial pressure for non reacting gaseous. mix. The total pressure is the sum of partial pressure of each gas.

$$1 + 2 + 3 \dots [P_T = P_1 + P_2 + P_3 + \dots]$$

$$P_{\text{gas}} = \chi_{\text{gas}} \times P_T$$

Kinetic Theory of GasesPostulates #

1. There is no force of attrⁿ blw gas molecules
2. Mass of gas molecule is negligible (Point mass)
3. The collisions blw gas molecules are perfectly elastic
4. The motion of gas molecules is chaotic (Random)
5. The avg. K.E. of gas molecules is directly proportional to absolute temp (Temp in K)

$$\text{Avg KE} \propto T$$

$$\frac{R}{N_A} = K$$

$$[\text{Avg KE per mole} = \frac{3}{2} RT] \quad K = 1.38 \times 10^{-23}$$

$$R = \frac{25}{3} \text{ J/mol/K}$$

$$\text{For } n \text{ moles Avg KE} = \frac{3}{2} nRT$$

$$\text{Avg KE per molecule} = \frac{3}{2} KT$$

Kinetic gas eqⁿ

$$PV = \frac{1}{3} mN V_{\text{rms}}^2$$

$$P = P_a \quad N = \text{No. of molecules}$$

$$V = m^3$$

$$m = \text{Kg}$$

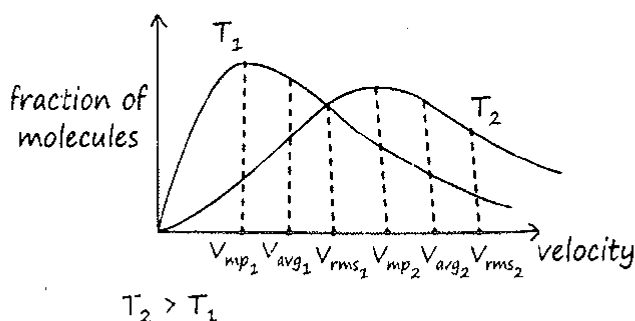
$$V_{\text{rms}} = \text{m/s}$$

Molecular Speeds

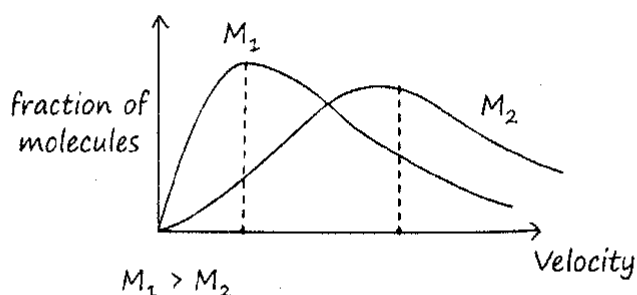
V_{rms} (root mean square)	V_{avg} (average)	V_{mp} (most probable)
$V_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3P}{d}}$ $V_{rms} = \text{m/s}$ $R = \frac{25}{3} \text{ J/mol/K}$ $T = \text{K}$ $M = \text{molar mass (kg)}$	$V_{avg} = \sqrt{\frac{8RT}{\pi M}} = \sqrt{\frac{8P}{\pi d}}$ $\frac{RT}{M} = \frac{KT}{m} \quad d = \frac{PM}{RT}$ $\frac{RT}{M} = \frac{P}{d}$	The velocity possessed by max fraction of molecules $V_{mp} = \sqrt{\frac{2RT}{M}} = \sqrt{\frac{2P}{d}}$ $\sqrt{3} > \sqrt{\frac{8}{\pi}} > \sqrt{2}$ $V_{rms} > V_{avg} > V_{mp}$ $R > A > M$

Maxwell Distribution Curve

Case 1: For same gas at diff temp.



Case 2: For diff gas at same temp.



Graham's Law of diffusion/Effusion

Diffusion # The spontaneous flow of gas molecules from higher concⁿ to Lower concⁿ.

Effusion # Diffusion from pin hole or small hole or orifice.

Law # The rate of diffusion/Effusion is inversely proportional to the square root of the density at constt T and P.

$$r \propto \frac{1}{\sqrt{d}} \propto \frac{1}{\sqrt{V \cdot D}} \propto \frac{1}{\sqrt{M \cdot W}}$$

$$\left[\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}} \right]$$

Q. Compare rate of diffusion?

	CH_4	CO_2	SO_2
MW	16	44	64
rate	r_1	r_2	r_3

$$r_1 > r_2 > r_3$$

Q. The r.O.D of methane at a given temp is twice that of a gas x.

Find MW^x?

$$\frac{r_{\text{CH}_4}}{r_x} = 2$$

$$\frac{\text{MW}_x}{16} = 4$$

$$\sqrt{\frac{\text{MW}_x}{16}} = 2$$

$$[\text{MW}_x = 64]$$

Rate

1. Volume

$$r = \frac{V}{t} \text{ (diffused)}$$

2. Length

$$r = \frac{l}{t} \text{ (travelled)}$$

3. Moles

$$r = \frac{n}{t} \text{ (diffused)}$$

	Gas 1	Gas 2
V	V_1	V_2
t	t_1	t_2
l	l_1	l_2
n	n_1	n_2
MW	M_1	M_2

$$1. \frac{V_1}{V_2} \times \frac{t_2}{t_1} = \sqrt{\frac{M_2}{M_1}}$$

$$2. \frac{l_1}{l_2} \times \frac{t_2}{t_1} = \sqrt{\frac{M_2}{M_1}}$$

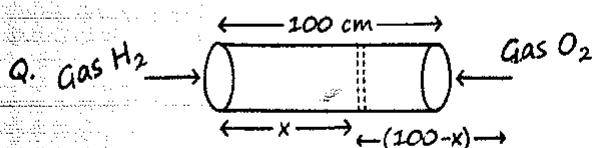
$$3. \frac{n_1}{n_2} \times \frac{t_2}{t_1} = \sqrt{\frac{M_2}{M_1}}$$

Q. Two gases A and B having same volume diffuse through a porous portion in 20 and 10 sec respectively. The M.W of A is 49 u. Find MW_B ?

$$\frac{V_A}{V_B} \times \frac{t_B}{t_A} = \sqrt{\frac{MW_B}{MW_A}}$$

$$\frac{10}{20} = \sqrt{\frac{MW_B}{49}}$$

$$\frac{1}{4} = \frac{MW_B}{49} \quad \text{i.e., } MW_B = \frac{49}{4}$$



Find the distance from H_2 end where they will meet first?

$$\frac{x}{100-x} \times \frac{1}{1} = \sqrt{\frac{32}{2}}$$

$$\frac{x}{100-x} = 4$$

$$x = 400 - 4x$$

$$5x = 400$$

$$x = 80 \text{ cm}$$

If Pressure is Variable at Const temp

$$r \propto \frac{P}{\sqrt{d}} \propto \frac{P}{\sqrt{MW}}$$

$$\frac{r_1}{r_2} = \frac{P_1}{P_2} \sqrt{\frac{M_2}{M_1}}$$

$$\left(\frac{n_1}{n_2} \right)_{\text{effused out}} = \left(\frac{n_1}{n_2} \right)_{\text{initially}} \sqrt{\frac{M_2}{M_1}}$$

Q. A football bladder contains equimolar proportions of H_2 and O_2 . The composition by mass of the mixture effusing out of punctured football is in the ratio.

$$\left(\frac{n_{H_2}}{n_{O_2}} \right)_{\text{effused out}} = \left(\frac{n_{H_2}}{n_{O_2}} \right)_{\text{initial}} \sqrt{\frac{M_{O_2}}{M_{H_2}}}$$

$$\frac{n_{H_2}}{n_{O_2}} = 1 \times \sqrt{\frac{32}{2}} = 4$$

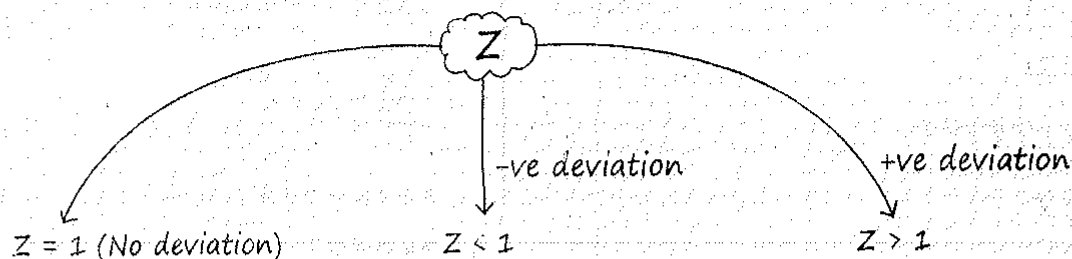
$$\frac{W_{H_2}}{W_{O_2}} \times \frac{MW_{O_2}}{MW_{H_2}} = 4 \quad \left[\frac{W_{H_2}}{W_{O_2}} = \frac{4 \times 2}{32} = \frac{1}{4} \right]$$

Real Gas

1. Compressibility factor (Z)

$$Z = \frac{V_{\text{real}}}{V_{\text{ideal}}} = \frac{V_{\text{obs}}}{n \cdot RT} \cdot P$$

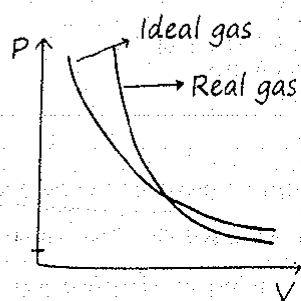
$$= \frac{P(V_m)_{\text{obs}}}{RT}$$



- | | | |
|--|--------------------------------------|--------------------------------------|
| ⇒ Real gas = Ideal gas | ⇒ Real gas | ⇒ Real gas |
| ⇒ No force of attraction and repulsion | ⇒ Attraction force > repulsive force | ⇒ Attraction force < repulsive force |
| ⇒ Can not be liquefied | ⇒ Easy to liquefy or compress | ⇒ Diff to liquefy or compress |

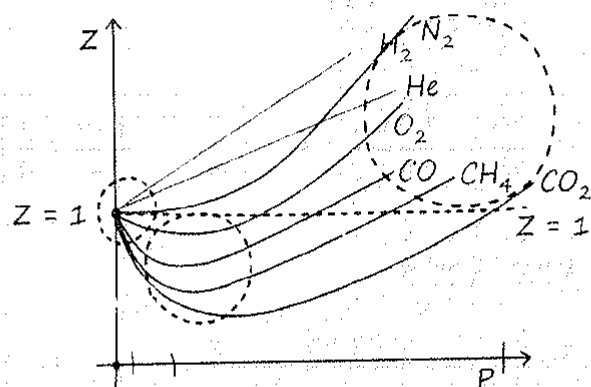
2. Graphical analysis

(1) P vs V



- # At very Low pressure real gas behaves like ideal gas.
- # At Low pressure and High temp Real gas behaves like ideal gas.

(2) Z vs P



Case 1: At very Low pressure

$$Z = 1$$

Case 2: At Low pressure or moderate

$$Z < 1$$

Case 3: At High pressure

$$Z > 1$$

Case 4: H_2 and He

(a) At very Low pressure

$$[Z = 1]$$

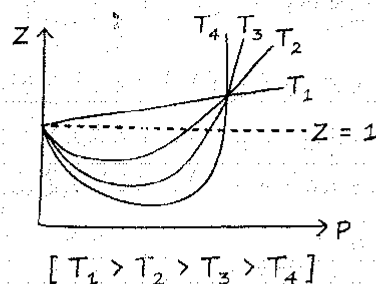
(moderate)

(b) At Low and High pressure

$$[Z > 1]$$

H_2 or He behave like a typical real gas at very low temp

(3) Z vs P for a gas at diff temp



3. Vander Waal's Eqⁿ

$$\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT$$

$$\text{P.C.T} = \text{Pressure correction term} = \frac{+n^2 a}{V^2}$$

$$\text{V.C.T} = \text{Volume correction term} = -nb$$

a = V.W const $\propto$ Intermolecular attraction forces

b = excluded volume = $4V_m$
or co. volume

$$= 4 \times \frac{4}{3} \pi r^3 \times N_A$$

Unit of ' a ' $\text{atm L}^2 \text{mol}^{-2}$

Unit of ' b ' L mol^{-1}

Case 1: At very Low pressure

$a = \text{neglected}$

$b = \text{neglected}$

$$\left(P + \frac{n^2 a}{V^2}\right)^0 (V - nb)^0 = nRT$$

$$PV = nRT$$

$$PV_m = RT$$

$$\frac{PV_m}{RT} = 1$$

$$\boxed{Z = 1}$$

Case 2: At Low or moderate pressure

$b = \text{neglected}$

$$\left(P + \frac{n^2 a}{V^2}\right) (V - nb)^0 = nRT$$

$$\left(P + \frac{a}{V_m^2}\right) (V_m) = RT$$

$$PV_m + \frac{a}{V_m} = RT$$

$$\frac{PV_m}{RT} + \frac{a}{RTV_m} = 1$$

$$\boxed{Z = 1 - \frac{a}{RTV_m}}$$

$$\boxed{Z < 1}$$

Case 3: At high pressure

$a = \text{neglected}$

$$\left(P + \frac{n^2 a}{V^2}\right) (V - nb) = nRT$$

$$P(V_m - b) = RT$$

$$PV_m - Pb = RT$$

$$\frac{PV_m}{RT} - \frac{Pb}{RT} = 1$$

$$\boxed{Z = 1 + \frac{Pb}{RT}}$$

$$\boxed{Z > 1}$$

Liquefaction of the Gases

Critical Parameters *

1. Critical temp (T_c) # The temp above which gas can not be liquified even at very high P.

$$\left[T_c = \frac{8a}{27Rb}\right]$$

2. Critical pressure (P_c)

$$\left[P_c = \frac{a}{27b^2}\right]$$

3. Critical volume (V_c)

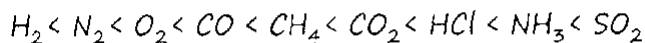
$$[V_c = 3b]$$

4. Compressibility factor at critical point

$$Z_c = \frac{P_c V_c}{RT_c} = \frac{3}{8} < 1$$

ease of liquefaction $\propto a \propto \frac{1}{b} \propto T_c$

Order of liquefaction



$\Rightarrow$ Boyle's Temp (T_B). The temp at which real gas mostly behaves like ideal gas

$$\left[T_B = \frac{a}{Rb}\right]$$

$\Rightarrow$ Inversion Temp (T_i)

$$\left[T_i = \frac{2a}{Rb}\right]$$

$$\boxed{T_i > T_B > T_c}$$

Practice

Q. The Z is Less than unity at STP therefore:

- (A) $V_m = 22.4 \text{ L}$ (B) $V_m < 22.4 \text{ L}$
 (C) $V_m > 22.4 \text{ L}$ (D) $V_m = 44.8 \text{ L}$

$$Z = \frac{PV_m}{RT} < 1 \quad ; \quad V_m < \frac{RT}{P}$$

$V_m < 22.4 \text{ L}$

Q. A real gas most closely approaches the ideal behaviour at?

- (A) 15 atm & 200 K
 (B) 1 atm and 273 K
 (C) 0.5 atm & 500 K
 (D) 15 atm and 500 K

Q. Gases	$a(\text{L}^2 \text{ atm mol}^{-2})$
O_2	1.36
N_2	1.39
NH_3	4.17
CH_4	2.25

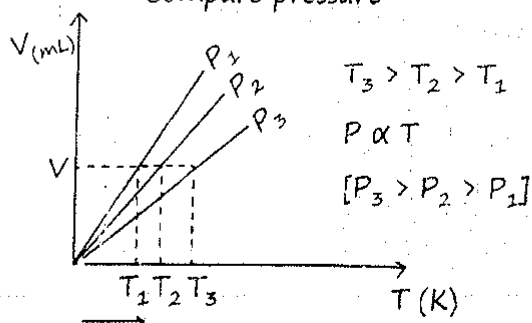
Write order of liquefaction?

liquefaction $\text{NH}_3 > \text{CH}_4 > \text{N}_2 > \text{O}_2$

Q. At High pressure, Vanderwaals equⁿ reduces to

- (A) $PV = RT$ (B) $PV = RT + \frac{a}{V}$
 (C) $PV = RT + Pb$ (D) $PV = RT - \frac{a}{V^2}$
 $\left[Z = 1 + \frac{Pb}{RT} \right]$

Q. Compare pressure



Q. The pressure of 1: 4 mixture (mole ratio) of H_2 and O_2 enclosed in a vessel is 1 atm. Find partial pressure of O_2 (N/m^2)?

$$P_{\text{O}_2} = x_{\text{O}_2} \times P_T$$

$$= \frac{n_{\text{O}_2}}{n_{\text{H}_2} + n_{\text{O}_2}} \times P_T = \left(\frac{1}{\frac{n_{\text{H}_2}}{n_{\text{O}_2}} + 1} \right) \times P_T$$

$$= \frac{1}{\left(\frac{1}{4} + 1 \right)} \times 1 = \frac{4}{5} \text{ atm}$$

$$P_{\text{O}_2} (\text{N/m}^2) = \frac{4}{5} \times 10^5$$

Q. A balloon is filled with H_2 at R.T. it will burst if pressure exceeds 0.2 bar.

If at 1 bar the gas occupies 2L volume, up to what volume can the balloon be expanded?

$$T = \text{const} \quad P_1 V_1 = P_2 V_2$$

$$1 \times 2 = 0.2 \times V_2$$

$$[V_2 = 10 \text{ L}]$$

Q. Calculate volume occupied by 8.8 gm CO_2 at 27°C and 1 atm pressure?

$$PV = nRT$$

$$1 \times V = \frac{8.8}{44} \times \frac{1}{12} \times 300 \quad [V_{(L)} = 5\text{L}]$$

$$V(L) = \frac{2}{10} \times \frac{1}{12} \times 300 = 5$$

Practice

Q. Consider three 1L flask A, B and C filled with gases NO , NO_2 and N_2O respectively each at 1atm and 273 K. In which flask do the molecules have the highest avg KE?

All have same avg KE [avg KE $\propto T$]

Q. V_{rms} of gas is $30\sqrt{R}$ m/s at 27°C . Find molar mass of gas?

$$30\sqrt{R} = \sqrt{\frac{3R \times 300}{M}}$$

$$900R = \frac{900R}{M} \quad ; \quad M = 1 \text{ Kg mol}^{-1}$$

Q. 6×10^{22} gas molecules each of mass 10^{-24} kg are taken in the vessel of 10 L.

Find pressure (Pa)? (Given $V_{rms} = 100 \text{ m/s}$)

$$P \times 10 \times 10^{-3}$$

$$= \frac{1}{3} \times 10^{-24} \times 6 \times 10^{22} \times 10000$$

$$[P = 2 \times 10^4 \text{ Pa}]$$

Q. Incorrect regarding KTG?

- (A) Gas particles are considered as point mass
- (B) The molecules are in random motion
- (C) When molecules collide, they lose energy
- (D) When gas is heated, the molecules move faster

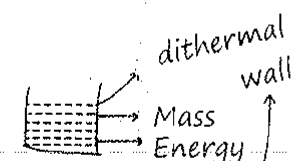
(1) System # The part of universe which is under observation is called system.

(2) Surroundings # The rest part of the universe (other than system)

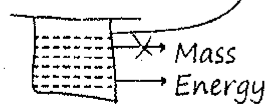
System + Surrounding = 'The Universe'

Types of System

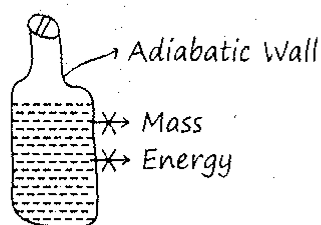
(1) Open



(2) Closed



(3) Isolated



Properties of System

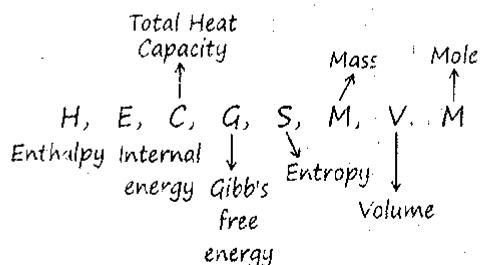
(1) Intensive # Mass or size independent

P , T , d (density), molar heat capacity, specific heat capacity, vapour pressure, refractive index

$$\left[\frac{E}{E} = 1 \right]$$

Intensive prop are non additive

(2) Extensive # Mass or size dependent



Extensive prop are additive in nature

State and Path Function

State Funcⁿ # Depends on initial (P_1, V_1, T_1) and final state. (P_2, V_2, T_2) e.g. $\Delta U, \Delta G, \Delta S$ or ΔE

Path Funcⁿ # Depends on path of the process. eg

q and w

↓ ↓
Heat Work

Internal Energy (ΔU)

(1) The sum of all energy like KE, PE, BE etc

(2) State Funcⁿ

$$\Delta U = U_2 - U_1$$

(3) Depends on Temp

Heat $\Rightarrow$ Energy transfer due to temp diff

Work $\Rightarrow$ The energy against external pressure

1st Law of Thermodynamics

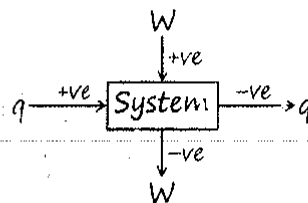
$$[\Delta U = q + w]$$

(1) Heat absorbed by the system = +ve

(2) Heat released by the system = -ve

(3) Work done by the system = -ve

(4) Work done on the system = +ve



$$[1 \text{ Cal} = 4.2 \approx 4 \text{ J}]$$

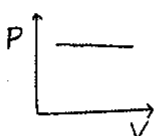
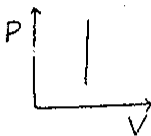
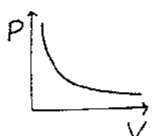
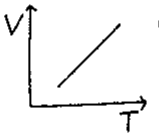
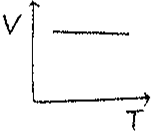
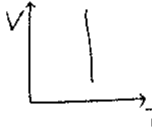
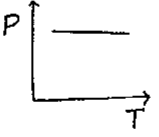
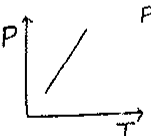
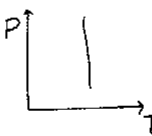
Q. Find ΔU , if a system absorbs 15kJ of heat and does 5kJ of work?

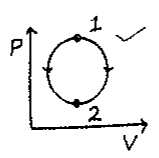
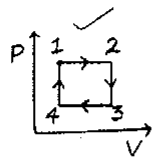
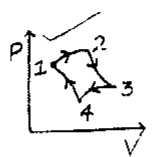
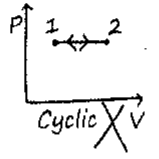
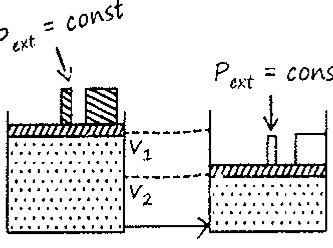
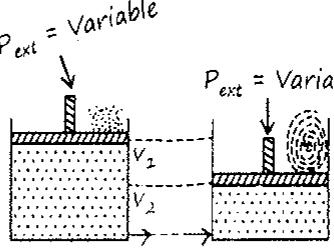
$$q = +15\text{kJ} \quad W = -5\text{kJ}$$

$$\Delta U = q + w$$

$$= 15 - 5 = 10\text{kJ}$$

Processes

Isobaric $\Delta P = 0$	Isochoric ($\Delta V = 0$)	($\Delta T = 0$) Isothermal
		 $PV = nRT$ $PV = \text{const}$ $P \propto \frac{1}{V}$
 $PV = nRT$ $[V \propto T]$		
	 $PV = nRT$ $P \propto T$	

Cyclic	Irreversible	Reversible
 In cyclic process, initial and final states are same with the formation of loop.    1, $\rightarrow$ 2 $\rightarrow$ Isobaric Exp (Heating) 2, $\rightarrow$ 3 $\rightarrow$ Isochoric Cooling 3, $\rightarrow$ 4 $\rightarrow$ Isobaric Cooling 4, $\rightarrow$ 1 $\rightarrow$ Isochoric Heating	 # at const. ext pressure # one step process # fast process	 # at variable ext pressure # multi step # slow

Heat Capacity

(1) C or C_T # The amount of Heat absorbed by a system to increase its temp by 1°C .

$$C = \frac{+q}{\Delta T}; \left(\frac{\text{J}}{\text{K}} \right)$$

$$(2) C_m; C_m = \frac{+q}{n\Delta T}; \left(\frac{\text{J}}{\text{mol-K}} \right)$$

$$(3) \text{ Specific heat capacity } C_s \text{ or } SC_s = \frac{+q}{m\Delta T} \left(\frac{J}{gm-K} \right)$$

$$(4) C_p ; C_p = \frac{+q_p}{\Delta T}$$

Heat change at const. pressure = ΔH

$$(5)^* C_{p,m} ; C_{p,m} = \frac{+q_p}{n\Delta T}$$

$$[\Delta H = nC_{p,m}\Delta T]$$

$$(6) C_{p,s} ; C_{p,s} = \frac{+q_p}{m\Delta T}$$

$$(7) C_v ; C_v = \frac{+q_v}{\Delta T}$$

$$(8)^* C_{v,m} ; C_{v,m} = \frac{+q_v}{n\Delta T}$$

isochoric process $w = 0$ [$\Delta U = q_v$]

$$[\Delta U = nC_{v,m}\Delta T]$$

$$(9) C_{v,s} : C_{v,s} = \frac{+q_v}{m\Delta T}$$

Imp. points related to heat capacity

(1) Mayer's relation [$C_{p,m} - C_{v,m} = R$]

(2) Heat Capacity C or $C_T \rightarrow$ path funcⁿ

C_p & $C_v \rightarrow$ are not path funcⁿ

(3) Heat capacity may vary from $-\infty$ to ∞ depending upon the process.

$$(4) C_T = nC_m = mC_s$$

$$[C_m = MW \times C_s]$$

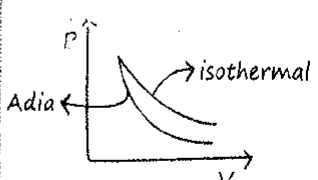
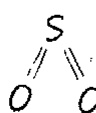
$$(5) \text{ Isothermal } C = \infty$$

$$\text{Isobaric } C = C_p$$

$$\text{Isochoric } C = C_v$$

$$\text{Adiabatic } (q = 0) C = 0$$

Adiabatic Process

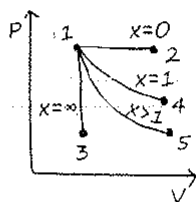
	Gas	f	$C_v = \frac{fR}{2}$ C_v	C_p	$\left(\chi = 1 + \frac{2}{f} \right)$ χ
$[q = 0]$ $\Delta U = q + w$ $\Delta U = 0 + W_{ad}$ $[\Delta U = W_{ad}]$ for Rev. adiabatic process $\left\{ \begin{array}{l} PV^\gamma = \text{Const} \\ T_v^{\gamma-1} = \text{Const} \\ P^{1-\gamma} T^\gamma = \text{Const} \end{array} \right\} \gamma = \frac{C_p}{C_v}$ Poisson's ratio Isothermal vs Adiabatic 	Mono atomic He, Ne, Ar etc	3T	$\frac{3R}{2}$	$\frac{5R}{2}$	5/3
	H_2, N_2, O_2, F_2 etc Diatomic or Triatomic linear $CO_2: O=C=O$	$3(T) + 2(R)$ = 5	$\frac{5R}{2}$	$\frac{7R}{2}$	7/5
	Triatomic non linear SO_2 	$3(T) + 2(R)$ = 6	$\frac{6R}{2} = 3R$	4R	4/3

$w, q, \Delta U$ and ΔH Calculation

Process	Work done by the system	q	$\Delta U = q + w$ i.e, $\Delta U (q_r)$	$\Delta H (q_p)$
(1) Reversible Isothermal $(P_1 V_1 = P_2 V_2)$ $R = \frac{1}{12}$; $W = L$ $= \frac{25}{3}$; $W = J$ $= 2$; $W = Cal$	$W = -nRT \ln \left(\frac{V_2}{V_1} \right)$ $W = -2.3 nRT \log_{10} \left(\frac{V_2}{V_1} \right)$ $W = -2.3 nRT \log_{10} \left(\frac{P_1}{P_2} \right)$	$q = -W$	0	0
(2) Irreversible isothermal	$W = -P_{ext} (V_2 - V_1)$ $= -P_{ext} \left(\frac{nRT}{P_2} - \frac{nRT}{P_1} \right)$ $P_{ext} \approx P_2$	$q = -W$	0	0
(3) Isobaric	$W = -P_{ext} (V_2 - V_1)$ $= -P_{ext} \left(\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1} \right)$ $= -nR\Delta T$	$q_p = \Delta H = nC_p \Delta T$	$q_v = \Delta U = nC_v \Delta T$	$q_p = \Delta H = nC_p \Delta T$
(4) Isochoric	$w = 0$	$q_v = \Delta U = nC_v \Delta T$	$\Delta U = nC_v \Delta T$	$\Delta H = nC_p \Delta T$
(5) Rev. Adiabatic $(T_2 < T_1)$	$W_{rev} = \frac{nR(T_2 - T_1)}{\gamma - 1}$ $= nC_v \Delta T$	$q = 0$ $PV^\gamma = \text{Const.}$ $TV^{\gamma-1} = \text{Const.}$ $P^{1-\gamma} T^\gamma = \text{Const.}$	$\Delta U = nC_v \Delta T$	$\Delta H = nC_p \Delta T$
(6) Irr. Adiabatic	$W_{irr} = \frac{nR(T_2 - T_1)}{\gamma - 1}$ $= nC_v \Delta T$	$q = 0$ $nC_v(T_2 - T_1) = -P_{ext}(V_2 - V_1)$ $= -P_{ext} \left(\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1} \right)$	$\Delta U = nC_v \Delta T$	$\Delta H = nC_p \Delta T$
(7) Cyclic Process All State Func ⁿ = 0	Area enclosed in PV diagram If C.W $\Rightarrow W = -ve$ If ACW $\Rightarrow W = +ve$	$q = -W$	0	0

(8) Polytropic Process

$$[PV^X = \text{Const}]$$



$$W = \frac{P_2 V_2 - P_1 V_1}{X - 1}$$

$$W = \frac{nR\Delta T}{X - 1}$$

$$\Delta U = nC_V \Delta T$$

$$q = nC \Delta T$$

$$\left[C = C_V + \frac{R}{1 - X} \right]$$

$$\Delta H = nC_P \Delta T$$

Relation b/w ΔH and ΔU

$$[\Delta H = \Delta U + \Delta n_g RT]$$

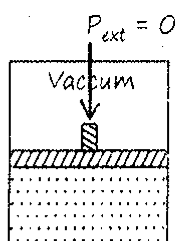
$$[q_p = q_v + \Delta n_g RT]$$

Case 1: If $\Delta n_g = 0$; $\Delta H = \Delta U$

Case 2: If $\Delta n_g < 0$; $\Delta H < \Delta U$

Case 3: If $\Delta n_g > 0$; $\Delta H > \Delta U$

Free Expansion



$$\Delta U = 0; q = 0, W = 0$$

Q. The workdone during the expansion of gas from a volume of 14dm^3 to 16dm^3 against a constant external pressure of 2atm at const temp is?

$$W_{\text{irr}} = -P_{\text{ext}} (V_2 - V_1)$$

$$= -2 \times (16 - 14) = -2 \times 2$$

$$= -4 \text{ L-atm}$$

$$1 \text{ L atm} = 100 \text{ J} = 24 \text{ cal} = 100 \times 10^7 \text{ erg}$$

$$[1 \text{ L-atm} > 1 \text{ cal} > 1 \text{ J} > 1 \text{ erg}]$$

Q. If ΔH for Reaction

$\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$ is -92kJ/mol at 300K , find ΔU (in kJ/mol) at 300K ?

$$\Delta H = \Delta U + \Delta n_g RT$$

$$-92000 = \Delta U + (-2) \times \frac{25}{8} \times 300$$

$$\Delta U = -92000 + 5000 = -87000$$

$$\Delta U = -87\text{kJ/mol}$$

Q. 3 moles of an ideal gas expanded spontaneously into Vacuum. The work done will be?

$$[W = 0]$$

Q. Which is intensive prop?

$$(A) \Delta H$$

$$(B) \Delta U$$

$$(C) \text{Volume}$$

$$(D) \text{Specific heat}$$

Q. Which is not state funcⁿ?

$$(A) q_P$$

$$(B) q_V$$

$$(C) W_{\text{ad}}$$

$$(D) W_{\text{isother}}$$

$$\Delta U = q_P$$

$$[\Delta U = W_{\text{ad}}]$$

Q. 1 mole of an ideal gas at 300K is expanded isothermally from initial volume of 1L to 10L find w , q and ΔU (in cal)?

$$W_{\text{rev}} = -2.3 nRT \log_{10} \left(\frac{V_2}{V_1} \right)$$

$$= -2.3 \times 1 \times 2 \times 300 \log_{10} \left(\frac{10}{1} \right)$$

$$= -2.3 \times 600 \text{ cal}$$

$$= -23 \times 60 \text{ cal}$$

$$\underline{-1380 \text{ cal}}$$

$$W_{\text{rev}} = -1.3 \text{ Kcal}$$

$$\Delta U = 0$$

$$q = -W = -(-1.3 \text{ Kcal})$$

$$= +1.3 \text{ Kcal}$$

Q. 5 mol of O_2 gas is heated at const volume from $10^\circ C$ to $20^\circ C$. Find ΔU

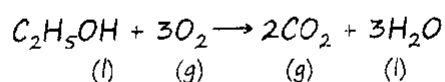
Given: $C_p = \frac{\text{Cal}}{\text{mol-K}}$; $R = 2 \frac{\text{Cal}}{\text{mol-K}}$

$$\Delta U = nC_v\Delta T \quad C_p - C_v = R$$

$$= 5 \times 5 \times 10 \quad 7 - C_v = 2$$

$$= 250 \text{ Cal} \quad C_v = 5$$

Q. Find Δn_g if we consider the combustion of 1 mol liquid ethanol if reactants and products are at 298K?



$$\Delta n_g = 2 - 3 = -1$$

Q. If a gas absorbs 100J of Heat and expands by 500cm^3 against a const pressure of $2 \times 10^5 \text{ N/m}^2$ find ΔV ?

$$q = +100 \quad W = -P\Delta V$$

$$= -2 \times 10^5 \times 500 \times 10^{-6} \frac{\text{N}}{\text{m}^2} \times \text{m}^3$$

$$= -100 \text{ J}$$

$$\Delta U = q + W = 100 - 100 = 0$$

Q. A gas expands adiabatically at const pressure such that $T \propto V^{-1/2}$ find γ ?

$$TV^{\gamma-1} = \text{const} \quad T = KV^{-1/2}$$

$$\gamma - 1 = \frac{1}{2} \quad TV^{1/2} = K$$

$$\gamma = \frac{1}{2} + 1 = \frac{3}{2}$$

Q. 1 mol of an ideal gas ($C_{v,m} = \frac{5R}{2}$) at 300K and 5 atm is expanded adiabatically to a final pressure of 2 atm against a constant pressure of 2 atm find final temp of gas?

$$nC_v(T_2 - T_1) = -P_{\text{ext}} \left(\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1} \right)$$

$$1 \times \frac{5R}{2} (T_2 - 300) = -2 \left(\frac{1 \times R \times T_2}{2} - \frac{1 \times R \times 300}{5} \right)$$

$$\frac{5RT_2}{2} - 750R = -RT_2 + 120R$$

$$\frac{7RT_2}{2} = 870R$$

$$T_2 = \frac{870 \times 2}{7} = 248.5 \text{ K}$$

Q. Calculate the final temp of a monoatomic ideal gas that is expanded Rev. and adia. from 2L to 16L at (300 K)?

$$TV^{\gamma-1} = \text{const} \quad \gamma - 1 = \frac{5}{3} - 1 = \frac{2}{3}$$

$$T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$$

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^{2/3}$$

$$\frac{T_2}{300} = \left(\frac{2}{16} \right)^{2/3} = \left(\frac{1}{2} \right)^{4/3} = \frac{1}{4}$$

$$T_2 = \frac{300}{4} = 75 \text{ K}$$

Entropy

(1) The degree of Randomness or disorderness is called Entropy.

(2) Entropy is a state function and Extensive prop.

$$(3) \Delta S = S_2 - S_1 = S_f - S_i$$

(4) Unit J/K or Cal/K

(5) Mathematically entropy can be defined as the heat absorbed at a particular temp in rev process

$$\left[\Delta S = \frac{q_{\text{rev}}}{T} \right]$$

Factors Affecting Entropy

(1) Physical state : ($S < L < G$) (order of S)

(2) For chemical Rxn: $\Delta n_g > 0$; $\Delta S > 0$
 $\Delta n_g < 0$; $\Delta S < 0$

(3) Temp : $\Delta S \propto T$

(4) Pressure : $\Delta S \propto \frac{1}{P}$

(5) Mixing : $S + S$
 $L + L$ [$\Delta S > 0$]
 $G + G$

Calculation of Entropy in diff process

General Formulae

$$(1) \Delta S = nC_V \ln \left(\frac{T_2}{T_1} \right) + nR \ln \left(\frac{V_2}{V_1} \right)$$

$$(2) \Delta S = nC_P \ln \left(\frac{T_2}{T_1} \right) + nR \ln \left(\frac{P_1}{P_2} \right)$$

Case 1: In Isobaric process

$$\Delta S = nC_P \ln \left(\frac{T_2}{T_1} \right)$$

Case 2: In Isochoric process

$$\Delta S = nC_V \ln \left(\frac{T_2}{T_1} \right)$$

Case 3: In Isothermal process

$$\Delta S = nR \ln \left(\frac{V_2}{V_1} \right) = nR \ln \left(\frac{P_1}{P_2} \right)$$

Case 4: In reversible adiabatic process ($q_{rev} = 0$)

$\Delta S = 0$; Rev adiabatic processes are also known as Isoentropic processes

Case 5: In irreversible adiabatic process, use

General formulae

Entropy Change During Phase Transition

(1) Entropy of fusion $S \rightleftharpoons L$

$$\Delta S_{fus} = \frac{\Delta H_{fus}}{T_m}$$

(2) Entropy of vap. $L \rightleftharpoons G$

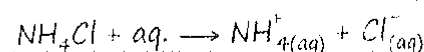
$$\Delta S_{vap} = \frac{\Delta H_{vap}}{T_b}$$

(3) Entropy of sublimation $S \rightleftharpoons G$

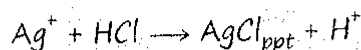
$$\Delta S_{sub} = \frac{\Delta H_{sub}}{T_s}$$

Some famous points related to Entropy

1. Entropy Graphite > Diamond
2. On rusting of Fe, Entropy increases
3. On dissolution, Entropy increases



4. Ag^+ on reaction with HCl, Entropy decreases



5. On boiling of egg, Entropy increases due to denaturation of protein.

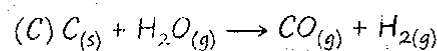
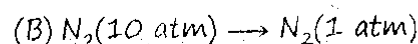
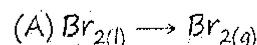
6. On stretching of rubber band, Entropy decreases.

Practice questions on Entropy

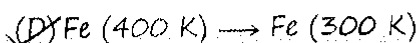
Q. The enthalpy change for transition of liquid to steam is 40 kJ/mol at 400K find ΔS for the process?

$$\Delta S_{vap} = \frac{\Delta H_{vap}}{T} = \frac{40 \times 10^3}{400} \text{ J/K} = 100 \text{ J/K}$$

Q. $\Delta S < 0$ for



$$\Delta n_g = 2 - 1 = 1 > 0$$



Q. 3 mol of an ideal gas expand rev from a volume of 8 dm³ to 80 dm³ at 300 K. Find ΔS (J/K)?

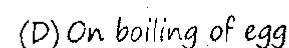
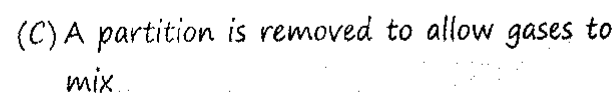
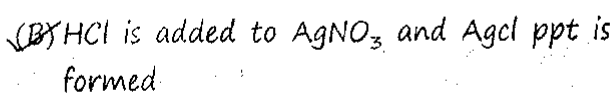
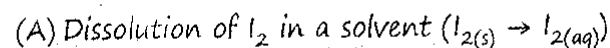
$$\Delta S = nR \ln \left(\frac{V_2}{V_1} \right)$$

$$= 3 \times \frac{25}{3} \times 2.3 \log_{10} \left(\frac{80}{8} \right)$$

$$= 25 \times 2.3$$

$$= 57.5 \text{ J/K}$$

Q. $\Delta S < 0$ for



2nd Law of Thermodynamics (SLOT)

- (1) The entropy of universe is always increasing.
- (2) All Natural process are irreversible and spontaneous process.

(3) $[\Delta S_{\text{Univ}} \geq 0]$

$$\begin{array}{cc} \Delta S > 0 & \Delta S = 0 \\ \text{Univ} & \text{Univ} \\ \text{Irr} & \text{Rev} \end{array}$$

(Natural)

ΔS_{Univ} in Irr.

System T_1 , q amount of heat is released by the system

Surr. T_2 , q amount of heat is absorbed by the surr.

$$\Delta S_{\text{Univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$$

$$= \left(\frac{-q}{T_1} \right) + \left(\frac{+q}{T_2} \right) > 0$$

ΔS_{Univ} in Rev ($T_1 = T_2$)

$$\Delta S_{\text{Univ}} = \frac{-q}{T} + \frac{q}{T} = 0$$

Gibb's Free Energy

- (1) It is the part of total energy of the system which can be converted to do useful work (non expansion work)

Energy = Useful work + Randomness energy

$$H = G + TS$$

$$(G = H - TS)$$

- (2) It is an extensive prop and state funcⁿ

$$\Delta G = G_2 - G_1 = G_f - G_i$$

- (3) Gibb's helmholtz eqn

$$[\Delta G = \Delta H - T\Delta S]$$

it is used to predict the spon.

- (4) The decrease in Gibb's free energy is equal to max useful work or non experiment

$$[W_{\text{useful}} = -\Delta G]$$

- (5) Relation b/w ΔG_{sys} and ΔS_{univ}

$$[\Delta G_{\text{sys}} = -T\Delta S_{\text{univ}}]$$

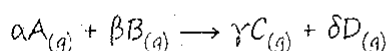
Case 1: $\Delta S_{\text{univ}} > 0$; $\Delta G_{\text{sys}} < 0$ spon

Case 2: $\Delta S_{\text{univ}} < 0$; $\Delta G_{\text{sys}} > 0$ non spon

Case 3: $\Delta S_{\text{univ}} = 0$; $\Delta G_{\text{sys}} = 0$ Eq^m

Criteria for spon		$[\Delta G_{\text{sys}} = \Delta H_{\text{sys}} - T\Delta S_{\text{sys}}]$
ΔH	ΔS	Criteria for spontaneity
+ve	+ve	$ T\Delta S > \Delta H $
+ve	-ve	Always non spontaneous
-ve	+ve	Always spontaneous
-ve	-ve	$ \Delta H > T\Delta S $

Relation b/w ΔG° and K_{eq} (eqm const)



$$\Delta G = \Delta G^\circ + RT \ln Q$$

ΔG = Gibb's free energy at Temp T_K

ΔG° = Standard Gibb's free energy

R = gas const,

T = Temp (K)

Q = Rxn quotient

At eq^m $\Delta G = 0$

$$Q = K_{\text{eq}}$$

$$0 = \Delta G^\circ + RT \ln K_{\text{eq}}$$

$$\boxed{\Delta G^\circ = -RT \ln K_{\text{eq}}}$$

Relation b/w ΔG° and E_{cell}° (standard Emf of Cell)

$$[\Delta G^\circ = -nF E_{\text{cell}}^\circ]$$

n = no. of e^- s involve in the balanced Rxn

F = Faradays constant = 96500 C

E_{cell}° = standard emf of cell

$$= (E_R^\circ)_C - (E_R^\circ)_A$$

$$-RT \ln K_{\text{eq}} = -nF E_{\text{cell}}^\circ$$

$$\left[E_{\text{cell}}^\circ = \frac{RT}{nF} \ln K_{\text{eq}} \right]$$

Q. For a rxn $\Delta H = 40.63 \text{ kJ/mol}$ and $\Delta S = 100 \text{ J/K}$ find ΔG at 27°C ?

$$\Delta G = \Delta H - T\Delta S$$

$$= 40.63 \times 1000 - 300 \times 100$$

$$= 40630 - 30000$$

$$\Delta G = 10630$$

$$\Delta G = 10.63 \text{ kJ/mol}$$

Q. For a rxn $N_2 + 3H_2 \rightarrow 2NH_3$ $\Delta H = -100 \text{ kJ/mol}$ and $\Delta S = -200 \text{ J/K}$. Calculate temp. at which rxn will proceed in forward dirⁿ?

at eq^m $\Delta G = 0$

$$\Delta H = T\Delta S$$

$$T = \frac{-100 \times 10^3}{-200} = 500 \text{ K}$$

[T < 500]

Q. The ΔG° for a rxn at 27°C is X Kcal/mol. If eq^m const is 100 then X is

$$\Delta G^\circ = -2.3 RT \log_{10} K$$

$$X \times 1000 = -2.3 \times 2 \times 300 \log 10^2$$

$$X = \frac{-2.3 \times 1200}{100} = \frac{-27.6}{10} = -2.76$$

Q. Favoured conditon for spontaneous reaction when ΔH and ΔS both are +ve?

$$\Delta G = \Delta H - T\Delta S$$

at $|\Delta H| < |T\Delta S|$ ie at $T > \Delta H/\Delta S$

$\Delta G < 0$ spontaneous.

Gibb's Free Energy

Third Law of thermodynamics

At 0K, the entropy of pure perfect crystalline solid is taken as zero

Exceptions (1) NO, N_2O

(2) CO, CO_2

(3) Mixture of isotopes

(4) Ice

It is the heat involved in a chemical Rxn.

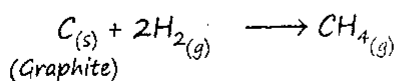
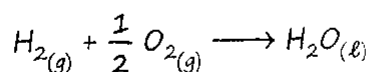
Endothermic $R + \text{Heat} \longrightarrow P$ $\Delta_r H = +ve$

Exothermic $R \longrightarrow P + \text{Heat}$ $\Delta_r H = -ve$

Different types of Heat of Reaction

1. Heat of formation ($\Delta_f H^\circ$)

The heat absorbed or released when 1 mol of a comp. is formed by most stable form of elements in standard state.



$\Delta_f H^\circ = 0$ for $H_{2(g)}$, $O_{2(g)}$, $N_{2(g)}$, $F_{2(g)}$, $Cl_{2(g)}$

$Br_{2(l)}$, $I_{2(s)}$, $C_{(graphite)}$

$P_{4(white)}$, $S_{8(Rhombic)}$

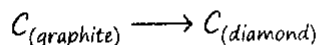
2. Heat of combustion ($\Delta_{comb}^\circ H$)

The heat released when 1 mol of a substance is combined with excess of O_2 to form products



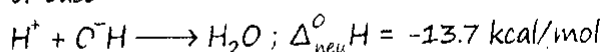
3. Heat of transition ($\Delta_{trans}^\circ H$)

The heat absorbed or released during conversion of one allotropic form to another form



4. Heat of neutralization ($\Delta_{neu}^\circ H$)

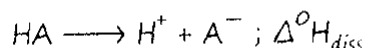
It is the heat released when 1 gram equivalent of an acid is neutralized by 1 gram equivalent of base



order of magnitude of $(SA + SB) > (WA + SB) > (WA + WB)$

5. Heat of ionisation (or dissociation) ($\Delta_{diss}^\circ H$ or $\Delta_{ion}^\circ H$)

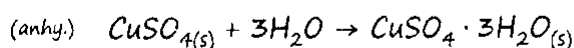
The heat required for the complete dissociation of 1 mol of acid (or Base)



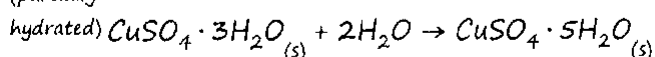
$\Delta_{diss}^\circ H \uparrow$, weaker is the acid

6. Enthalpy of Hydration ($\Delta_{hyd}^\circ H$)

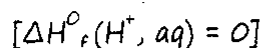
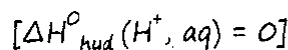
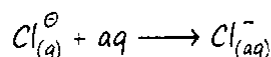
Case 1: For salt



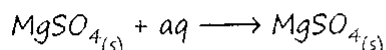
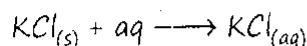
(partially



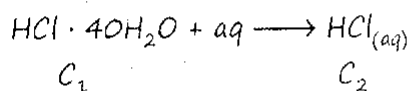
Case 2: For ion



7. Enthalpy of solution ($\Delta_{sol}^\circ H$)

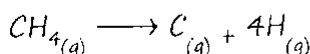
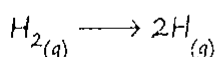


8. Enthalpy of dilution ($\Delta_{dil}^\circ H$)



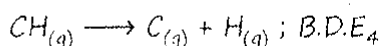
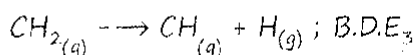
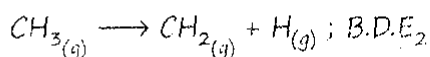
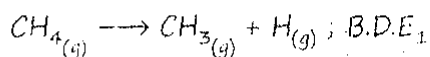
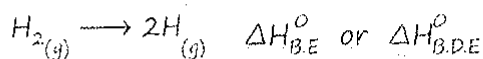
9. Heat of atomisation ($\Delta_a H^\circ$)

It is the enthalpy change when 1 mol of given molecule is dissociated into free atoms in gas phase



10. Bond energy (B.E or E or $\Delta H_{B.E}^\circ$)

It is the avg of enthalpies when 1 mol of a particular bond between two atoms is broken in gaseous phase



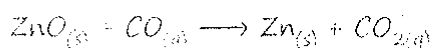
$$B.E = \frac{B.D.E_1 + B.D.E_2 + B.D.E_3 + B.D.E_4}{4}$$

Calculation of $\Delta_r H^\circ$ when $\Delta_f H^\circ$ is given



$$\Delta_r H^\circ = [\gamma \times \Delta_f H^\circ(C) + \delta \times \Delta_f H^\circ(D)] - [\alpha \times \Delta_f H^\circ(A) + \beta \times \Delta_f H^\circ(B)]$$

Q. Find $\Delta_r H^\circ$ for the reaction



Given: Comp. $\Delta_f H^\circ$ (kJ/mol)

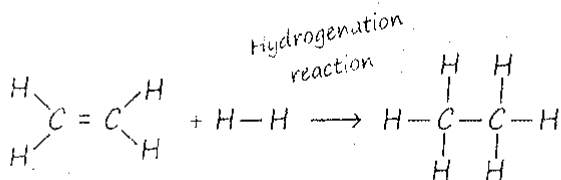
$$ZnO_{(s)} \quad -350$$

$$CO_{2(g)} \quad -390$$

$$CO_{(g)} \quad -110$$

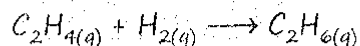
$$\begin{aligned} \Delta_r H^\circ &= \Delta_f H^\circ(CO_2) - (\Delta_f H^\circ(ZnO) + \Delta_f H^\circ(CO)) \\ &= -390 - (-350 - 110) \\ &= +70 \text{ kJ/mol (Here } \Delta_f H^\circ(Zn)(s) = 0) \end{aligned}$$

Calculation of $\Delta_r H^\circ$ when B.E is given



$$\begin{aligned} \Delta_r H^\circ &= (B.E)_R - (B.E)_P \\ &= [B.E_{C=C} + 4B.E_{C-H} + B.E_{H-H}] \\ &\quad - [6B.E_{C-H} + B.E_{C-C}] \end{aligned}$$

Q. Find $\Delta_r H^\circ$ for the Rxn



Given: Bonds BE (kJ/mol)

$$C-H \quad 414$$

$$C-C \quad 347$$

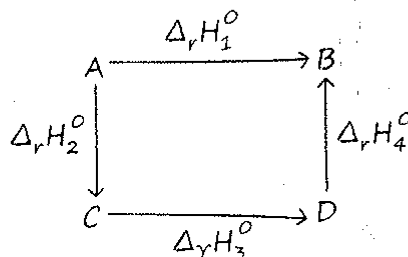
$$C=C \quad 615$$

$$H-H \quad 435$$

$$\begin{aligned} \Delta_r H^\circ &= [615 + 4 \times 414 + 435] \\ &\quad - [6 \times 414 + 347] \\ &= -125 \text{ kJ/mol} \end{aligned}$$

Hess's Law

The total change of heat content of a reaction is same whether the reaction is carried out by direct path or indirect path



$$[\Delta_r H_1^° = \Delta_r H_2^° + \Delta_r H_3^° + \Delta_r H_4^°]$$

- $A \longrightarrow B \quad \Delta_r H_1^°$
 $B \longrightarrow A \quad \Delta_r H_2^° = -\Delta_r H_1^°$
- $A \longrightarrow B \quad \Delta_r H_1^°$
 $aA \longrightarrow aB \quad \Delta_r H_2^° = a \times \Delta_r H_1^°$
- $A \longrightarrow B \quad \Delta_r H_1^°$
 $C \longrightarrow D \quad \Delta_r H_2^°$
 $A + C \longrightarrow B + D \quad \Delta_r H_3^° = \Delta_r H_1^° + \Delta_r H_2^°$

Q. Reaction $\Delta_r H$ (kJ/mol)

$$\textcircled{1} \quad \frac{1}{2} A \longrightarrow B \quad +150$$

$$\textcircled{2} \quad 3B \longrightarrow 3C + D \quad -125$$

$$\textcircled{3} \quad E + A \longrightarrow 2D \quad +350$$

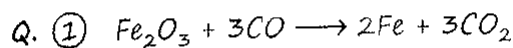
Find ΔH for ④ $B + D \longrightarrow E + 3C$

$$2 \times \textcircled{1} - \textcircled{3} + \textcircled{2} = \textcircled{4}$$

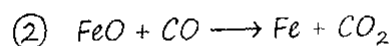
$$2 \times 150 + (-350) + (-125) = \Delta H$$

$$300 - 350 - 125 = \Delta H$$

$$\Delta H = -175 \text{ kJ/mol}$$



$$\Delta H = -27 \text{ kJ/mol}$$



$$\Delta H = -16.5 \text{ kJ/mol}$$

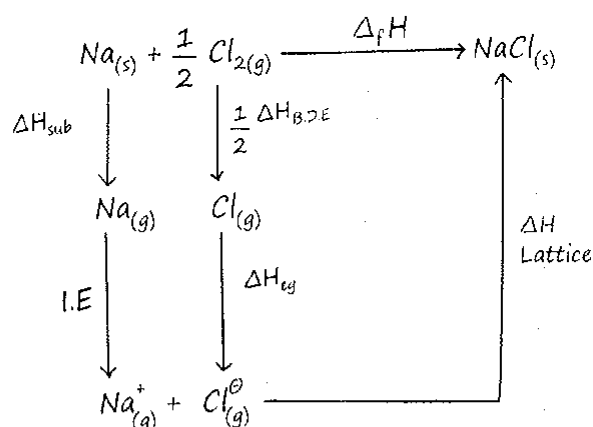
Find ΔH for ③ $\text{Fe}_2\text{O}_3 + \text{CO} \longrightarrow 2\text{FeO} + \text{CO}_2$

$$\textcircled{1} - 2 \times \textcircled{2} = \textcircled{3}$$

$$-27 + (+16.5 \times 2) = \Delta H$$

$$[\Delta H = +6 \text{ kJ/mol}]$$

Born Habers Cycle



$$\Delta_f H = \Delta H_{\text{sub}} + \text{I.E.} + \frac{1}{2} \Delta H_{\text{BDE}} + \Delta H_{\text{eg}} + \Delta H_{\text{lattice}}$$

Kirchoff's Equation

At const. pressure

$$(\Delta H)_{T_2} - (\Delta H)_{T_1} = \Delta C_p (T_2 - T_1)$$

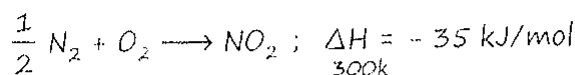
$$\Delta C_p = (C_p)_P - (C_p)_R$$

At const. volume

$$(\Delta U)_{T_2} - (\Delta U)_{T_1} = \Delta C_v (T_2 - T_1)$$

$$\Delta C_v = (C_v)_P - (C_v)_R$$

Q. Find ΔH at 400 K for



Given: Gas C_p ($\text{JK}^{-1} \text{mol}^{-1}$)

$$\text{NO}_2 \quad 35$$

$$\text{N}_2 \quad 30$$

$$\text{O}_2 \quad 30$$

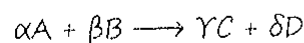
$$\Delta C_p = 35 - \frac{1}{2} \times 30 - 30$$

$$= 5 - 15 = -10 \text{ JK}^{-1} \text{mol}^{-1}$$

$$(\Delta H)_{400} - (-35) = \frac{-10 \times 100}{1000} = -1$$

$$[(\Delta H)_{400} = -36 \text{ kJ/mol}]$$

Entropy of chemical Reaction

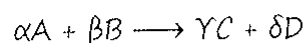


$$\Delta_r S^\circ = (S_m^\circ)_P - (S_m^\circ)_R$$

$$= [\gamma \times S_{m(C)}^\circ + \delta \times S_{m(D)}^\circ]$$

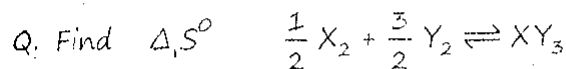
$$- [\alpha \times S_{m(A)}^\circ + \beta \times S_{m(B)}^\circ]$$

Gibb's free energy of a Reaction



$$\Delta_r G^\circ = [\gamma \times \Delta_f G_{(C)}^\circ + \delta \times \Delta_f G_{(D)}^\circ]$$

$$- [\alpha \times \Delta_f G_{(A)}^\circ + \beta \times \Delta_f G_{(B)}^\circ]$$



$$\text{Comp.} \quad S_m^\circ (\text{JK}^{-1} \text{mol}^{-1})$$

$$\text{X}_2 \quad 60$$

$$\text{Y}_2 \quad 40$$

$$\text{XY}_3 \quad 50$$

$$\Delta_r S^\circ = S_{m(\text{XY}_3)}^\circ - \frac{1}{2} \times S_{m(\text{X}_2)}^\circ - \frac{3}{2} \times S_{m(\text{Y}_2)}^\circ$$

$$= 50 - \frac{1}{2} \times 60 - \frac{3}{2} \times 40$$

$$= 20 - 60$$

$$= -40 \text{ JK}^{-1} \text{mol}^{-1}$$

Practice

Q. Which is correct for free expansion?

- (A) $q = 0, \Delta T \neq 0, w = 0$
 (B) $q \neq 0, \Delta T = 0, w = 0$
☒ (C) $q = 0, \Delta T = 0, w = 0$
 (D) $q = 0, \Delta T < 0, w \neq 0$

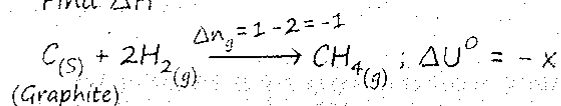
Q. Which is intensive property?

- (A) ΔH (B) ΔS
☒ (C) Specific heat (D) Volume

Q. For $A_{2(g)} \rightarrow 2A_{(g)}$ correct option is

- | | ΔH | ΔS |
|---|------------|------------|
| (A) | + | - |
| ☒ (B) | + | + |
| (C) | - | - |
| (D) | - | + |

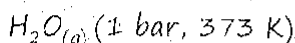
Q. For formation of $CH_{4(g)}$; ΔU^0 is $-x \frac{\text{kJ}}{\text{mol}}$
 Find ΔH^0



$$\Delta H^0 = \Delta U^0 + \Delta n_g RT = -x \times 10^3 + (-1)RT$$

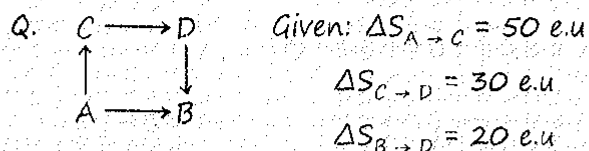
$$= -x \times 10^3 - RT$$

Q. For $H_2O_{(l)} (1 \text{ bar}, 373K) \rightarrow$



Correct option is

- | | ΔG | ΔS | $H_2O(l) \rightleftharpoons H_2O_{(g)}$ |
|---|------------|------------|---|
| ☒ (A) | 0 | + | $\Delta G = 0$ |
| (B) | 0 | - | $\Delta S = +ve$ |
| (C) | + | 0 | |
| (D) | - | + | |



Find $\Delta S_{A \rightarrow B}$

$$\Delta S_{A \rightarrow B} = \Delta S_{A \rightarrow C} + \Delta S_{C \rightarrow D} + \Delta S_{D \rightarrow B}$$

$$= 50 + 30 - 20 = 60 \text{ e.u.}$$

Q. Find $\Delta_f H^0$ (HCl)

Given: (kJ/mol)

Bond	B.E
H-H	434
Cl-Cl	242
H-Cl	430

$$\frac{1}{2} H_{2(g)} + \frac{1}{2} Cl_{2(g)} \longrightarrow H-Cl_{(g)}$$

$$\Delta_f H^0 = \Delta_f H^0 = \frac{1}{2} B.E_{H-H} + \frac{1}{2} B.E_{Cl-Cl} - B.E_{H-Cl}$$

$$= \frac{1}{2} \times 434 + \frac{1}{2} \times 242 - 430$$

$$= -92 \text{ kJ/mol}$$

Q. $X_2O_{4(l)} \rightarrow 2 XO_{2(g)}; \Delta U = 2 \text{ kcal/mol}$

$\Delta S = 20 \text{ Cal/K/mol}$ at 300K. find ΔG

$$\Delta G = \Delta H - T\Delta S = \Delta U + \Delta n_g RT - T\Delta S$$

$$= 2000 + 2 \times 2 \times 300 - 300 \times 20$$

$$= 2000 + 1200 - 6000$$

$$= 3200 - 6000$$

$$= -2800 = -2.8 \text{ kcal/mol}^{-1}$$

Q. 200 mL of 0.1 M H_2SO_4 is mixed with 150 mL of 0.2 M KOH. Find the value of heat evolved (in kJ)

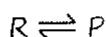
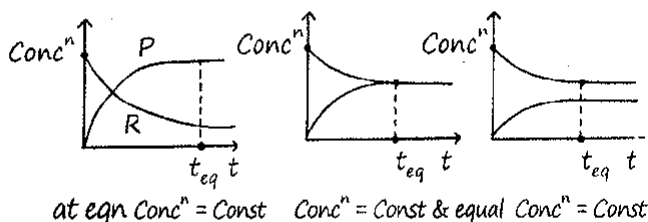
$$\Delta_{neut} H = -57.1 \text{ kJ mol}^{-1}$$

$$\text{Eq. of } H_2SO_4 = 0.1 \times 2 \times 200 \times 10^{-3} = 0.04$$

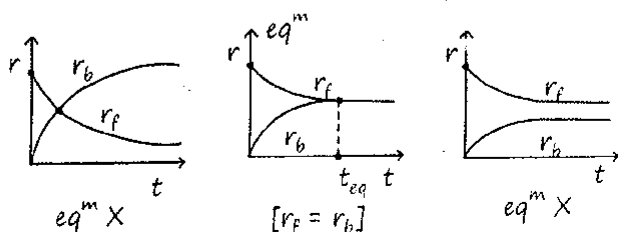
$$\text{Eq. of KOH} = 0.2 \times 1 \times 150 \times 10^{-3} = \boxed{0.03}$$

$$1 \text{ eq} \Rightarrow 57.1 \text{ kJ}$$

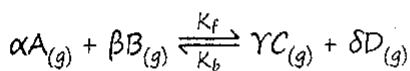
$$0.03 \text{ eq} \Rightarrow 0.03 \times 57.1 \text{ kJ} = 1.7 \text{ kJ}$$

1. Concⁿ vs time

2. Rate vs time

at eq^m [r_f = r_b]r_f = rate of forward Rxnr_b = rate of Backward RxnLaw of Mass Action

The rate of rxn is directly proportional to the product of active masses of reactants to raised the power of respective stoichiometric co-efficient.

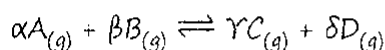
For solids active masses = 1 $\left[C = \frac{d}{M} \right]$ 

$$r_f \propto [A]^\alpha [B]^\beta ; r_b \propto [C]^\gamma [D]^\delta$$

$$r_f = K_f [A]^\alpha [B]^\beta ; r_b = K_b [C]^\gamma [D]^\delta$$

$$\text{at equilibrium } r_f = r_b ; K_P = \frac{P_C^\gamma P_D^\delta}{P_A^\alpha P_B^\beta}$$

$$\text{eq}^m \text{ const } K_C = \frac{K_f}{K_b} = \frac{[C]^\gamma [D]^\delta}{[A]^\alpha [B]^\beta}$$

eq^m const

$$K_{eqb} = \frac{K_f}{K_b}$$

$$K_C = \frac{[C]^\gamma [D]^\delta}{[A]^\alpha [B]^\beta}$$

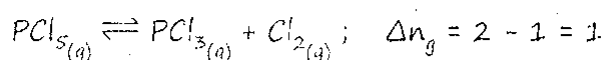
$$K_P = \frac{P_C^\gamma P_D^\delta}{P_A^\alpha P_B^\beta}$$

$$\text{Unit ; } K_C = \left(\frac{\text{mol}}{\text{L}} \right)^{\Delta n_g}$$

$$K_P = (\text{atm})^{\Delta n_g}$$

$$\Delta n_g = (\gamma + \delta) - (\alpha + \beta)$$

Q. 2 moles of PCl₅, 4 moles of PCl₃ and 1 mole of Cl₂ are present in 10 L container at eq^m find K_C of



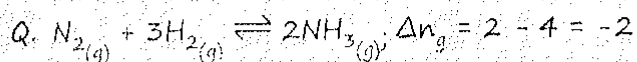
$$K_C = \frac{\frac{2}{10} \times \frac{4}{10} \times \frac{1}{10}}{\frac{2}{10}} = 0.2 (M)^1$$

Relation b/w K_P and K_C

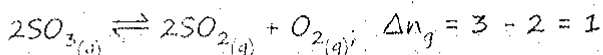
$$[K_P = K_C (RT)^{\Delta n_g}]$$

Case 1: Δn_g > 0 ; K_P > K_CCase 2: Δn_g < 0 ; K_P < K_CCase 3: Δn_g = 0 ; K_P = K_CCase 4: If T = $\frac{1}{R} = 12 \text{ K}$ K_P = K_C for any value of Δn_g

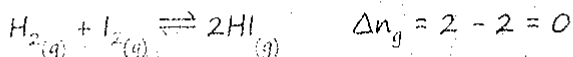
$$R = 0.0821 \frac{\text{L atm}}{\text{mol-k}} = \frac{1}{12}$$



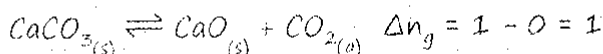
$$K_p < K_c$$



$$K_p > K_c$$

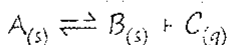


$$K_p = K_c$$



$$K_p > K_c$$

Q. Find K_p of following rxn at 480 K
if $K_c = 10$ M

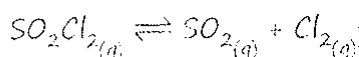


$$K_p = K_c (RT)^{\Delta n_g}$$

$$= 10 \left(\frac{1}{12} \times 480 \right)^1$$

$$= 10 \times 40 = 400 \text{ atm}$$

Q. Find temp at which $\frac{K_p}{K_c} = 3$ for



$$K_p = K_c (RT)^{\Delta n_g}$$

$$\frac{K_p}{K_c} = (RT)^1$$

$$RT = 3$$

$$T = \frac{3}{R}$$

$$T = \frac{3}{\frac{1}{12}} = 36 \text{ K}$$

Apps of K

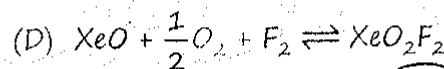
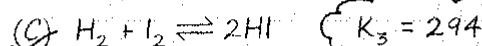
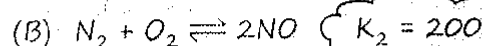
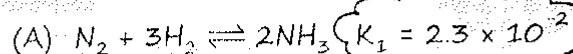
$K \uparrow$ Product stable

$K \downarrow$ Reactant stable

$$K = \frac{[P]}{[R]} \quad K \uparrow (P) \text{ more stable at eqb.}$$

$$K \downarrow (R) \text{ more stable at eqb.}$$

Q. Product is more stable in case of



$$K_4 = 1.4 \times 10^{-3}$$

Factors Affecting K

Van't Hoff eqⁿ

1. Temp

$$\log_{10} \left(\frac{K_2}{K_1} \right) = \frac{\Delta H}{2.303R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

ΔH = Heat of Rxn

R = gas const

$$= 8.314$$

$$= \frac{25}{3} \frac{J}{\text{mol-K}}$$

$$= 2 \frac{\text{Cal}}{\text{mol-K}}$$

Endo ($\Delta H = +ve$)

Exo ($\Delta H = -ve$)

$$T_1 < T_2$$

$$T_1 < T_2$$

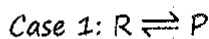
$$K_1 < K_2$$

$$K_2 < K_1$$

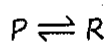
$$T \uparrow K \uparrow$$

$$T \uparrow K \downarrow$$

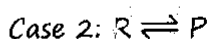
2. Stoichiometry of Rxn



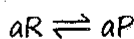
$$K_1$$



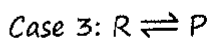
$$K_2 = \frac{1}{K_1}$$



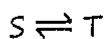
$$K_1$$



$$K_2 = K_1^a$$

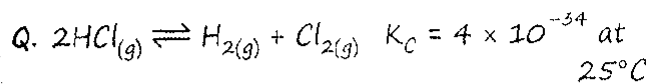


$$K_1$$



$$K_2$$





find K_C for $\frac{1}{2} \text{H}_{2(g)} + \frac{1}{2} \text{Cl}_{2(g)} \rightleftharpoons \text{HCl}_{(g)}$

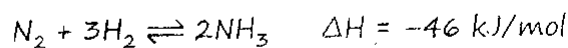
$$K_2 = \left(\frac{1}{K_1}\right)^{\frac{1}{2}} = \left(\frac{1}{4 \times 10^{-34}}\right)^{\frac{1}{2}}$$

$$= \frac{1}{2 \times 10^{-17}} = \frac{10}{2} \times 10^{16}$$

$$= 5 \times 10^{16}$$

E_{eq}^m const depends only on the temp.
and the stoichiometry of reaction

Q. Find K_{eqb} at 127°C if eq^m const is 4 at 27°C for the Rxn



$$\log_{10} \left(\frac{K_2}{K_1}\right) = \frac{-46 \times 10^3}{2.3 \times 8.3} \left[\frac{1}{300} - \frac{1}{400}\right]$$

$$= -2400 \times \frac{1}{100} \left[\frac{1}{3} - \frac{1}{4}\right]$$

$$= -24 \times \frac{1}{12}$$

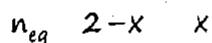
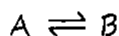
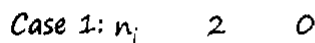
$$\log_{10} \left(\frac{K_2}{4}\right) = -2$$

$$\frac{K_2}{4} = 10^{-2}$$

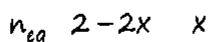
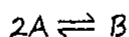
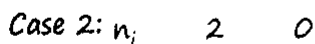
$$K_2 = 4 \times 10^{-2}$$

Degree of Dissociation

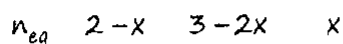
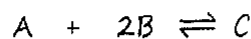
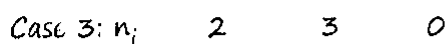
$$\text{D.O.D} = \frac{\text{No. of moles dissociated}}{\text{Initial moles taken}}$$



$$\text{D.O.D of A} = \frac{x}{2}$$

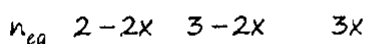
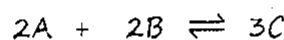
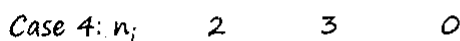


$$\text{D.O.D of A} = \frac{2x}{2} = x$$



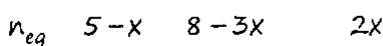
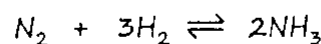
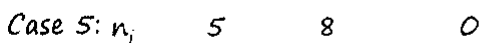
$$\text{D.O.D of A} = \frac{x}{2}$$

$$\text{D.O.D of B} = \frac{2x}{3}$$



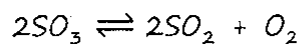
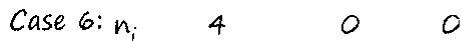
$$\text{D.O.D of A} = \frac{2x}{2} = x$$

$$\text{D.O.D of B} = \frac{2x}{3}$$

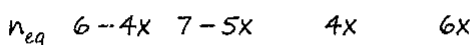
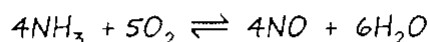


$$\text{D.O.D of N}_2 = \frac{x}{5}$$

$$\text{D.O.D of H}_2 = \frac{3x}{8}$$



$$\text{D.O.D of SO}_3 = \frac{2x}{4} = \frac{x}{2}$$



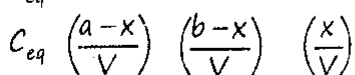
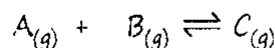
$$\text{D.O.D of NH}_3 = \frac{4x}{6} = \frac{2x}{3}$$

$$\text{D.O.D of O}_2 = \frac{5x}{7}$$

Apps of D.O.D

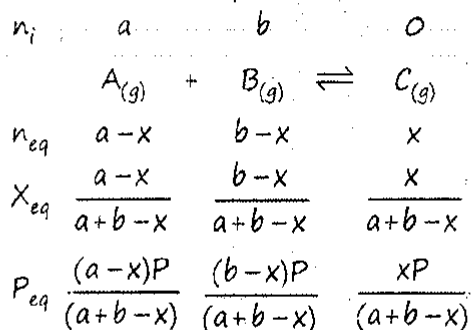
To Find E_{eq}^m constants

Case 1: Calculation of K_C



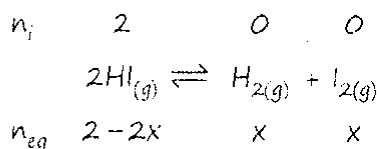
$$K_c = \frac{\left(\frac{x}{V}\right)}{\left(\frac{a-x}{V}\right)\left(\frac{b-x}{V}\right)}$$

Case 2: Calculation of K_p

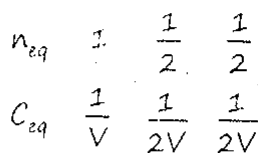


$$K_p = \frac{\left(\frac{xP}{a+b-x}\right)}{\left(\frac{(a-x)P}{a+b-x}\right) \times \left(\frac{(b-x)P}{a+b-x}\right)}$$

Q. $2HI_{(g)} \rightleftharpoons H_{2(g)} + I_{2(g)}$ at certain temp only 50% HI is dissociated find K_c

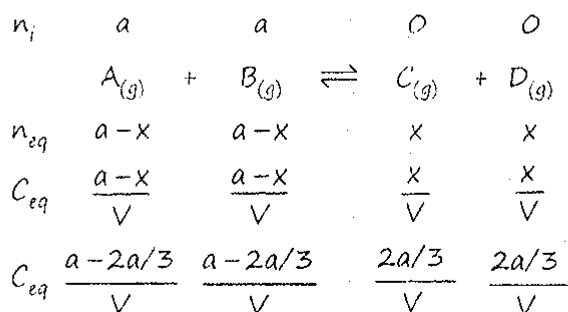


D.O.D of HI = $\frac{2x}{2} = x = \frac{1}{2}$



$$K_c = \frac{\frac{1}{2V} \times \frac{1}{2V}}{\left(\frac{1}{V}\right)^2} = \frac{1}{4}$$

Q. $A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)} + D_{(g)}$ if initially A & B both are taken in equal amount but at eq_m $[D]_{eq} = 2[A]_{eq}$ find K_c ?



$$K_c = \frac{\frac{2a}{3} \times \frac{2a}{3}}{\frac{a}{3} \times \frac{a}{3}} = 4$$

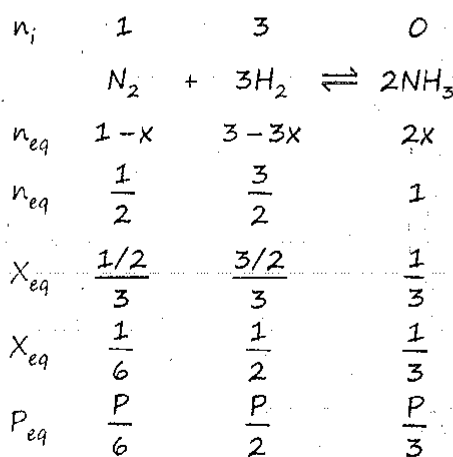
$$\frac{x}{V} = 2 \left(\frac{a-x}{V}\right)$$

$$x = 2a - 2x$$

$$3x = 2a$$

$$x = \frac{2a}{3}$$

Q. $N_2 + 3H_2 \rightleftharpoons 2NH_3$ for rxn initially the mole ratio was 1 : 3 of $N_2 : H_2$. At eq^m 50% of each has reacted, if the eq^m pressure is P. Find K_p ?



$$\Delta n_g = 2 - 4 = -2$$

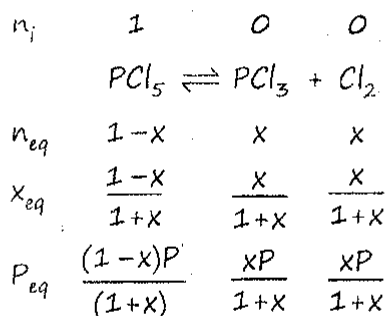
$$D.O.D \text{ of } N_2 = \frac{x}{1} = \frac{1}{2}$$

$$K_p = \frac{(P/3)^2}{(P/6)(P/2)^3}$$

$$= \frac{\frac{1}{9} P^2}{\frac{1}{6} \times \frac{1}{8} P^3} = \frac{1}{3} P^{-2} = \frac{16}{3P^2} (\text{atm})^{-2}$$

Q. Write expression of K_p for $PCl_5 \rightleftharpoons PCl_3 + Cl_2$

Given D.O.D = x. eq^m total pressure is P



$$K_p = \frac{\frac{x^2 P^2}{(1+x)^2}}{\frac{(1-x)P}{(1+x)}} = \frac{x^2 P}{1-x^2}$$

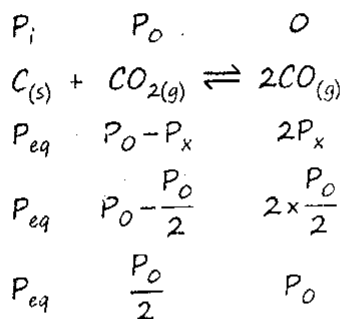
$$1 \gg x$$

$$K_p = x^2 P$$

$$x = \sqrt{\frac{K_p}{P}} \quad \boxed{x \propto \frac{1}{\sqrt{P}}}$$

Calculation of K_p if pressure is given.

Q. In the Rxn the eq^m pressure is 12 atm. If 50% CO_2 reacts find K_p



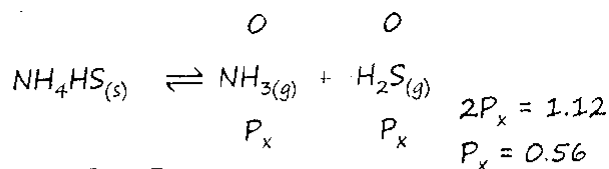
$$\text{D.O.D of } \text{CO}_2 = \frac{P_x}{P_o} = \frac{1}{2}$$

$$P_x = \frac{P_o}{2}$$

$$\frac{P_o}{2} + P_o = 12; \frac{3P_o}{2} = 12; P_o = 8$$

$$K_p = \frac{(P_o)^2}{(P_o/2)} = 2P_o = 2 \times 8 = 16 \text{ (atm)}$$

Q. For the rxn, the observed pressure is 1.12 atm find K_p ?



$$\begin{aligned}
 K_p &= P_x \times P_x \\
 &= 0.56 \times 0.56 = 0.3136 \text{ (atm)}^2
 \end{aligned}$$

Relation b/w D.O.D and Vapour density

$$\left[\alpha = \frac{D-d}{d(n-1)} \right] (n \neq 1)$$

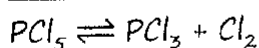
$$\alpha = \text{D.O.D}$$

$$D = \text{V.D of Reactant}$$

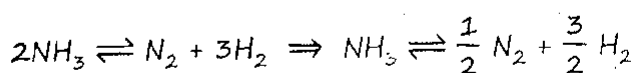
$$\text{V.D} = D = \frac{M.W_{\text{Reactant}}}{2}$$

$$d = \text{V.D of Rxn mix}$$

Significance of n

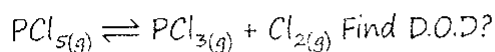


n = moles of products = 2



$$n = \frac{1}{2} + \frac{3}{2} = \frac{4}{2} = 2$$

Q. The V.D of gaseous mix is 58 at 250°C for the Rxn



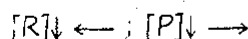
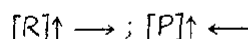
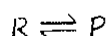
$$d = 58 \quad D = \frac{208.5}{2} = 104.25$$

$$\alpha = \frac{D-d}{d(n-1)} = \frac{104.25-58}{58(2-1)}; \alpha = 0.8$$

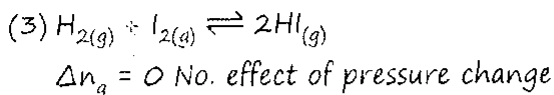
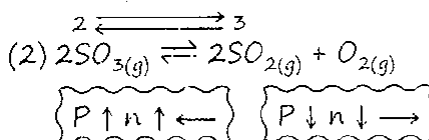
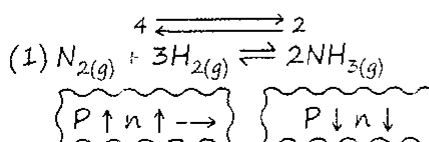
$$[\% \text{ D.O.D} = 80\%]$$

Le- chatelier's principle

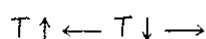
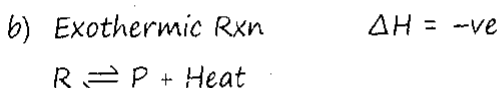
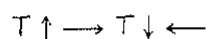
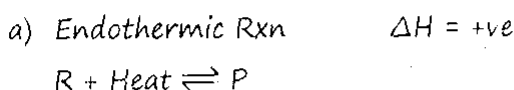
1. Effect of concⁿ



2. Effect of Pressure



3. Effect of temp



4. Effect of Catalyst

Does not affect the eq^m

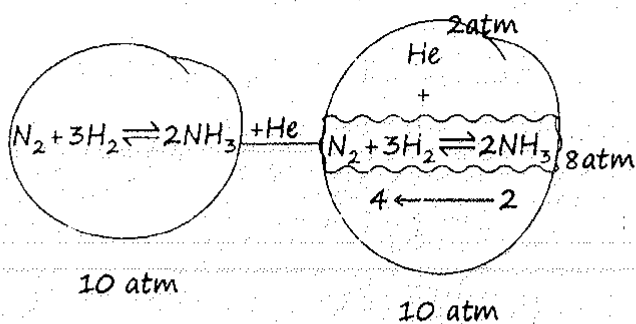
5. Effect of addition of inert gas

Case 1: at const volume

Does not affect the eq^m

Case 2: at const pressure

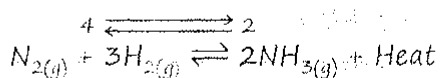
Rxn will shift in that direcⁿ where no. of moles increases.



$P \downarrow n \downarrow$

the rxn will proceed
in the dirⁿ where
no. of moles increases

Q. Predict the direction of eq^m?



(1) $[N_2] \uparrow \rightarrow$

(2) $[NH_3] \uparrow \leftarrow$

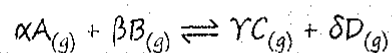
(3) $P \uparrow n \uparrow \rightarrow$

(4) $T \uparrow \leftarrow$

(5) By adding catalyst

(6) Addition of Ne at const pressure $\leftarrow$

Reaction Quotient



$$Q_c = \frac{[C]^\gamma [D]^\delta}{[A]^\alpha [B]^\beta}$$

[A], [B]

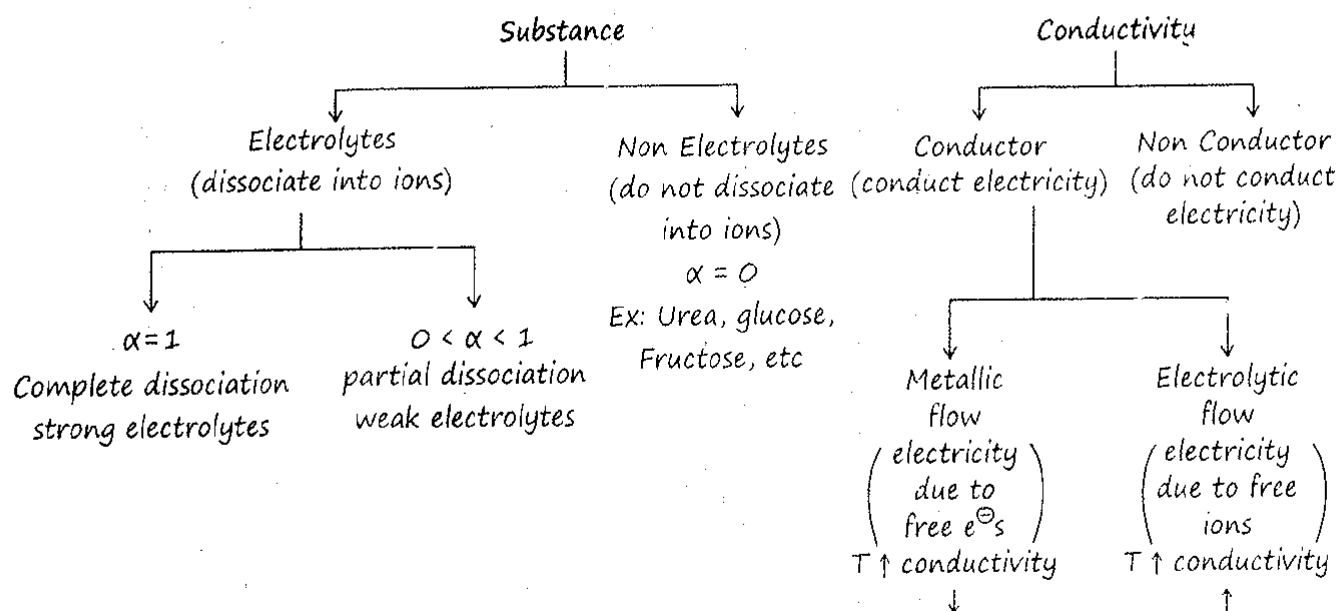
[C], [D]

not at eq^m

Case 1: $Q_c = K_c$; eq^m

Case 2: $Q_c > K_c$; B.W.R

Case 3: $Q_c < K_c$; F.W.R



Strong Electrolytes		Weak Electrolytes	
SA	SB	WA	WB
$H_2SO_4, HClO_4, HClO_3, HNO_3, HCl, HBr, HI$ ** All types of salts	All alkali metal hydroxides & $Ba(OH)_2$	$H_2SO_3, HClO_2, HClO, HNO_2$, All oxyacids of P, All organic carboxylic acids, HCN, H_3BO_3, H_2CO_3, HF	All alkaline earth metal hydroxides except $Ba(OH)_2$, $Al(OH)_3$, All Organic Bases, NH_3 , d-block metal hydroxides

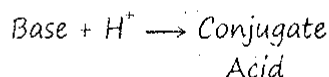
Acid Base theory

Arrhenius theory		Bronsted theory	
AA	AB	BA	BB
Donate H^+ ion in aq. solution; $HX, H_2SO_4, H_3PO_4, HClO_4, HCN$ etc. $HX \xrightarrow{H_2O} H^+ + X^-$ $\downarrow H_2O$ H_3O^+ $\downarrow H_2O$ $H_5O_2^+$ $\downarrow H_2O$ $H_7O_3^+ \text{ etc}$ ** H_3BO_3 is not AA.	OH donor in aq. solution. Ex: $MOH, M(OH)_2, M(OH)_3$ $MOH \xrightarrow{H_2O} M^+ + OH^-$ $\downarrow H_2O$ $H_5O_3^+ \xleftarrow{H_2O} H_3O_2^+ \text{ etc}$	H^+ donor $HX \longrightarrow H^+ + X^-$ * All AA are BA	H^+ acceptor $X^O \xrightarrow{H^+} HX$ $HCO_3^- \xrightarrow{H^+} H_2CO_3$ $H_2PO_4^- \xrightarrow{H^+} H_3PO_4$ $HCO_3^- \xrightarrow{H^+} H_2CO_3$ $HO^- \xrightarrow{H^+} H_2O$

Lewis Theory

LA	L.B
Lone pair acceptor H^+ , CO_2 , SF_4 , PF_5 , BCl_3 , $AlCl_3$, $FeCl_3$ etc	Lone pair donor $X^\ominus$, $OH^\ominus$, NH_3 , H_2O etc
$L.B: \curvearrowright O=C=\curvearrowright O$	

Acid Base Theory

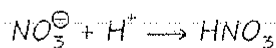


WB

SCA

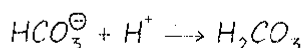
SB

WCA



WB

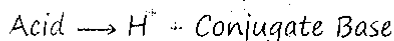
SCA



SB

WCA

Conjugate acid-base pair

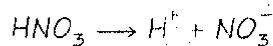


SA

W.C.B

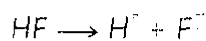
WA

S.C.B



SA

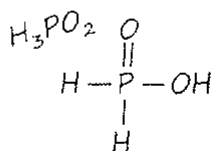
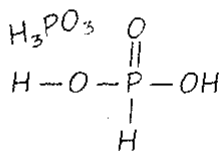
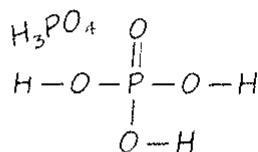
WCB



WA

SCB

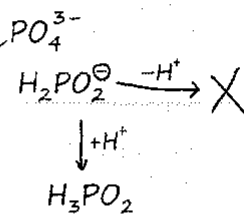
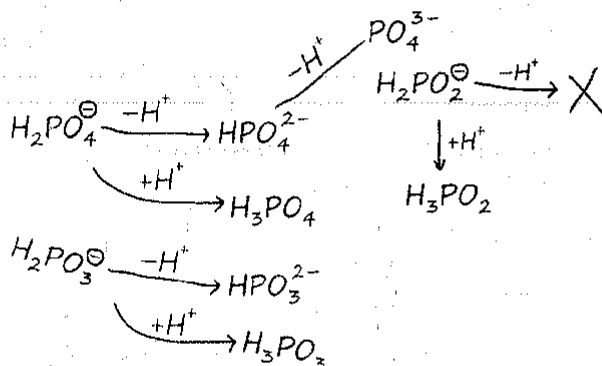
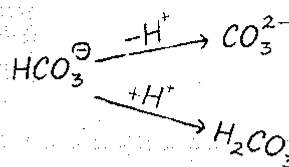
1. Acid base theory



HPO_3^{2-}
Amphiprotic
X

2. Amphiprotic Species

H^+ donor as well as H^+ acceptor



pH Calculation

$$pH = -\log[H^+]$$

$$pOH = -\log_{10}[OH^-]$$

$$\# \quad pH + pOH = 14 \text{ at } 25^\circ C$$

$$[H^+] = 10^{-7} = [OH^-]$$

$$\# \quad pH + pOH = 12 \text{ at } 90^\circ C$$

$$[H^+] = 10^{-6} = [OH^-]$$

$$T \uparrow \quad k_w \uparrow$$

Case 1: Strong acid/Base

$$(1) [H^+] \geq 10^{-6} M$$

Q. Find pH of $10^{-3} M HCl$

$$[H^+] = 10^{-3} M$$

$$pH = -\log 10^{-3} = 3$$

$$pH = 3$$

$$-\log(a \times 10^b) = b - \log a$$

$$(2) [H^+] < 10^{-6}$$

Q. Find pH of $10^{-8} M HCl$

$$[H^+] = 10^{-8} M$$

$$pH = -\log 10^{-8} = 8 \quad \text{X}$$

$$[H^+]_{\text{net}} = [H^+]_{\text{from acid}} + [H^+]_{\text{from } H_2O}$$

$$= 10^{-8} + 10^{-7}$$

$$= 10^{-8} [1 + 10] = 11 \times 10^{-8}$$

$$pH = -\log(11 \times 10^{-8}) = 8 - \log 11$$

$$pH = 6.9$$

Case 2: pH of mix of two SA & SB

$$[H^+]/[OH^-] = \frac{[N_1V_1 \pm N_2V_2]}{V_1 + V_2}$$

$$= \frac{[M_1V_1nf_1 \pm M_2V_2nf_2]}{V_1 + V_2}$$

Q. Find pH

50 mL of $\frac{M}{5}$ HCl + 50 mL of $\frac{M}{10}$ H_2SO_4

$$[H^+] = \frac{M_1V_1nf_1 + M_2V_2nf_2}{V_1 + V_2}$$

$$= \frac{\frac{1}{5} \times 50 \times 1 + \frac{1}{10} \times 50 \times 2}{50 + 50}$$

$$= \frac{10 + 10}{100} = \frac{20}{100} = 2 \times 10^{-1}$$

$$P^H = -\log(2 \times 10^{-1}) = 1 - \log 2 = 1 - 0.3$$

$$P^H = 0.7$$

Q. Find pH

75 mL of $\frac{M}{2}$ $Ba(OH)_2$ + 25 mL of $\frac{M}{1}$ HCl

$$[OH^-] = \frac{\left| \frac{1}{2} \times 75 \times 2 - 1 \times 25 \times 1 \right|}{100}$$

$$[OH^-] = \frac{|75 - 25|}{100} = \frac{50}{100} = \frac{1}{2}$$

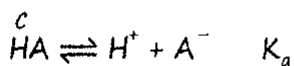
$$P^{OH} = -\log \frac{1}{2} = \log 2 = 0.3$$

$$P^H = 14 - 0.3 = 13.7$$

Case 3: pH of weak monobasic acid

or

Weak monoacidic base



$$\alpha = \sqrt{\frac{K_a}{C}} ; [H^+] = \sqrt{K_a C}$$

$$pH = \frac{1}{2} [pK_a - \log C]$$

+ Degree of dissociation of weak electrolyte is inversely proportional to square root of

the concⁿ or directly proportional to square root of dilution $\left[\alpha = \frac{1}{\sqrt{C}} \text{ or } \alpha \propto \sqrt{V} \right]$

Q. Calculate pH of 0.1 M CH_3COOH ($K_a = 10^{-5}$)

$$pH = \frac{1}{2} (pK_a - \log C) \quad P^{K_a} = -\log K_a$$

$$= \frac{1}{2} (5 - \log 10^{-1}) = -\log 10^{-5} = 5$$

$$= \frac{1}{2} (5 + 1)$$

$$= \frac{6}{2} \quad \alpha = \sqrt{\frac{10^{-5}}{10^{-1}}} = \sqrt{10^{-4}} = 10^{-2}$$

$$pH = 3$$

$$[H^+] = \sqrt{K_a C}$$

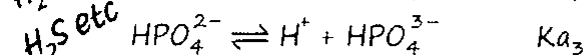
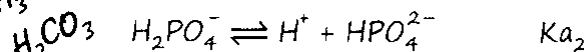
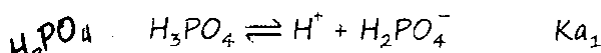
$$= \sqrt{10^{-5} \times 10^{-1}}$$

$$= \sqrt{10^{-6}} = 10^{-3}$$

Case 4: pH of weak poly basic acid

or

Weak poly acidic base



$$K_{a1} > K_{a2} > K_{a3}$$

$$\alpha = \sqrt{\frac{K_{a1}}{C}} ; [H^+] = \sqrt{K_{a1} C} = [H_2PO_4^-]$$

$$pH = \frac{1}{2} [pK_{a1} - \log C] \quad [HPO_4^{2-}] = K_{a2}$$

$$[PO_4^{3-}] = \frac{K_{a2} K_{a3}}{[H^+]}$$

Q. Find pH of 0.1 M H_3PO_4

$$K_{a1} = 10^{-3}, K_{a2} = 10^{-8}, K_{a3} = 10^{-13}$$

$$pH = \frac{1}{2} [pK_{a1} - \log C]$$

$$= \frac{1}{2} (3 - \log 10^{-1})$$

$$= \frac{4}{2} \quad \alpha = \sqrt{\frac{10^{-3}}{10^{-1}}} = \sqrt{10^{-2}} = 10^{-1}$$

$$pH = 2 \quad [H^+] = [H_2PO_4^-] = \sqrt{10^{-3} \times 10^{-1}} = 10^{-2}$$

$$[\text{HPO}_4^{2-}] = 10^{-8}$$

$$[\text{PO}_4^{3-}] = \frac{10^{-8} \times 10^{-13}}{10^{-2}} = 10^{-19}$$

Case 5: PH of mix of two weak acid

or

two weak bases

$$[\text{H}^+] = \sqrt{K_{a1}C_1 + K_{a2}C_2}$$

or

$$[\text{OH}^-] = \sqrt{K_{b1}C_1 + K_{b2}C_2}$$

C_1 & C_2
concⁿ after
mixing

Q. Find P^H of Soln obtained by mixing equal volumes of 0.02 M HOCl ($K_{a1} = 2 \times 10^{-4}$) and 0.2 M CH_3COOH ($K_{a2} = 2 \times 10^{-5}$)

$$C_1 = \frac{0.02}{2} = 0.01; C_2 = \frac{0.2}{2} = 0.1$$

$$\begin{aligned} [\text{H}^+] &= \sqrt{2 \times 10^{-4} \times 10^{-2} + 2 \times 10^{-5} \times 10^{-1}} \\ &= \sqrt{2 \times 10^{-6} + 2 \times 10^{-6}} = \sqrt{4 \times 10^{-6}} \\ &= 2 \times 10^{-3} \end{aligned}$$

$$P^H = -\log(2 \times 10^{-3}) = 3 - \log 2 = 3 - 0.3$$

$$P^H = 2.7$$

Case 6: P^H of mix of WA and SA

or

WB and SB

WA C_1 K_{a1}

SA C_2

$$[\text{H}^+] = \frac{-C_2 \pm \sqrt{C_2^2 + 4K_{a1}C_1}}{2}$$

Q. Find P^H of mix of 10^{-4} M HCl & 10^{-2} M CH_3COOH ($K_a = 2 \times 10^{-5}$)

$$\begin{aligned} [\text{H}^+] &= \frac{-10^{-4} \pm \sqrt{(10^{-4})^2 + 4 \times 2 \times 10^{-5} \times 10^{-2}}}{2} \\ &= \frac{-10^{-4} + \sqrt{10^{-8} + 8 \times 10^{-7}}}{2} \end{aligned}$$

$$= \frac{-10^{-4} + \sqrt{10^{-8} + 80 \times 10^{-8}}}{2}$$

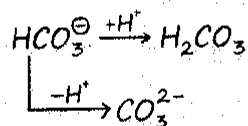
$$= \frac{-10^{-4} + \sqrt{81 \times 10^{-8}}}{2} = \frac{-10^{-4} + 9 \times 10^{-4}}{2}$$

$$= \frac{8 \times 10^{-4}}{2} = 4 \times 10^{-4}$$

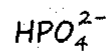
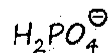
$$P^H = -\log(4 \times 10^{-4})$$

$$= 4 - \log 4 = 4 - 0.6 = 3.4$$

Case 7: P^H of Amphiprotic species



$$[P^H = \frac{P^{K_{a1}} + P^{K_{a2}}}{2}]$$



$$[P^H = \frac{P^{K_{a1}} + P^{K_{a2}}}{2}] \quad [P^H = \frac{P^{K_{a2}} + P^{K_{a3}}}{2}]$$

Q. Find P^H of a) 0.01 M NaH_2PO_4

b) 0.1 M Na_2HPO_4

for H_3PO_4 $K_{a1} = 10^{-3}$

$K_{a2} = 10^{-8}$

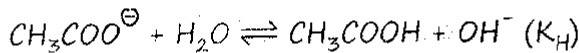
$K_{a3} = 10^{-13}$

$$a) P^H = \frac{3 + 8}{2} = \frac{11}{2} = 5.5$$

$$b) P^H = \frac{8 + 13}{2} = \frac{21}{2} = 10.5$$

Salt hydrolysis

WA + SB (CH_3COONa)



(1) Anionic hydrolysis

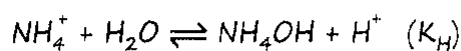
$$(2) K_H = \frac{K_w}{K_a}$$

$$(3) h = \sqrt{\frac{K_H}{C}}$$

$$(4) P^H = 7 + \frac{1}{2} [PK_a + \log C]$$

$P^H > 7$ Nature of solⁿ after hydrolysis is basic

SA + WB (NH_4Cl)



(1) Cationic

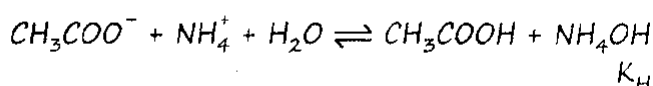
$$(2) K_H = \frac{K_w}{K_b}$$

$$(3) h = \sqrt{\frac{K_H}{C}}$$

$$(4) P^H = 7 - \frac{1}{2} [PK_b + \log C]$$

$P^H < 7$ Nature of solⁿ after hydrolysis is Acidic

WA + SB ($\text{CH}_3\text{COONH}_4$)



(1) Cationic and anionic both

$$(2) K_H = \frac{K_w}{K_a K_b}$$

$$(3) h = \sqrt{K_H} \quad (\text{D.O. hydrolysis is independent of conc}^n \text{ of salt})$$

$$(4) P^H = 7 + \frac{1}{2} [PK_a - PK_b]$$

$K_a > K_b$; $PK_a < PK_b$; $pH < 7$ Acidic

$K_a < K_b$; $PK_a > PK_b$; $pH > 7$ Basic

$K_a = K_b$; $PK_a = PK_b$; $pH = 7$ Neutral

* Salt of SA + SB does not hydrolyse ($P^H = 7$)

Q. Which salts will have highest P^H ?

(A) KCl
SA + SB
 $P^H = 7$

(B) Na_2CO_3
WA + SB
 $P^H > 7$

(C) CuSO_4
SA + WB
 $P^H < 7$

(D) N.O.T

Q. PK_b for CN^- is 4.7. The P^H of 0.5M aq. NaCN solⁿ is

$$(PK_a + PK_b = 14; PK_a = 14 - 4.7 = 9.3)$$

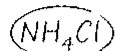
$$\text{WA} + \text{SB} \quad P^H = 7 + \frac{1}{2} [PK_a + \log C]$$

$$= 7 + \frac{1}{2} [9.3 + \log \frac{1}{2}]$$

$$= 7 + \frac{1}{2} [9.3 - 0.3]$$

$$= 7 + 4.5 = 11.5$$

Q. K_b for NH_4OH is 2×10^{-5} find P^H of



SA + WB

$$K_H = \frac{K_w}{K_b} = \frac{10^{-14}}{2 \times 10^{-5}} \\ = 0.5 \times 10^{-9} \\ = 5 \times 10^{-10}$$

Q. In correct match $P^H < 7$ Acidic
SA + WB

(A) FeCl_3 in H_2O - Basic

(B) NH_4Cl " " - Acidic
SA + WB

(C) $\text{CH}_3\text{COONH}_4$ " " $\rightarrow K_a = K_b$
WA + WB
 $P^H = 7$ Neutral

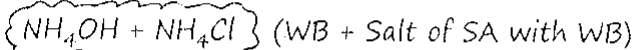
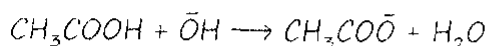
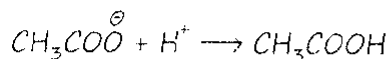
(D) Na_2CO_3 " " - Basic
WA + SB
 $P^H > 7$

Buffer Solutions

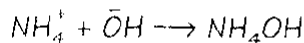
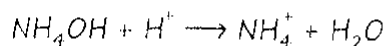
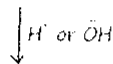
The solutions which resist the change in P^H by adding small amount of strong acid or strong base



Acidic buffer



Basic Buffer

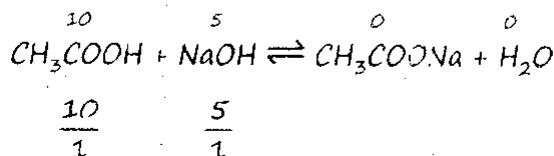


Acidic buffer can be prepared by three ways

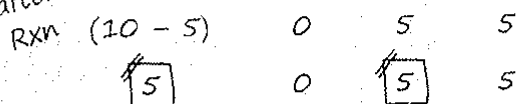
1. $\text{CH}_3\text{COOH} + \text{CH}_3\text{COONa}$

WA (Salt of WA + SB)

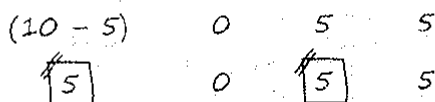
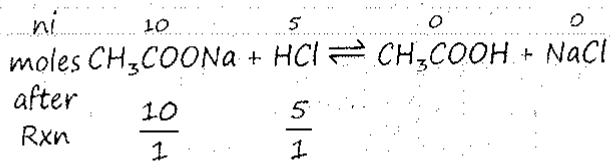
2. WA is mixed with SB ($n_{\text{WA}} > n_{\text{SB}}$)



moles
after
Rxn



3. Salt of (WA + SB) with SA ($n_{\text{salt}} > n_{\text{SA}}$)



Examples of Buffer Solutions

→ Acidic

WA Salt of WA + SB

1. $\text{HCN} + \text{NaCN}$
2. $\text{H}_3\text{BO}_3 + \text{Na}_2\text{B}_4\text{O}_7$
3. $\text{H}_2\text{CO}_3 + \text{NaHCO}_3 \Rightarrow$ Used to maintain p^{H} of blood (7.26 - 7.42)

4. $\text{H}_3\text{PO}_4 + \text{NaH}_2\text{PO}_4$
5. $\text{CH}_3\text{COOH} + \text{CH}_3\text{COONa}$
6. $\text{HNO}_2 + \text{NaNO}_2$

→ Basic

WB Salt of WB + SA

1. $\text{CH}_3\text{NH}_2 + \text{CH}_3\text{NH}_3^+\text{Cl}^-$
2. $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$
3. $\text{PhNH}_2 + \text{PhNH}_3^+\text{Cl}^-$
4. $\text{C}_5\text{H}_5\text{N} + \text{C}_5\text{H}_5\text{NH}^+\text{Cl}^-$

→ Salt of WA + WB

$\text{CH}_3\text{COONH}_4$, NH_4CN , AgCN etc.

Calculation of p^{H} of Buffer Solutions

Henderson's Acidic

$$\text{Eq}^{\text{n}} \quad \text{p}^{\text{H}} = \text{p}^{\text{K}_a} + \log_{10} \frac{[\text{Salt}]}{[\text{Acid}]}$$

$$= \text{p}^{\text{K}_a} + \log_{10} \frac{[\text{Conjugate base}]}{[\text{Acid}]}$$

Basic

$$\text{p}^{\text{OH}} = \text{p}^{\text{K}_b} + \log_{10} \frac{[\text{Salt}]}{[\text{Base}]}$$

$$= \text{p}^{\text{K}_b} + \log_{10} \frac{[\text{Conjugate acid}]}{[\text{Base}]}$$

Calculation of p^{H} of Buffer Solutions

Q. 50 mL of 2M CH_3COOH mixed with 10 mL of 1M CH_3COONa will have an approx p^{H} of ($\text{K}_a = 10^{-5}$)

$$\text{p}^{\text{H}} = 5 + \log \left(\frac{n_{\text{Salt}}}{\frac{V_{\text{solution}}}{n_{\text{acid}}}} \right)$$

$$= 5 + \log \left(\frac{10 \times 1}{50 \times 2} \right)$$

$$= 5 + \log \frac{1}{10}$$

$$= 5 - \log 10$$

$$= 5 - 1$$

$$\text{p}^{\text{H}} = 4$$

Q. 0.05M NH_4OH solution is dissolved in 0.001M NH_4Cl solution find p^{H} ($\text{K}_b = 10^{-5}$)

$$\text{p}^{\text{OH}} = 5 + \log \left(\frac{0.001}{0.05} \right)$$

$$= 5 + \log \left(\frac{1}{50} \right)$$

$$= 5 - \log 50$$

$$= 5 - 1.7$$

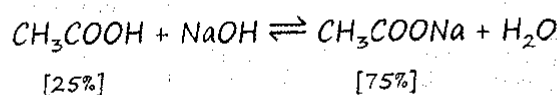
$$\text{p}^{\text{OH}} = 3.3$$

$$\text{p}^{\text{H}} = 14 - 3.3$$

$$\text{p}^{\text{H}} = 10.7$$

Q. $\frac{N}{10}$ CH_3COOH was titrated with $\frac{N}{10}$ NaOH .

When 75% of titration is over then find p^{H} ? ($\text{K}_a = 10^{-5}$)



$$\text{p}^{\text{H}} = \text{p}^{\text{K}_a} + \log \left(\frac{75}{25} \right)$$

$$= 5 + \log 3$$

$$= 5 + 0.48$$

$$\text{p}^{\text{H}} = 5.48$$

Q. $P^H - P^{Ka} = 1$ will be applicable for an acidic buffer when

- (A) $[Acid] = [Conju\ base]$
 (B) $[Acid] \times 10 = [Conju\ base]$
 (C) $[Acid] = [Conju\ base] \times 10$
 (D) N.O.T

$$P^H = P^{Ka} + \log \frac{[Conju\ base]}{[Acid]}$$

$$\frac{[Conju\ base]}{[Acid]} = 10$$

$$P^H - P^{Ka} = \log \frac{[Conju\ base]}{[Acid]}$$

$$1 = \log \frac{[Conju\ base]}{[Acid]}$$

Buffer Range

$$\frac{1}{10} \leq \frac{[Conju\ base]}{[Acid]} \leq 10$$

$$[(P^{Ka} - 1) \leq P^H \leq (P^{Ka} + 1)]$$

Maximum buffer action $[P^H = P^{Ka}]$

Buffer Capacity

Buffer capacity = $\frac{\text{No. of moles of strong acid or strong base added per L of Sol}^n}{\text{Change in } P^H \text{ of buffer}}$

Q. When 2 moles of HCl is added to 1 L of an acidic buffer Solⁿ its P^H changes from 3.9 to 3.4 find its buffer capacity

$$B.C = \frac{2}{0.5} = 4$$

Solubility and Solubility Product

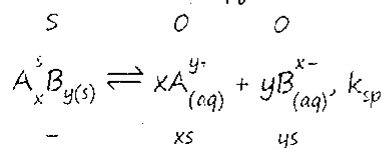
Solubility

Max. no. of moles of solute dissolved in a solvent to obtain 1 L solⁿ.

$$\text{Solubility} \rightleftharpoons \frac{\text{mol}}{L}$$

$$\left[\text{Solubility} \left(\frac{g}{L} \right) = \text{Solubility} \left(\frac{\text{mol}}{L} \right) \times MW \right]$$

Relation between k_{sp} & s_o



$$k_{sp} = (xs)^x (ys)^y$$

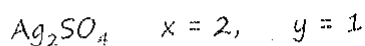
$$k_{sp} = x^x y^y s^{x+y}$$

$$NaCl, \quad x = 1 \quad y = 1$$

$$k_{sp} = 1^1 \times 1^1 \times s^{1+1} = s^2$$

Compound	x	y	Relation between s and k_{sp}
$Ba^{2+} SO_4^{2-}$	1	1	$k_{sp} = 1^1 \times 1^1 \times s^{1+1} = s^2$
$(NH_4)_2 SO_4^{2-}$	2	1	$k_{sp} = 2^2 \times 1^1 \times s^{2+1} = 4s^3$
$Hg^{2+} Cl_2^-$	1	2	$k_{sp} = 1^1 \times 2^2 \times s^{1+2} = 4s^3$
$K_3 PO_4^{3-}$	3	1	$k_{sp} = 3^3 \times 1^1 \times s^{3+1} = 27s^4$
$Ca_3^{2+} (PO_4^{3-})_2$	3	2	$k_{sp} = 3^3 \times 2^2 \times s^{3+2} = 108s^5$
$Hg - Hg \quad (Hg_2^{2+} Cl_2) **$	1	2	$k_{sp} = 1^1 \times 2^2 \times s^{1+2} = 4s^3$
$Ag_2^+ CrO_4^{2-}$	2	1	$k_{sp} = 2^2 \times 1^1 \times s^{2+1} = 4s^3$
			# $Na^+ K^+ Rb^+ PO_4^{3-}$
			$k_{sp} = 1^1 \times 1^1 \times 1^1 \times 1^1 \times s^{1+1+1+1} = s^4$

Q. If solubility of Ag_2SO_4 is $10^{-2} \frac{\text{mol}}{\text{L}}$ then find its K_{sp} ?

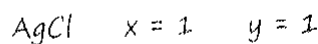


$$K_{sp} = 2^2 \times 1^1 \times S^{2-1}$$

$$= 4S^1$$

$$= 4 \times (10^{-2})^3 = 4 \times 10^{-6} \left(\frac{\text{mol}}{\text{L}} \right)^3$$

Q. If K_{sp} of AgCl is $10^{-10} \left(\frac{\text{mol}}{\text{L}} \right)^2$ then find its solubility in gm/L ?



$$K_{sp} = 1^1 \times 1^1 \times S^{1+1} = S^2$$

$$10^{-10} = S^2$$

$$S = 10^{-5} \frac{\text{mol}}{\text{L}}$$

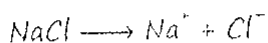
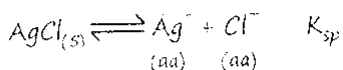
$$s \left(\frac{\text{gm}}{\text{L}} \right) = s \left(\frac{\text{mol}}{\text{L}} \right) \times \text{MW}_{\text{salt}}$$

$$= 10^{-5} \times 143.5$$

$$s \left(\frac{\text{gm}}{\text{L}} \right) = 1.435 \times 10^{-3}$$

Solubility in common ion effect

By adding common ion, solubility of Salt decreases. Solubility ↓



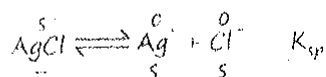
Q. Find solubility of AgCl ($K_{sp} = 10^{-10} \left(\frac{\text{mol}}{\text{L}} \right)^2$)

(a) In H_2O

(b) In 0.1 M AgNO_3

(c) In 0.1 M CaCl_2

(a) In H_2O

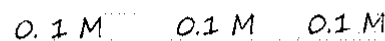
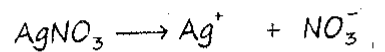
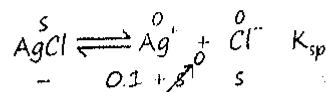


$$K_{sp} = (S)^1 \times (S)^1 = S^2$$

$$10^{-10} = S^2$$

$$S = 10^{-5} \text{ M}$$

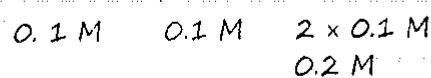
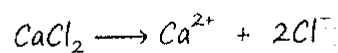
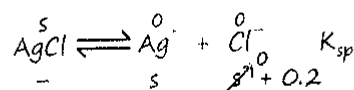
(b) In 0.1 M AgNO_3



$$K_{sp} = (0.1)^1 \times (S)^1$$

$$S = \frac{10^{-10}}{0.1} = 10^{-9} \frac{\text{mol}}{\text{L}}$$

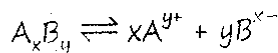
(c) In 0.1 M CaCl_2



$$K_{sp} = (S)^1 \times (0.2)^1$$

$$S = \frac{10^{-10}}{0.2} = 5 \times 10^{-10} \left(\frac{\text{mol}}{\text{L}} \right)$$

Ionic product of salt



$$K_{IP} = [\text{A}^{y+}]^x [\text{B}^{x-}]^y$$

Case (1) $K_{IP} = K_{sp}$

Saturated solⁿ

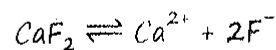
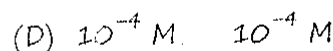
Case (2) $K_{IP} > K_{sp}$

ppt occurs

Case (3) $K_{IP} < K_{sp}$

Unsaturated solⁿ

Q. The ppt of CaF_2 ($K_{sp} = 1.7 \times 10^{-10}$) is obtained when following are mixed.



$$K_{IP} = [\text{Ca}^{2+}] [\text{F}^-]^2$$

(A) $K_{IP} = 10^{-3} \times (10^{-5})^2 = 10^{-13} ; K_{sp} > K_{IP}$

(B) $K_{IP} = 10^{-5} \times (10^{-3})^2 = 10^{-11} ; K_{sp} > K_{IP}$

(C) $K_{IP} = 10^{-2} \times (10^{-3})^2 = 10^{-8} ; K_{sp} > K_{IP}$

Some Imp points

① Types of Salts

(a) Normal salts

NaCl , Na_2SO_4 , Na_3PO_4 etc.

(b) Acid salts

NaHCO_3 , NaHSO_4

(c) Basic salts

$\text{Zn}(\text{OH})\text{Cl}$, $\text{Mg}(\text{OH})\text{Cl}$ etc.

(d) Double salts

$\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$ Potash alum

(e) Complex salts

$[\text{Ag}(\text{NH}_3)_2]\text{Cl}$, $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$ etc.

(f) Mixed salts

NaKS , NaKRBPO_4 etc.

② Isohydric Solⁿ

Solⁿ having same $[\text{H}^+]$ Concⁿ

$$[\text{H}^+]_1 = [\text{H}^+]_2$$

$$\sqrt{K_{a1}c_1} = \sqrt{K_{a2}c_2}$$

$$[K_{a1}c_1 = K_{a2}c_2]$$

③ Relative strength of acids

$$\text{R.S} = \frac{[\text{H}^+]_1}{[\text{H}^+]_2} = \frac{\sqrt{K_{a1}c_1}}{\sqrt{K_{a2}c_2}}$$

WA + SB Mixing ($K_a(\text{CH}_3\text{COOH}) = 10^{-5}$)

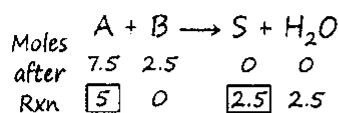
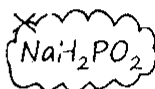
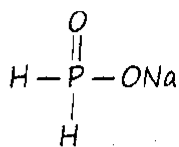
Case 1: 75mL of 0.1 M CH_3COOH

+

25mL of 0.1 M NaOH

$$n_{\text{WA}} = 75 \times 0.1 = 7.5 \text{ mmol}$$

$$n_{\text{SB}} = 25 \times 0.1 = 2.5 \text{ mmol}$$



$$\text{pH} = 5 + \log \left(\frac{2.5}{5} \right)$$

$$\text{pH} = 5 + \log \frac{1}{2} = 5 - \log 2 = 5 - 0.3 = 4.7$$

Case 2: 50 mL of 0.1 M CH_3COOH

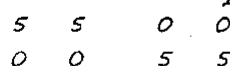
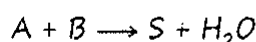
+

50 mL of 0.1 M NaOH

$$n_{\text{WA}} = 5$$

$$n_{\text{SB}} = 5$$

$$C = \frac{5}{100} = \frac{1}{20}$$



$$\text{pH} = 7 + \frac{1}{2} \left[5 + \log \frac{1}{20} \right]$$

$$= 7 + \frac{1}{2} [5 - \log 20] = 7 + \frac{1}{2} [5 - 1.3]$$

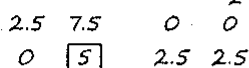
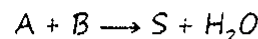
$$= 7 + \frac{1}{2} \times 3.7 = 7 + 1.85 = 8.85$$

Case 3: 25mL of 0.1 M CH_3COOH

+

75 mL of 0.1 M NaOH

$$n_{\text{WA}} = 2.5 ; n_{\text{SB}} = 7.5$$



$$[\text{OH}^-] = \frac{5}{100} = \frac{1}{20}$$

$$\text{pOH} = -\log \frac{1}{20} = \log 20 = 1.3$$

$$[\text{pH} = 14 - 1.3 = 12.7]$$

Concept of Oxidation Number

Real or imaginary charge present on an atom.

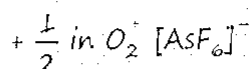
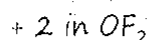
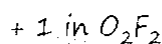
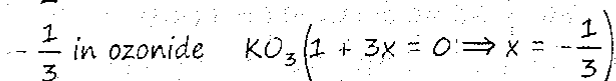
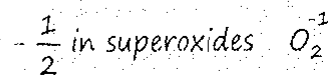
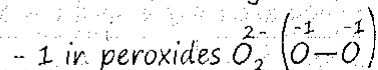


RULES

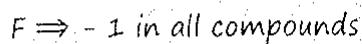
1. O.N of H is generally +1 but in Metal hydrides it is -1



2. O.N of 'O' is usually -2 but



3. O.N of Halogens (Except F) is usually -1 to +7

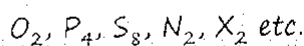


4. Alkali metals $\Rightarrow +1$ always

Alkaline earth metals $\Rightarrow +2$ always } in their compounds

5. O.N of N is usually -3 to +5

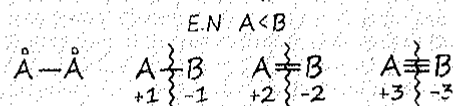
6. O.N of element in elemental state is always zero



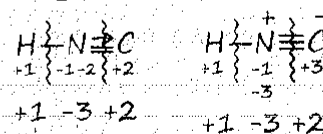
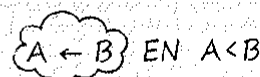
7. The sum of O.N of all the elements in a compound is always zero.

8. The sum of O.N of all the elements in an ion is always equal to charge on that ion

9. O.N in covalent bond



10. O.N in coordinate bond



Find average O.N. of underlined atom

1. KClO_3

$$1 + x + 3 \times (-2) = 0$$

$$x = +5$$

2. $\text{Fe}_{0.94}\text{O}$

$$0.94x - 2 = 0$$

$$x = \frac{2}{0.94} = \frac{200}{94}$$

3. $\text{S}_4\text{O}_6^{2-}$

$$4x - 12 = -2$$

$$4x = 10$$

$$x = 2.5$$

4. $\text{Na}_2\text{S}_2\text{O}_3$

$$2 + 2x - 6 = 0$$

$$x = 2$$

5. H_5IO_6

$$5 + x - 12 = 0$$

$$x = 7$$

6. HN_3

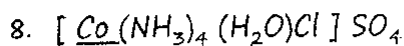
$$1 + 3x = 0$$

$$x = -\frac{1}{3}$$

7. $[\text{Ni}(\text{CO})_4]$

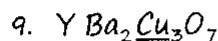
$$x + 4 \times 0 = 0$$

$$x = 0$$



$$x + 4 \times 0 + 0 - 1 - 2 = 0$$

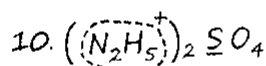
$$x = +3$$



$$3 + 4 + 3x - 14 = 0$$

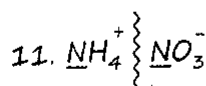
$$3x = 7$$

$$x = \frac{7}{3}$$



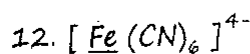
$$2 + x - 8 = 0$$

$$x = +6$$



$$x + 4 = 1 \quad x - 6 = -1$$

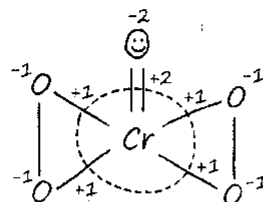
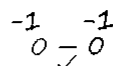
$$x = -3 \quad x = +5$$



$$x + 6 \times (-1) = -4$$

$$x = +2$$

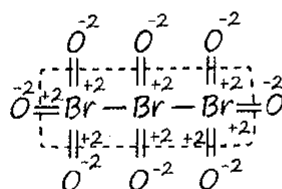
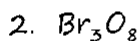
Calculation of O.S. (Individual O.N)



$$\text{O.N of 'Cr'} = +6$$

$$\text{O.N of 'O'} = -1$$

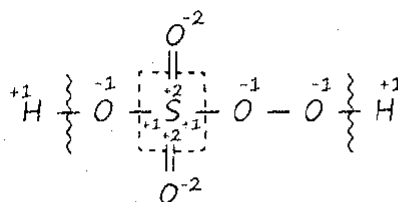
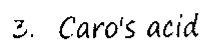
$$\text{O.N. of O in Peroxy bond (P.L)} = 2$$



$$\text{O.N of 'Br'} = +4 \text{ \& } +6$$

$$\text{O.N of 'O'} = -2$$

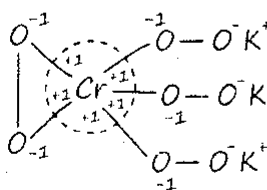
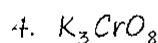
$$\text{P.L} = 0$$



$$\text{O.N of 'S'} = +6$$

$$\text{O.N of 'O'} = -1 \text{ \& } -2$$

$$\text{P.L} = 1$$

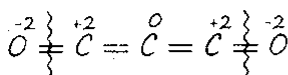
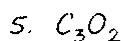


$$\text{O.N of 'Cr'} = +5$$

$$\text{O.N of 'O'} = -1$$

$$\text{O.N of 'K'} = +1$$

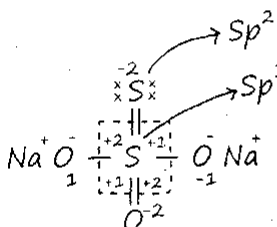
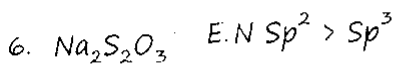
$$\text{P.L} = 4$$



$$\text{O.N of 'C'} = 0 \text{ \& } +2$$

$$\text{O.N of 'O'} = -2$$

$$\text{P.L} = 0$$

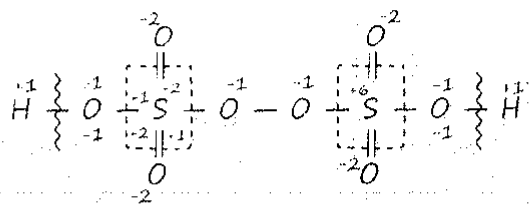
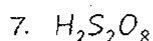


$$\text{O.N of 'S'} = -2 \text{ \& } +6$$

$$\text{O.N of 'O'} = -2$$

$$\text{O.N of 'Na'} = +1$$

$$\text{P.L} = 0$$

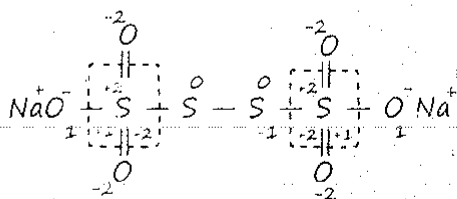
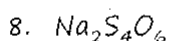


O.N of 'S' = +6

O.N of 'O' = -1 & -2

O.N of 'H' = +1

$$\text{P.L} = 1$$



O.N of 'S' = +5 & +6

O.N of 'O' = -2

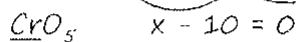
O.N of 'Na' = +1

$$\text{P.L} = 0$$

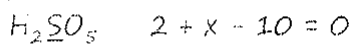
How to Find P.L without drawing the Structure

P.L is present when

$$\text{Avg O.N} > \text{Max O.N} \rightarrow \text{Group no.}$$



Yes, P.L is present



$$x = +8 > +6$$

Yes P.L is present

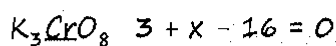
How to find no. of P.L without drawing the structure

Case 1: When only one central atom is present

$$\text{P.L} = \frac{\text{Avg O.N} - \text{Max O.N}}{2}$$



$$\text{P.L} = \frac{10 - 6}{2} = \frac{4}{2} = 2$$



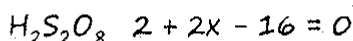
$$x = 13 > 6$$

$$\text{P.L} = \frac{13 - 6}{2} = \frac{7}{2} = 3.5$$

$$\text{P.L} = 4$$

Case 2: When two C.A are present

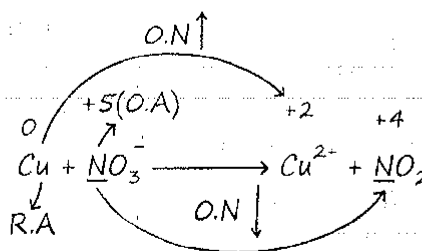
$$\text{P.L} = \text{Avg O.N} - \text{Max O.N}$$



$$x = 7 > 6$$

$$\text{P.L} = 7 - 6 = 1$$

Oxidising and Reducing agent

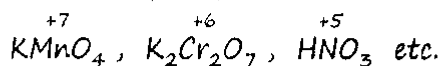


O.N ↑, Oxidises itself, R.A

O.N ↓, Reduces itself, O.A

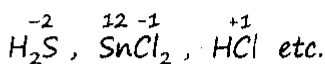
Case 1: If C.A is in its max O.N

Reduces itself, O.A



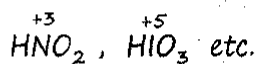
Case 2: If C.A is in its min O.N

Oxidises itself, R.A



Case 3: If C.A is in its intermediate O.N

Oxidises and Reduces itself, O.A & R.A both



Normality (N)

$$N = \frac{\text{No. of gm equivalent}}{V_{(L)}}$$

$$\text{No. of gm equivalent} = \frac{W}{E.W}$$

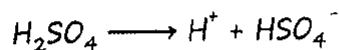
$$E.W = \frac{\text{Atomic or Molecular weight}}{n. \text{ factor}}$$

$$N = M \times n_f$$

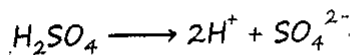
n.factor

Case (1) for acids

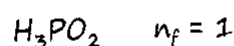
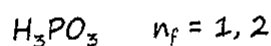
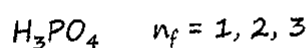
n_f = no. of H^+ ions furnished in the solⁿ (or, no. of lone pairs accepted) per moleate



$$n_f = 1$$

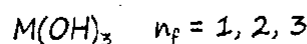


$$n_f = 1$$



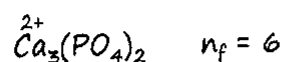
Case (2) for Bases

n_f = no. of $\bar{O}H$ ions furnished in the solⁿ per molecule



Case (3) for Salts

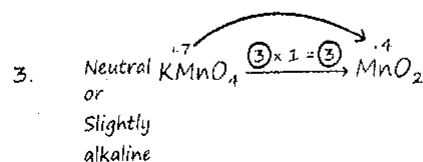
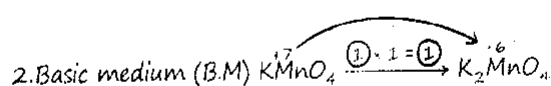
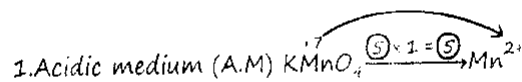
n_f = Total +ve charge or total -ve charge per molecule



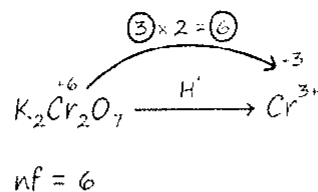
Case (4) Redox Reaction

- ① When only one atom can undergo either oxidation or reduction

n_f = Total change in O.N. per molecule

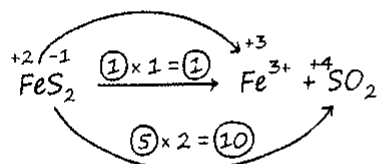


135
BNA

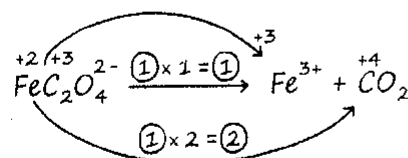


- ② When two or more atoms can undergo either oxidation or reduction

n_f = total change in O.N. per molecule



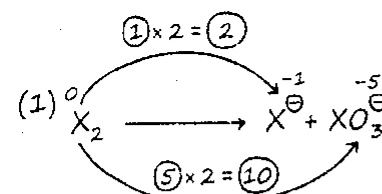
$$n_f = n_{f1} + n_{f2} = 1 + 10 = 11$$



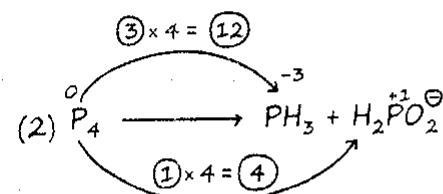
$$n_f = 1 + 2 = 3$$

- ③ Disproportionation Reaction

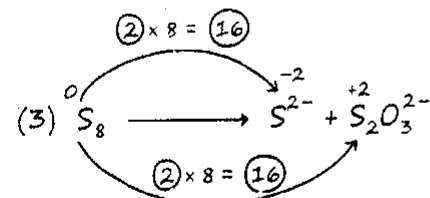
$$n_f = \frac{n_1 n_2}{n_1 + n_2}$$



$$n_f = \frac{2 \times 10}{2 + 10} = \frac{20}{12} = \frac{5}{3}$$



$$n_f = \frac{12 \times 4}{12 + 4} = \frac{48}{16} = 3$$



$$n_f = \frac{16 \times 16}{16 + 16} = 8$$

Titration

1. Acid - Base titration

No. of gram equivalent of Acid = no. of gram equivalent of Base

No. of gram equivalent = $n \times n_f$

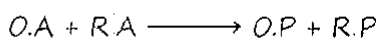
$$= N \times V_{(L)} = M \times N_f \times V_{(L)}$$

$$(1) n_1 \times n_{f_1} = n_2 \times n_{f_2}$$

$$(2) n_1 \times n_{f_1} = M_2 \times n_{f_2} \times V_{(L)}$$

$$(3) M_1 \times n_{f_1} \times V_1 = M_2 \times n_{f_2} \times V_2$$

2. Redox titration



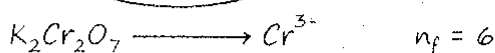
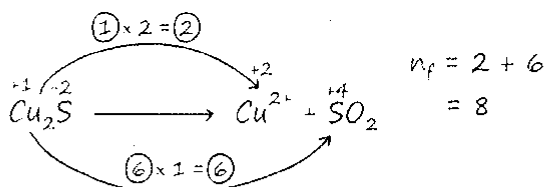
gram equivalent of O.A = gram eq. of R.A

$$(1) n_1 \times n_{f_1} = n_2 \times n_{f_2}$$

$$(2) n_1 \times n_{f_1} = M_2 \times n_{f_2} \times V_{2(L)}$$

$$(3) M_1 \times n_{f_1} \times V_1 = M_2 \times n_{f_2} \times V_2$$

Q. Find the no. of moles of $K_2Cr_2O_7$ needed to oxidise one mole of Cu_2S in A.M?



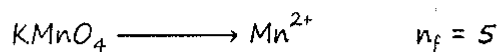
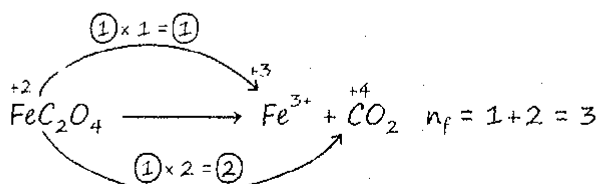
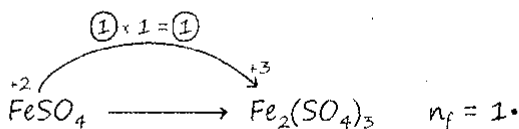
$$n_{eqv.}(Cu_2S) = n_{eqv.}(K_2Cr_2O_7)$$

$$n_1 \times n_{f_1} = n_2 \times n_{f_2}$$

$$1 \times 8 = n_2 \times 6$$

$$n_2 = \frac{8}{6} = \frac{4}{3}$$

Q. How many moles of $KMnO_4$ are needed to oxidised a mixture of 1 mol of each $FeSO_4$ and FeC_2O_4 in A.M?



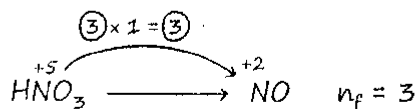
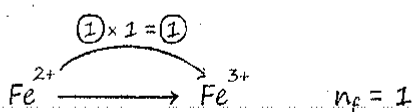
$$n_{eqv.}(KMnO_4) = n_{eqv.}(FeSO_4) + n_{eqv.}(FeC_2O_4)$$

$$n_1 \times n_{f_1} = n_2 \times n_{f_2} + n_3 \times n_{f_3}$$

$$n_1 \times 5 = 1 \times 1 + 1 \times 3$$

$$\left[n_1 = \frac{4}{5} \right]$$

Q. Find volume of 6M HNO_3 needed to oxidize 5.6 gm of Fe^{2+} to Fe^{3+} where HNO_3 gets converted to NO ?



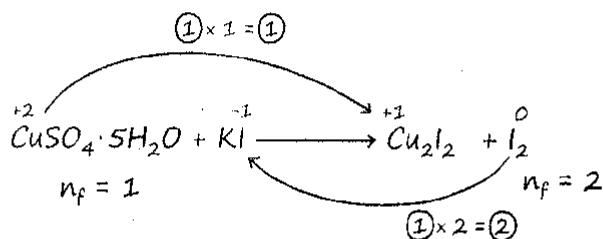
$$n_{eqv.}(HNO_3) = n_{eqv.}(Fe^{2+})$$

$$M_1 \times V_1 \times n_{f_1} = n_2 \times n_{f_2}$$

$$6 \times V_1 \times 3 = \frac{5.6}{56} \times 1$$

$$V_1 = \frac{0.1}{18} = \frac{1}{180} \text{ L}$$

Q. How much amount of $CuSO_4 \cdot 5H_2O$ is required for liberation of 2.54 gm I_2 when titrated with KI?

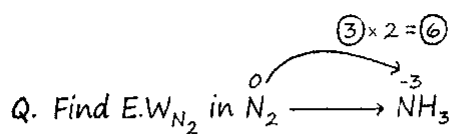


$$n_{eqv.}(CuSO_4 \cdot 5H_2O) = n_{eqv.}(I_2)$$

$$n_1 \times n_{f_1} = n_2 \times n_{f_2}$$

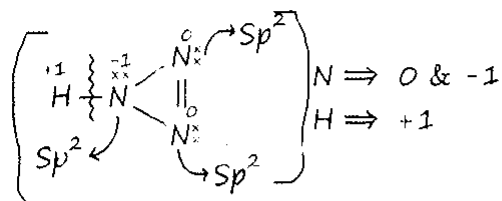
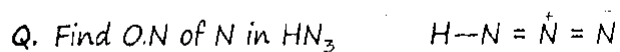
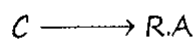
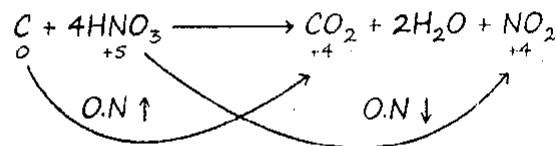
$$\frac{W_1}{249.5} \times 1 = \frac{2.54}{254} \times 2$$

$$W_1 = \frac{499}{100} = 4.99 \text{ gm}$$

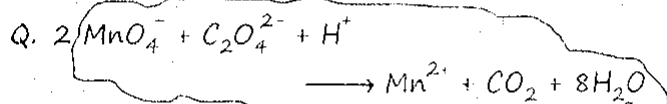


$$E.W_{N_2} = \frac{MW_{N_2}}{n_f} = \frac{28}{6} = \frac{14}{3} \text{ g}$$

Q. Find O.A and R.A



Q. O.N of Fe in Haemoglobin? Fe^{2+}



the balance equation coeff. for MnO_4^- , $C_2O_4^{2-}$ and H^+ are respectively?

(A) 2, 5, 16 (B) 16, 5, 2

(C) 2, 16, 5 (D) 5, 2, 16

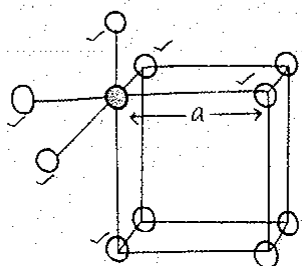
Unit Cell and Lattice

The simplest unit which can repeat itself over again and again to form Lattice

Types of cubic unit cell

1. Simple cubic (S.C) primitive
2. Body centered cubic (B.C.C) nonprimitive
3. Face centered cubic (F.C.C) nonprimitive

1. Simple cubic (primitive)



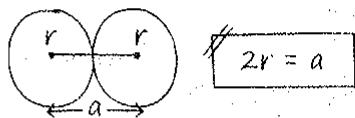
8 corners $\Rightarrow$ atoms
Sharing of corners
 $= \frac{1}{8}$ per unit cell

(1) Z_{eff} = Effective no. of atoms per unit cell

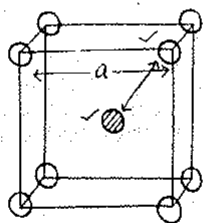
$$= 8 \times \frac{1}{8} = 1$$

(2) Nearest neighbours or C.N = 6

(3) Relation b/w radius of atom & edge length



2. B.C.C



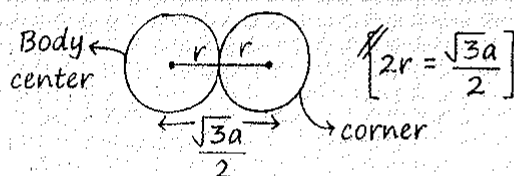
1 B.C $\Rightarrow$ 1 atom
8 corners $\Rightarrow$ 8 atoms
Sharing of corners
 $= \frac{1}{8}$ per unit cell

(1) Z_{eff} = Effective no. of atoms per unit cell

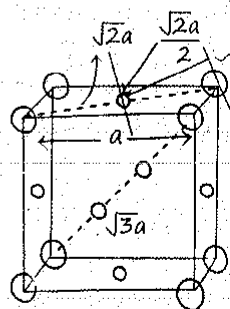
$$= 8 \times \frac{1}{8} + 1 = 2$$

(2) Nearest neighbours or C.N = 8

(3) Relation b/w radius of atom & edge length



3. F.C.C



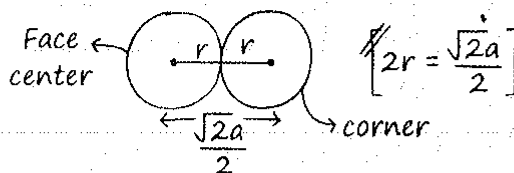
6 F.C $\Rightarrow$ 6 atoms
8 corners $\Rightarrow$ 8 atoms
Sharing of corners $= \frac{1}{8}$
Sharing of F.C $= \frac{1}{2}$ } per unit cell

(1) Z_{eff} = Effective no. of atoms per unit cell

$$= 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 1 + 3 = 4$$

(2) Nearest neighbours or C.N = 12

(3) Relation b/w radius of atom & edge length



Packing Fraction or Packing Efficiency

The space or fraction occupied by atoms in unit cell

$$\left[\% \text{ P.F.} = \frac{Z \times V_{\text{atom}}}{V_{\text{unit cell}}} \times 100 \right]$$

Case 1: In S.C

$$\% \text{ PF} = \frac{1 \times \frac{4}{3} \pi r^3}{a^3} \times 100$$

$$(a = 2r)$$

$$\text{PF} = \frac{\pi}{6} \times 100$$

$$\text{PF} = 52.4\%$$

$$\% \text{ void} = 100 - \% \text{ PF}$$

$$= 47.6\%$$

Case 2: In B.C.C

$$\%PF = \frac{2 \times \frac{4}{3} \pi r^3}{a^3} \times 100$$

$$\left(2r = \frac{\sqrt{3}a}{2} \right)$$

$$PF = \frac{\sqrt{3}\pi}{8} \times 100$$

$$PF = 68\%$$

$$\% \text{ void} = 32\%$$

Case 3: In F.C.C

$$\%PF = \frac{4 \times \frac{4}{3} \pi r^3}{a^3} \times 100$$

$$\left(2r = \frac{\sqrt{2}a}{2} \right)$$

$$PF = \frac{\pi}{\sqrt{2}} \times 100$$

$$= 74\%$$

$$\% \text{ void} = 26\%$$

Points

1. Z_{eff}
2. C.N
3. Relation blw a and r
4. % PF
5. % void

Case 1: In S.C

$$Z_{\text{eff}} = 1$$

$$C.N = 6$$

$$\text{Relation between } r \text{ \& } a = 2r = a$$

$$\% P.F = 52.4\% = \left(\frac{\pi}{6} \right)$$

$$\% \text{ void} = 47.6\%$$

Case 2: In B.C.C

$$Z_{\text{eff}} = 2$$

$$C.N = 8$$

$$\text{Relation between } r \text{ \& } a = 2r = \frac{\sqrt{3}a}{2}, r = \frac{\sqrt{3}}{4} a$$

$$\% pF = 68\% = \left(\frac{\sqrt{3}\pi}{8} \right)$$

$$\% \text{ Void} = 32\%$$

Case 3: In F.C.C

$$Z_{\text{eff}} = 4$$

$$C.N = 12$$

$$\text{Relation between } a \text{ \& } r = 2r = \frac{\sqrt{2}a}{2}, r = \frac{a}{2\sqrt{2}}$$

$$\% PF = 74\% = \left(\frac{\pi}{3\sqrt{2}} \right)$$

$$\% \text{ Void} = 26\%$$

Q. A $\longrightarrow$ Each corner

B $\longrightarrow$ Alternate Face center

Find simplest formula?

$$A_{8 \times \frac{1}{8}} \quad B_{2 \times \frac{1}{2}}$$

$$[AB]$$

Q. Na crystallizes in BCC unit cell of Edge length 4.3 Å find radius of Na

$$2r = \frac{\sqrt{3}a}{2}$$

$$r = \frac{\sqrt{3} \times 4.3}{4}$$

$$= \frac{1.73 \times 4.3}{4} \text{ Å}$$

Q. A $\longrightarrow$ each Corner $\left(\frac{1}{8} \right)$

B $\longrightarrow$ B.C (1)

C $\longrightarrow$ each F.C $\left(\frac{1}{2} \right)$

D $\longrightarrow$ each E.C $\left(\frac{1}{4} \right)$

Find simplest formula

$$A_{8 \times \frac{1}{8}} \quad B_1 \quad C_{6 \times \frac{1}{2}} \quad D_{12 \times \frac{1}{4}}$$

$$[A B C_3 D_3]$$

Q. A $\longrightarrow$ each Corner B $\longrightarrow$ each F.C

When one of the B is missing then find new E.F?

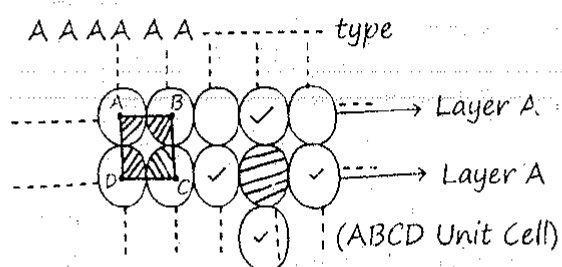
$$A_{8 \times \frac{1}{8}} \quad B_{5 \times \frac{1}{2}}$$

$$A \quad B_{\frac{5}{2}} [A_2 B_5]$$

Closed Packing Case 1: In 2D

1. Square closed packing

(a) Arrangement



$$(b) Z_{eff} = 4 \times \frac{1}{4} = 1$$

$$(c) C.N = 4$$

(d) Packing fraction

$$\%PF = \frac{Z_{eff} \times A_{disc}}{A_{unit\ cell}} \times 100$$

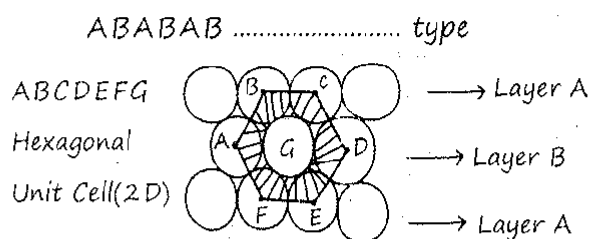
$$\%PF = \frac{1 \times \pi r^2}{a^2} \times 100$$

$$\%PF = \frac{\pi}{4} \times 100 = 78.5\%$$

$$\%void = 100 - 78.5 = 21.5\%$$

2. Hexagonal closed packing

(a) Arrangement



$$(b) Z_{eff} = 1 + 6 \times \frac{1}{3} = 1 + 2 = 3$$

$$(c) C.N = 6$$

(d) Packing fraction

$$\%PF = \frac{3 \times \pi r^2}{6 \times \frac{\sqrt{3}}{4} a^2} \times 100$$

$$\%PF = \frac{\pi}{2\sqrt{3}} \times 100 = 91\%$$

$$2r = a$$

$$\%void = 9\%$$

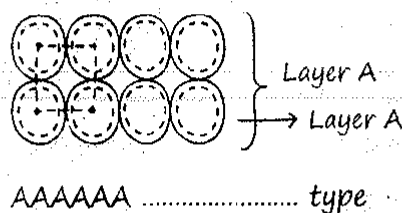
Stability $\propto$ PF

HCP(2D) > SCP(2D)

Closed Packing Case 2: In 3D

1. Simple cubic packing

(a) Arrangement



$$(b) Z_{eff} = 1$$

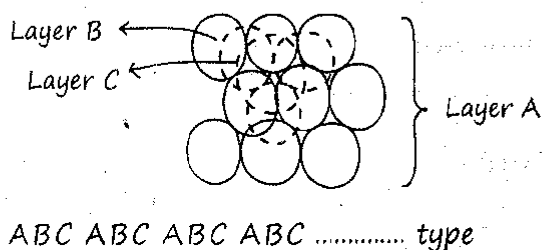
$$(c) C.N = 6$$

$$(d) \%PF = 52.4\%$$

$$\%void = 47.6\%$$

2. Cubic closed packing (CCP) or (FCC)

(a) Arrangement



(1) Layer A is HCP (2D)

(2) Layer B is placed over voids of Layer A

(3) Layer C is placed over unoccupied voids of Layer A & B.

$$(b) Z_{eff} = 4$$

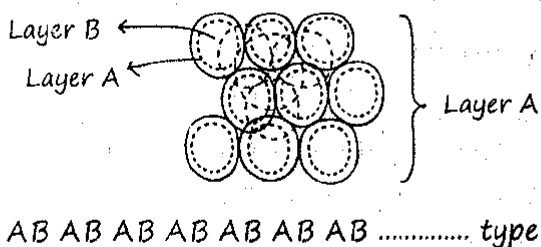
$$(c) C.N = 12$$

$$(d) \%PF = 74\%$$

$$\%void = 26\%$$

3. Hexagonal closed packing (HCP 3D)

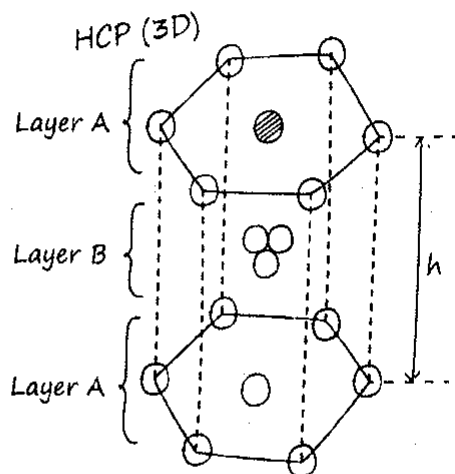
(a) Arrangement



(1) Layer A is HCP (2D)

(2) Layer B is placed over voids of Layer A

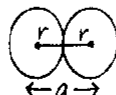
(3) 3rd Layer A is same as that of first Layer A



$$(b) Z_{\text{eff}} = 3 + 2 \times \frac{1}{2} + 12 \times \frac{1}{6} = 6$$

$$(c) C.N = 12$$

$$(d) 2r = a$$



(e) Height of HCP (3D)

$$= 4 \sqrt{\frac{2}{3}} r$$

(f) Base Area of HCP

$$= 6 \times \frac{\sqrt{3}}{4} a^2$$

$$(g) \left(V_{\text{HCP}} \right)_{3D \text{ unit cell}} = 24 \sqrt{2} r^3$$

$$(h) \% \text{PF} = \frac{6 \times \frac{4}{3} \pi r^3}{24 \sqrt{2} r^3} \times 100$$

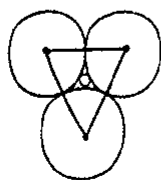
$$= \frac{\pi}{3\sqrt{2}} \times 100$$

$$= 74 \%$$

$$(i) \% \text{void} = 26 \%$$

Types of Voids

1. Triangular void

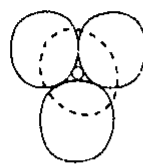


$$\# C.N = 3$$

of
void

$$\# \text{H.C.P (2D)}$$

2. Tetrahedral voids (T.V or T.H.V or Td void)

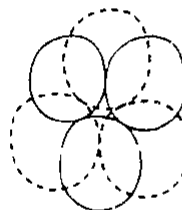


$$\# C.N = 4$$

$$\# \text{CCP or FCC}$$

$$\# \text{H.C.P (3D)}$$

3. Octahedral void (O.H.V or O.V or Od void)

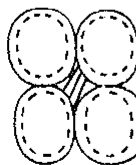


$$\# C.N = 6$$

$$\# \text{CCP or FCC}$$

$$\# \text{H.C.P (3D)}$$

4. Cubical void



$$\# C.N = 8$$

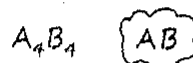
$$\# \text{Simple cubic}$$

$$\# \text{Size of void Triangular} < \text{T.H.V} < \text{O.H.V} < \text{cubical}$$

$$(\text{***})^{\infty} \begin{cases} \text{If No. of atoms } (Z_{\text{eff}}) = N \\ \text{Valid in } \begin{cases} \text{T.H.V} = 2N; \text{ O.H.V} = N \\ \text{HCP(3D) \& (Closest packed store)} \\ \text{CCP(fcc)} \end{cases} \end{cases}$$

$$Q. A \longrightarrow \text{HCP (3D) 4}$$

$$B \longrightarrow 100 \% \text{OHV 4}$$



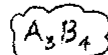
$$Q. A \longrightarrow \text{CCP (fcc) 6}$$

$$B \longrightarrow 50 \% \text{OHV } 6 \times \frac{1}{2}$$



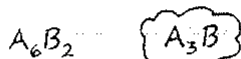
$$Q. A \longrightarrow \text{CCP (fcc) 4}$$

$$B \longrightarrow \frac{2}{3} \text{THV } \frac{2}{3} \times 8 = \frac{16}{3}$$



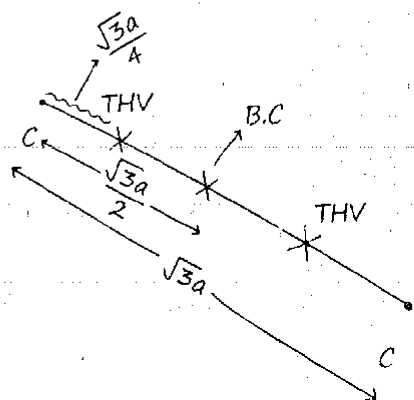
Q. A $\longrightarrow$ HCP (3D) 6

B $\longrightarrow \frac{1}{3}$ OHV $6 \times \frac{1}{3} = 2$



T.H.V and O.H.V in fcc unit cell

Case 1: T.H.V

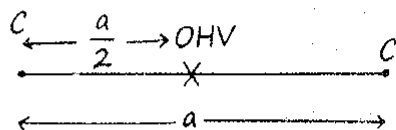


Two THV on each body diagonal per FCC unit cell

Total THV = 8

Case 2: O.H.V

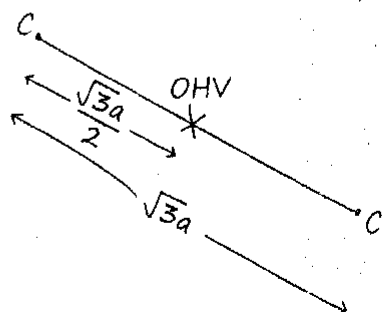
1. At Edge center



Total number of OHV at edge center FCC per unit cell

$$= 12 \times \frac{1}{4} = 3$$

2. At Body center



OHV at body centre = 1

Total OHV = 3 + 1 = 4

Density of Unit Cell

$$d = \frac{Z \times M}{N_A \times a^3} = \frac{Z \times W}{N \times a^3}$$

d = density (gm/cc)

Z = No. of atoms per unit cell

M = Molar mass

N_A = avogadro's No = 6×10^{23}

a = edge length (cm)

W = Weight (gm)

N = No. of atoms

Q. An element with density 2.8 gm/cc forms a fcc unit cell with edge length 400 pm. Calculate Molar mass of element

$$d = \frac{Z \times M}{N_A \times a^3}; M = \frac{d N_A a^3}{Z}$$

$$= \frac{2.8 \times 6 \times 10^{23} \times (400 \times 10^{-10})^3}{4}$$

$$= 26.8 \text{ g/mol}$$

Q. An element has BCC structure with a cell edge of 300 pm. the density of element is 8 gm/cc. How many atoms are present in 200 gm of element?

$$N = \frac{Z \times W}{d \times a^3}$$

$$= \frac{2 \times 200}{8 \times (300 \times 10^{-10})^3}$$

$$= \frac{50}{27 \times 10^{-24}}$$

$$= \frac{50}{27} \times 10^{24} \text{ atoms}$$

Radius Ratio	Radius Ratio	C.N	Examples
$R.R = \frac{r_{\text{void}}}{r_{\text{atom}}}$	$0.155 \leq \frac{r_c}{r_a} < 0.225$	3	B_2O_3
$= \frac{r_c}{r_a}$	$0.225 \leq \frac{r_c}{r_a} < 0.414$	4	ZnS
$= \frac{r^+}{r^-}$	$0.414 \leq \frac{r_c}{r_a} < 0.732$	6	NaCl
	$0.732 \leq \frac{r_c}{r_a} < 1$	8	CsCl

Q. If the radii of A^+ and B^- are 95 pm and 181 pm respectively, then find C.N. of ' A^+ '?

$$R.R = \frac{r^+}{r^-} = \frac{95}{181} = 0.52$$

$$0.414 \leq \frac{r_c}{r_a} < 0.732$$

$$C.N \text{ of } A^+ = 6$$

Q. The ionic radii of Rb^+ and I^- are respectively 1.46 Å and 2.16 Å. The most probable type of structure exhibited by it is?

- (A) CsCl (B) NaCl
(C) ZnS (D) N.O.T

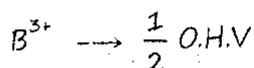
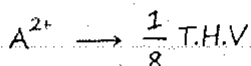
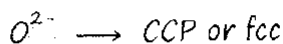
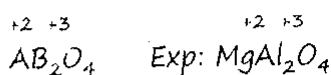
$$R.R = \frac{r^+}{r^-} = \frac{1.46}{2.16} = 0.67$$

$$0.414 \leq \frac{r_c}{r_a} < 0.732$$

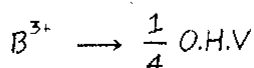
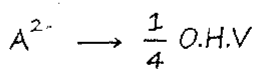
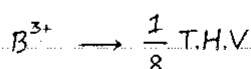
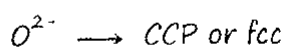
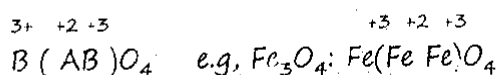
Some Imp Ionic Comp

Comp	Location of Ions	Formula Unit	C.N (C:A)	Relation b/w a and r
(1) NaCl (Rock salt structure)	$Cl^{\ominus} \rightarrow CCP \text{ (fcc)}$ $Na^+ \rightarrow 100\% \text{ OHV}$	4	6 : 6	$r_c + r_a = \frac{a}{2}$
(2) CsCl	$Cl^{\ominus} \rightarrow S.C$ $Cs^+ \rightarrow \text{Body center}$	1	8 : 8	$r_c + r_a = \frac{\sqrt{3}a}{2}$
(3) ZnS (Zinc blende)	$S^{2-} \rightarrow CCP \text{ or fcc}$ $Zn^{2+} \rightarrow 50\% \text{ THV or}$ Alternate T.H.V	4	4 : 4	$r_c + r_a = \frac{\sqrt{3}a}{4}$
(4) CaF_2 (Fluorite structure)	$Ca^{2+} \rightarrow CCP \text{ or fcc}$ $F^{\ominus} \rightarrow 100\% \text{ THV}$	4	8 : 4	$r_c + r_a = \frac{\sqrt{3}a}{4}$
(5) Na_2O (Anti fluorite structure)	$O^{2-} \rightarrow fcc \text{ or CCP}$ $Na^+ \rightarrow 100\% \text{ THV}$	4	4 : 8	$r_c + r_a = \frac{\sqrt{3}a}{4}$

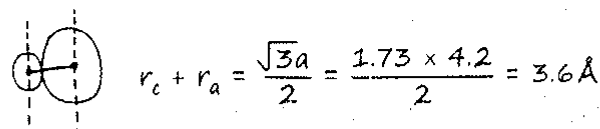
(6) Spinel



(7) Inverse Spinel



Q. CsBr has cubic structure with edge length 4.2 Å find the shortest inter ionic distance?



Q. Edge length of M^+X^- (fcc structure) is 7.2 Å find radius of $X^\ominus$ (Given $r_{M^+} = 1.6 \text{ Å}$)

$$r_c + r_a = \frac{a}{2}$$

$$1.6 + r_a = \frac{7.2}{2} = 3.6$$

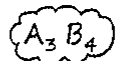
$$a = 2 \text{ Å}$$

Q. In a solid 'AB' having the 'NaCl' structure. 'A' atoms occupy the corners of cubic unit cell. If all the face centered atoms along one of the axes are removed then find E.F?

$$B \rightarrow \text{Corner} + \text{f.C} \quad 8 \times \frac{1}{8} + 4 \times \frac{1}{2}$$

$$A \rightarrow \text{E.C} + \text{B.C} \quad 12 \times \frac{1}{4} + 1$$

$$A_{1+2} B_{3+1}$$



cations and anions are missing from the crystal in stoichiometric ratio

High Radius Ratio

High C.N

Density decreases

Exp

NaCl, KCl

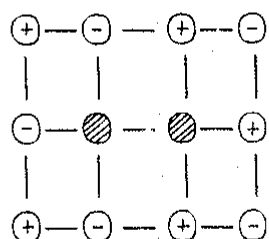
CsCl, KBr

etc

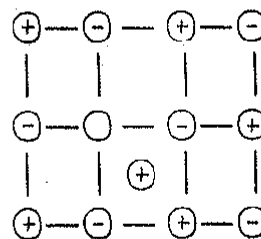
Defects

Stoichiometric

Schottky



Frenkel (Dislocation)



AgBr

Exhibit both schottky and Frenkel defect

One of the ion (usually cation) is dislocated from its usual Lattice point to some other interstitial site

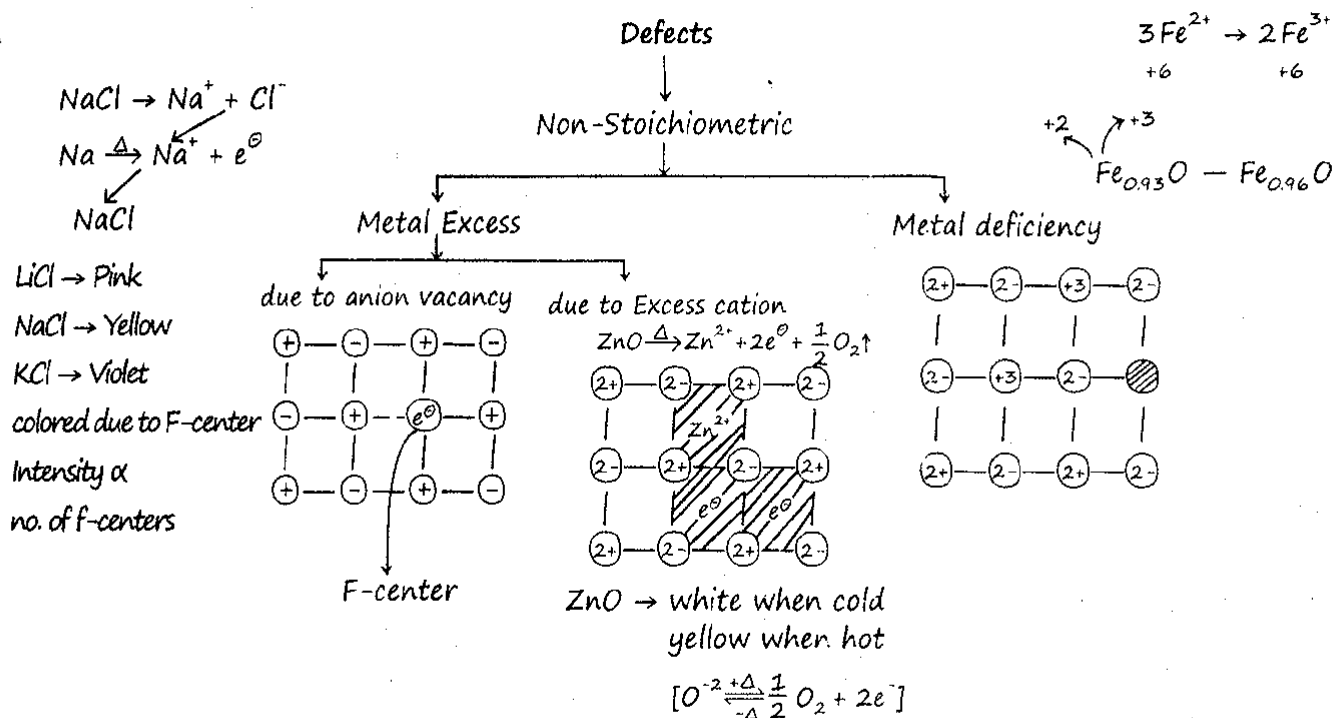
Density remains same

Low C.N

Exp

ZnS, AgCl

AgI etc

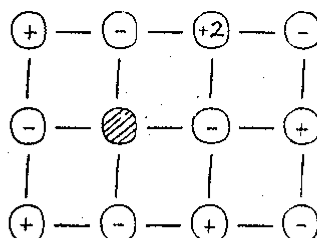
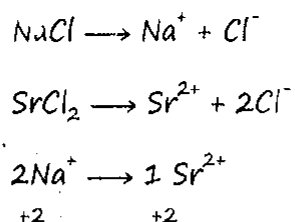


Defects

↓

Impurity defect

When NaCl is doped with SrCl_2
When AgCl is doped with CdCl_2



No. of cation vacancy
=
No. of Sr^{2+} ions

Some Imp. Points

1. Different b/w crystalline and Amorphous solid

Crystalline

- (1) Regular arrangement
- (2) Anisotropic
- (3) Sharp M.P
- (4) Long range order

Exp: SiO_2 (quartz), NaCl, KCl, CsCl etc

Amorphous

- (1) Irregular arrangement
- (2) Isotropic
- (3) Range of M.P
- (4) Short range order

Exp: SiO_2 (Silica), Rubber, plastic

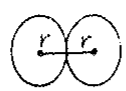
2. Types of crystalline solids

	Molecular	Ionic	Metallic	Covalent (Network)
Constituent particles	Molecules	ions	+ve charged kernel & delocalised e^- s	Covalently bonded atoms
forces	Hydrogen bond & V.W forces	Electrostatic	Metallic bond	Covalent bond
Exp	Ar, I_2 , CO_2 , CCl_4 , H_2O (ice)	NaCl, MgO, CsCl, CaF_2 etc	Cu, Ag, Au etc	SiC, SiO_2 , Si, $C_{diamond}$ graphite, AlN etc
Electrical conductivity	Insulator	Insulator in solid but conductor in molten & aq sol ⁿ	Conductor	Insulator & conductor
M.P	Low	High	Fairly high	Very high

3. P.F of Diamond cubic struc.

$C \rightarrow$ CCP or fcc (4)

$C \rightarrow$ 50% THV $\left(8 \times \frac{1}{2} = 4\right)$ $Z_{eff} = 8$



$$\% PF = \frac{8 \times \frac{4}{3} \pi r^3}{a^3} \times 100$$

$$\left(2r = \frac{\sqrt{3}a}{4}\right) \quad \% PF = \frac{\frac{32}{3} \pi r^3}{\left(\frac{8r}{\sqrt{3}}\right)^3} \times 100$$

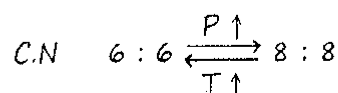
$$= \frac{\sqrt{3}\pi}{16} \times 100 = 34\%$$

$$a = \frac{8r}{\sqrt{3}}$$

4. Effect of Temp and pressure on C.N of comp.

$P \uparrow$ $C.N \uparrow$

$T \uparrow$ $C.N \downarrow$



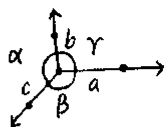
NaCl CsCl

Type Type

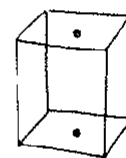
5. Coordination No.

	Type of Neighbour	Distance	Number
SC	Nearest Neighbour	a	6
	Next	$\sqrt{2}a$	12
BCC	Nearest	$\frac{\sqrt{3}a}{2}$	8
	Next	a	6
FCC	Nearest	$\frac{a}{\sqrt{2}}$	12
	Next	a	6

6. Types of crystals



Crystal	Interaxial distance	Interaxial angles	Bravais Lattice (14)
Cu Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90^\circ$	3(SC, BCC, FCC) E.C.
Rh Rhombohedral	$a = b = c$	$\alpha = \beta = \gamma \neq 90^\circ$	1(SC) End Center
Te Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	2(SC, BCC)
He Hexagonal	$a = b \neq c$	$\alpha = \beta = 90^\circ; \gamma = 120^\circ$	1(SC)
aur Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	4(SC, BCC, FC, EC*)
mona Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90^\circ; \beta \neq 90^\circ$	2(SC, EC)
Tere triclinic par	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^\circ$	1(SC)



Practice

Q. Find % of Fe^{2+} ions in $\text{Fe}_{0.93}\text{O}$?

($\text{Fe}_{93}\text{O}_{100}$) Let Fe^{2+} ions = x

Fe^{3+} ions = $93 - x$

$$2x + 3(93 - x) = 200$$

$$2x + 279 - 3x = 200$$

$$x = 79$$

$$\% \text{ of } \text{Fe}^{2+} \text{ ions} = \frac{79}{93} \times 100$$

Q. Which of the following is Pseudo Solid or Super cooled liq?

- (A) Si (B) CO_2
(C) Ag (D) Glass

Q. If NaCl is doped with 10^{-3} mol % SrCl_2 , what is the concⁿ of cation vacancies?

$$\begin{aligned} \text{No. of cation vacancies} &= \text{No. of } \text{Sr}^{2+} \text{ ions} \\ &= 10^{-3} \times 10^{-2} \times N_A \\ &= 10^{-5} \times 6 \times 10^{23} \\ &= 6 \times 10^{18} \text{ per mole} \end{aligned}$$

1. Vapour Pressure

The pressure exerted by the vapour over the surface of liq. at eqb. in closed container is called V.P of liq.

[at eqb. Rate of evaporation = Rate of condensation]

Factors affecting V.P

1. Temp.

$$[V.P \propto T]$$

$$2. V.P \propto \frac{1}{I.M.F}$$

Q. Compare V.P and B.P of the given halides

Hydrogen Bonding HF , HCl , HBr , HI

$$B.P. \text{ HCl} < \text{HBr} < \text{HI} < \text{HF}$$

$$\left[V.P \propto \frac{1}{B.P} \right]$$

$$V.P. \text{ HCl} > \text{HBr} > \text{HI} > \text{HF}$$

2. Raoult's Law

Case 1: For two volatile liq

A + B
Solvent Solute

$$P_S = P_T = P_A + P_B$$

$$[P_T = P_A^\circ X_A + P_B^\circ X_B]$$

If A is more volatile than B

P_T or P_S = Total pressure

P_A = V.P of A in solⁿ

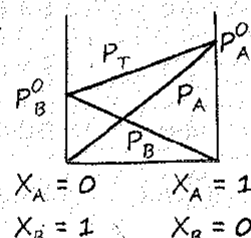
P_B = V.P of B in solⁿ

P_A° = V.P of pure A

P_B° = V.P of pure B

X_A = mole fraction of A

X_B = mole fraction of B } in Solution phase



Case 2: When solute is non volatile

$$P_T = P_A + P_B$$

$$P_T = P_A$$

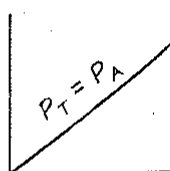
$$[P_T = P_A^\circ X_A]$$

$$P_T = P_A^\circ (1 - X_B) \quad P_A^\circ X_B = P_A^\circ - P_T$$

$$P_T = P_A^\circ - P_A^\circ X_B$$

$$\left[X_B = \frac{P_A^\circ - P_T}{P_A^\circ} \right]$$

RLVP



$$X_A = 0$$

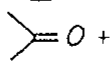
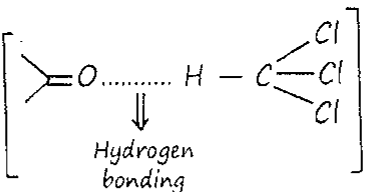
$$X_A = 1$$

$$X_B = 1$$

$$X_B = 0$$

B is solute

Types

Ideal	Non ideal	
	+ve deviation	-ve deviation
1. Obeys R.L 2. $(P_T)_{obs} = (P_T)_{cal}$ 3. Inter molecular force blw A-A and B-B are similar to Intermolecular force blw A-B 4. $\Delta H_{mix} = 0$ $\Delta V_{mix} = 0$ $\Delta S_{mix} > 0$ $\Delta G_{mix} < 0$	1. Does not Obeys R.L 2. $(P_T)_{obs} > (P_T)_{cal}$ 3. $IMF(A-B) < IMF(A-A)$ or $IMF(B-B)$ 4. $\Delta H_{mix} > 0$ $\Delta V_{mix} > 0$ $\Delta S_{mix} > 0$ $\Delta G_{mix} < 0$	1. Does not Obeys R.L 2. $(P_T)_{obs} < (P_T)_{cal}$ 3. $I.M.F(A-B) > IMF(A-A)$ & $IMF(B-B)$ 4. $\Delta H_{mix} < 0$ $\Delta V_{mix} < 0$ $\Delta S_{mix} > 0$ $\Delta G_{mix} < 0$
<u>Examples</u> 1. Bromo ethane & chloro ethane 2. Benzene + Toluene 3.  & 	1. $CCl_4 + C_6H_6$ 2. $CCl_4 + CH_3OH$ 3. $CCl_4 + C_2H_5OH$ 4. $C_2H_5OH + H_2O$ 1. $CCl_4 + \dots = +ve$ 2. $CHCl_3 + \dots = -ve$ 3. $C_6H_6 + \dots = +ve$ 4. acid + $\dots = -ve$	1. $HNO_3 + H_2O$ * 2.  + $CHCl_3$ 3. $CHCl_3 + CH_3COOH$ 4. $HCl + H_2O$ 

Q. Which is not equal to zero for an ideal solⁿ?

- (A) ΔH_{mix} (B) ΔS_{mix}
 (C) ΔV_{mix} (D) None of these

Q. Which is Temp. dependent?

- (A) Molality
 (B) Molarity
 (C) Mole fraction
 (D) Weight percentage

Q. If 3 moles of A is mixed with 2 moles of B find P_A , P_B and P_T ?

(Given: $P_B^0 = 100$ mm of Hg
 $n_A = 3$; $P_A^0 = 200$ mm of Hg)

$$n_B = 2; P_A = P_A^0 X_A = \frac{40}{5} \times \frac{3}{5} = 120 \text{ mm of Hg}$$

$$X_A = \frac{3}{5}; X_B = 1 - \frac{3}{5} = \frac{2}{5}$$

$$P_B = P_B^0 X_B = \frac{20}{5} \times \frac{2}{5} = 40 \text{ mm of Hg}$$

$$P_T = P_A + P_B = 120 + 40 = 160 \text{ mm of Hg}$$

Q. Which exhibit negative deviation?

- (A) $CCl_4 + C_6H_6$
 (B) $C_2H_5OH + H_2O$
 (C) acetone + $CHCl_3$
 (D) $CCl_4 + CH_3OH$

3. Colligative properties

M.I*

The property which depends on no. of solute particles irrespective of its nature

Van't Hoff factor (i)

$$= \frac{\text{Normal molar mass}}{\text{abnormal molar mass}}$$

$$= \frac{\text{Observed C.P.}}{\text{Normal C.P.}}$$

$$= \frac{\text{No. of moles after association/dissociation}}{\text{No. of moles before association/dissociation}}$$

Calculation of (i)

Case 1: For dissociation $\text{KCl} \rightarrow \text{K}^+ + \text{Cl}^-$

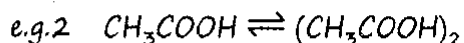
$$(i > 1) \quad [i = 1 + (n - 1)x] \quad n = 2$$

n = no. of ions produced per molecule x = D.O.D

Case 2: For association

$$(i < 1) \quad i = 1 + \left(\frac{1}{n} - 1\right)x$$

n = no. of associated to form one molecule x = D.O.A



$$n = 2$$

Case 3: $i = 1$ for neither association nor dissociation

Q. Calculate D.O.D for $\text{Ba}(\text{NO}_3)_2$ if $i = 2.74$

$$i = 1 + (n - 1)x \quad \text{Ba}^{2+} + 2\text{NO}_3^-$$

$$2.74 = 1 + (3 - 1)x \quad n = 3$$

$$\frac{1.74}{2} = x \quad x = 0.87$$

$$\% \text{ D.O.D} = 87\%$$

Q. Find i for $\text{Ca}_3(\text{PO}_4)_2$ if D.O.D is 100 %

$$[i = 1 + (n - 1)x] \quad x = 1$$

$$i = 1 + n - 1$$

$$i = n$$

$$i = 5$$

Note:
Glucose
Urea
 $i = 1$

Colligative prop formulae :

1. R.L.V.P $\Rightarrow \frac{P_A^\circ - P_s}{P_A^\circ} = i x_B$
(Relative lowering in V.P) $i \left(\frac{n_B}{n_B + n_A} \right)$

$$\left[\frac{P_A^\circ - P_s}{P_A^\circ} = i \frac{n_B}{n_A} \right]$$

2. E.B.P

(Elevation in Boiling point)

$$[\Delta T_b = i K_b m]$$

$$T_b - T_b^\circ \quad T_b = \text{B.P of sol}^n$$

$$T_b^\circ = \text{B.P of solvent}$$

$$K_b = 0.52 \text{ K kg mol}^{-1} \text{ for H}_2\text{O}$$

m = molality

3. D.F.P (Depression in freezing point)

$$\Delta T_f = i K_f m$$

$$T_f^\circ - T_f \quad T_f = \text{F.P of sol}^n$$

$$T_f^\circ = \text{F.P of solvent}$$

$$K_f = 1.86 \text{ K kg mol}^{-1} \text{ for H}_2\text{O}$$

m = molality

4. Osmotic pressure :

$$C = \text{conc}^n \text{ i.e. } \left(\frac{\text{mol}}{\text{L}} \right)$$

R = gas const

$$\Pi = i CRT$$

$$\frac{1}{\Pi_1} \quad \frac{2}{\Pi_2}$$

$$T = K$$

1. $\Pi_1 = \Pi_2$ (isotonic) Π = pressure (atm)

2. $\Pi_1 > \Pi_2$ (1 is hypertonic or 2 is hypotonic)

3. $\Pi_1 < \Pi_2$ (1 is hypo or 2 is hypertonic)

Practice

Q. Compare freezing point of given compounds.

- (A) K_2SO_4 $i=3$ (B) $NaCl$ $i=2$
 (C) Urea $i=1$ (D) $AlCl_3$ $i=4$
 $i \uparrow \Delta T_f \uparrow T_f \downarrow$
 $(C) > (B) > (A) > (D)$

Q. Compare B.P of given compounds.

- (A) $0.1 M KCl$ $i=2$ (B) $0.1 M BaCl_2$ $i=3$
 (C) $0.1 M FeCl_3$ $i=4$ (D) $0.1 M Fe_2(SO_4)_3$ $i=5$
 $i \uparrow \Delta T_b \uparrow T_b \uparrow$
 $(D) > (C) > (B) > (A)$

Q. The van't Hoff factor $K_3[Fe(CN)_6]$ is 4. Which of the following has same van't Hoff factor?

- (A) $NaCl$ $i=2$ (B) Na_2SO_4 $i=3$
 (C) $Al_2(SO_4)_3$ $i=5$ (D) $Al(NO_3)_3$ $i=4$

Q. $0.5 M$ solⁿ of Urea $i=1$ is isotonic with

- (A) $0.5 M NaCl$ (B) $0.5 M$ Sugar
 (C) $0.5 M BaCl_2$ (D) $0.5 M$ Benzoic acid in benzene

$$\Pi = i c R T$$

i same

Π same

Q. 0.2 molal acid H A $i=2$ is 20% ionized in aq. solⁿ ($K_f = 1.86 K/molality^{-1}$)

Find F.P of solⁿ? $i = 1 + (n - 1)x$

$$\Delta T_f = i K_f m \quad i = 1 + (2 - 1) \times 0.2$$

$$i = 1.2$$

$$0 - T_f = 1.2 \times 1.86 \times 0.2$$

$$[T_f = -0.45^\circ C]$$

Q. $2 \times 10^{-3} m$ aq. solⁿ of an ionic comp

$[Co(NH_3)_5NO_2]Cl$ freezes at $-0.00732^\circ C$

Find i ? ($K_f = 1.86^\circ C/m$) $[Co(NH_3)_5NO_2]Cl$

$$\Delta T_f = i K_f m$$

$$0 - (-0.00732) = i \times 1.86 \times 2 \times 10^{-3}$$

$$[i = 2]$$

Q. The V.P of pure $CH_3OH(A)$ is 0.15 bar at $25^\circ C$. The V.P of this liq. in solⁿ is 0.09 bar calculate mole fraction of B?

$$P_A^\circ = 0.15 \text{ bar}; \quad P_A = 0.09 \text{ bar}$$

$$P_A = P_A^\circ X_A \Rightarrow \frac{0.09}{0.15} = X_A$$

$$X_A = \frac{3}{5}; \quad X_B = 1 - \frac{3}{5} \Rightarrow \frac{2}{5}$$

Q. At $40^\circ C$ the vap. pressure of $CH_3OH(A)$ and $EtOH(B)$ solⁿ is represented by

$$P = 120x_A + 138 \text{ torr} \text{ where } x_A \text{ is mole}$$

fraction of MeOH. Find values of $\left(\frac{P_B^\circ}{X_B}\right)_{X_A \rightarrow 0}$ and $\left(\frac{P_A^\circ}{X_A}\right)_{X_A \rightarrow 1}$

$$X_B = 1; \quad X_A = 0 \quad P_B^\circ = 138 \text{ torr}$$

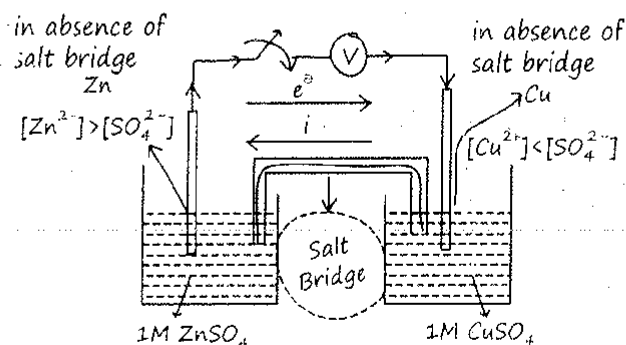
$$X_B = 0; \quad X_A = 1 \quad P_A^\circ = 258 \text{ torr}$$

$$\left(\frac{P_B^\circ}{X_B}\right)_{X_A \rightarrow 0} = 138 \text{ torr}$$

$$\left(\frac{P_A^\circ}{X_A}\right)_{X_A \rightarrow 1} = 258 \text{ torr}$$

1. Electrochemical Cell (Galvanic or Voltaic cell)

Daniel Cell

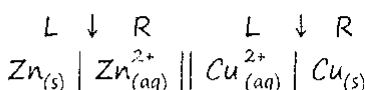


Anode Oxi: $\text{Zn} \rightarrow \text{Zn}^{2+} + 2e^-$

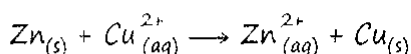
Red Cathode: $\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}$

Anode [L O A N] (RRCP)
Left Oxidation Anode

Cell representation



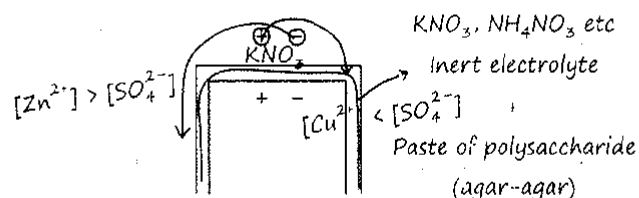
Cell Rxn



Standard Cell potential (E°_{cell})

$$E^\circ_{\text{cell}} = (E^\circ_R)_C - (E^\circ_R)_A$$

Salt Bridge & its funcⁿ



+ KCl is not used as electrolyte in case of:

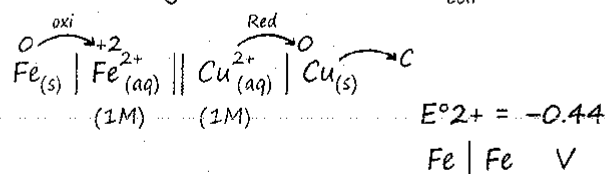
Ag, Tl, Pb, Hg
Aji Thailand Pyar Hogya
Main

+ Mobility of Cation $\approx$ Mobility of anion

Main Function of salt bridge.

- (1) To maintain electrical neutrality
- (2) To complete inner circuit without mixing of two solⁿ
- (3) To reduce liq-liq junc. potential

Q. For following Galvanic cell find E°_{cell}



Given $E^\circ \text{Fe} \mid \text{Fe}^{2+} = +0.44 \text{ V}$

$$E^\circ \text{Cu} \mid \text{Cu} = +0.34 \text{ V}$$

$$\begin{aligned} E^\circ_{\text{cell}} &= (E^\circ_R)_C - (E^\circ_R)_A \\ &= (+0.34) - (-0.44) \\ &= 0.78 \text{ V} \end{aligned}$$

ΔG° and K_{eq} for Galvanic Cell

$$(1) \Delta G^\circ = -nFE^\circ_{\text{cell}}$$

n = No. of electrons involved in balanced cell reaction

$$F = 96500 \text{ C}$$

$$(2) \Delta G^\circ = -RT \ln K_{\text{eq}}$$

$$R = 2 \text{ cal} \left(= \frac{25}{3} \right) \text{ J/mol/K}$$

Q. Find ΔG° & K_{eq} for Daniel cell

$$E^\circ_{\text{cell}} = 1.1 \text{ V}$$

$$\Delta G^\circ = -2 \times 96500 \times 1.1 = -212.3 \times 10^3$$

$$\Delta G^\circ = -RT \ln K_{\text{eq}} = \frac{-25}{3} \times 298 \ln K_{\text{eq}}$$

$$[\Delta G^\circ = -212.3 \text{ kJ}] \quad [K_{\text{eq}} = 10^{37}]$$

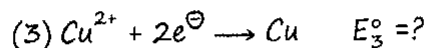
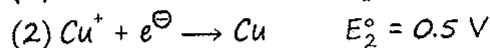
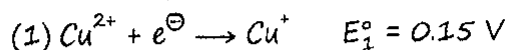
ΔG° is Extensive (additive)

E°_{cell} is Intensive (non additive)

Q. Given $E^\circ_{\text{Cu}^{2+} \mid \text{Cu}^+} = +0.15 \text{ V}$

$$E^\circ_{\text{Cu}^+ \mid \text{Cu}} = +0.5 \text{ V}$$

Find $E^\circ_{\text{Cu}^{2+}|\text{Cu}}$



$(3) = (1) + (2)$

$\Delta G_3^\circ = \Delta G_1^\circ + \Delta G_2^\circ$

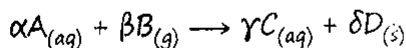
$-n_3 FE_3^\circ = -n_1 FE_1^\circ - n_2 FE_2^\circ$

$n_3 FE_3^\circ = n_1 FE_1^\circ + n_2 FE_2^\circ$

$2E_3^\circ = 1 \times (0.15) + 1 \times (0.5)$

$E_3^\circ = \frac{0.65}{2} = 0.325 \text{ V}$

Nernst Eqⁿ

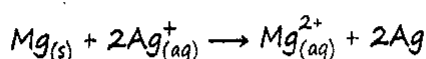
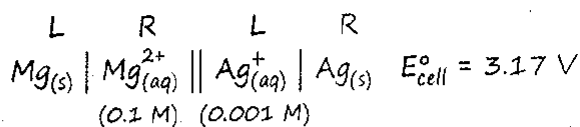


$E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{0.06}{n} \log \frac{[C]^\gamma}{[A]^\alpha P_B^\beta}$

At Eq^m $E_{\text{cell}} = 0 \quad Q = K_{\text{eq}} \Rightarrow E_{\text{cell}}^\circ - \frac{RT}{nf} \ln K_{\text{eq}}$

$[E_{\text{cell}}^\circ = \frac{0.06}{n} \log \text{at } 25^\circ\text{C } K_{\text{eq}}]$

Q. Find EMF of cell



$E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{0.06}{n} \log \frac{[\text{Mg}^{2+}]}{[\text{Ag}^+]^2}$

$= 3.17 - \frac{0.06}{2} \log \frac{10^{-1}}{(10^{-3})^2}$

$= 3.17 - 0.03 \times \log 10^5$

$= 3.17 - 0.03 \times 5$

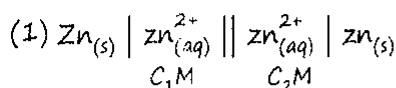
$= 3.17 - 0.15$

$= 3.02 \text{ V}$

Concentration Cell

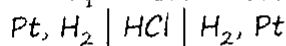
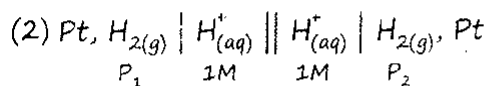
Same half electrode is present with diff concⁿ or pressure

$E_{\text{cell}}^\circ = 0$



$E_{\text{cell}} = \frac{0.06}{n} \log \left(\frac{C_2}{C_1} \right)$

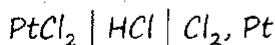
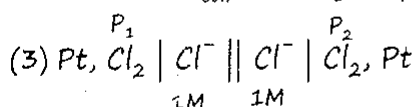
Condⁿ for spontaneous $\Delta G < 0 \Rightarrow E_{\text{cell}} > 0 \Rightarrow C_2 > C_1$



$E_{\text{cell}} = \frac{0.06}{n} \log \left(\frac{P_1}{P_2} \right)$

Condⁿ for spontaneous

$\Delta G < 0 ; E_{\text{cell}} > 0 ; P_1 > P_2$



$E_{\text{cell}} = \frac{0.06}{n} \log \left(\frac{P_2}{P_1} \right)$

Condⁿ for spontaneous

$\Delta G < 0 ; E_{\text{cell}} > 0 ; P_2 > P_1$

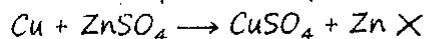
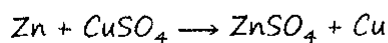
2. Electrochemical Series

Le	(Li)	Ni	Neeche	kahe
kar	K	Sn	Suno	$E_R^\circ = -ve$
Barah	Ba	Pb	Pyar	$\uparrow$
Sardar	Sr	(H)	hai	$E_R^\circ = 0$
Ka	Ca	Cu	Kyun	$\downarrow$
Naam	Na	I	aai	$E_R^\circ = +ve$
Maonge	Mg	Ag	Aaj	
ali	Al	Hg	Hongkong main	
mahan	Mn	Br	Britney	
zan	Zn	Pt	patai	
kar	Cr	O	aur	
Fen'ke	Fe	Cl	kilobar	
C.D.	Cd	Au	Sona Laai	
loi	Co	(F)	free main	

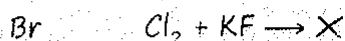
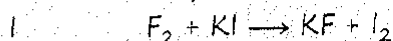
Apps

(1) Reactivity of metals and Reducing power decreases top to bottom.

(2) Metal displacement



(3) Non metal displacement



Cl

F

Electrochemical Series

Q. Metal $E_R^\circ (M^{2+}/1M)$

Li $-3.05 V$

Ba $-2.9 V$

Na $-2.71 V$

Mg $-2.37 V$

Compare reducing power

Li > Ba > Na > Mg

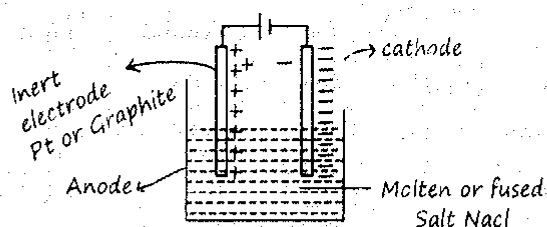
Q. Compare reactivity

Ba, Zn, K, Cu

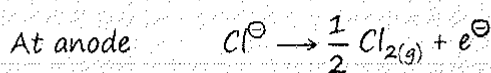
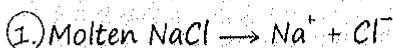
[K > Ba > Zn > Cu]

Reactivity

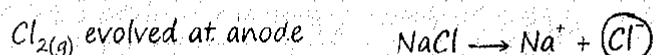
3. Electrolytic Cell



Products of electrolysis



Na(s) deposited at cathode

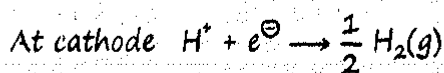


P.D.T. (Preferential discharge theory)

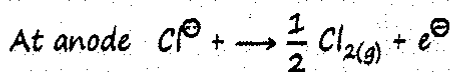
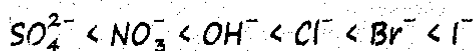
Case 1: For cation

$E_R^\circ \uparrow$ top to bottom

that ion will migrate first



Case 2: Ease of Migration



(1) $H_{2(g)}$ evolved at Cathode
(2) $Cl_{2(g)}$ evolved at Anode

Nature after Electrolysis is Basic medium

Electrolyte	Electrode	Product at cathode	Product at anode
Aq $CuSO_4$	Pt or graphite	Cu	O_2
Aq $AgNO_3$	Pt or graphite	Ag	O_2
Aq NaOH	Pt or graphite	H_2	O_2
Aq $AgNO_3$	Ag electrode	$Ag^+ + e^- \rightarrow Ag$	$Ag \rightarrow Ag^+ + e^-$
Conc H_2SO_4	Pt or graphite	$2H^+ + 2e^- \rightarrow H_2$	$2SO_4^{2-} \rightarrow S_2O_8^{2-} + 2e^-$
Aq NaCl	Hg electrode	$Na^+ + e^- \rightarrow Na \xrightarrow{Hg} Na_{(Hg)}$	Cl_2

Faradays Law

1st Law $W \propto Q$

$W = ZQ$ (Z = electrochemical equivalent)

$$W = \frac{E \cdot It}{F}$$

$$\frac{W}{E} = \frac{It}{F}$$

2nd Law

$$[n_1 \times n_{f1} = n_2 \times n_{f2}]$$

$$\left[n \times n_f = \frac{It}{F} \right]$$

Q. How long should a current of 0.25 Amp be passed through a molten metal salt to deposit that much weight of metal which is equal to its Electrochemical equivalent.

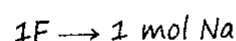
$$W = Z ; W = ZQ$$

$$Q = 1 ; it = 1$$

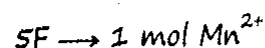
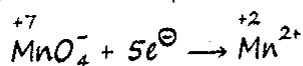
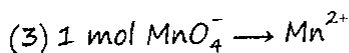
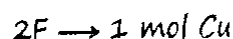
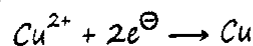
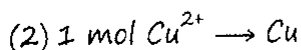
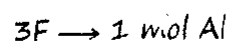
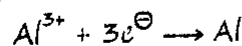
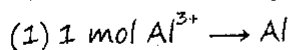
$$0.25 \times t = 1$$

$$[t = 4 \text{ amp}]$$

3rd Low



Q. How much charge is req for following Rxns



4. Resistance, Conductance, resistivity conductivity of a cell

(1) Resistance (R) (ohm)

$$[R_{\text{unit}} = \frac{\rho l}{A}] \Omega \text{ (ohm)}$$

ρ = Resistivity or sp. resistance

l = Sep. b/w electrodes (cm)

A = Area of cross-section (cm²)

(2) Resistivity (unit)

$$[\rho_{\text{unit}} = \frac{RA}{l}] \text{ ohm-cm } (\Omega\text{-cm})$$

(3) Conductance (G)

(ohm)⁻¹ or mho

((Ω)⁻¹ or $\mathcal{U}$)

$$[G = \frac{1}{R}]$$

(4) Conductivity or sp. conductance (K) (unit)

$$K_{\text{unit}} = \frac{1}{\rho}$$

$$[K = \frac{l}{RA}] \quad \begin{matrix} \text{ohm}^{-1} \text{ cm}^{-1} \\ \text{Mho cm}^{-1} \end{matrix}$$

(5) Cell const (σ or G^*) (Ω)⁻¹ cm⁻¹

$$[\sigma = \frac{l}{A}] \text{ cm}^{-1} \quad \begin{matrix} \mathcal{U} \text{ cm}^{-1} \\ \text{S cm}^{-1} \end{matrix}$$

Q. If K of 0.1 N KCl is $1.2 \times 10^{-2} \text{ S cm}^{-1}$ and

$R = 60 \Omega$ find G^*

$$K = 1.2 \times 10^{-2} \text{ S cm}^{-1}$$

$$R = 60$$

$$R = \frac{\rho l}{A}$$

$$R = \frac{1}{K} \times G^*$$

$$G^* = RK = 60 \times 1.2 \times 10^{-2}$$

$$= 72 \times 10^{-2}$$

$$= 0.72 \text{ cm}^{-1}$$

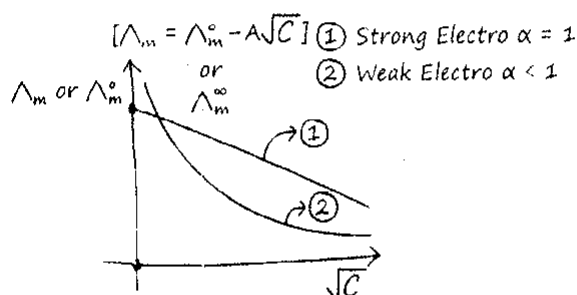
Molar Conductivity

$$[\Lambda_m = \frac{K \times 1000}{M}] \quad (N = M \times n.f.)$$

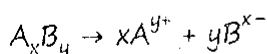
Equivalent Conductivity

$$[\Lambda_{eq} = \frac{K \times 1000}{N}] \quad [\Lambda_{eq} = \frac{\Lambda_m}{n.f.}]$$

Variation of Λ_m & Λ_{eq} with Concⁿ



5. KOHLRAUSCH LAW



$$\Lambda_m^\circ (A_x B_y) = x \times \lambda_m^\circ A^{y+} + y \times \lambda_m^\circ B^{x-}$$

$$\lambda_{eq}^\circ (A_x B_y) = \lambda_{eq}^\circ A^{y+} + \lambda_{eq}^\circ B^{x-}$$



$$\Lambda_m^\circ = \lambda_m^\circ(\text{Ca}^{2+}) + 2 \times \lambda_m^\circ(\text{Cl}^-)$$

$$\Lambda_{eq}^\circ = \lambda_{eq}^\circ(\text{Ca}^{2+}) + \lambda_{eq}^\circ(\text{Cl}^-)$$

App of K.L

$$(1) \alpha = \frac{\Lambda_m}{\Lambda_m^\circ}$$

$$(2) K_a = \frac{C\alpha^2}{1-\alpha}$$

$$(3) \Lambda_m = \frac{K \times 1000}{S} \quad K_{sp} = x^x y^y s^{x+y}$$

$$(4) \text{CH}_3\text{COONa}, \text{CH}_3\text{COOH}, \text{HCl}, \text{NaCl}$$

$$\Lambda_{m\text{CH}_3\text{COOH}}^\circ = \Lambda_{m\text{CH}_3\text{COONa}}^\circ + \Lambda_{m\text{HCl}}^\circ - \Lambda_{m\text{NaCl}}^\circ$$

Q. The conductivity of $10^{-2} \text{ M CH}_3\text{COOH}$ is $3.9 \times 10^{-5} \text{ S cm}^{-1}$ find K_a & α (Given $\Lambda_{m\text{CH}_3\text{COONa}}^\circ = 390 \text{ S cm}^2 \text{ mol}^{-1}$)

$$C = 10^{-2}, K = 3.9 \times 10^{-5}$$

$$\Lambda_m^C = \frac{3.9 \times 10^{-5} \times 1000}{10^{-2}}$$

$$= 3.9$$

$$\alpha = \frac{3.9}{390} = 10^{-2}$$

$$K_a = \frac{10^{-2} \times (10^{-2})^2}{1 - 10^{-2}} = 10^{-6}$$

Q. The conductivity of saturated soln of $\text{Ba}_3(\text{PO}_4)_2$ is $1.2 \times 10^{-5} \text{ S cm}^{-1}$ & Λ_m° of $\text{Ba}_3(\text{PO}_4)_2$ is $1200 \text{ S cm}^2 \text{ mol}^{-1}$ find K_{sp}

$$\Lambda_m^\circ = \frac{K \times 1000}{S}$$

$$1200 = \frac{1.2 \times 10^{-5} \times 1000}{S \times 10^3}$$

$$S = 10^{-5}$$

$$K_{sp} = 3^3 \times 2^2 \times S^{3+2} = 27 \times 4 \times S^5 = 108 \times (10^{-5})^5 = 108 \times 10^{-25}$$

Q. Calculate $\Lambda_{m\text{CH}_3\text{COOH}}^\circ$

Given Electrolyte $\text{S cm}^2 \text{ mol}^{-1}$

CH_3COONa 91

HCl 426

NaCl 126

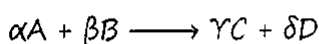
$$\Lambda_{m\text{CH}_3\text{COOH}}^\circ = \Lambda_{m\text{CH}_3\text{COONa}}^\circ + \Lambda_{m\text{HCl}}^\circ - \Lambda_{m\text{NaCl}}^\circ$$

$$= 91 + 426 - 126$$

$$= 391 \text{ S cm}^2 \text{ mol}^{-1}$$

Effect of dilution on K & Λ_m

dilution $\uparrow K \downarrow \Lambda_m \uparrow$

Rate of Reaction

Unit of R.O.R
 $\left(\frac{\text{mol l}^{-1} \text{ time}^{-1}}{\text{atm time}^{-1}} \right)$

Avg

$$\text{R.O.R} = -\frac{\Delta[A]}{\alpha \Delta t} = -\frac{\Delta[B]}{\beta \Delta t} = +\frac{\Delta[C]}{\gamma \Delta t} = +\frac{\Delta[D]}{\delta \Delta t}$$

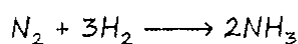
$$\text{R.O. disappearance of A} = -\frac{\Delta[A]}{\Delta t} \quad \text{R.O.D of B} = -\frac{\Delta[B]}{\Delta t}$$

$$\text{R.O. Appearance of C} = +\frac{\Delta[C]}{\Delta t} \quad \text{R.O.A of D} = +\frac{\Delta[D]}{\Delta t}$$

$$\text{Instantaneous R.O.R} = -\frac{d[A]}{\alpha dt} = -\frac{d[B]}{\beta dt} = +\frac{d[C]}{\gamma dt} = +\frac{d[D]}{\delta dt}$$

At $\Delta t \longrightarrow 0$; Avg rate = Instantaneous rate

Q. Write Expression of Rate of Rxn



$$\text{Avg R.O.R} = \frac{-\Delta[\text{N}_2]}{1 \Delta t} = \frac{-\Delta[\text{H}_2]}{3 \Delta t} = \frac{+\Delta[\text{NH}_3]}{2 \Delta t}$$

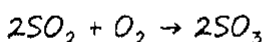
$$\text{Instantaneous R.O.R} = \frac{-d[\text{N}_2]}{1 dt} = \frac{-d[\text{H}_2]}{3 dt} = \frac{+d[\text{NH}_3]}{2 dt}$$

$$\text{R.O.D of N}_2 = \frac{-\Delta[\text{N}_2]}{\Delta t}$$

$$\text{R.O.D of H}_2 = \frac{-\Delta[\text{H}_2]}{\Delta t}$$

$$\text{R.O.A of NH}_3 = \frac{+\Delta[\text{NH}_3]}{\Delta t}$$

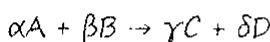
Q. If R.O.D of SO_2 is $2 \times 10^{-3} \text{ M sec}^{-1}$ then find R.O.A of SO_3



$$\text{R.O.R} = \frac{-\Delta[\text{SO}_2]}{2 \Delta t} = \frac{-\Delta[\text{O}_2]}{1 \Delta t} = \frac{+\Delta[\text{SO}_3]}{2 \Delta t}$$

$$\frac{\text{R.O.D of SO}_2}{2} = \frac{\text{R.O.A of SO}_3}{2}$$

$$\text{R.O.A of SO}_3 = 2 \times 10^{-3} \text{ M sec}^{-1}$$

Rate Law

$$\gamma \alpha [A]^x [B]^y \Rightarrow \begin{cases} \gamma = K[A]^x [B]^y \\ \gamma = \text{R.O.R} \end{cases}$$

K = rate const. or velocity const

$\alpha = x, \beta = y$ for elementary reaction
or

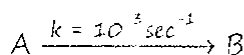
x & $y \Rightarrow$ orders i^{th} step reaction
or

x = order w.r.t A Single step reaction

y = order w.r.t B R.D.S

$x + y = P$ overall order [Unit of K =
or of reaction $(\text{M})^{1-P} \text{ time}^{-1}$]
 n

Q. Find order of the Reaction



$$\text{Unit of } k = (\text{M})^{1-P} \text{ sec}^{-1}$$

$$[P=1] \quad 1^{\text{st}} \text{ order}$$

Q. For $r = k[A]^n[B]^m$ o. doubling the concⁿ of A and halving the concⁿ of B, the ratio of new rate to the earlier rate will be?

$$r^1 = k 2^n [A]^n \frac{1}{2^m} [B]^m = k 2^{n-m} [A]^n [B]^m$$

$$\left[\frac{r^1}{r} = 2^{n-m} \right]$$

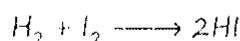
Q. Find the unit of rate const for zero and 2nd order reaction?

$$\text{Unit of } k \Rightarrow (\text{M})^{1-P} \text{ time}^{-1}$$

$$P = 0 \quad \text{M time}^{-1}$$

$$P = 2 \quad \text{M}^{-1} \text{ time}^{-1}$$

Q. Write R.O.R for



Instantaneous R.O.R

$$= \frac{-d[\text{H}_2]}{1 dt} = \frac{-d[\text{I}_2]}{1 dt} = \frac{+d[\text{HI}]}{2 dt}$$

Order	Molecularity
(1) Experimental quantity	(1) Theoretical quantity
(2) Order can be zero, -ve, +ve fractional	(2) Molecularity can be 1, 2 and 3
(3) Order = Stoichiometric coefficient for 1 step or Elementary Rxn	(3) Molecularity = Stoichiometric Coefficient
(4) For complex Rxn order can be calculated by mech.	(4) There is no meaning of molecularity for complex reaction. R.D.S → Molecularity can be defined

Calculation of order of Reaction

Case 1: Find order of given reaction: $2A + 3B \rightarrow 4C$

Exp	[A]	[B]	Initial rate
1	0.1 M	0.1 M	$2 \times 10^{-4} \text{ M sec}^{-1}$
2	0.1 M	0.2 M	$4 \times 10^{-4} \text{ M sec}^{-1}$
3	0.2 M	0.2 M	$16 \times 10^{-4} \text{ M sec}^{-1}$

$$r = k[A]^x [B]^y \quad [r = k[A]^2 [B]^1]$$

Find (k)

$$2 \times 10^{-4} = k(0.1)^x (0.1)^y \quad \text{---(1)} \quad 2 \times 10^{-4} = k(0.1)^x (0.1)$$

$$4 \times 10^{-4} = k(0.1)^x (0.2)^y \quad \text{---(2)} \quad k = 0.2 \text{ (M)}^{-2} \text{ sec}^{-1}$$

$$16 \times 10^{-4} = k(0.2)^x (0.2)^y \quad \text{---(3)}$$

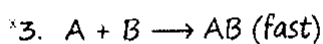
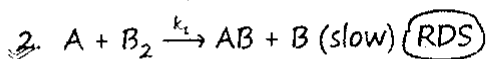
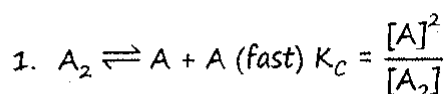
$$\frac{\textcircled{1}}{\textcircled{2}} \quad \frac{1}{2} = \left(\frac{1}{2}\right)^y \quad \frac{\textcircled{2}}{\textcircled{3}} \quad \frac{1}{4} = \left(\frac{1}{2}\right)^x \quad \text{Overall order}$$

$$y = 1$$

$$x = 2$$

$$= 3$$

Case 2: Find order of given reaction $A_2 + B_2 \xrightarrow{k} 2AB$; follows the mech below $r = k[A_2]^x [B_2]^y$



$$r = k_1[A]^1 [B_2]^1$$

$$= k_1 (K_c[A_2])^{\frac{1}{2}} [B_2]$$

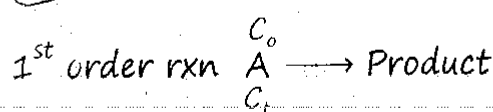
$$r = k[A_2]^{\frac{1}{2}} [B_2]$$

$$\text{Order w.r.t } A_2 = \frac{1}{2}$$

$$\text{Order w.r.t } B_2 = 1$$

$$\text{Overall order} = \frac{1}{2} + 1 = \frac{3}{2}$$

M.I.*



$$1. \frac{-d[A]}{dt} = k[A]$$

$$2. kt = \ln\left(\frac{C_o}{C_t}\right) = \ln\left(\frac{a}{a-x}\right)$$

$$3. t_{50\%} = \frac{\ln 2}{k} = \frac{0.693}{k} \quad (\text{Independent of } C_o)$$

$$4. t_{75\%} \quad t_{90\%} \quad t_{99\%} \quad t_{99.9\%} \quad t_{99.99\%}$$

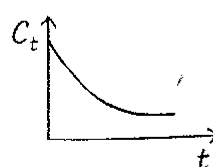
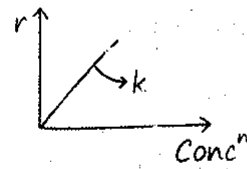
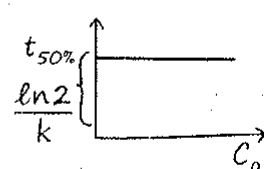
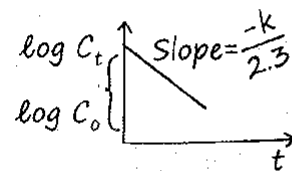
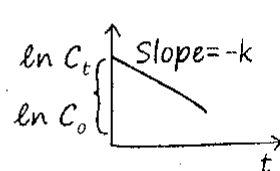
$$2t_{50\%} \quad 3.33t_{1/2} \quad 6.66t_{1/2} \quad 10t_{1/2} \quad 13.34t_{1/2}$$

$$[kt_{x\%} = \ln\left(\frac{100}{100-x}\right)]$$

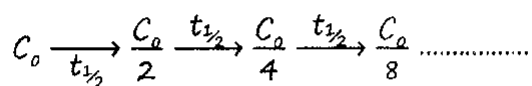
$$5. \text{D.O.D} \quad \alpha = 1 - e^{-kt}$$

$$6. C_t = C_o e^{-kt}$$

7. Graphs



Q. For 1st order rxn the concⁿ of reactant decreases from 0.8 M to 0.4 M in 15 min. The time taken for the concⁿ to change from 0.1 M to 0.025 M is



$$0.8 \text{ M} \xrightarrow{t_{1/2}=15} 0.4 \text{ M} \left\{ \begin{array}{l} 0.1 \xrightarrow{t_{1/2}} 0.05 \xrightarrow{t_{1/2}} 0.025 \\ t = 2t_{1/2} = 2 \times 15 = 30 \text{ min} \end{array} \right.$$

Q. In the given eqⁿ $kt = \ln C_0 - \ln C_t$ the curve b/w $\ln C_t$ vs t is

$$[\ln C_t = -kt + \ln C_0] \quad \text{Straight line}$$

$$[y = -mx + C]$$

Q. $t = 0 \quad A_{(g)} \rightarrow 2B_{(g)} + C_{(g)}$ Initial pressure = P_i
 $t = t; P_i - P_x \quad 2P_x \quad P_x$ Pressure at time $t = P_t$
 Write expression of k ?

$$P_t = P_i - P_x + 2P_x + P_x \quad (kt = \ln \frac{a}{a-x})$$

$$P_t = P_i + 2P_x$$

$$P_x = \left(\frac{P_t - P_i}{2} \right)$$

$$(kt = \ln \left(\frac{P_i}{P_i - P_x} \right))$$

$$kt = \ln \left(\frac{P_i}{P_i - \left(\frac{P_t - P_i}{2} \right)} \right)$$

$$= \ln \left(\frac{2P_i}{2P_i - P_t + P_i} \right)$$

$$[kt = \ln \left(\frac{2P_i}{3P_i - P_t} \right)]$$

n^{th} order rxn

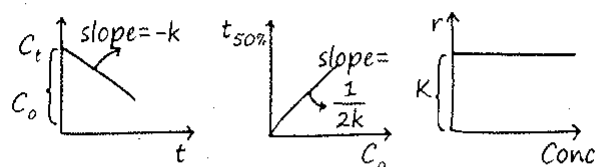
$$\left[kt = \frac{C_0^{1-n} - C_t^{1-n}}{1-n} \right] \quad n \neq 1$$

$$t_{1/2} \propto C_0^{1-n}$$

Zero order $n = 0$	2 nd order $n = 2$
$C_t = -kt + C_0$	$\frac{1}{C_t} = kt + \frac{1}{C_0}$
$t_{50\%} = \frac{C_0}{2k} [t_{1/2} \propto C_0]$	$t_{50\%} = \frac{1}{kC_0} t_{50\%} \propto \frac{1}{C_0}$
$[T_{75\%} = 1.5 t_{50\%}]$	$[t_{75\%} = 3 t_{50\%}]$

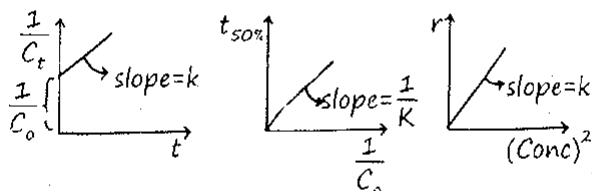
1. Zero order rxn

$$r = k[A]^0 = k$$



2. 2nd order rxn

$$r = k[A]^2$$



Q. For a zero order rxn, $k = 0.01 \text{ M sec}^{-1}$. Find concⁿ of reactant after 10 sec if initial concⁿ is 1 M.

$$C_t = -kt + C_0$$

$$C_t = -10^{-2} \times 10 + 1$$

$$= 1 - 0.1$$

$$[C_t = 0.9 \text{ M}]$$

Q. For 2nd order rxn find C_0 & k

$$\begin{aligned} \frac{1}{C_t} &= kt + \frac{1}{C_0} \\ \text{Slope} &= \frac{1}{100} \\ \frac{1}{C_0} &= 0.5 \Rightarrow [C_0 = 2 \text{ M}] \\ k &= \frac{1}{100} \\ &= 10^{-2} \text{ M}^{-1} \text{ sec}^{-1} \end{aligned}$$

Temp. dependence on rate & rate const

$$\frac{r_{T+10}}{r_T} = (\mu)^{\frac{\Delta T}{10}} \quad (\mu = \text{Temp. coeff. (2 to 3)})$$

$$= (2)^{\frac{\Delta T}{10}} \quad (\mu = 2 \text{ generally})$$

Q. If Temp. Coeff. is 2 then calculate rate of rxn at 60°C if at 20°C it is r ?

$$\frac{r_{60^\circ\text{C}}}{r_{20^\circ\text{C}}} = (2)^{\frac{40}{10}} = 2^4$$

$$[r_{60^\circ\text{C}} = 16r_{20^\circ\text{C}}]$$

Arrhenius Eqⁿ

$$[k = Ae^{-E_a/RT}]$$

k = rate const

A = Arrhenius const or frequency factor

E_a = Activation energy

R = Universal gas const

$$= 2 \text{ Cal } \text{K}^{-1} \text{ mol}^{-1} = \frac{25}{3} \left(= 8.3 \right) \text{ J } \text{K}^{-1} \text{ mol}^{-1}$$

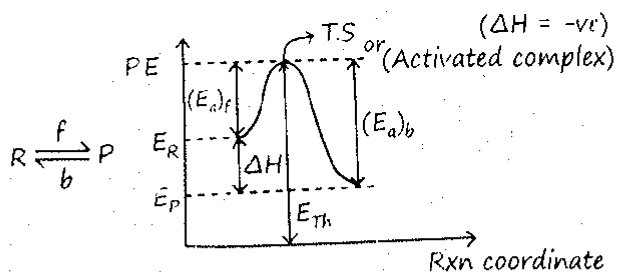
T = Temp (K)

$$\ln\left(\frac{k_2}{k_1}\right) = \frac{E_a}{R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

$$\log\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.3R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

Energy profile diagrams

Case 1: For Exothermic Rxn

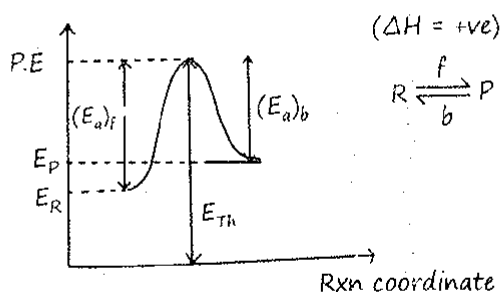


$$\Delta H = E_P - E_R = (E_a)_f - (E_a)_b$$

E_{Th} = Threshold energy

E_a = Activation energy

Case 2: For Endothermic Rxn



$$\Delta H = E_P - E_R = (E_a)_f - (E_a)_b$$

Practice

Q. If concⁿ of reactants is increased by 'X' the rate const K becomes

- (A) $e^{K/X}$ (B) K/X
(C) K (D) X/K

Q. For rxn $XA \longrightarrow YB$ ($\log 2 = 0.3$)

$$\log\left(\frac{-d[A]}{dt}\right) = \log\left(\frac{-d[B]}{dt}\right) + 0.3$$

Find $\frac{X}{Y}$

$$\log\left(\frac{-d[A]}{dt}\right) = \log\left(\frac{-d[B]}{dt}\right) + \log 2$$

$$\frac{-d[A]}{dt} = 2 \left(\frac{-d[B]}{dt} \right)$$

$$X = 2$$

$$y = 1 \Rightarrow \frac{X}{Y} = \frac{2}{1}$$

Q. For the rxn $A \xrightarrow{k} \text{products}$, the rate of rxn is doubled when the concⁿ of A is increased by 4 times find order of reaction

$$r = k[A]^x \quad 2 = 2^{2x}$$

$$2r = k(4[A])^x \quad 2x = 1$$

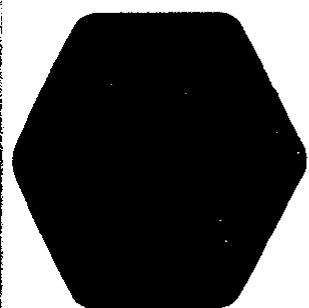
$$2 = 4^x \quad x = \frac{1}{2}$$

Q. Find $(E_a)_b$ if $\Delta H = +54 \text{ kJ/mol}$ for Rxn $N_2O_4 \longrightarrow 2NO_2$ and $E_a(f) = +57 \text{ kJ/mol}$

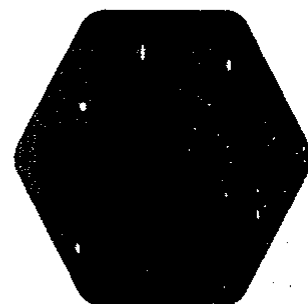
$$\Delta H = (E_a)_f - (E_a)_b$$

$$54 = 57 - (E_a)_b$$

$$[(E_a)_b = +3 \text{ kJ/mol}]$$



Inorganic Chemistry



1. The first part of the
document is a list of
the names of the
persons who were
present at the
meeting. The names
are listed in
alphabetical order.
The names are
as follows:

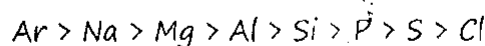
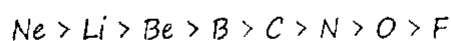
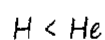
1. The first part of the
document is a list of
the names of the
persons who were
present at the
meeting. The names
are listed in
alphabetical order.
The names are
as follows:

Atomic size

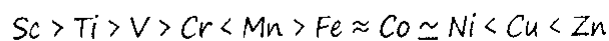
- + In period (L to R) $\div z_{\text{eff}} \uparrow$ Attrcⁿ $\uparrow$ size $\downarrow$
 + In group (T to B) $\div$ shell no $\uparrow$ size $\uparrow$

Exceptions

1. Inert gas has highest radii in respective period



2. 3d series



3. $r_{Al} \approx r_{Ga}$ (due to poor screening of 3d $e^{\ominus}$ s)

Q. Compare size

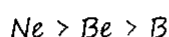
1. Li, K, Na



2. V, Fe, Mn



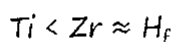
3. Ne, B, Be



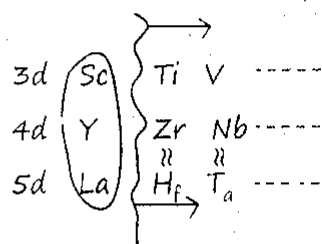
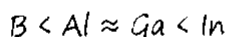
4. Sc, Y, La



5. Ti, Zr, Hf



6. Al, B, Ga, In



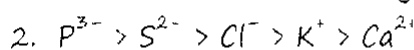
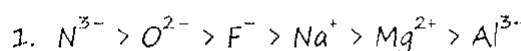
1. Ionic Radius

1. L to R

Isoelectronic species

$$\left[\text{Size} \propto \frac{1}{\text{Atomic no.}} \right]$$

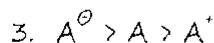
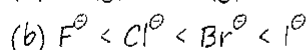
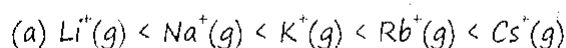
M.I*



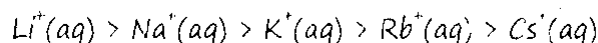
2. T to B

Shell no $\uparrow$ Size $\uparrow$

M.I*



2. Hydrated radius



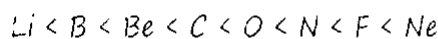
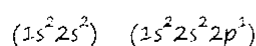
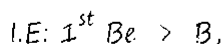
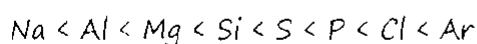
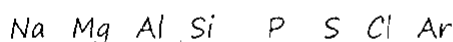
Imp.

Hydrated radius $\uparrow$ conductance or ionic mobility $\downarrow$

3. Ionisation Energy

(ΔH or IE or IP)

1. Inert gas has highest I.E. in respective period due to fully filled configuration.

2. 2nd period3. 3rd period

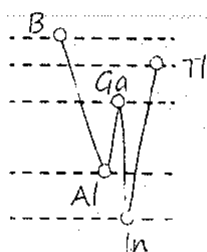
2nd I.E. $\text{Be} < \text{B}$

I.E. 1st $\text{N} > \text{O}$

$(1s^2 2s^2 2p^3) (1s^2 2s^2 2p^4)$

2nd I.E. $\text{N} < \text{O}$

4. Boron family



$\text{B} > \text{Tl} > \text{Ga} > \text{Al} > \text{In}$

$\text{Tl} > \text{In}$ (Lanthanoid Contraction)

$\text{Ga} > \text{Al}$ (poor screening of $3d$ electrons)

5. $[5d > 3d > 4d]$



(due to L.C.)

Q. Compare I.E.

1. $\text{Be} < \text{C} < \text{N}$

$\text{Be} < \text{C} < \text{N}$

2. $\text{O} < \text{N} < \text{Ne}$

$\text{Ne} > \text{N} > \text{O}$

3. $\text{P} < \text{S} < \text{Ar}$

$\text{Ar} > \text{P} > \text{S}$

4. $\text{Ti} < \text{Zr} < \text{Hf}$

$\text{Hf} > \text{Ti} > \text{Zr}$

5. $\text{Mg} < \text{Ca} < \text{Sr}$

$\text{Mg} > \text{Ca} > \text{Sr}$

6. $\text{Al} < \text{Ga} < \text{In}$

$\text{Ga} > \text{Al} > \text{In}$

4. Electron gain enthalpy ($\Delta_{eg}H$)

Exceptions

1. 2nd period

$\text{Be} < \text{Ne} < \text{N} < \text{B} < \text{Li} < \text{C} < \text{O} < \text{F}$

Beta ne na Bolkar Ladki Ko Fasaya

(zero or +ve)

2. 3rd period

$\text{Mg} < \text{Ar} < \text{Al} < \text{Na} < \text{P} < \text{Si} < \text{S} < \text{Cl}$

maine aur alka ne pyari si shadi karli

(zero or +ve)

3. 3 period $>$ 2 period

(due to vacant 'd' orbitals)

4. $\text{Cl} > \text{F} > \text{Br} > \text{I}$

5. $\text{S} > \text{Se} > \text{Te} > \text{Po} > \text{O}$

due to compact nature

6. Inert gases have zero or +ve electron gain enthalpy.

Q. Compare $\Delta_{eg}H$

1. $\text{Li} < \text{Be} < \text{O}$

$\text{Be} < \text{Li} < \text{O}$

2. $\text{Na} < \text{Ar} < \text{Si}$

$\text{Ar} < \text{Na} < \text{Si}$

3. $\text{Se} < \text{O} < \text{S}$

$\text{S} > \text{Se} > \text{O}$

4. $\text{F} < \text{Cl} < \text{Br}$

$\text{Cl} > \text{F} > \text{Br}$

5. $\text{Si} < \text{C}$

$\text{Si} > \text{C}$

6. $\text{N} < \text{B} < \text{F}$

$\text{N} < \text{B} < \text{F}$

5. Electronegativity

The tendency of an atom to attract shared pair e^- s towards itself.

$\text{A} \xrightarrow{+} \text{B} \quad (\text{B} > \text{A}) \text{ EN}$

$\text{A} \xleftarrow{+} \text{B} \quad (\text{B} < \text{A}) \text{ EN}$

Trends

1. L to R $\text{EN} \uparrow$

2. T to B $\text{EN} \downarrow$

1 1.5 2 2.5 3 3.5 4

Li Be B C N O F

H, P (2.1)

C, I, S (2.5)

N, Cl

3 3.1

$\text{A} < \text{A}^+ < \text{A}^{2+}$

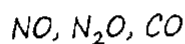
$\text{A} > \text{A}^- > \text{A}^{2-}$

6. Nature of oxides

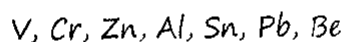
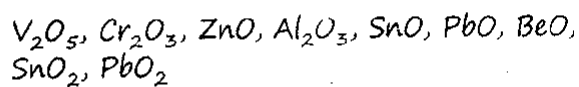
1. Non metallic oxides → Acidic

2. Metallic oxides → Basic

3. Neutral oxides



4. Amphoteric oxides

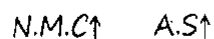


5. Amphoteric Metals

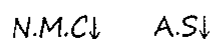
Vema car zanabe aali snak pade bey
main

Acidic Strength

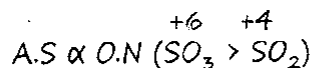
1. L to R



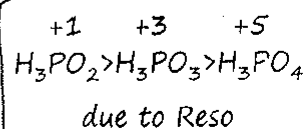
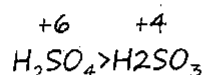
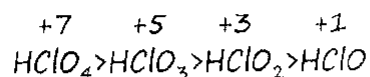
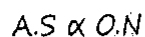
2. T to B



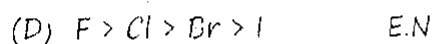
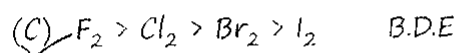
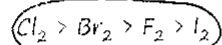
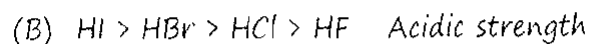
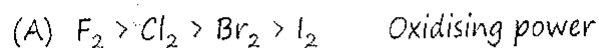
3. Oxides of same element



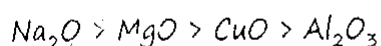
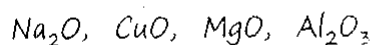
4. A.S of oxyacids



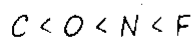
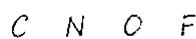
Q. Which of the following order is incorrect according to the prop against it?



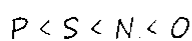
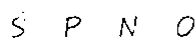
Q. Compare basic strength



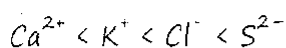
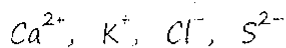
Q. Compare I.P



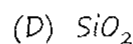
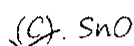
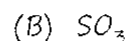
Q. Compare E.N



Q. Compare ionic radii



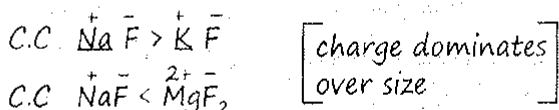
Q. Which is an amphoteric oxide



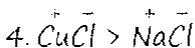
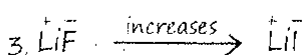
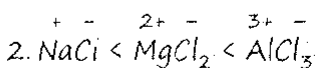
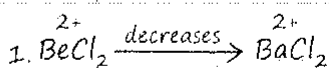
1. Fajan's rule

Small cation, polarising power ↑ bigger anion
Polarisability ↑

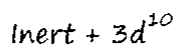
& high charge on cation or anion causes
polarisation ↑ covalent character ↑



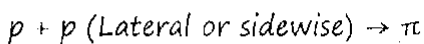
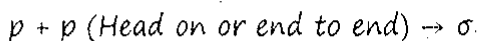
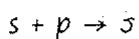
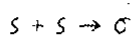
Q. Compare covalent character?



Pseudo inert gas config. > Inert gas config.



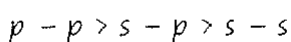
2. V.B.T (Valence bond theory)



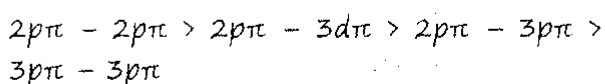
Imp. Points

1. s can not form π

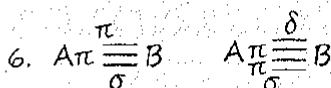
2. Strength $\propto$ overlapping area for same principal Q.N



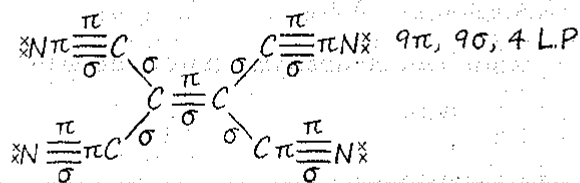
3. Strength of π bond



4. π bond is weaker than σ

5. $A \equiv A > A = A > A - A$ (Bond Strength)

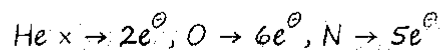
Q. Find σ and π bonds in $\text{C}_2(\text{CN})_4$



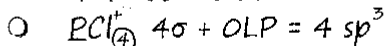
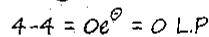
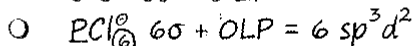
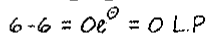
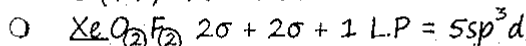
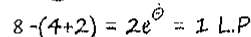
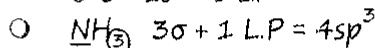
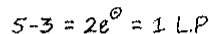
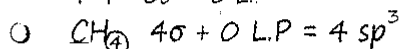
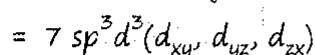
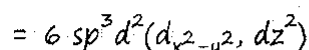
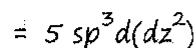
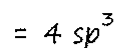
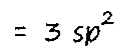
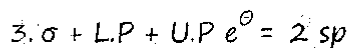
3. Hybridisation

1 2 3 4 5 6 7 8
Li Be B C N O F Ne

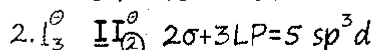
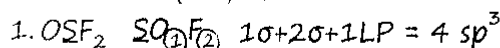
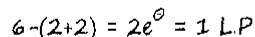
1. Find valence e^- s of central atom



2. Find sharing e^- s of side atom



Q. Find hybridisation of underlined atom



$$6-6 = 0e^{\ominus} = 0 \text{ L.P.}$$

3. N_2 $\text{NH}_2^{\ominus}$ $2\sigma + 0\text{LP} = 2 \text{ sp}$

$$8-6 = 2e^{\ominus} = 1 \text{ L.P.}$$

4. $\text{BrO}_3^{\ominus}$ $3\sigma + 1\text{LP} = 4 \text{ sp}^3$

$$8-(2+2) = 4e^{\ominus} = 2 \text{ L.P.}$$

5. XeO_3F_2 $1\sigma + 2\sigma + 2\text{LP} = 5 \text{ sp}^3 d$

$$4-4 = 0e^{\ominus} = 0 \text{ L.P.}$$

6. NO_2^+ $2\sigma + 0\text{LP} = 2 \text{ sp}$

$$6-6 = 0e^{\ominus} = 0 \text{ L.P.}$$

7. $\text{NO}_3^{\ominus}$ $3\sigma + 0\text{LP} = 3 \text{ sp}^2$

Species	Cationic part	Anionic part
$\text{PCl}_5 (\text{s})$	$\text{PCl}_4^+ (\text{sp}^3)$	$\text{PCl}_6^- (\text{sp}^3 d^2)$
$\text{PBr}_5 (\text{s})$	$\text{PBr}_4^+ (\text{sp}^3)$	Br^-
$\text{N}_2\text{O}_5 (\text{s})$	$\text{NO}_2^+ (\text{sp})$	$\text{NO}_3^- (\text{sp}^2)$
$\text{XeF}_6 (\text{s})$	$\text{XeF}_5^+ (\text{sp}^3 d^2)$	F^-
$\text{Cl}_2\text{O}_6 (\text{s})$	$\text{ClO}_2^+ (\text{sp}^2)$	$\text{ClO}_4^- (\text{sp}^3)$

Odd $e^{\ominus}$ molecules

$$5-4 = 1e^{\ominus} = 1 \text{ unpaired } e^{\ominus}$$

NO_2

$$2\sigma + 1 \text{ unpaired } e^{\ominus} = 3 \text{ sp}^2$$

$\dot{\text{C}}\text{H}_3 \text{ sp}^2$

$\dot{\text{C}}\text{F}_3 \text{ sp}^3$

4. V.S.E.P.R theory

$$\text{L.p-L.p} > \text{L.p-B.p} > \text{B.p-B.p}$$

Geometry or structure

$$\text{BP} (\sigma) + \text{L.P} = 2 \text{ Linear}$$

= 3 Trigonal planar

= 4 Tetrahedral

= 5 T.B.P

= 6 Octahedral

= 7 P.B.P

B.P (σ)	L.P	Shape
2 ~~~~~	0 ~~~~~	Linear
2 ~~~~~	1 ~~~~~	Angular or v-shape or Bent
2 ~~~~~	2 ~~~~~	Angular or v-shape or Bent
2 ~~~~~	3 ~~~~~	Linear
3 ~~~~~	0 ~~~~~	Trigonal planar
3 ~~~~~	1 ~~~~~	Trigonal pyramidal
3 ~~~~~	2 ~~~~~	T-shape
4 ~~~~~	0 ~~~~~	Tetrahedral
4 ~~~~~	1 ~~~~~	see-saw
4 ~~~~~	2 ~~~~~	sq. planar
5 ~~~~~	0 ~~~~~	T.B.P (Trigonal bi pyramidal)
5 ~~~~~	1 ~~~~~	Sq. pyramidal
6 ~~~~~	0 ~~~~~	octahedral
6 ~~~~~	1 ~~~~~	distorted octahedral
7 ~~~~~	0 ~~~~~	P.B.P (pentagonal bi pyramidal)

Hybridisation	Molecules	B.P (σ)	L.P	Shape	(σ + LP) Geometry
sp^3d	$8-2=6e^\ominus=3LP$ XeF_2 2σ	2	3	Linear	5 T.B.P
sp^3d	$8-(4+2)=2e^\ominus=1LP$ XeO_2F_2 4σ	4	1	see-saw	5 T.B.P
sp^3	$8-6=2e^\ominus=1LP$ ClO_2 3σ	3	1	Trigonal pyramidal	4 Tetrahedral
sp^3	I_3^- $6-2=4e^\ominus=2LP$ II_2 2σ	2	2	Bent	4 Tetrahedral

5. Dipole moment (μ)

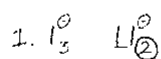
$\mu = 0$ non polar

$\mu \neq 0$ polar

1. (B.P)	σ	L.P	$\mu = 0$
2	0		BeCl_2
3	0		BF_3
4	0		CH_4
5	0		PCl_5
6	0		SF_6
7	0		IF_7
2. (B.P)	σ	L.P	$\mu = 0$
2	3		XeF_2
4	2		XeF_4

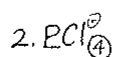
Q. Polar or non polar?

$$8-2=6e^\ominus=3LP$$



$$2\sigma + 3LP \quad \mu = 0 \text{ non polar}$$

$$4-4=0e^\ominus=0LP$$



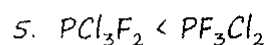
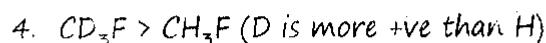
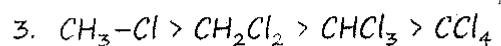
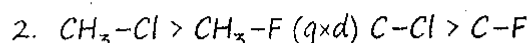
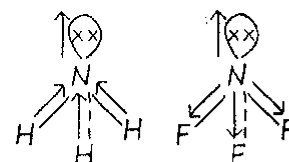
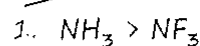
$$4\sigma + 0LP \quad \mu = 0 \text{ non polar}$$

$$6-4=2e^\ominus=1LP$$

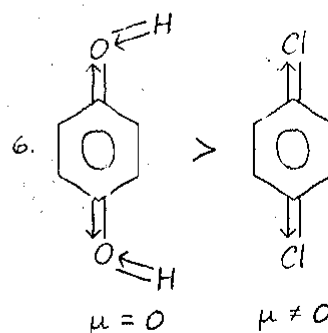
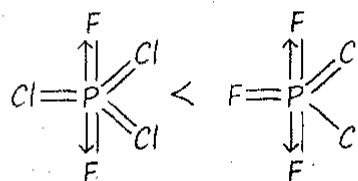


$$4\sigma + 1LP \quad \mu \neq 0 \text{ non polar}$$

Some Imp. Points

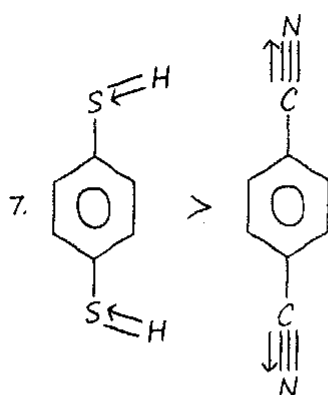


$$\mu = 0 \quad \mu \neq 0$$



$$\mu = 0$$

$$\mu \neq 0$$



M.O.T Bond order

Case 1: If e^- s in a molecule ion

$$8 \leq \text{no. of } e^- \leq 20$$

$\boxed{8}$	$\boxed{9}$	$\boxed{10}$	$\boxed{11}$	$\boxed{12}$	$\boxed{13}$
0	0.5	1	1.5	2	2.5

$$D \boxed{14}^3$$

$\boxed{15}$	$\boxed{16}$	$\boxed{17}$	$\boxed{18}$	$\boxed{19}$	$\boxed{20}$
2.5	2	1.5	1	0.5	0

Case 2: If no. of e^- s ≤ 8

$$[BABA] \text{ B.O} = \left(\frac{N_b - N_a}{2} \right)$$

1	2	3
H_2	H_2^+	H_2^-
2	1	3
2	①	2①
BABA	BABA	BABA
		B.O =
B.O =	B.O =	$\frac{2-1}{2}$
$\frac{2-0}{2}$	$\frac{1-0}{2}$	$= \frac{1}{2} = 0.5$
= 1	= 0.5	

1. Stability $\propto$ B.O

If B.O is same, then last e^- in B.M.O is stable.

$$\text{①} > \text{②} > \text{③}$$

2. B.E $\propto$ B.O $\propto$ $\frac{1}{B.L}$

If no. of e^- s = even (except 10 and 16) $\rightarrow$ dia

3. If no. of e^- s = odd $\rightarrow$ para

Q. Find B.O and Magnetic nature

B_2	$10e^-$	1	P	B_2	$10e^-$	1	P
B_2^-	$11e^-$	1.5	P	B_2^+	$9e^-$	0.5	P
② C_2	$12e^-$	2	D	C_2^-	$13e^-$	2.5	P
C_2^+	$11e^-$	1.5	P	N_2	$14e^-$	3	D
O_2^+	$15e^-$	2.5	P	O_2	$16e^-$	2	P
O_2^-	$17e^-$	1.5	P	F_2	$18e^-$	1	D
O_2^{2-}	$18e^-$	1	D	NO	$15e^-$	2.5	P
O_2^{2+}	$14e^-$	3	D	NO^+	$14e^-$	3	D
F_2^+	$17e^-$	1.5	P	CO	$14e^-$	3	D
Li_2	$6e^-$		D				

$$\left[CO^+ 13 \right] \quad \begin{matrix} 2/2/2 \\ B/A B A \end{matrix}$$

$$B.O = 3.5 \text{ P} \quad B.O = \frac{2}{2} = 1$$

Q. Which of the following would have a permanent dipole moment?

(A) BF_3 (B) SiF_4

$$6 - 4 = 2e^- = 1LP$$

(C) SF_4 (D) XeF_4

Q. Compare covalent character.

$$MF < MCl < MBr < MI$$

Q. Which is paramagnetic?

(A) NO^+	(B) N_2^+	(C) CO	(D) C_2
$14e^-$	$13e^-$	$14e^-$	$12e^-$
D	P	D	D

Q. Which of the following two are isostructural?

(A) XeF_2 & IF_2
⑤ T.B.P ⑤ T.B.P

(B) NH_3 & BF_3
3BP+1LP 3BP+0LP
= 4 Tetra = 3 Trigonal planar

(C) CO_3^{2-} & SO_3^{2-}
3BP+0LP 3BP+1LP
= 3 Trigonal planar = 4 Tetrahedral

(D) PCl_5 & ICl_5
 T.B.P. $5\text{BP} + 1\text{LP} = 6$ octahedral

Q. Which species does not exist?

- (A) Be_2^+ (B) Be_2 (C) B_2 (D) N_2
 $7e^-$ $8e^-$ $10e^-$ $14e^-$

B.O = 0

Q. In which molecule central atom has sp^2 ?

- (A) BeCl_2 (B) BF_3
 $2\text{BP} + 0\text{LP}$ $3\text{BP} + 0\text{LP}$
 $2 = sp$ $3 = sp^2$
 (C) C_2H_2 (D) NH_3
 $\text{CH} \equiv \text{CH}$ $3\text{BP} + 1\text{LP}$
 $sp \quad sp$ $= 4 = sp^3$

Q. Compare bond length

N_2^0 N_2 N_2^{2-} B.O $\text{N}_2 > \text{N}_2^- > \text{N}_2^{2-}$
 15 (2.5) 14 (3) 16 (2) B.L $\text{N}_2 < \text{N}_2^- < \text{N}_2^{2-}$

Q. Compare covalent character.

$1+$ $2+$ $3+$ $4+$
 $\text{LiCl} < \text{BeCl}_2 < \text{AlCl}_3 < \text{CCl}_4$

Q. In which molecule central atom has sp^3d ?

- (A) XeF_2 (B) H_3O^+
 $2\text{BP} + 3\text{LP}$ $3\text{BP} + 1\text{LP}$
 (C) XeF_4 (D) XeF_6
 $4\text{BP} + 2\text{LP}$ $6\text{BP} + 1\text{LP}$

$3\text{BP} + 1\text{LP} = 4$

Q. In chemical change $\text{PCl}_3 \rightarrow \text{PCl}_5$

Hybridisation of P changes from?

sp^3 sp^3d

Q. Which has linear shape but not sp hybridisation

- (A) XeF_2 (B) I_3^+
 $2\text{BP} + 3\text{LP}$ $2\text{BP} + 3\text{LP}$

Linear

Linear

(5) sp^3d

(5) sp^3d

- (C) CO_2 (D) Both A and B

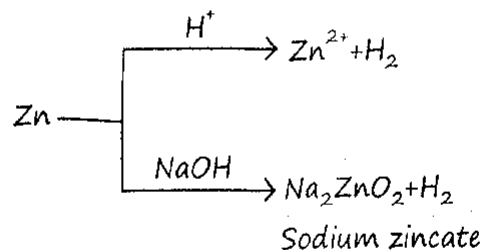
$2\text{BP} + 0\text{LP}$

2 Linear

sp

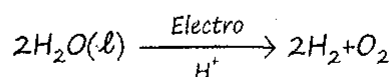
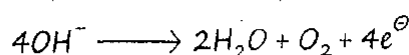
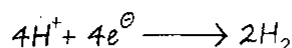
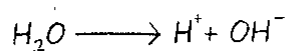
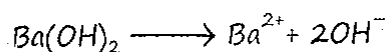
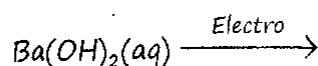
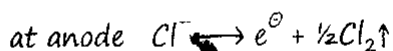
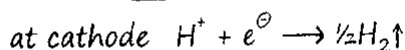
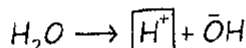
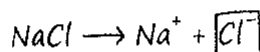
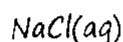
M.O.P OF H_2

1. Lab Methods

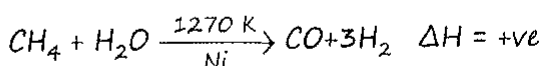
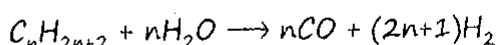


2. Commercial Methods

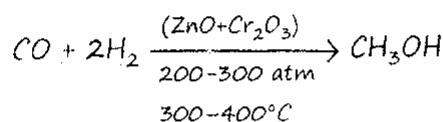
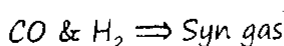
(a) Electrolysis of acidified water

(b) For high purity H_2 (> 99.95%)(c) It is obtained as by product in the production of $NaOH$ & Cl_2 by electrolysis of Brine solution.

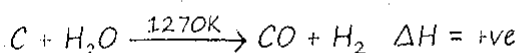
(d) Reaction of steam on hydrocarbon & coke



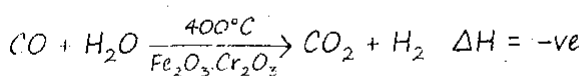
Steam methane reforming reaction



Coal gasification reaction

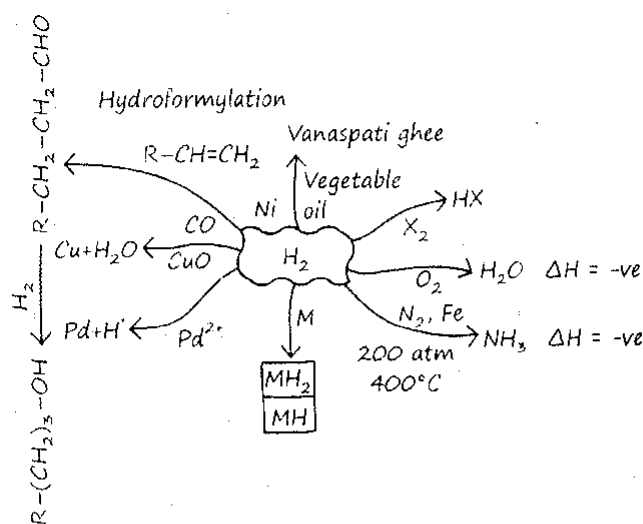


Water gas shift reaction

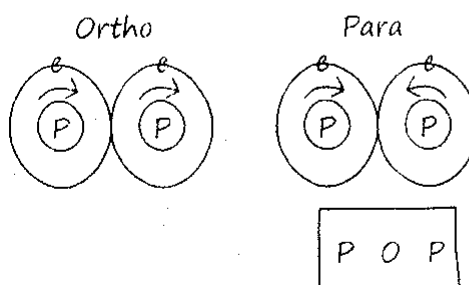
Physical prop of H_2

1. It is colourless, tasteless, odourless, combustible gas.
2. It is lighter than air.
3. Insoluble in H_2O .

Chemical prop



Ortho and para Hydrogen



1. Also known as nuclear spin isomers

2. Net spin para $\Rightarrow 0$

ortho $\Rightarrow$ double

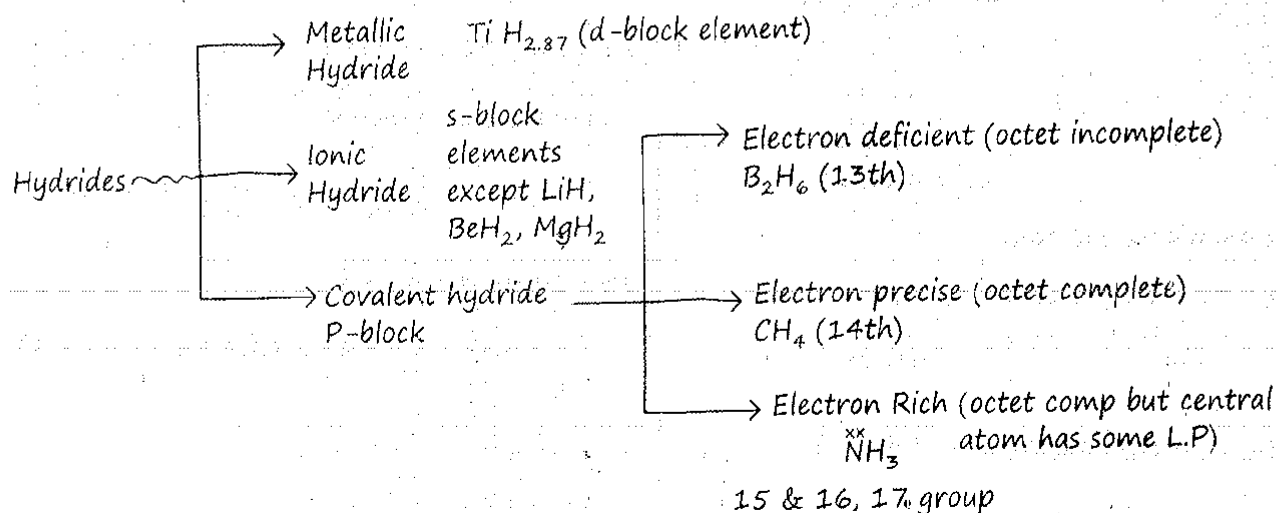
3.	T	Ortho	Para
	OK	-	100%
	at liquefaction temp	50%	50%

at room temp 75%
or
high temp (3 : 1)

4. Chemical prop are same

5. Physical prop different

MP & BP $O > P$

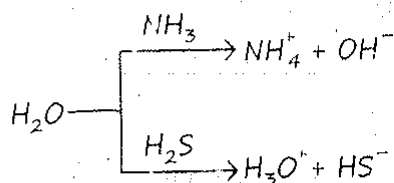


Electrical conductance

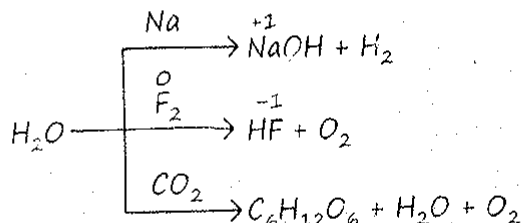
I.H $>$ M.H $>$ C.H

Water Chemical Reaction

1. Amphoteric nature



2. Redox reaction

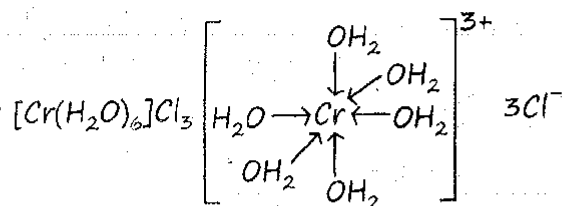


3. Autoprotolysis reaction

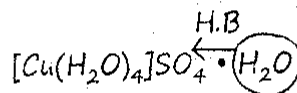


4. Hydrated H_2O

(a) Coordinated H_2O

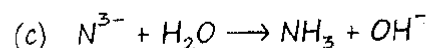
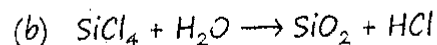
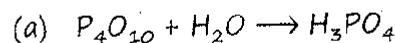


(b) Hydrogen bonded

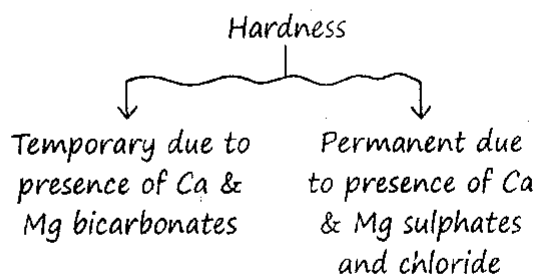
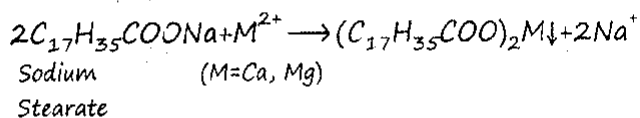
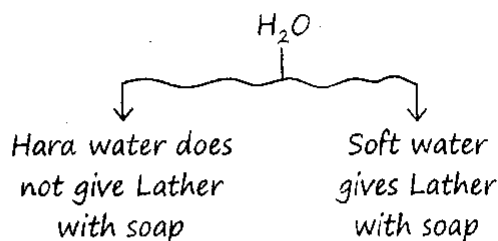


(c) Interstitial H_2O ; $BaCl_2 \cdot 2H_2O$

5. Hydrolysis reaction



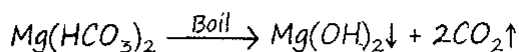
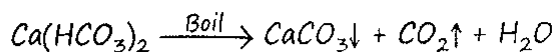
Hardness of H₂O



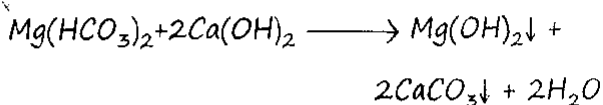
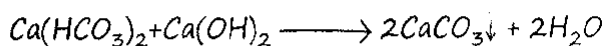
B	Sc
T	P

Removal of Hardness

- By boiling (for removal of temp hardness)

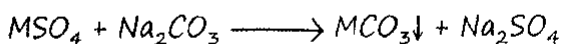
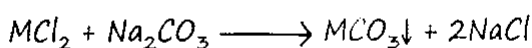


- Clark's method



For removal of permanent Hardness

- By washing soda (Na₂CO₃)

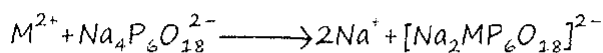
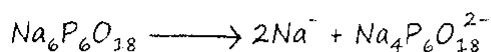


(M = Ca & Mg)

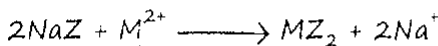
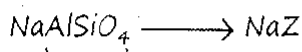
- Calgon's method



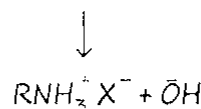
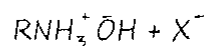
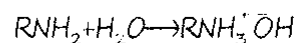
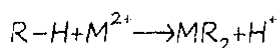
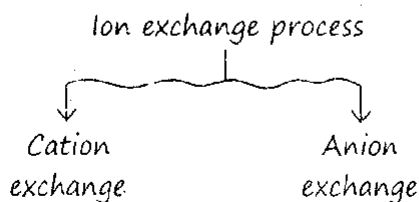
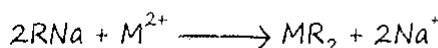
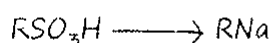
Sodium hexameta phosphate



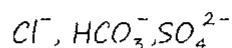
- Zeolite/permutit process



- Synthetic resin method



OH exchange for



PPm

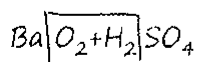
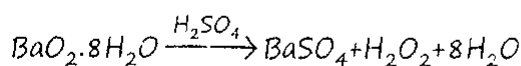
$$PPm = \frac{\text{wt. of solute}}{\text{wt. of solution}} \times 10^6$$

xPPm $\Rightarrow$ x gm solute is present in 10⁶ gm H₂O

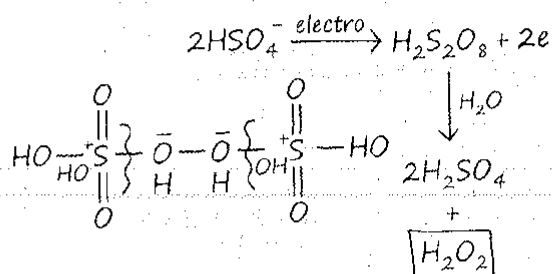
x gm solute is present in 10⁶ mL H₂O

Hydrogen peroxide

- From BaO₂·8H₂O



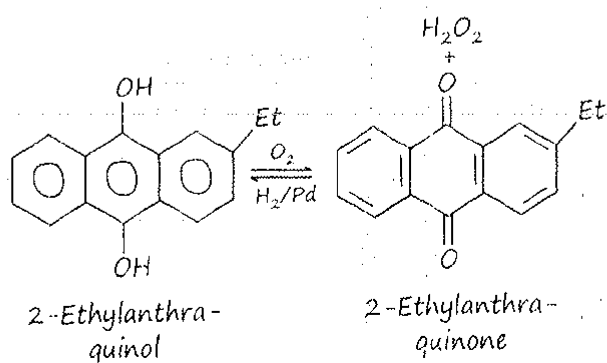
2. From electrolysis of conc H_2SO_4



Lab method for D_2O_2



3. Industrial method

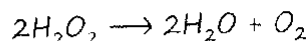


Volumetric strength of H_2O_2

'10' V H_2O_2

1 L H_2O_2 solution gives 10L O_2 at STP

According to the reaction:



1. $[V.S. = 11.2 \times M]$

2. $[V.S. = 5.6 \times N]$

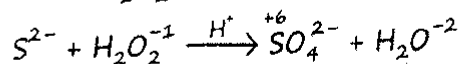
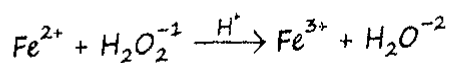
3. $[\text{Strength (g/L)} = \frac{68}{22.7} \times V.S.]$

4. $\% \frac{W}{V} = \frac{S}{10}$

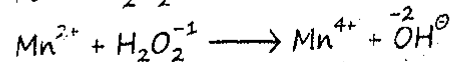
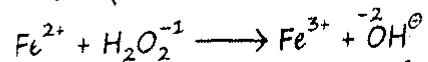
Chemical prop of H_2O_2 (Perhydrol)

Case 1 : Acts as an O.A

1. In A.M (Acidic Medium)

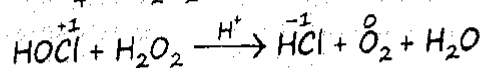
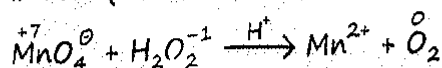


2. In B.M (Basic Medium)

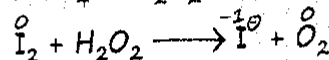
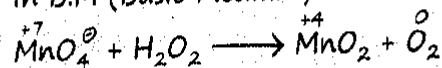


Case 2 : Acts as a R.A

1. In A.M (Acidic Medium)

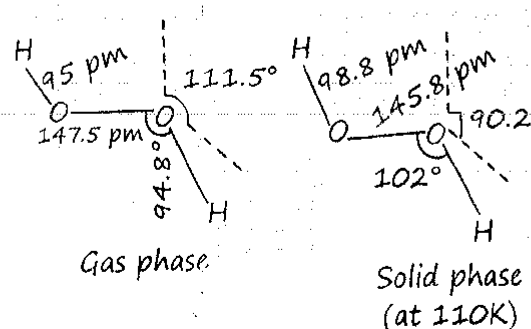


2. In B.M (Basic Medium)



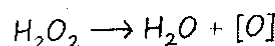
Structure

Open book structure

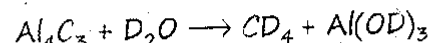
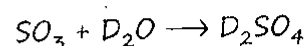
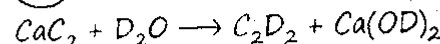


Bleaching action of H_2O_2

Oxidative bleaching



Coloured matter $\xrightarrow{[\text{O}]}$ colourless (permanent)



Practice

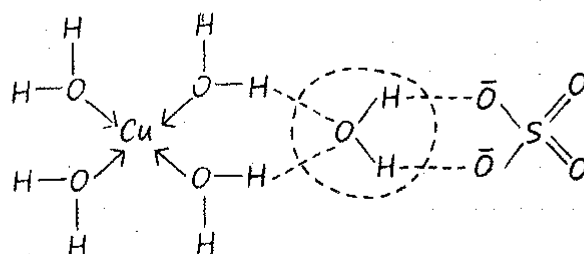
Q. PH_5 does not exist why?

High value of $\Delta_a H$ (Heat of atomisation)

& $\Delta_{eg} H$ (electron gain enthalpy) of hydrogen.

Q. How many hydrogen bonded H_2O molecules are associated in $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (Blue vitriol)?

Ans. 1



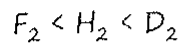
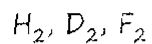
Q. Find strength of 10 V solution of H_2O_2
 $\left(\frac{\%W}{V}\right)$

$$VS = 10$$

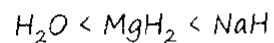
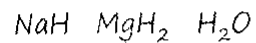
$$(S) \text{ strength (g/L)} = \frac{68}{22.7} \times 10$$
$$= 30$$

$$\frac{\%W}{V} = \frac{S}{10} = \frac{30}{10} = 3$$

Q. Compare B.D.E



Q. Compare reducing prop?

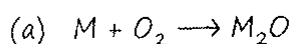
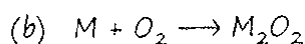
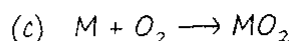


Alkali Metals

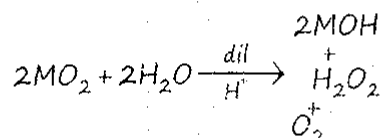
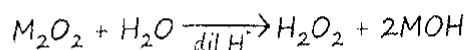
Size	Li	I.E
Metallic character	Na	E.N
Ionic nature of compounds	K	M.P
Sofness	Rb	B.P
	Cs	
	↓ increases	↓ decreases

Density $Li < K < Na < Rb < Cs$

Chemical property:-

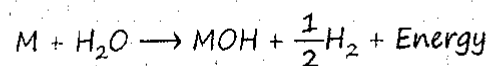
1. Reaction with O_2  $M = Li$ (some peroxide, (Li_2O_2) also) $M = Na$ (some superoxide also NaO_2) $(M = K, Rb, Cs)$ Stability $M_2O > M_2O_2 > MO_2$ Oxidising power $M_2O < M_2O_2 < MO_2$

Hydrolysis of peroxides & superoxides

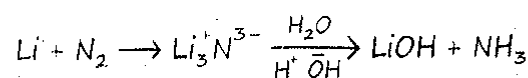
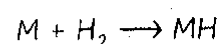


Oxides

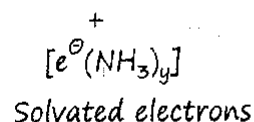
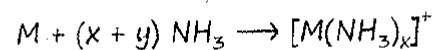
Li_2O	Basicity & solubility increases
Na_2O	
K_2O	
Rb_2O	
Cs_2O	

Ionic Radii $Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$
(g) (g) (g) (g) (g)Hydrated Radii $Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$
(aq) (aq) (aq) (aq) (aq)2. Reaction with H_2O 

LiOH	Basicity & solubility
NaOH	
KOH	
RbOH	
CsOH	
	↓ Increases

3. Reaction with N_2 4. Reaction with H_2 

LiH	Stability
NaH	
KH	
RbH	
CsH	
	↓ decreases

5. Reaction with aq NH_3 

Reducing agent

Paramagnetic

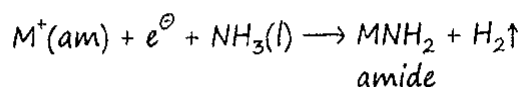
Conducting

Blue colour

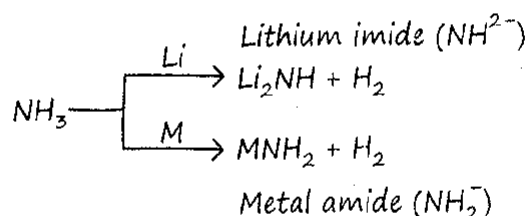
When conc increases Paramagnetic $\rightarrow$ diamagnetic

Blue $\rightarrow$ Bronze

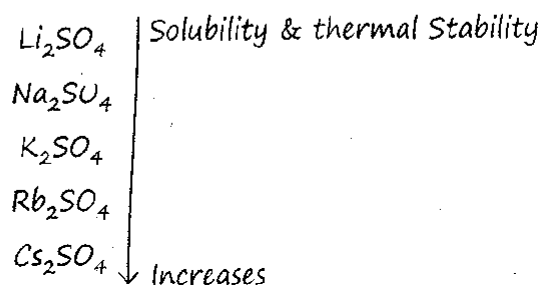
On standing



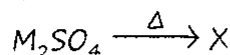
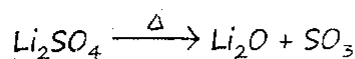
6. Reaction with dry NH_3



7. Sulphates (M_2SO_4)

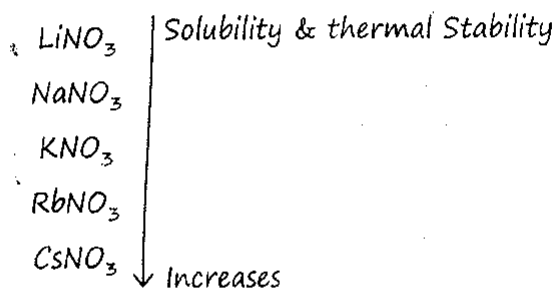


Li_2SO_4 is insoluble

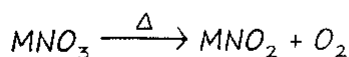
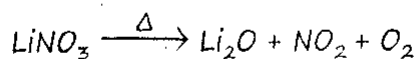


($M = Na, K, Rb, Cs$)

8. Nitrates (MNO_3)

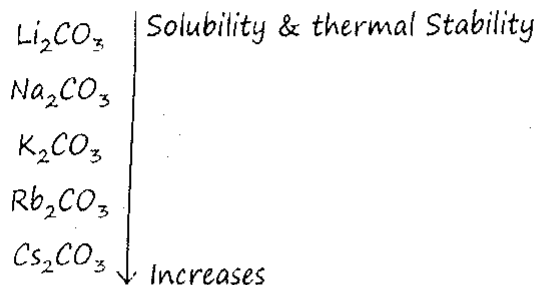


All nitrates are soluble

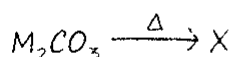
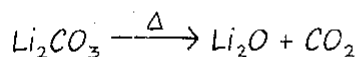


($M = Na, K, Rb, Cs$)

9. Carbonates (M_2CO_3)

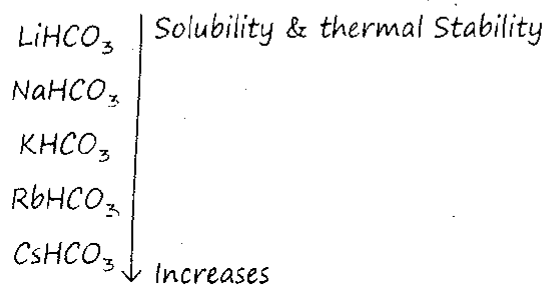


Li_2CO_3 is insoluble



($M = Na, K, Rb, Cs$)

10. Bicarbonates ($MHCO_3$)



$LiHCO_3$ does not exist as solid solution

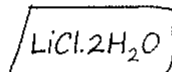
Rest all exist as solid $MHCO_3$

($M = Na, K, Rb, Cs$)

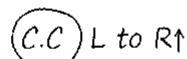
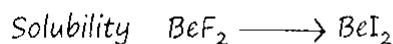
11. Halides (MX)

M.P & B.P $F > Cl > Br > I$

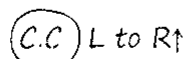
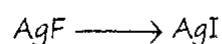
Covalent character $LiF < LiCl < LiBr < LiI$



(Rest all chlorides do not form hydrates)



Solubility L to R $\downarrow$

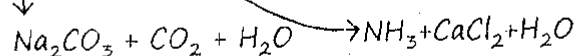
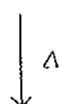
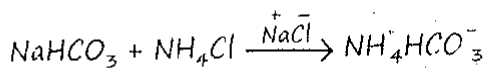
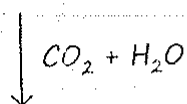
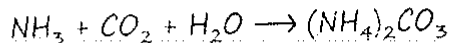


Solubility L to R $\downarrow$

Some Important Comp of Na

1. Sodium carbonate (Na_2CO_3)

Solvay ammonia process



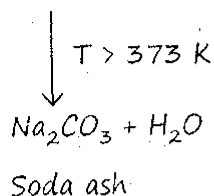
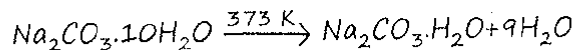
Solvay ammonia process is not used for preparation of K_2CO_3 due to more solubility of KHCO_3

Prop

(a) Hydrated form is called washing soda
 $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$

(b) White crystalline solid $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$

(c) Heating effect (Glauber's Salt)



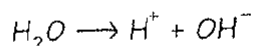
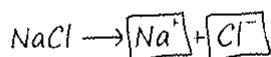
2. Sodium hydroxide (NaOH)

Castner-Kellner cell

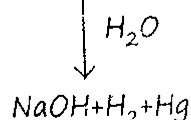
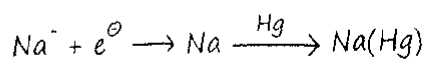
Brine solution (aq NaCl)

$\text{Hg} \rightarrow$ cathode

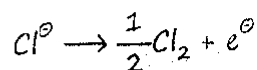
$\text{C} \rightarrow$ anode



At Cathode



At Anode



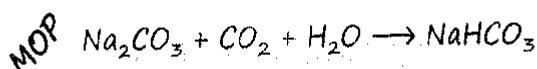
Prop

(a) White solid

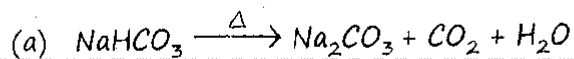
(b) Melts at 591 K

(c) Strong alkaline solution

3. Sodium hydrogen carbonate (NaHCO_3) (Baking soda)



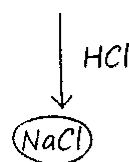
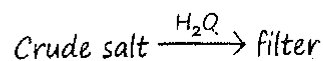
Property:-



(b) Mild antiseptic

(c) Fire extinguishers

4. NaCl (Common salt)



Flame Test

Li	Na	K	Rb	Cs
Crimson red	Yellow	Violet	Red violet	Blue
Cr	Y	Vi	Re	Bablu

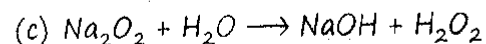
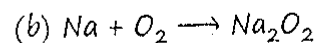
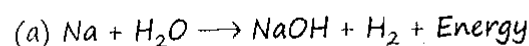
Practice

Q. LiF & CsI are very less soluble why?

$\text{LiF} \rightarrow$ due to High L.E

$\text{CsI} \rightarrow$ due to Less hydration of Cs^+

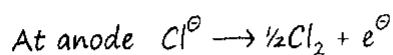
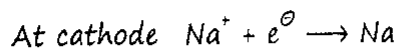
Q. Find products



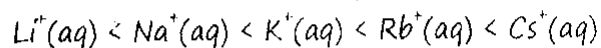
Q. Prepare Na from NaCl

Molten NaCl

Down's cell

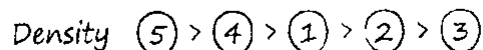
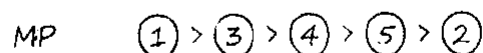


Q. Compare ionic mobility in aq solution



Alkaline earth metals

Size metallic charac Ionic nature of compounds softness Increase ↓	(1) Be	I.E & E.N ↓ Decreases
	(2) Mg	
	(3) Ca	
	(4) Sr	
	(5) Ba	



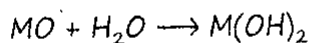
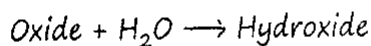
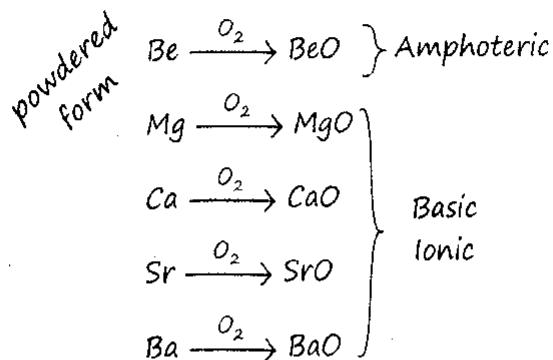
M.P (Madhya pradesh) ko

B.P (Bhopal)

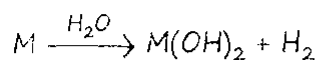
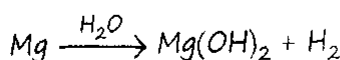
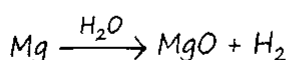
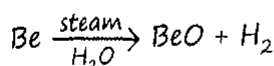
Density (Denge)

Chemical prop

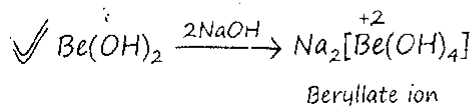
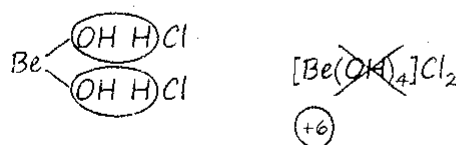
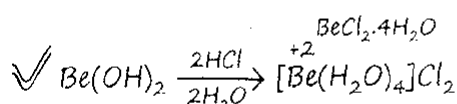
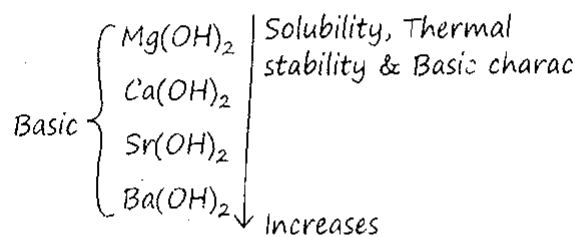
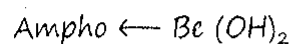
1. Reaction with O_2 (max covalent charac)



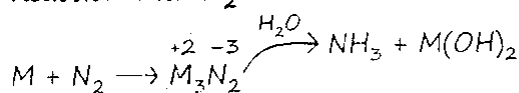
2. Reaction with H_2O



Hydroxides

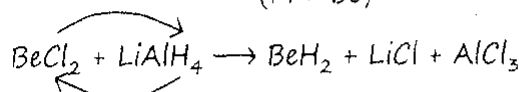
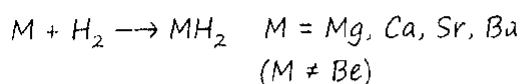


3. Reaction with N_2



M = all alkaline earth metals

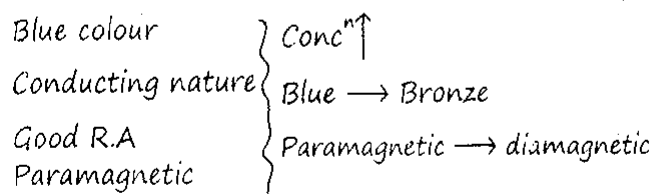
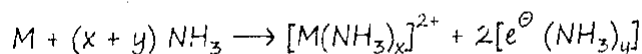
4. Reaction with H_2



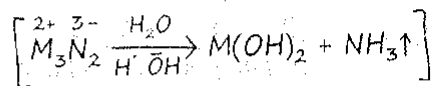
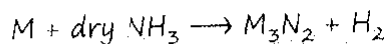
$(\text{BeH}_2)_n$ & $(\text{MgH}_2)_n$ are polymeric covalent hydride

Rest all MH_2 are ionic in nature.

5. Reaction with aq. NH_3

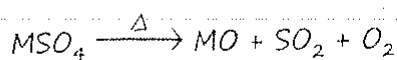


6. Reaction with dry NH_3



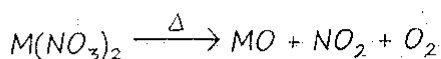
7. Sulphates (MSO_4)

Thermal stability (T.S)	BeSO_4	Solubility
	MgSO_4	
	CaSO_4	
	SrSO_4	
	BaSO_4	
Increases $\downarrow$		$\downarrow$ Decreases



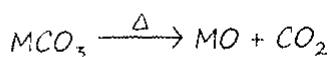
8. Nitrates $\text{M}(\text{NO}_3)_2$

T.S	$\text{Be}(\text{NO}_3)_2$	Solubility
	$\text{Mg}(\text{NO}_3)_2$	
	$\text{Ca}(\text{NO}_3)_2$	
	$\text{Sr}(\text{NO}_3)_2$	
	$\text{Ba}(\text{NO}_3)_2$	
Increases $\downarrow$		$\downarrow$ Decreases



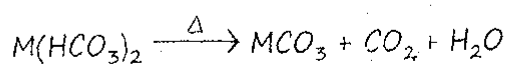
9. Carbonates (MCO_3)

T.S	BeCO_3	Solubility
	MgCO_3	
	CaCO_3	
	SrCO_3	
	BaCO_3	
Increases $\downarrow$		$\downarrow$ Decreases



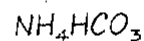
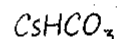
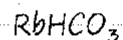
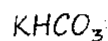
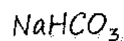
10. Bicarbonates $\text{M}(\text{HCO}_3)_2$

All bicarbonates exist in solⁿ



Note:

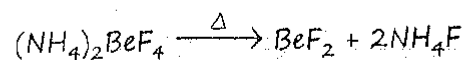
5 bicarbonates exist in solid form



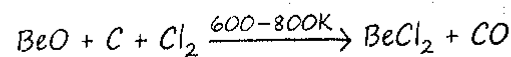
11. Halides



Best method for BeF_2

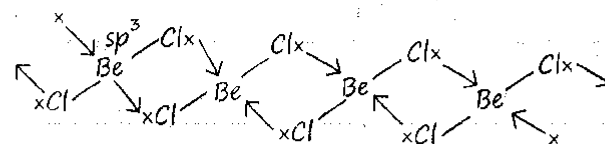


Best method for BeCl_2

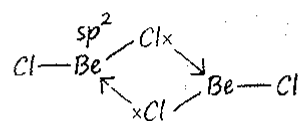


Structures of BeCl_2

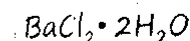
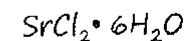
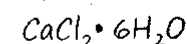
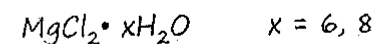
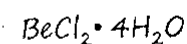
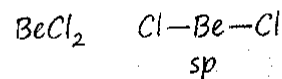
(a) In solid phase (BeCl_2)_n



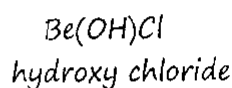
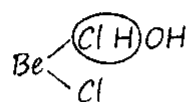
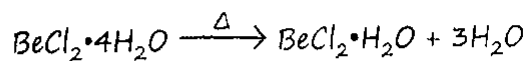
(b) In vapour phase (BeCl_2)₂



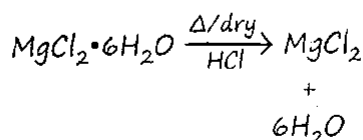
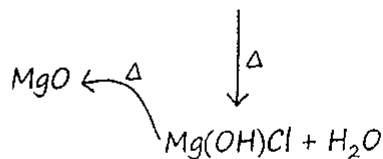
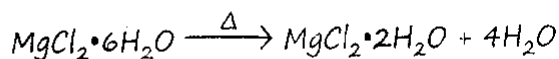
(c) When T is nearly 1200K



1. Heating of $\text{BeCl}_2 \cdot 4\text{H}_2\text{O}$



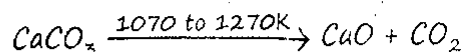
2. Heating of $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$



Some Imp. comp. of Ca

1. Quick lime (CaO)

MOP (a) From limestone (CaCO_3)

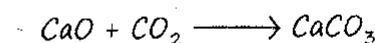
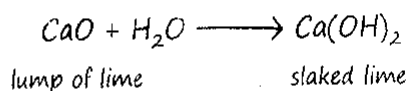


Prop (a) White amorphous solid

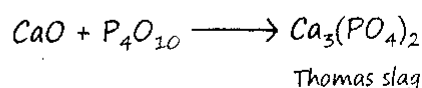
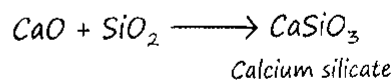
(b) High M.P

(c) It absorbs moisture & CO_2 from atmosphere

Slaking of lime



(d) It is basic in nature



2. Slaked lime Ca(OH)_2

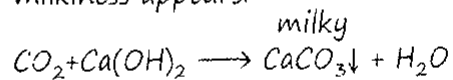
MOP (a) $\text{CaO} + \text{H}_2\text{O} \longrightarrow \text{Ca(OH)}_2$

Prop (a) White amorphous powder

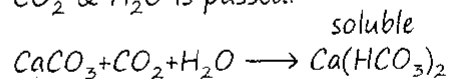
(b) aq solⁿ is called lime water

(c) Suspension of slaked lime is called milk of lime

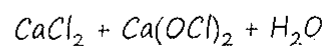
(d) When CO_2 is passed in lime water, milkiness appears.



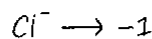
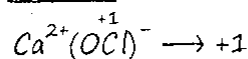
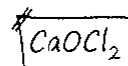
Milkiness disappears when excess CO_2 & H_2O is passed.



(e) $\text{Ca(OH)}_2 + \text{Cl}_2 \longrightarrow$



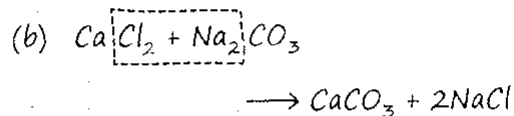
Calcium
hypochlorite



3. Lime stone (CaCO_3)

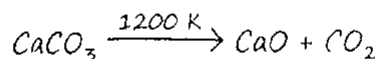
MOP (a) $\text{Ca(OH)}_2 + \text{CO}_2 \longrightarrow \text{CaCO}_3 + \text{H}_2\text{O}$

Excess $\text{CO}_2 + \text{H}_2\text{O}$ should be avoided

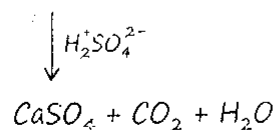
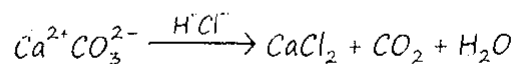


Prop (a) White fluffy powder

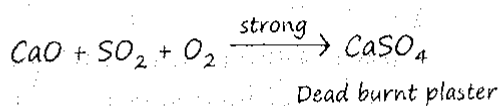
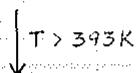
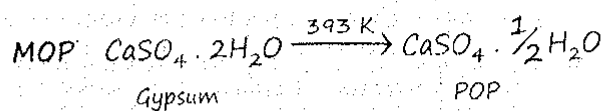
(b) Heating effect



(c) Rxn with HCl & H_2SO_4



4. Plaster of paris (POP) $\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O}$



Flame Test

Be & Mg do not show flame test

Ca	Sr	Ba
Brick	Crimson	apple
Red	Red	Green

Name	Formula
1. Indian salt petre	KNO_3
2. Chile salt petre	NaNO_3
3. Milk of magnesia	Mg(OH)_2
4. Caustic soda	NaOH
5. Caustic potash	KOH
6. Plaster of paris	$\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O}$
7. Gypsum	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
8. Dead burnt plaster	CaSO_4
9. Caliche	$\text{NaNO}_3 + \text{NaIO}_3$
10. Epsom's salt	$\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$
11. Glauber's salt	$\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$
12. Slaked lime	Ca(OH)_2
13. Lime stone (Marble)	CaCO_3

Name	Formula
14. Quick lime	CaO
15. Soda ash	Na_2CO_3
16. Washing soda	$\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$
17. Bleaching powder	CaOCl_2
18. Brine	aqNaCl
19. Nitrolim	$\text{CaC}_2 + \text{N}_2 \xrightarrow{\Delta} \text{CaCN}_2 + \text{C}$
20. Baking soda	NaHCO_3

Practice

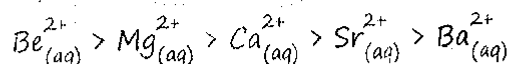
Q. (a) BeO is almost insoluble but BeSO_4 is soluble in H_2O ?

(b) BaO is soluble but BaSO_4 is insoluble in H_2O ?

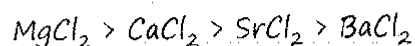
For solubility $\text{HE} > \text{LE}$

For insolubility $\text{HE} < \text{LE}$

Q. Compare hydrated radii



Q. Compare solubility

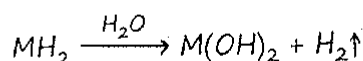
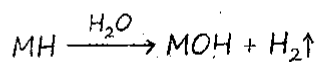


due to $\text{HE} > \text{LE}$

Q. Which comp is ionic in nature?



Hydrolith



Density	B	B.P
	Al	
	Ga	
	In	
Increases ↓	Tl	↓ Decreases

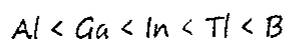
Exceptions

1. Atomic Radius



due to poor screening of $3d^{10} e^0$ in Ga

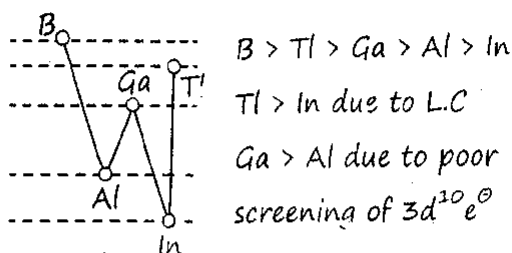
2. E.N



3. M.P

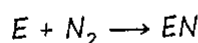
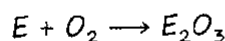


4. Ionisation energy (I.E)

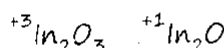
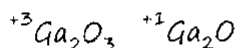


Chemical Properties

1. Reactivity towards air

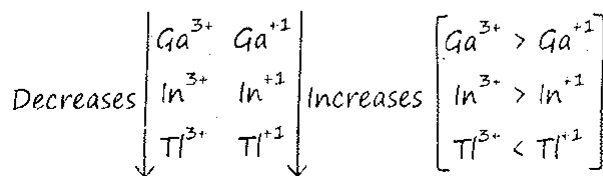


- (1) B does not react in crystalline form
- (2) Al forms very thin oxide layer (Al_2O_3)
- (3) Amorphous B and Al can form oxides and nitrides

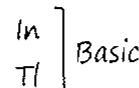
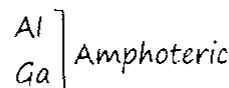


Stability of +3 O.N decreases down the group

Stability of +1 O.N increases down the group

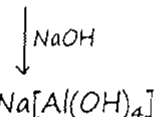
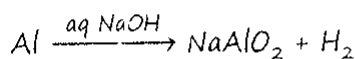
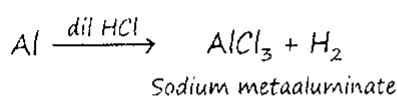


Oxides of B Acidic



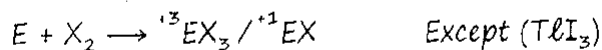
2. Rxn with acid and alkalis

B does not react



Sodium tetrahydroxoaluminate (III)

3. Reactivity towards Halogen

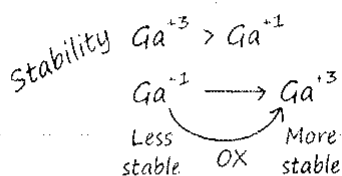


Stability

- (1) +3 Halides decreases down the group
- (2) +1 Halides increases down the group

Q. Compare Reducing power

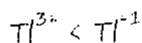
(1) GaCl_3 & GaCl



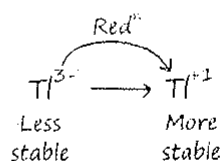
GaCl is better reducing agent than GaCl_3 .

Q. Compare Oxidising power

(2) TlCl_3 & TlCl

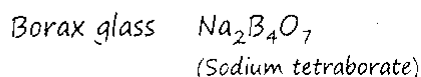
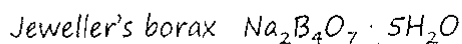
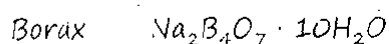


TlCl_3 is better O.A than TlCl .



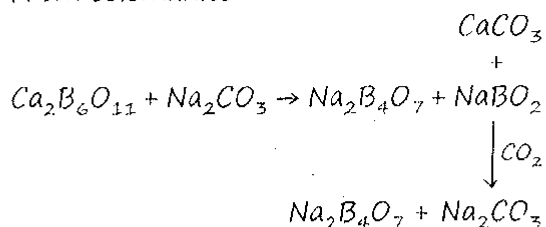
Some Imp. comp. of Boron

1. **Borax** # Tincal or suhaga



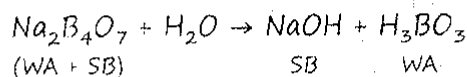
MOP

(1) From colemanite



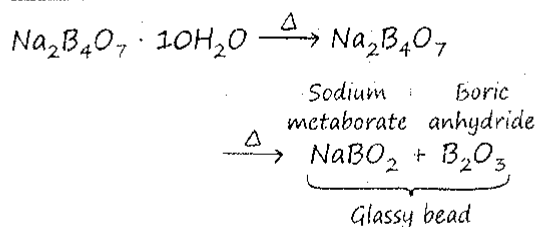
Prop

(1) Borax ^{aq soln} is alkaline in nature

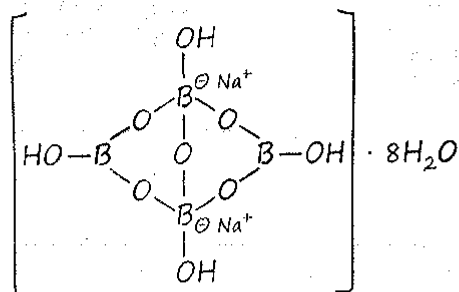
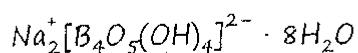


(2) Borax can act as acidic buffer

(3) Heating effect



(4) Structure $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$



(1) Hybridisation of B $\Rightarrow sp^2$ & sp^3

(2) Total B—O—B Bonds = 5

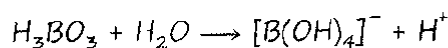
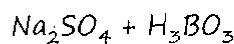
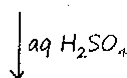
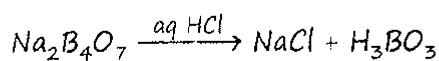
(3) Total B—O bonds = 14

(4) Total L.P = 34

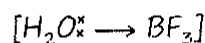
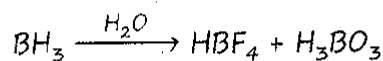
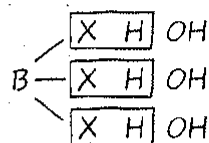
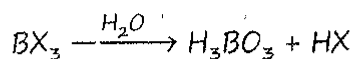
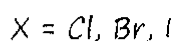
2. H_3BO_3 (Boric acid)

MOP

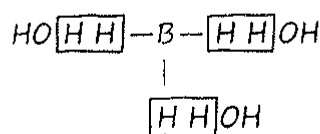
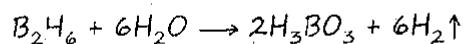
(1) From borax



(2) From boron trihalides

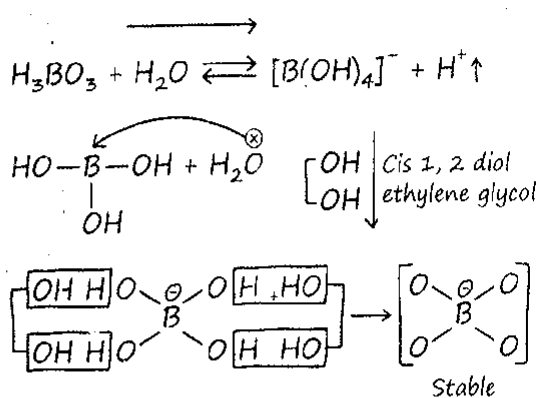


(3) From diborane

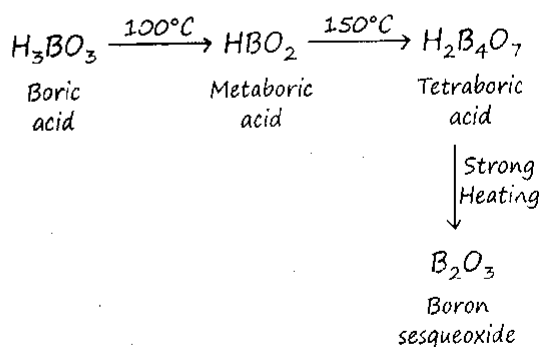


Prop

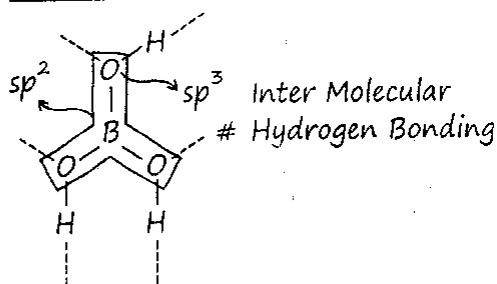
(1) It is a weak monobasic non protic Lewis acid



(2) Heating effect



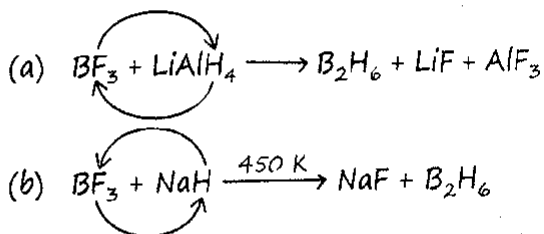
(3) Structure



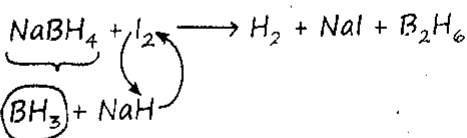
3. B_2H_6 (Diborane)

MOP

(1) From BF_3

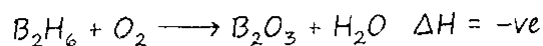


(2) From $NaBH_4$

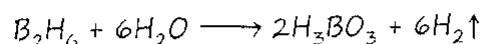


Prop

(1) Rxn with O_2

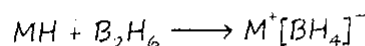


(2) Rxn with H_2O

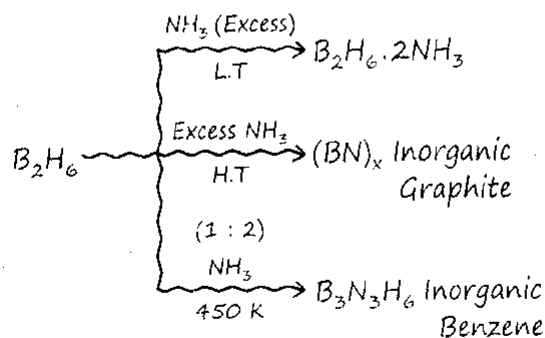


(3) Rxn with metal hydride (MH)

$M = Li \text{ or } Na$



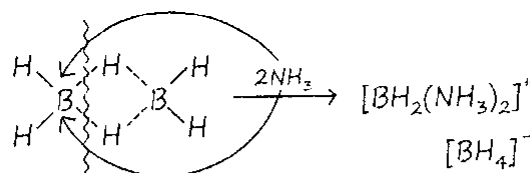
(4) Rxn with NH_3



RNH_2 R_2NH

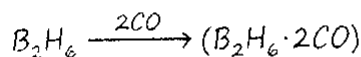
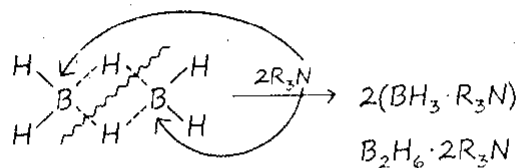
(5) Rxn with NH_3 , 1° amine, 2° amine

(Unsymmetrical bond breaking)

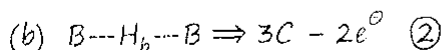
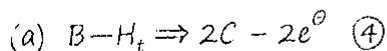
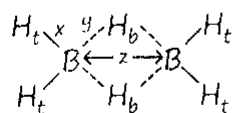
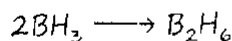


(6) Rxn with 3° amine (R_3N) & CO

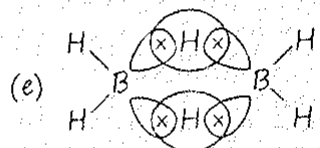
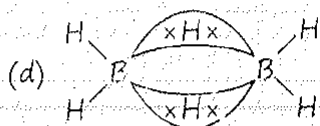
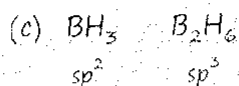
(Symmetrical bond breaking)



(7) Structure of B_2H_6



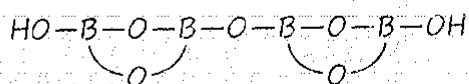
Banana Bond



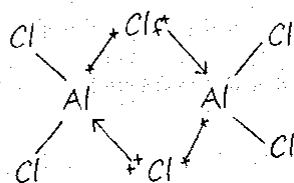
(f) Bond length $x < y < z$

Some Imp. Points of Boron Family

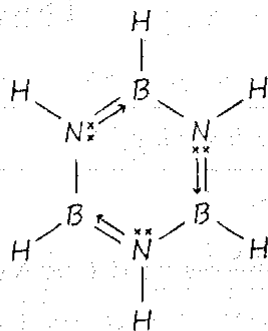
1. Structure of $\text{H}_2\text{B}_4\text{O}_7$ (Tetraboric acid)



2. Structure of Al_2Cl_6



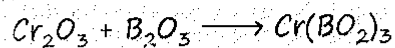
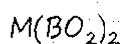
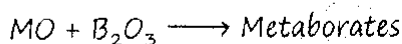
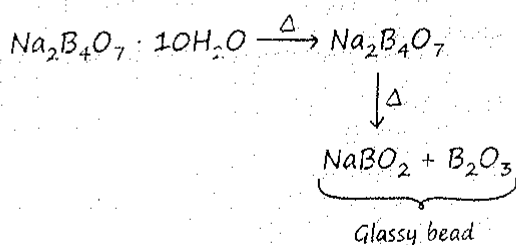
3. Structure of $\text{B}_3\text{N}_3\text{H}_6$



Aromatic

More reactive than C_6H_6

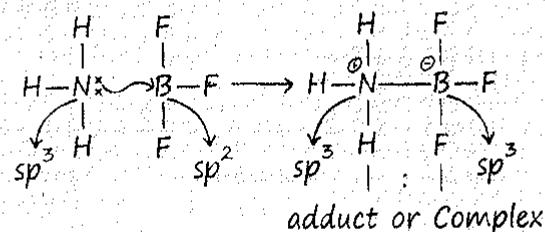
4. Borax bead test



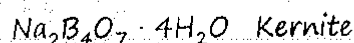
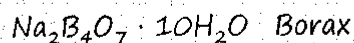
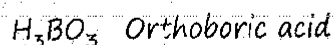
Metaborates of

Cr	Mn	Fe	Co	Ni	Cu
Green	Pink	Green	Blue	Brown	Blue
Gopal	prasad	Gupta	ki	badi	barbaad
Biwi					

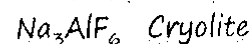
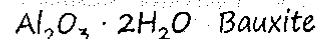
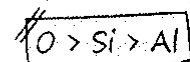
5. Rxn of NH_3 with BF_3



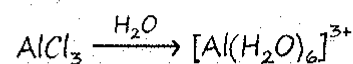
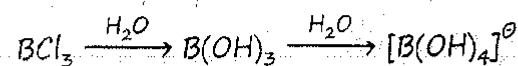
6. Occurance of B



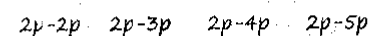
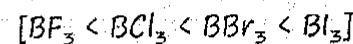
7. Occurance of Al



8. Hydrolysis of BCl_3 and AlCl_3



9. Acidic strength of trihalides of B

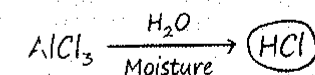


Practice

Q. Boron is unable to form BF_6^{3-} , why?

due to non availability of vacant 'd' orbitals.

Q. White fumes appear around the bottle of anhydrous AlCl_3 , why?



Q. $E_{\text{Al}^{3+}/\text{Al}}^\circ = -1.66 \text{ V}$

$E_{\text{Ti}^{3+}/\text{Ti}}^\circ = +1.26$

Compare

(a) electropositive charac of Al & Tl

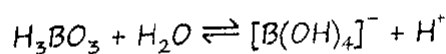
(b) oxidising power of Al^{3+} and Tl^{3+}

$$E_R^\circ \downarrow \quad E \cdot P \uparrow \quad E \cdot P \quad Al > Tl$$

$$O \cdot P \quad Tl^{3+} > Al^{3+}$$

Q. Why H_3BO_3 is a weak acid?

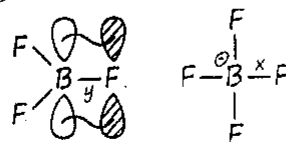
Because it is a non protic acid



Q. Compare stability

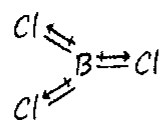
$${}^3+ BCl_3 > {}^3+ TlCl_3$$

Q. Compare Bond length B-F



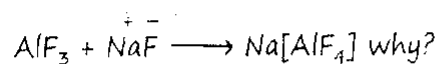
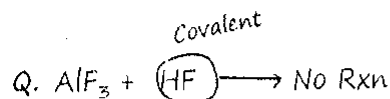
$$x > y$$

Q. BCl_3 is non polar why?



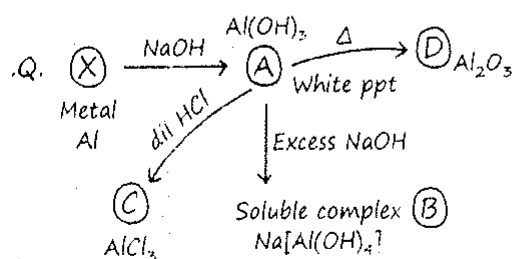
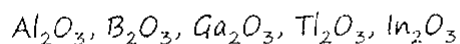
$$3 \text{ BP} + 0 \text{ LP}$$

$$D.M = 0$$



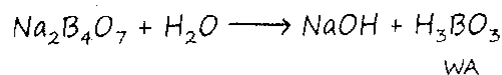
Q. Identify acidic basic and Amphoteric oxides.

Am Ac Am B B



Q. Aq solⁿ of Borax is alkaline why?

SB

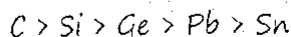


$$p^H > 7$$

Size	C	B.P
	Si	
	Ge	
	Sn	
	Pb	
Increases ↓		↓ Decreases

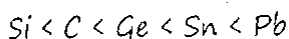
Exceptions

1. M.P & I.E

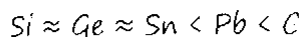


↓
due to L.C

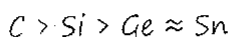
2. Density



3. E.N



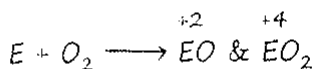
#Catenation# Self linking prop



Pb does not show catenation due to bigger size.

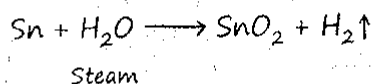
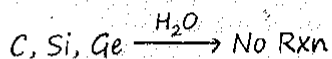
Chemical Prop #

1. Reactivity towards oxygen

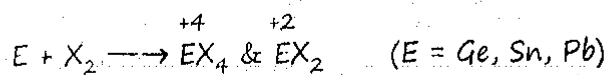


Neutral	CO	CO ₂	
Exist at High Temp	SiO	SiO ₂	Acidic
	GeO	GeO ₂	
	SnO	SnO ₂	Amphoteric
	PbO	PbO ₂	

Acidic strength $EO_2 > EO$

2. Reactivity towards H₂O.

due to formation of protective film of oxide (PbO)

3. Reactivity towards X₂

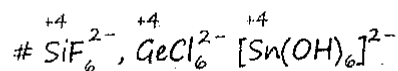
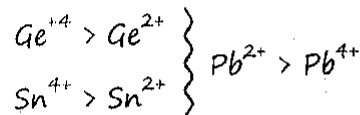
Stability

⁺⁴
E Top to bottom decreases

⁺²
E Top to bottom increases

due to inert pair effect

#	Ge ⁺⁴	Ge ⁺²	
Stability decreases ↓	Sn ⁺⁴	Sn ⁺²	Stability increases ↓
	Pb ⁺⁴	Pb ⁺²	



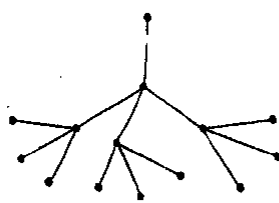
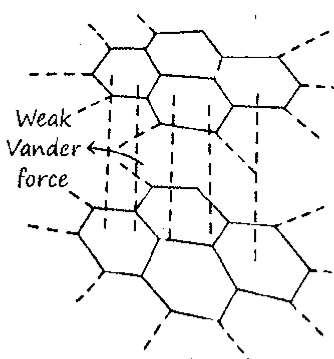
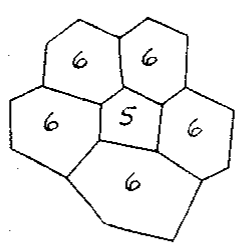
Exist due to vacant 'd' orbitals.

[SnCl₆]²⁻ does not exist due to bigger size of Cl.

PbI₄ does not exist due to I.P.E.

PbF₄ & SnF₄ are ionic in nature.

Allotropes of Carbon

Diamond	Graphite	Fullerenes
1. 3D network solid	1. Layered struc & Layers are held together by weak vander waal forces	1. Graphite $\xrightarrow[\text{He or Ne}]{\text{electric arc}}$ Fullerene Buck ball struc. $C_{60} \text{---} C_{350}$ $C_{60} \rightarrow$ Buck minster fullerene (Soccer ball)
2. sp^3	2. sp^2	2. sp^2
3. Non aromatic	3. Aromatic	3. Aromatic
4. dangling Bonds are present	4. dangling bonds are present	4. No dangling bond
5. Good thermal conductor but poor electrical conductor	5. Good conductor of heat and electricity	5. Good thermal and electrical conductor.
6. Diamond	6. Graphite	6. C_{60}
		
7. abrasive, hardest subs on earth	7. Lubricant, soft, slippery	7. $C_{70} \rightarrow$ Rugby ball

Other forms of elemental Carbon

1. Carbon black
2. Coke
3. Charcoal

These are impure form of Graphite or fullerenes

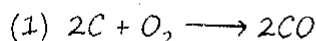
Hydrocarbon $\xrightarrow[\text{air}/\Delta]{\text{In limited Supply of}}$ Carbon black

Wood or coal $\xrightarrow[\text{air}/\Delta]{\text{In absence of}}$ Coke or Charcoal

Some Imp. Comp. of Carbon

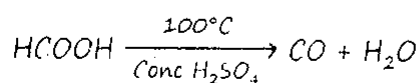
1. CO (Carbon mono oxide)

M.O.P

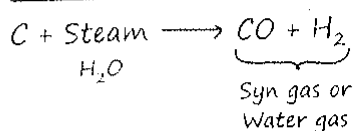


limited supply

- (2) Lab method



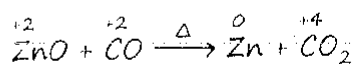
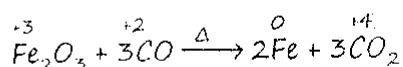
(3) Industrial method



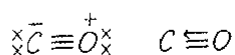
Prop

(1) Colourless, odourless, neutral, poisonous gas.

(2) It is a powerful reducing agent



Struc of CO

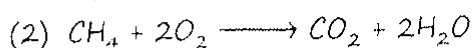
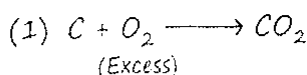


1σ, 2π, 2 L.P

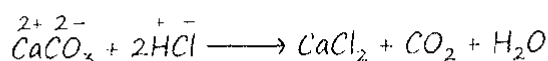
$$B.O = 3$$

2. Carbon dioxide (CO₂)

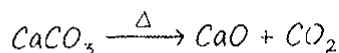
MOP



(3) Lab method



(4) Industrial method

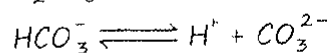
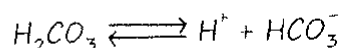


Prop of CO₂

(1) It is colourless, odourless acidic gas

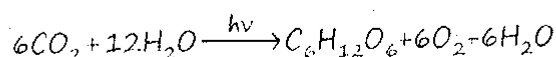
(2) Solid CO₂ → dry ice

(3) Dissolve in H₂O to form H₂CO₃

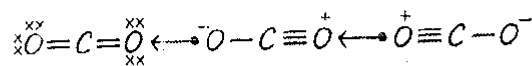


Which maintain the pH of blood

(4) Photosynthesis



Struc



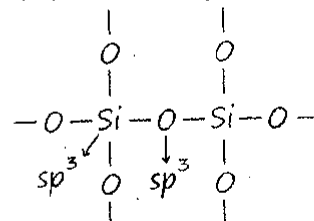
$$B.O = 2$$

2σ, 2π, 4 L.P

Some Imp. Comp of Silicon

1. SiO₂ or (SiO₂)_n

3D Network Solid

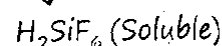


Prop

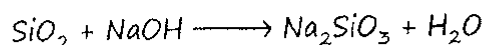
(1) Rxn with HF



Excess HF



(2) Rxn with NaOH



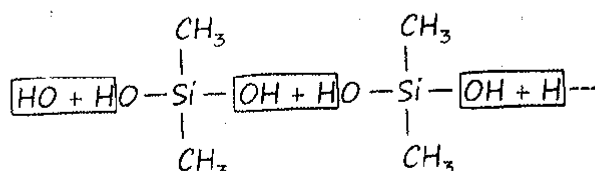
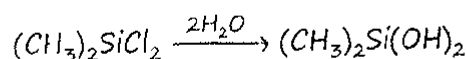
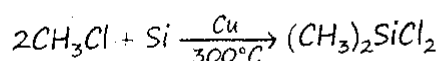
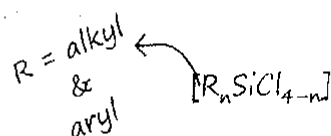
Sodium Silicate

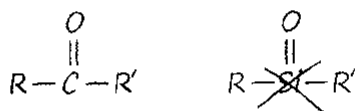
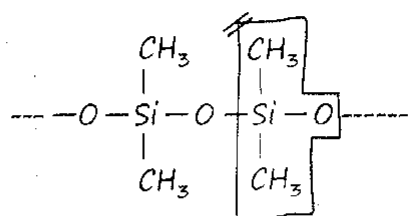
(3) SiO₂ → Crystalline quartz

↓
Amorphous
Silica

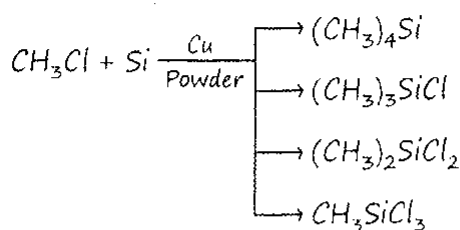
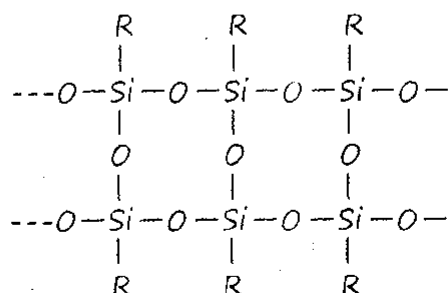
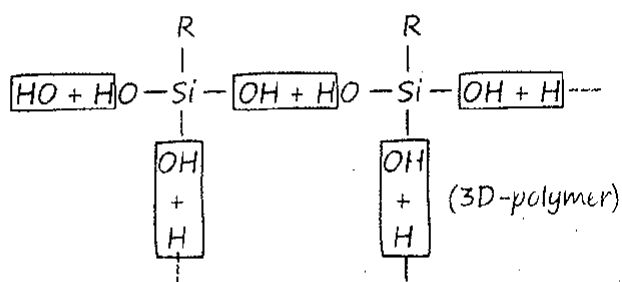
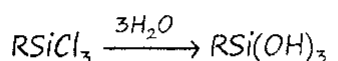
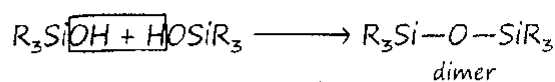
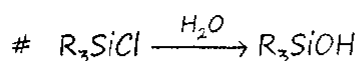
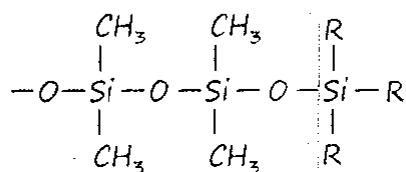
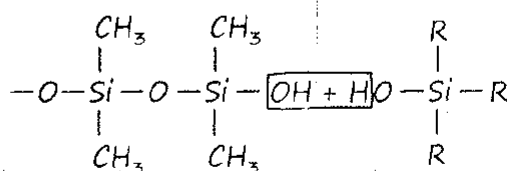
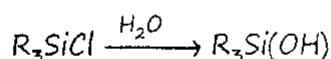
2. Silicones # [R₂SiO]_n

Organo silicon polymer



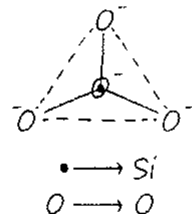
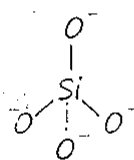


Polymer chain can be controlled by adding R_3SiCl



3. Silicates

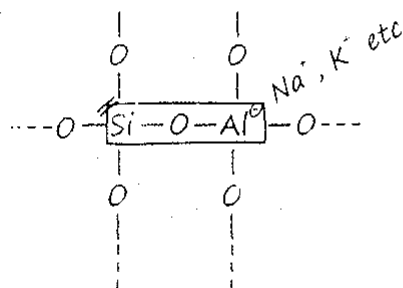
Silicates have SiO_4^{4-} unit



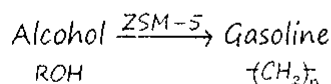
Silicates	Gen. formula	Sharing of 'O' atom
Orthosilicate	SiO_4^{4-}	0
Pyrosilicate	$Si_2O_7^{6-}$	1
Chain or cyclic silicate	$(SiO_3)_n^{2n-}$	2
2D	$(Si_2O_5)_n^{2n-}$	3
3D	$(SiO_2)_n$	4

4. Zeolites

Honeycomb like structure



ZSM-5

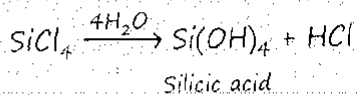


Used to remove hardness

Used in petrochemical industries

Some Imp. points of Carbon family

(1) Hydrolysis of SiCl_4



(2) Carbon has unique ability to form multiple bonds ($\text{C}=\text{C}$, $\text{C}=\text{O}$, $\text{C}\equiv\text{N}$) due to small size & high E.N.

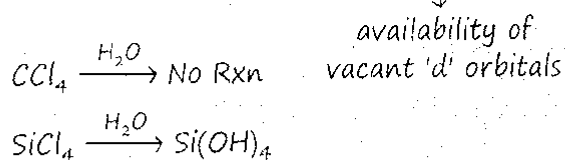
(3) GeX_4 is more stable than GeX_2 due to stability of $\text{Ge}^{+4} > \text{Ge}^{2+}$ due to inert pair effect.

(4) PbX_2 is more stable than PbX_4 due to stability of $\text{Pb}^{2+} > \text{Pb}^{4+}$ due to inert pair effect.

(5) SiF_6^{2-} , GeCl_6^{2-} , $[\text{Sn(OH)}_6]^{2-}$
 sp^3d^2 sp^3d^2 sp^3d^2

Practice

Q. CCl_4 does not hydrolyse, while SiCl_4 does, why?



Q. Find acidic, amphoteric & basic oxides?

$\boxed{\text{CO}}$	$\boxed{\text{CO}_2, \text{SiO}_2}$	$\boxed{\text{PbO, SnO}}$
Neutral	Acidic	Amphoteric

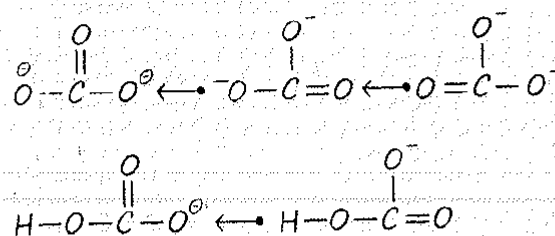
Q. Thermodynamically most stable allotrope of C is _____

Graphite

$G > F > D$

Q. $\text{PbCl}_4 \xrightarrow{\Delta} \text{PbCl}_2 + \text{Cl}_2$
 Less stable more stable

Q. Draw Resonating struc of CO_3^{2-} & HCO_3^-



Nitrogen family

E.N		Size & B.P
↓	N	↓
	P	
	As	
	Sb	
	Bi	
↓		↓
decreases		increases

Exception M.P

$\text{N} < \text{P} < \text{As} > \text{Sb} > \text{Bi}$

$\text{N} < \text{P} < \text{Bi} < \text{Sb} < \text{As}$

[MP $\propto$ C.L.E]

Oxidation number

+5 & +3

Stability of +5 O.N decreases down the group

Stability of +3 O.N increases down the group

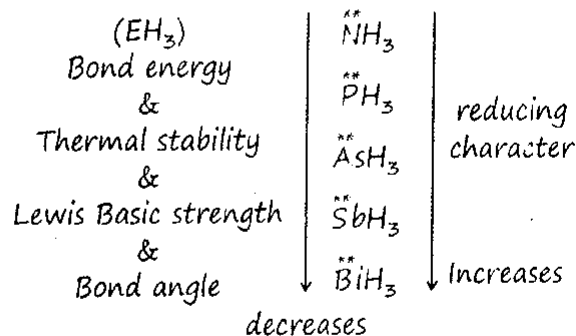
Due to I.P.E

Stability		Stability		Stability
As^{+5}	↓	As^{+3}	↓	$\text{As}^{+5} > \text{As}^{+3}$
Sb^{+5}		Sb^{+3}		$\text{Sb}^{+5} > \text{Sb}^{+3}$
Bi^{+5}		Bi^{+3}		$\text{Bi}^{+5} < \text{Bi}^{+3}$
Decreases		Increases		

NITROGEN FAMILY

Chemical properties:

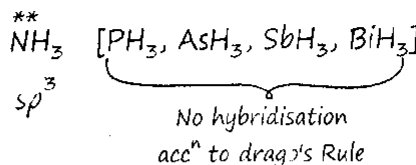
1. Reactivity towards Hydrogen



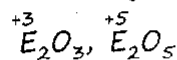
Exceptions

due to Inter n.olecular
H-Bonding

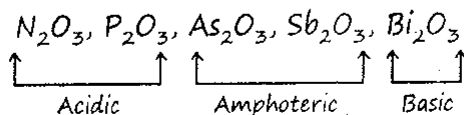
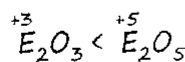
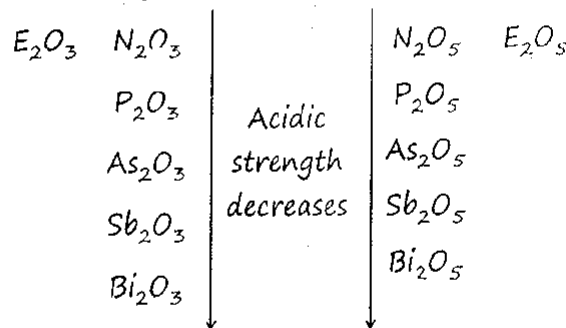
1. B.P # PH₃ < AsH₃ < NH₃ < SbH₃ < BiH₃
2. M.P # PH₃ < AsH₃ < SbH₃ < BiH₃ < NH₃

Hybridisation

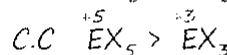
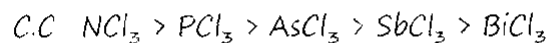
2. Reactivity towards Oxygen



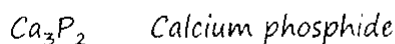
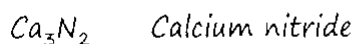
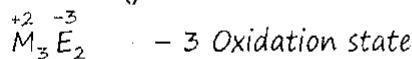
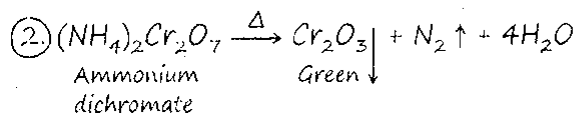
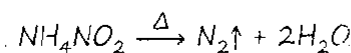
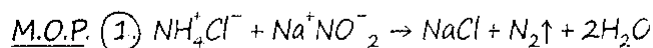
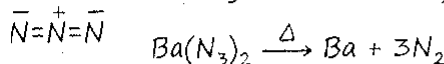
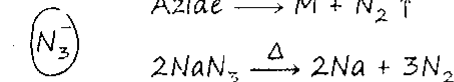
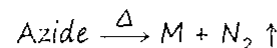
Acidic strength

3. Reactivity towards Halogen ⁺³EX₃, ⁺⁵EX₅

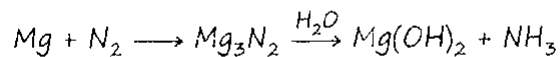
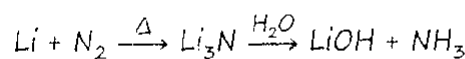
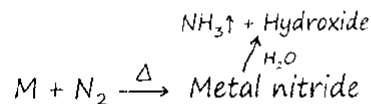
covalent charac # Fajan's rule



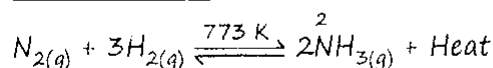
4. Reactivity towards Metals.

Some Imp. comp. of Nitrogen1. Dinitrogen (N₂)(3) For very N₂Property:

1. N₂ is inert at room temp due to High Bond diss. energy (N≡N)

2. Rxn with N₂

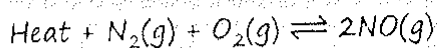
3. Haber's process



Fav. condⁿ for NH₃ production.

Low Temp, High pressure

4. Birkland Eyde process

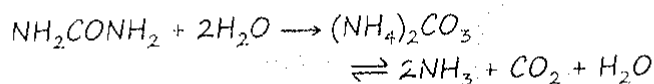


Fav. condⁿ for NO production

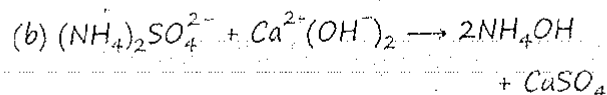
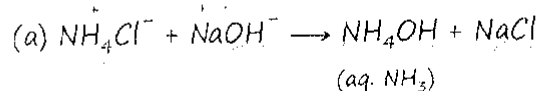
High temp, not affected by pressure

Ammonia (NH₃)

M.O.P 1. From Urea



2. From ammonium salts

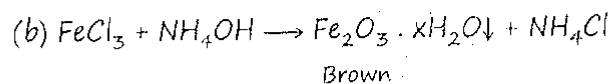
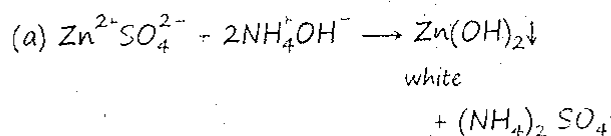


3. Haber's process

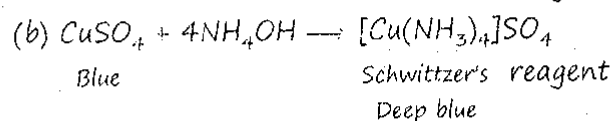
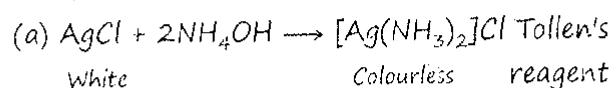
Prop

1. Colourless, Pungent smelling gas

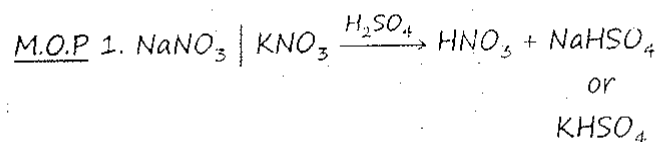
2. Rxn with metal salt



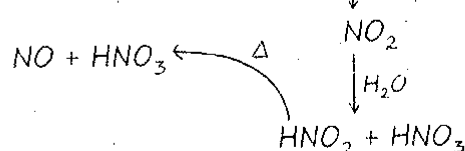
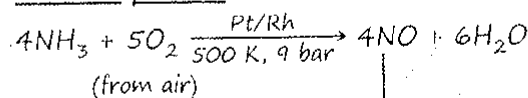
3. Complex formation



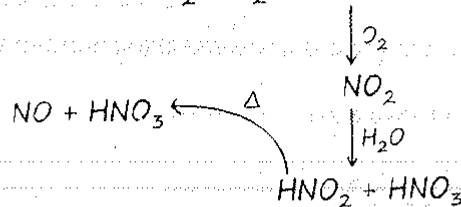
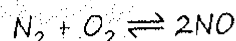
Nitric acid (HNO₃)



2. Ostwald process



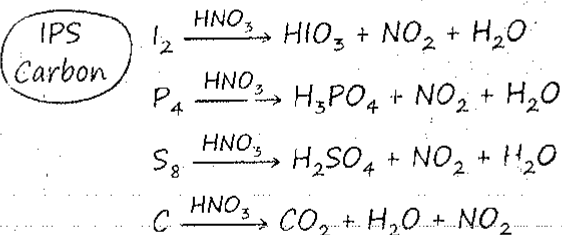
3. Birkland Eyde process



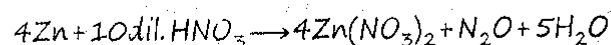
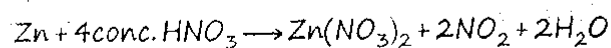
$\Delta H = +ve$

Prop

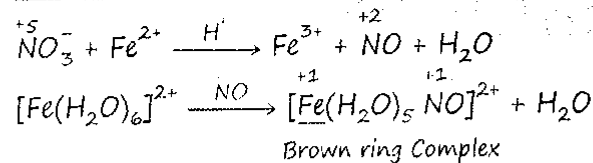
1. Rxn with non metals



2. Rxn with Metals



3. Brown ring Test (for NO₃⁻)

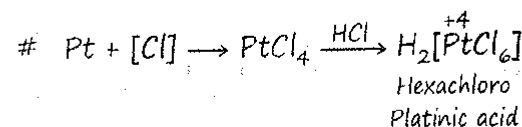
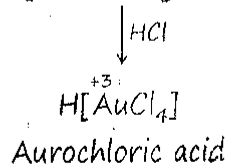
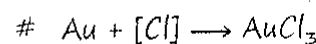
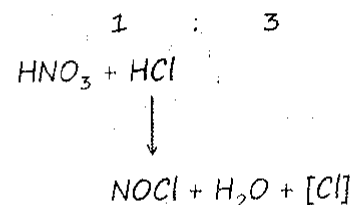


$$x + 5 \times 0 + 1 = 2$$

$$\boxed{x = 1}$$

4. Rxn of Au & Pt with aqua regia

(Conc HNO₃ + Conc HCl)



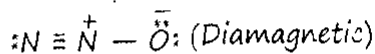
Oxides of Nitrogen

1. N_2O (Nitrous oxide)

O.N = +1

Colourless, neutral gas (laughing gas)

Struc # $N \equiv N \rightarrow O$ or



M.O.P $NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$
Ammonium nitrate

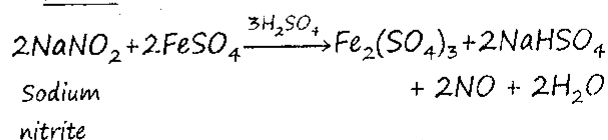
2. NO (Nitric oxide)

O.N = +2

Colourless, neutral gas

struc # $:\dot{N} = \ddot{O}: \text{ (Paramagnetic)}$

M.O.P

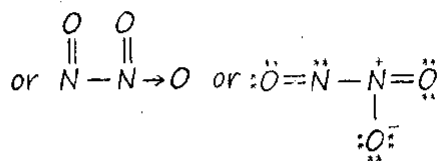


3. N_2O_3 (Nitrogen sesqui oxide)

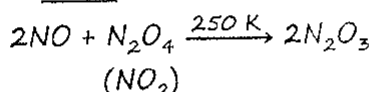
O.N = +3

Blue solid, acidic

struc # $\ddot{O} = \ddot{N} - \ddot{O} - \ddot{N} = \ddot{O}$
(Diamagnetic)



M.O.P



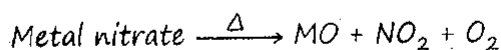
4. NO_2 (nitrogen dioxide)

O.N = +4

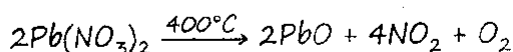
Brown acidic gas

struc # $O = \dot{N} \rightarrow O$ or $O = \overset{+}{N} - O^-$
(Para)

M.O.P



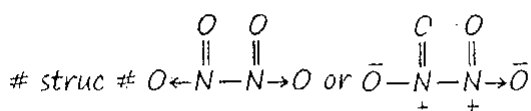
M = Alkaline earth metal, Pb, Zn, Cu, Li



5. N_2O_4 (di Nitrogen tetra oxide)

O.N = +4

Colourless acidic solid/liq.

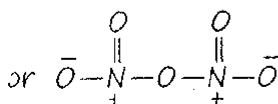
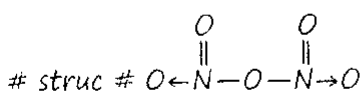


M.O.P # $2NO_2 \xrightleftharpoons[\text{Heat}]{\text{Cool}} N_2O_4$
Brown Colourless

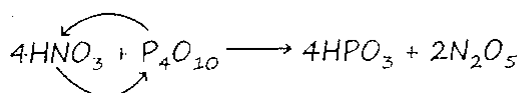
6. N_2O_5 (dinitrogen penta oxide)

O.N = +5

Colourless acidic solid



M.O.P

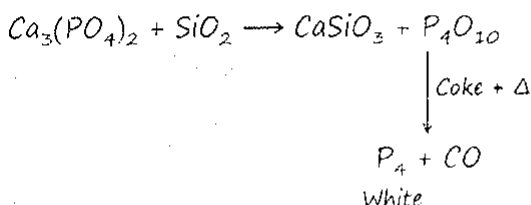


Solid $N_2O_5 \Rightarrow NO_2^+ NO_3^-$
sp sp²

Phosphorus

White phosphorus

M.O.P 1. From thomas slag

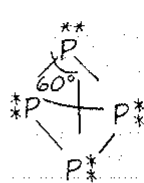


Prop

1. White waxy solid
2. Insoluble in H_2O but soluble in CS_2
3. Glows in dark
4. Kept under water

Structure

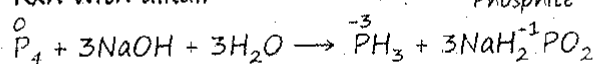
P_4 (White)



- (a) sp^3
- (b) Trigonal pyramid
- (c) $\angle PPP = 60^\circ$
- (d) Total P-P = 6 Bond
- (e) $Lp = 4$

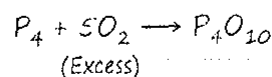
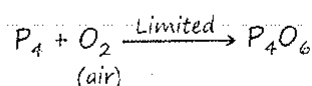
Chemical Rxns

1. Rxn with alkali



Disproportion.

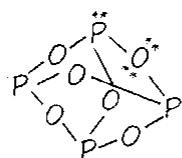
2. Rxn with oxygen



Sodium
Hypo
Phosphite

Struc

1. P_4O_6



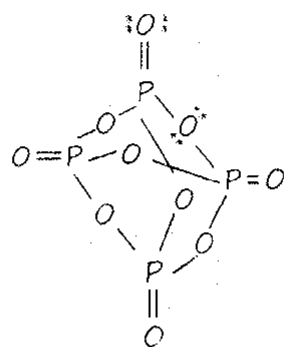
Sigma Bond = 12

π Bond = 0

$Lp = 16$

Hybridisation of P = sp^3

2. P_4O_{10}



Sigma Bonds = 16

π Bond = 4

Hybri of P = sp^3

$P_{\pi} - d_{\pi}$ Bonds = 4

$Lp = 20$

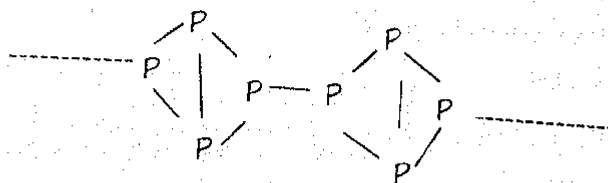
Red Phosphorus

M.O.P # White $P_4 \xrightarrow{200^\circ C}$ Red P_4

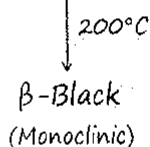
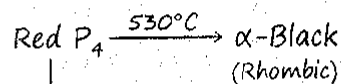
Prop

1. non poisonous in nature
2. does not glow in dark
3. Insoluble in both H_2O & CS_2

Structure # Polymeric form



Black Phosphorus



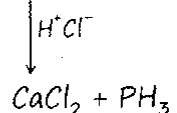
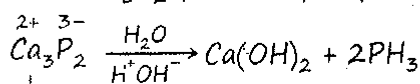
Reactivity $W > R > B$

Stability $W < R < B$

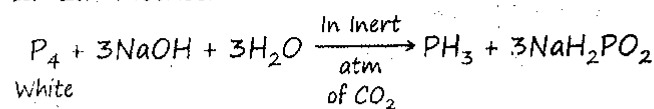
Phosphorus (PH_3)

M.O.P

1. From Ca_3P_2 (Calcium phosphide)

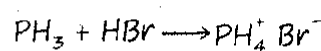


2. Lab Method

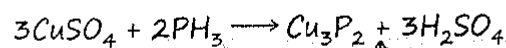


Prop

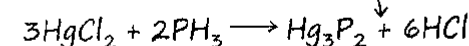
1. Colourless gas
2. Poisonous in nature
3. Basic in nature



4. Rxn with metal salts



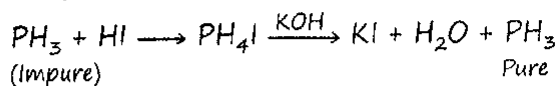
Black



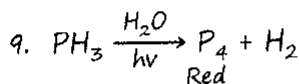
5. Rotten fish smell

6. non inflammable when pure, but becomes inflammable in presence of P_2H_4 or P_4 vapours.

7. For purification of PH_3

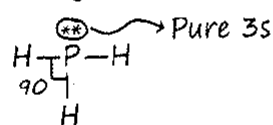


8. It explodes in presence of O.A like HNO_3 , Cl_2 & Br_2 vap



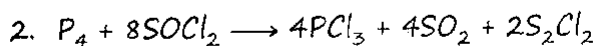
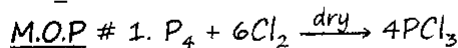
10. Hybridisation of P

No hybridisation accⁿ to drago's rule

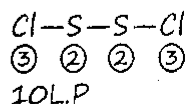


Halides of Phosphorus

PCl_3 #

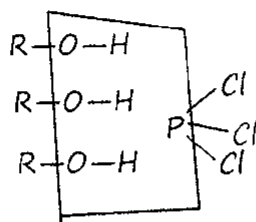
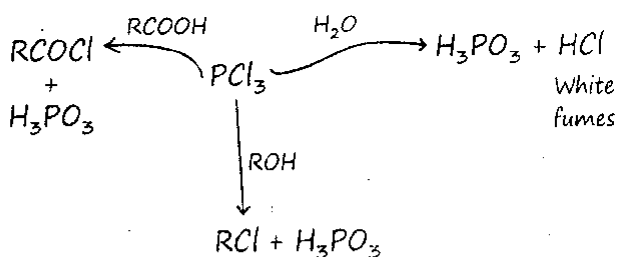


Thionyl
Chloride



Prop. # 1. Colourless oily liq

2.



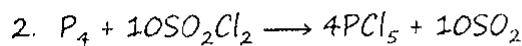
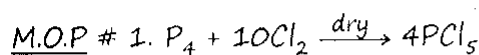
3. sp^3 , Trigonal Pyramidal

Oxyacids of phosphorus

Name	Formula	Structure	M.O.P	Basicity
(1) Hypophosphorous acid (Phosphinic acid)	H_3PO_2	$\begin{array}{c} \text{O} \\ \\ \text{H} \text{---} \text{P} \text{---} \text{OH} \\ \\ \text{H} \end{array}$	White $\text{P}_4 + \text{NaOH}$ $\downarrow$ $\text{PH}_3 + \text{NaH}_2\text{PO}_2$ $\downarrow \text{H}^+$ H_3PO_2	1

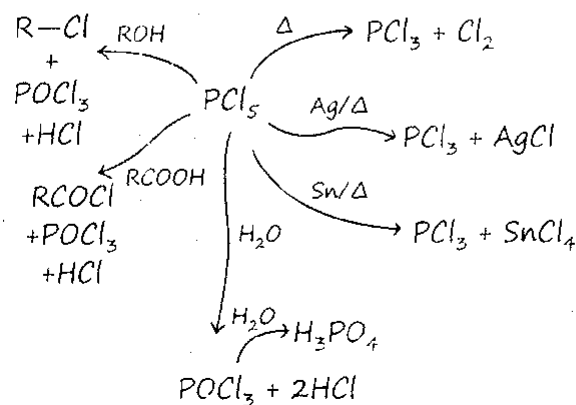
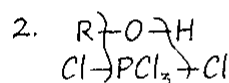
Halides of Phosphorus

PCl_5 #

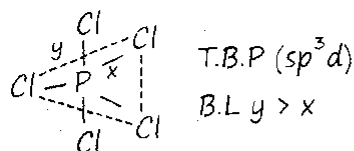


(Sulphuryl Chloride)

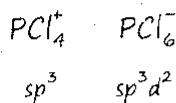
Prop. # 1. Yellowish powder



3. Struc



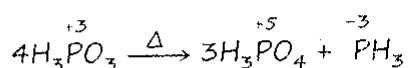
4. Solid PCl_5



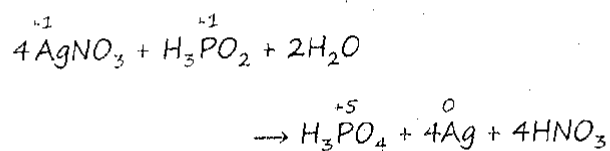
(2) Ortho phosphorous acid (Phosphonic acid)	$\overset{+3}{\text{H}_3\text{PO}_3}$	$\begin{array}{c} \text{O} \\ \parallel \\ \text{H}-\text{P}-\text{OH} \\ \\ \text{OH} \end{array}$	$\text{P}_2\text{O}_3 + \text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_3$	2
(3) Pyro phosphorous acid	$\overset{+3}{\text{H}_4\text{P}_2\text{O}_5}$	$\begin{array}{c} \text{O} \quad \text{O} \\ \parallel \quad \parallel \\ \text{HO}-\text{P}-\text{O}-\text{P}-\text{OH} \\ \quad \\ \text{H} \quad \text{H} \end{array}$	$\text{PCl}_5 + \text{H}_3\text{PO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_4\text{P}_2\text{O}_5$	2
(1) Hypophosphoric acid	$\overset{+4}{\text{H}_4\text{P}_2\text{O}_6}$ $4 + 2x - 12 = 0$ $x = 4$	$\begin{array}{c} \text{O} \quad \text{O} \\ \parallel \quad \parallel \\ \text{HO}-\text{P}-\text{P}-\text{OH} \\ \quad \\ \text{OH} \quad \text{OH} \end{array}$	$\text{Red P}_4 + \text{NaOH} \downarrow \text{H}_4\text{P}_2\text{O}_6$	4
(5) Ortho phosphoric acid	$\overset{+5}{\text{H}_3\text{PO}_4}$	$\begin{array}{c} \text{O} \\ \parallel \\ \text{HO}-\text{P}-\text{OH} \\ \\ \text{OH} \end{array}$	$\text{P}_2\text{O}_5 + \text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4$	3
(6) Pyro phosphoric acid	$\overset{+5}{\text{H}_4\text{P}_2\text{O}_7}$	$\begin{array}{c} \text{O} \quad \text{O} \\ \parallel \quad \parallel \\ \text{HO}-\text{P}-\text{O}-\text{P}-\text{OH} \\ \quad \\ \text{OH} \quad \text{OH} \end{array}$	$2\text{H}_3\text{PO}_4 \xrightarrow{\Delta} \text{H}_4\text{P}_2\text{O}_7$	4
(7) Poly meta phosphoric acid	$(\text{HPO}_3)_n$	$\begin{array}{c} \text{O} \quad \text{O} \\ \parallel \quad \parallel \\ \text{O}-\text{P}-\text{O}-\text{P}-\text{O} \\ \quad \\ \text{OH} \quad \text{OH} \end{array}$	$\text{H}_3\text{PO}_3 + \text{Br}_2$	-

Some Imp. points of N-family

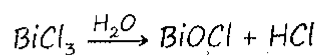
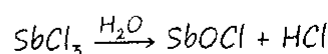
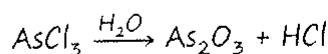
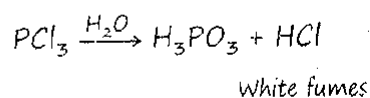
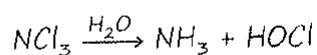
1. Dispro of H_3PO_3



2. Rxn of AgNO_3 with H_3PO_2

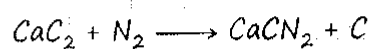


3. Hydrolysis of trichlorides

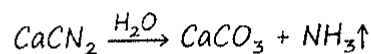


Pearl white
Bismuth oxy chloride

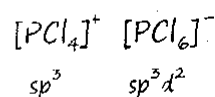
4. Nitrolim $\text{CaC}_2 + \text{N}_2$



Calcium Carbide Calcium Cyanamide



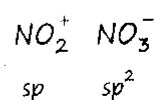
5. Solid PCl_5 exist as a dimer



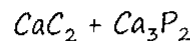
6. Solid PBr_5 exists as $[\text{PBr}_4]^+ \text{Br}^-$

sp^3

7. Solid N_2O_5 exist as Nitronium nitrate

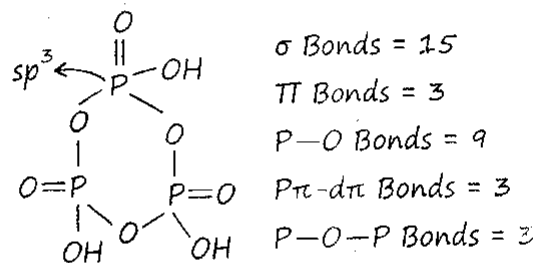


8. Holme's signal

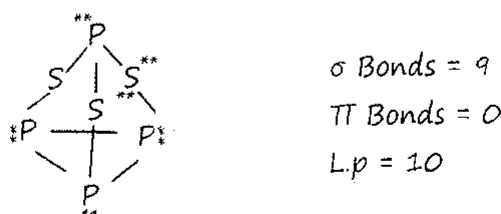


9. Structure of $(\text{HPO}_3)_3$

Cyclic trimeric metaphosphoric acid

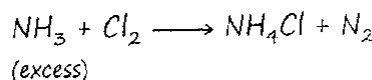
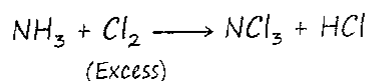


10. Structure of P_4S_3



11. Ammonia can be dried over quick lime (CaO)

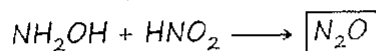
12. Rxn of NH_3 with Cl_2



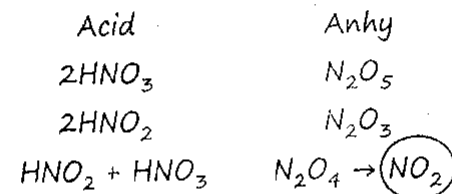
13. Rxn of NH_3 with CaOCl_2



14. Rxn of NH_2OH with HNO_2



15. Acid Anhydride concept



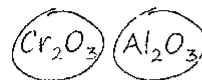
16. N does not form pentahalides due to non availability of vacant 'd' orbitals.

17. All trihalides of N except NF_3 are unstable. Only NF_3 is stable due to great extent of overlapping.

18. All trihalides except BiF_3 are covalent in nature, BiF_3 is ionic due to Fajan's rule.

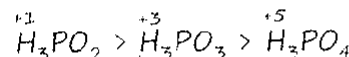
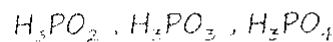
19. BiCl_5 does not exist due to I.P.E

20. Cr & Al do not react with HNO_3 due to formation of passive film of oxide

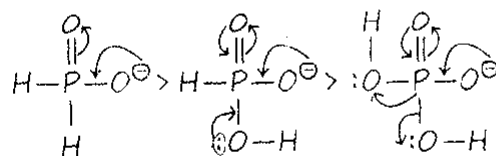


Practice

Q. Compare acidic strength



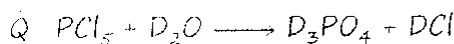
Stability



Q. Why does $\text{R}_3\text{P}=\text{O}$ exist but $\text{R}_3\text{N}=\text{O}$ does not? due to non availability of vacant 'd' orbitals in N

Q. N show catenation less than P?

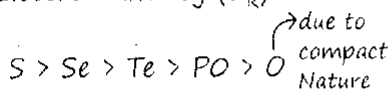
$\text{N}-\text{N}$ due to LP-LP repulsion in N



Oxygen Family

Size	O	I.E
&	S	&
MP & BP	Se	E.N
& density	Te	
	Po	
	Increases	decreases

Electron affinity (E_R):



Oxidation No.

+4 & +6

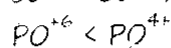
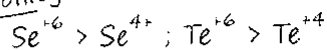
Stability of +6 O.N decreases

+4 O.N increases

Stability	Stability
Se ⁺⁶	Se ⁺⁴
Te ⁺⁶	Te ⁺⁴
PO ⁺⁶	PO ⁺⁴
decreases	Increases

due to I.P.E

Stability



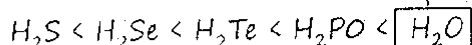
O.N of oxygen

O_2	zero
O^{2-}	Oxide -2
O_2^{2-}	Peroxide -1
O_2^-	Superoxide -1/2
$\text{O}_2^+ [\text{AsF}_6]^-$	+1/2
KO_3	-1/3
OF_2	+2
O_2F_2	+1

Prop

1. Reactivity Towards Hydrogen

BE & T.S & Bond angle	H_2O H_2S H_2Se H_2Te H_2Po	Acidic strength & Reducing Nature & Bond length
decreases (B.P & M.P)		Increases due to intermolecular H-Bonding



2. Reactivity towards oxygen

Oxidising power	EO_2 O_3 SO_2 SeO_2 TeO_2 PoO_2	EO_3 - SO_3 SeO_3 TeO_3 PoO_3	O.P
Increases			Increases

O.P $\text{EO}_3 > \text{EO}_2$

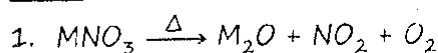
3. Reactivity towards halogen

EX_4	EX_6
Covalent charac.	$\text{EX}_6 > \text{EX}_4$
(Fajan's rule)	
Small cation, Bigger anion.	

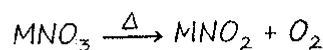
Some Imp. comp. of oxygen

1. Dioxygen

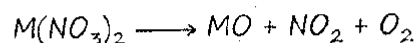
M.O.P



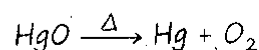
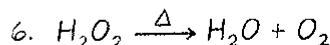
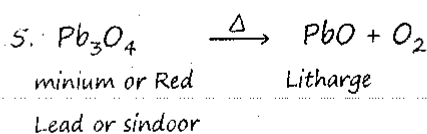
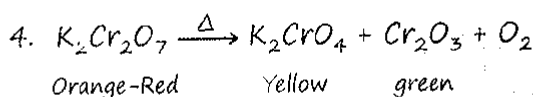
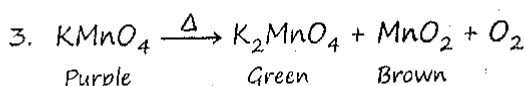
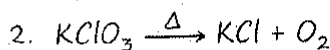
M = Li



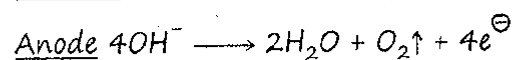
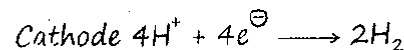
M = all alkali except Li



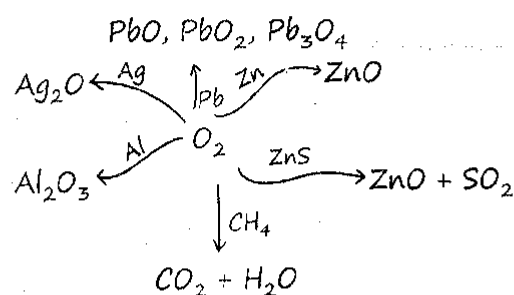
M = all alkaline earth metal Zn, Cu, Pb



8. Electro of H_2O



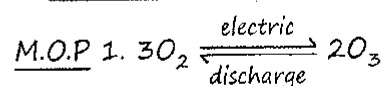
Prop. of dioxygen



Colourless paramagnetic gas (two unpaired e's),

B.O = 2

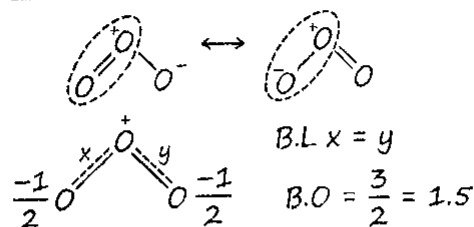
O_3 (Ozone)



Prop

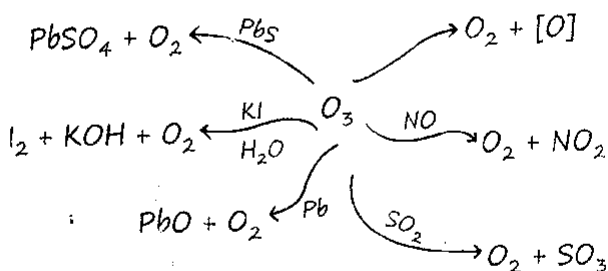
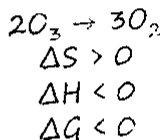
1. O_3 is pale blue gas, dark blue liq, violet black solid

2. Struc



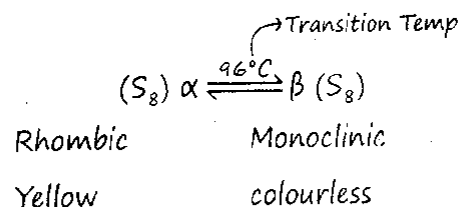
3. Chemical Rxn

O_3 is a powerful O.A.



Allotropes of S

α -Sulphur # It is formed by evaporating the solⁿ of S in CS_2 .



(1) at 96°C

Both are stable

(2) $T < 96^\circ\text{C}$

α stable

(3) $T > 96^\circ\text{C}$

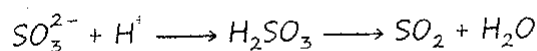
β stable

Some Imp. comp. of oxygen

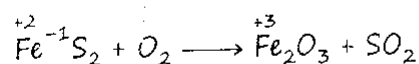


M.O.P 1. $\text{S} + \text{O}_2 \longrightarrow \text{SO}_2$

2. Lab Method



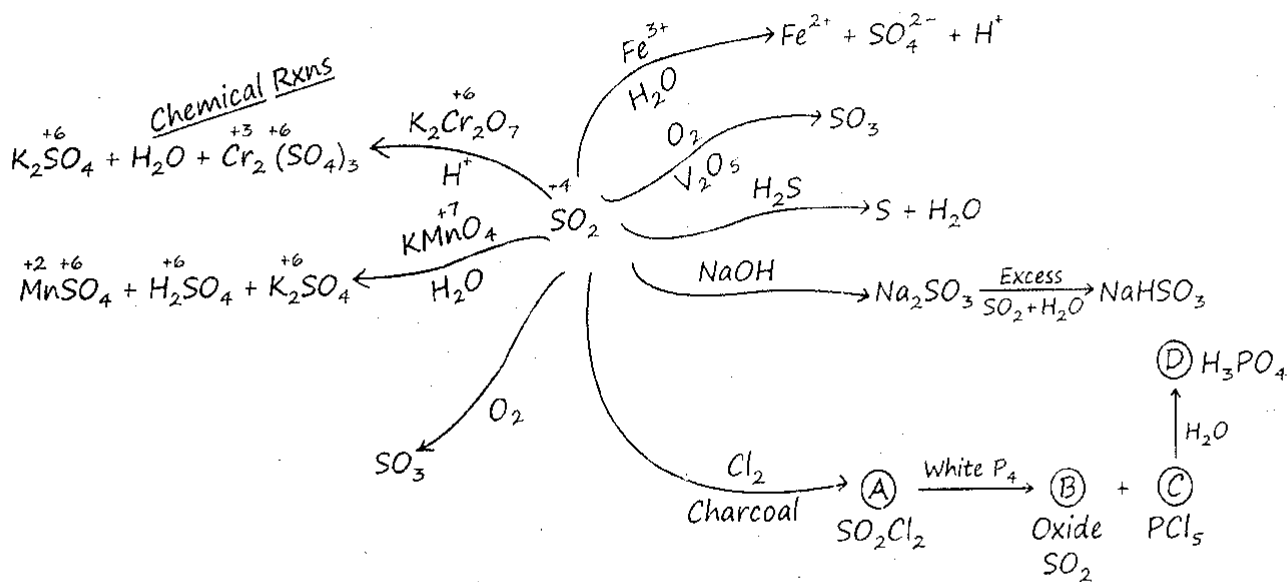
3. Industrial method



4. $\text{Cu} + \text{Conc H}_2\text{SO}_4 \longrightarrow \text{CuSO}_4 + \text{SO}_2 + \text{H}_2\text{O}$

Prop

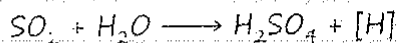
1. Colourless gas
2. Pungent smell
3. Soluble in H_2O
4. liq at Room temp



Some Imp. comp. of Sulphur



SO₂ acts as bleaching agent



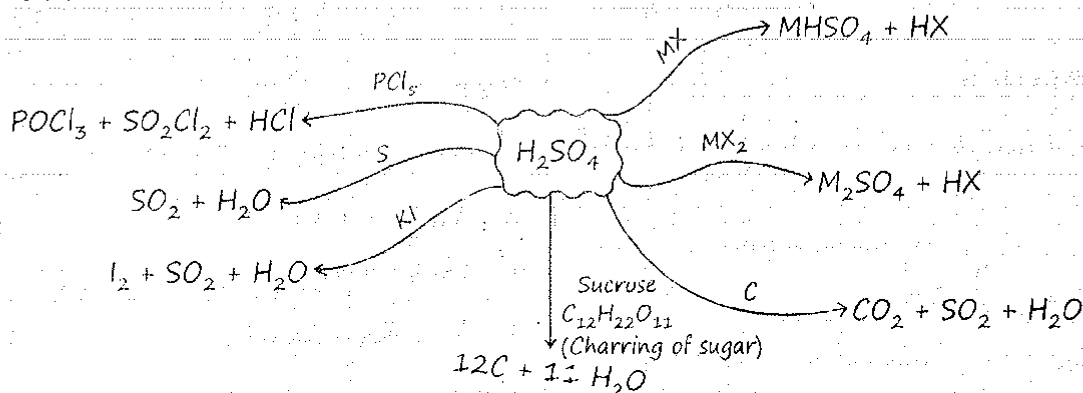
Coloured matter
↓
Colourless [O]

Reductive bleaching

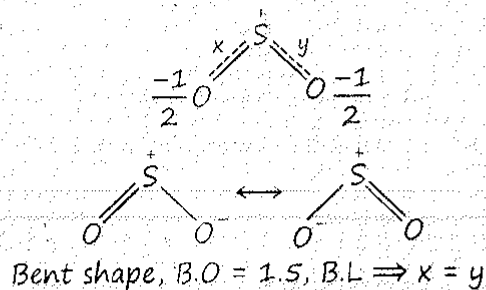
Temporary bleaching

Prop

Chemical Rxns

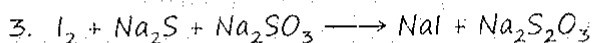
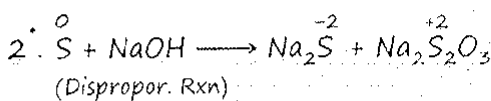
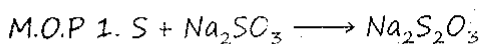
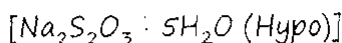


Struc.



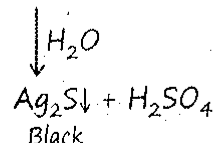
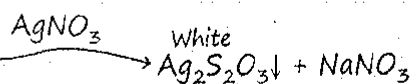
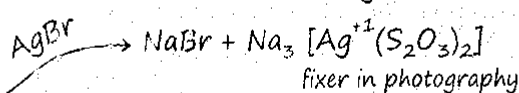
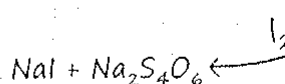
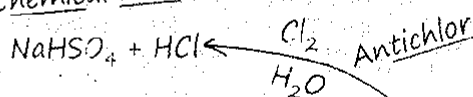
H₂SO₄ King of chemicals

Na₂S₂O₃ Sodium thio sulphate



Na₂S₂O₃ (Sodium thiosulphate)

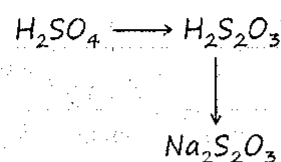
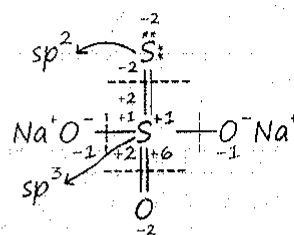
Chemical Rxns



Sodium argento thiosulphate

Prop

1. Structure

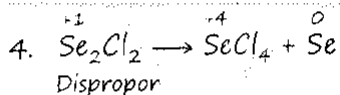


Oxyacids of Sulphur

Formula	O.N	Name	Structure
(1) H_2SO_3	+4	Sulphurous acid	$\begin{array}{c} O \\ \\ HO-S-OH \\ \cdot \cdot \end{array}$
(2) H_2SO_4	+6	Sulphuric acid	$\begin{array}{c} O \\ \\ H-O-S-O-H \\ \\ O \end{array}$
(3) H_2SO_5	+6	Caro's acid or peroxy mono sulphuric acid	$\begin{array}{c} \begin{array}{c} O \\ \\ H-O-S-O-O-H \\ \quad \quad \\ O \quad -1 \quad -1 \\ \cdot \cdot \end{array} \\ \downarrow \text{Peroxy Linkage} \\ O-O \\ \cdot \cdot \end{array}$
(4) $H_2S_2O_3$	-2 & +6	Thiosulphuric acid	$\begin{array}{c} S^{-2} \\ \\ H-O-S-O-H \\ \\ O \end{array}$
(5) $H_2S_2O_4$	+3	dithionous acid	$\begin{array}{c} O \quad O \\ \quad \\ HO-S-S-OH \\ \cdot \cdot \end{array}$
(6) $H_2S_2O_5$	+5 & +3	Pyrosulphurous acid	$\begin{array}{c} O \quad O \\ \quad \\ HO-S-S-OH \\ \cdot \cdot \end{array}$
(7) $H_2S_2O_6$	+5	Dithionic acid	$\begin{array}{c} O \quad O \\ \quad \\ HO-S-S-OH \\ \cdot \cdot \end{array}$
(8) $H_2S_2O_7$	+6	Oleum or pyrosulphuric acid	$\begin{array}{c} O \quad O \\ \quad \\ H-O-S-O-S-O-H \\ \cdot \cdot \end{array}$
(9) $H_2S_2O_8$	+6	Marshall's acid or peroxy disulphuric acid	$\begin{array}{c} \begin{array}{c} O \\ \\ HO-S-O-O-S-O-H \\ \quad \quad \\ O \quad -1 \quad -1 \\ \cdot \cdot \end{array} \\ \downarrow \text{Peroxy Bond} \\ O-O \\ \cdot \cdot \end{array}$

Some Imp. points of chalcogens

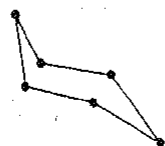
1. OF_6 does not exist due to absence of vacant 'd' orbitals.
2. All hexahalides of S are unstable except SF_6 .
3. SF_6 does not hydrolyse



5. Struc of S_8 , S_6
 S_8 Puckered Ring
Crown shape

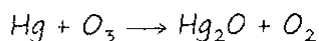


S_6 Chair form

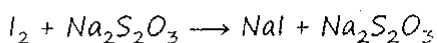
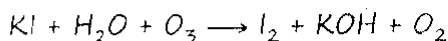


S_2 paramagnetic just like O_2
Exist at 1000 K

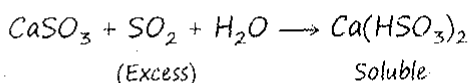
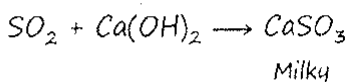
6. Tailing of Hg # Disturbance of Meniscus of Hg by passing ozone



7. Estimation of ozone $\div$



8. SO_2 turns lime water milky



Practice

Q. H_2O is liq, while H_2S is gas why?

Due to intermolecular H Bonding in H_2O .

Q. Which of the following does not react with oxygen directly?

(A) Zn (B) Ti

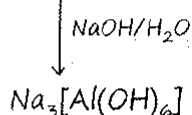
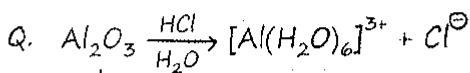
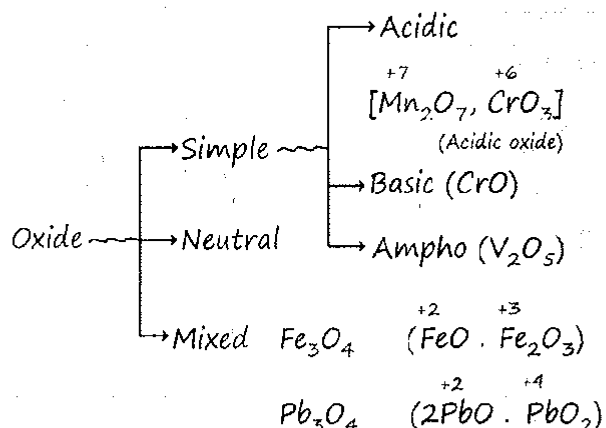
(C) Pt (D) Fe

Au & Pt Noble metal

Q. Which is mixed oxide?

(A) Mn_2O_7 (B) PbO_2

(C) Fe_3O_4 (D) ZnO



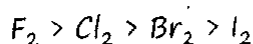
Size	F	IE
BP	Cl	EN
MP		
density	Br	
Increases ↓	I	↓ Decreases

Electron gain enthalpy

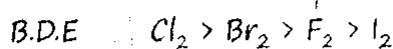
due to $\swarrow$
vacant 'd' orbitals $\text{Cl} > \text{F} > \text{Br} > \text{I}$

F_2	Yellow gas
Cl_2	Greenish yellow gas
Br_2	Reddish brown liq
I_2	Violet black solid

1. Oxidising power



2. Exception



→ due to L.p-L.p repulsion

Chemical Prop #

1. Reactivity towards Hydrogen

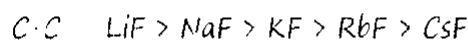
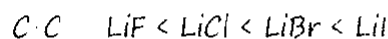
Acidic strength	HF	B.D.E
&		&
Reducing power	HCl	T.S
&		&
Bond length	HBr	Stability
	HI	
↓		↓
Increases		Decreases

Exception

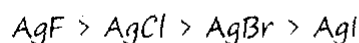
(1) B.P $\text{HCl} < \text{HBr} < \text{HI} < \text{HF}$

(2) M.P $\text{HCl} < \text{HBr} < \text{HF} < \text{HI}$

2. Reactivity towards Metals

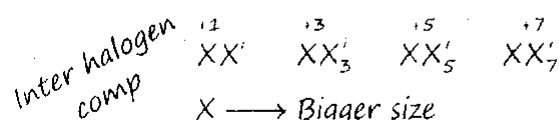


Solubility of silver halides



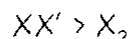
Soluble	White ppt	Pale yellow ppt	Yellow ppt
---------	-----------	-----------------	------------

3. Reactivity of halogen with other halogens


$$X \longrightarrow \text{Bigger size}$$

$X' \longrightarrow$ Small size

Inter halogen comp are usually more reactive than halogens

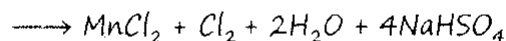
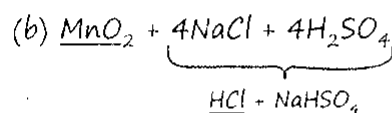
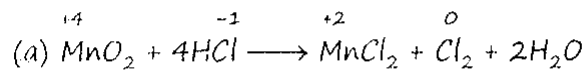


Some Imp. Comp. of CL

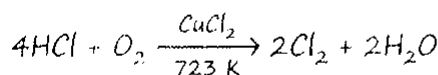


M.O.P

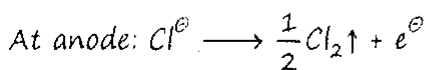
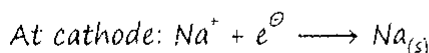
1. From pyrolusite

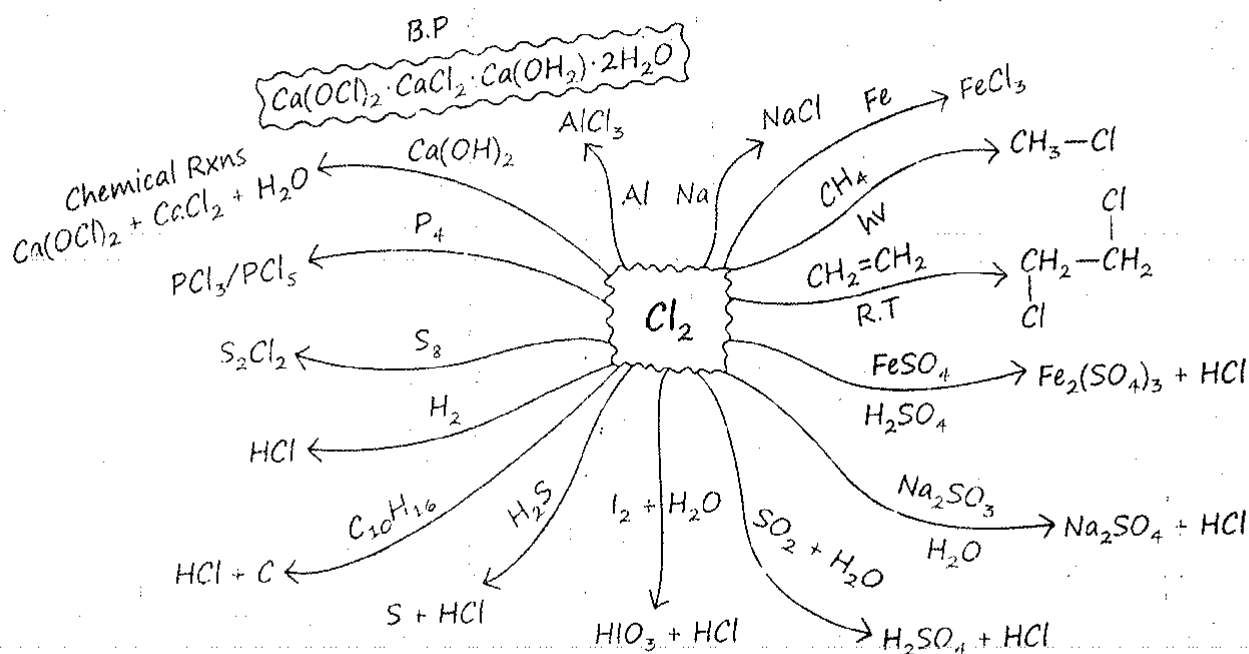


2. Deacon's process

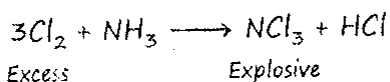
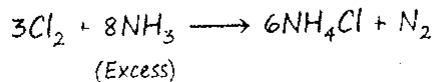


3. Electrolysis of Molten NaCl

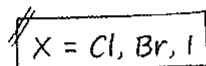
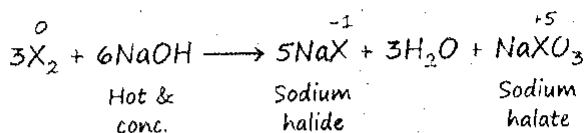
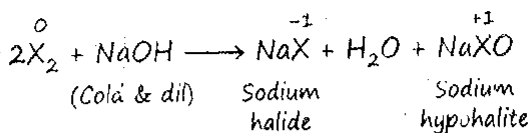




2. Rxn with NH_3

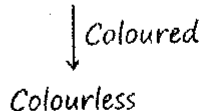
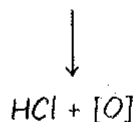
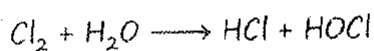


3. Rxn with alkali



Disproportion Rxn

4. Bleaching prop

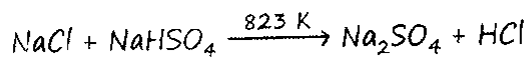
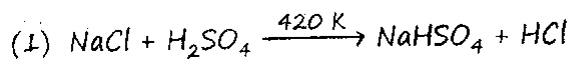


oxidative bleaching

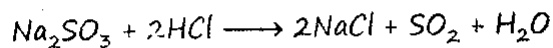
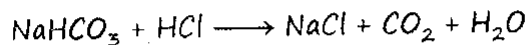
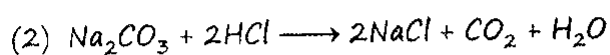
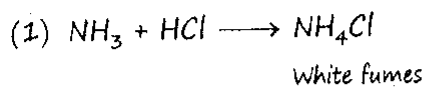
permanent bleaching

2. HCl

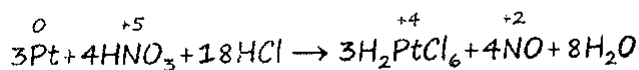
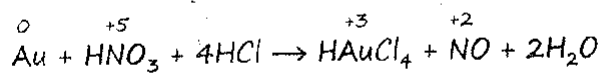
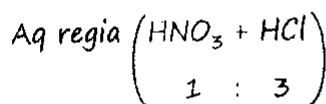
M.O.P



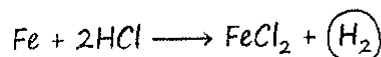
Prop



(3) Rxn of Au and Pt with aq. regia



(4) Rxn of Fe with HCl

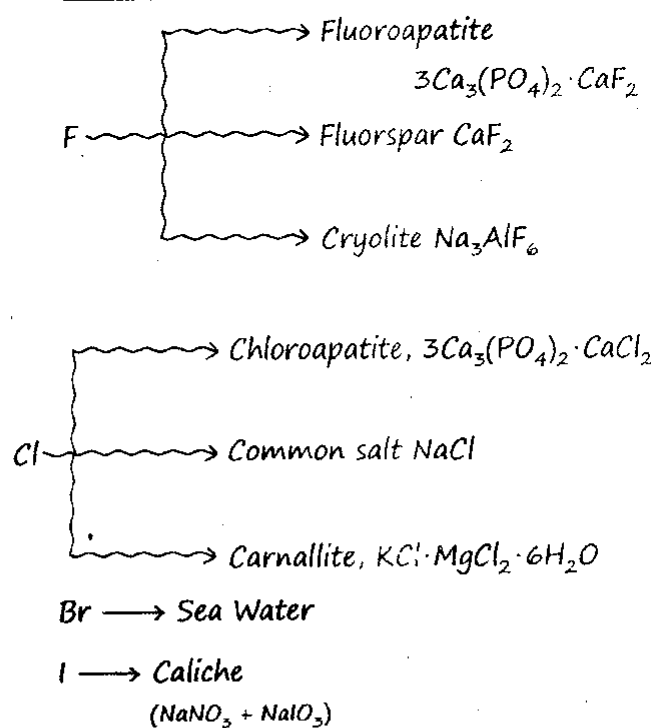


Oxy acids of Chlorine

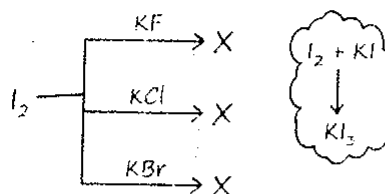
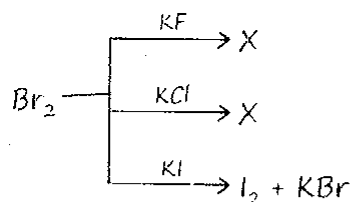
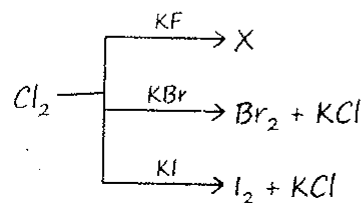
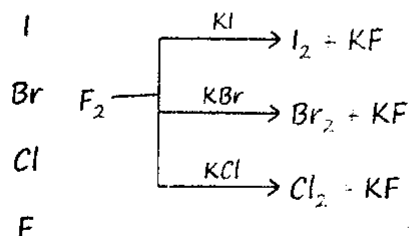
Formula	Name	Structure	O.N
Halic (I) acid HClO	Hypo chlorous acid	$\text{Cl}-\text{OH}$	+1
Halic (III) acid HClO_2	Chlorous acid	$\text{O}=\text{Cl}-\text{OH}$	+3
Halic (V) acid HClO_3	Chloric acid	$\begin{array}{c} \text{O} \\ \\ \text{O}=\text{Cl}-\text{OH} \end{array}$	+5
Halic (VII) acid HClO_4	Perchloric acid	$\begin{array}{c} \text{O} \\ \\ \text{O}=\text{Cl}-\text{OH} \\ \\ \text{O} \end{array}$	+7

Some Imp. points of Halogens

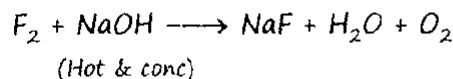
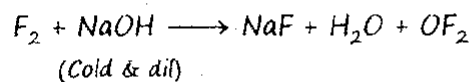
1. Occurrence



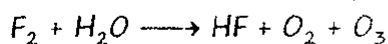
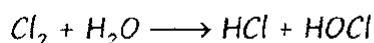
2. Non metal displacement Rxn



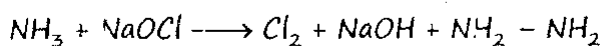
3. Rxn of F_2 with alkali



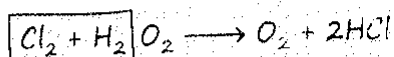
4. Rxn of F_2 & Cl_2 with H_2O



5. Rxn of NH_3 with NaOCl



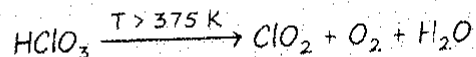
6. Rxn of Cl_2 with H_2O_2



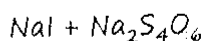
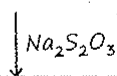
7. Rxn of HF with SiO_2 (Etching of glass)
(Excess)



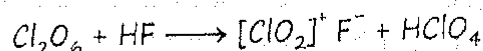
8. Heating of HClO_3



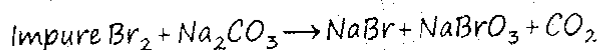
9. Rxn of I_2O_5 with CO



10. Rxn of Cl_2O_6 with HF

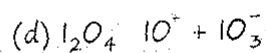
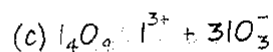
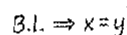
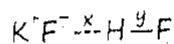
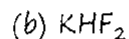
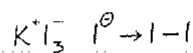
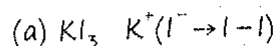


11. Purification of Br_2

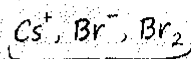


Pure

12. Structures



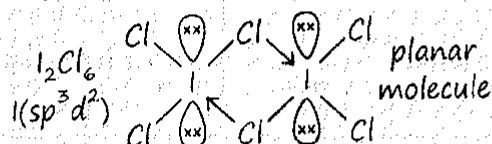
(e) CsBr_3



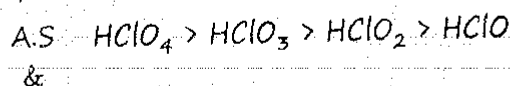
(f) CsI_3



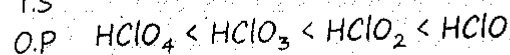
(g) ICl_3 dimer



13. Acidic strength, Thermal stability & oxidising power of oxyacids of Cl .

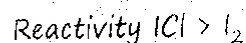


T.S

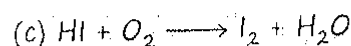
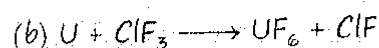
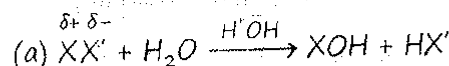


Practice

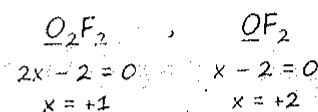
Q. Why is ICl more reactive than I_2 ?



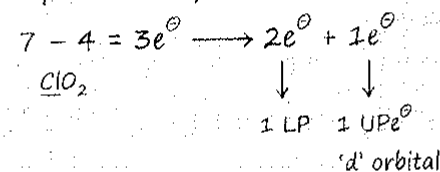
Q. Write the products formed



Q. Find O.N of 'O' in



Q. Unpaired e^- is present in which orbital in ClO_2



I.E	He	Size
&	Ne	B.P
En	Ar	M.P
	Kr	density
	Xe	
	Rn	
Decreases ↓		↑ Increases

Electron gain enthalpy ΔH_{eg}

$\Delta H_{eg} = +ve$ for all noble gases

$Ne < Ar \approx Kr < Xe < Rn < He$

Electron affinity

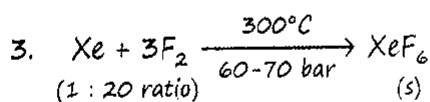
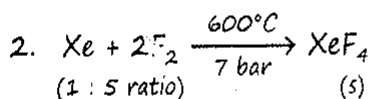
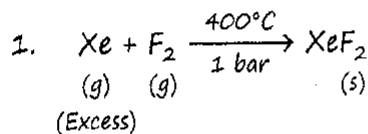
1st comp of Noble gas

$O_2^+[Pt F_6]^-$ 1st I.E(O_2) $\approx$ 1st I.E(Xe)

Red $Xe^+[Pt F_6]^-$
 KrF_2 RnF_2 Red

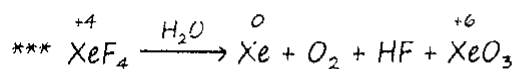
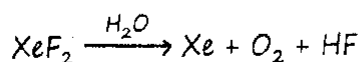
Fluorides of Xe

M.O.P #

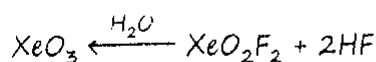
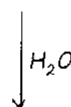
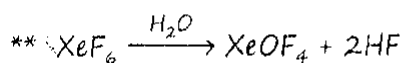


Prop

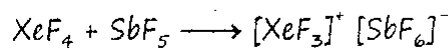
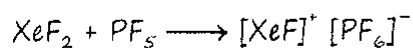
1. Hydrolysis



Disproportionation



2. Rxn of Xenon fluorides with PF_5 , SbF_5 and MF



M = Na, K, Rb, Cs

(Li) small size

Some Imp. Points of Noble Gases

1. Rxn of XeF_4 with O_2F_2

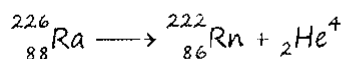


2. He & Ne do not form clathrates due to small size

3. Solubility in H_2O increases from top to bottom

4. Ar is used in arc welding

5. Decay of Ra



6.	O-N	Comp of Xe	Hybridisation	Geometry	Shape	LP on Xe	Total L-P
	+2	XeF_2	$sp^3d(2BP + 3LP)$	T.B.P	Linear	3	9
	+4	XeF_4	$sp^3d^2(4BP + 2LP)$	Octahedral	sq planar	2	14
	+4	XeF_3^+	$sp^3d(3BP + 2LP)$	T.B.P	T-shape	2	11
	+6	XeO_2F_2	$sp^3d(4BP + 1LP)$	T.B.P	See-Saw	1	11
	+6	XeO_3 (Explosive)	$sp^3(3BP + 1LP)$	Tetrahedral	Trigonal pyramidal	1	7

7. Rxn of XeO_3 with base

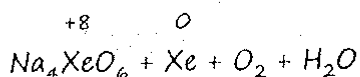
$$1 + 1 + x - 8 = 0$$

$$x = +6$$



$$4 + x - 12 = 0$$

$$x = +8$$



8. Inert gases are least reactive due to high I.E, fully filled config & more +ve electron gain enthalpy.

9. Inert gases can be separated by adsorption and desorption on charcoal.

Practice

Q. Noble gases have very Low B.P why?
due to weak vanderwaal forces

Q. Why has it been difficult to study the chemistry of Rn?
due to small half life.

Q. Why is He used in diving apparatus?
because of Low solubility in blood.

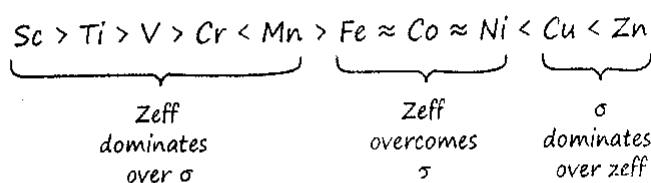
$$(n-1)d^{1-10} ns^{0-2}$$

3d series	Sc	Ti	V	*Cr	Mn	Fe	Co	Ni	*Cu	(Zn)
Z	21	22	23	24	25	26	27	28	29	30
4s	2	2	2	1	2	2	2	2	1	2
3d	1	2	3	5	5	6	7	8	10	10
4d series	Y	Zr	Nb*	Mo*	Tc*	Ru*	Rh*	Pd*	Ag*	(Cd)
Z	39	40	41	42	43	44	45	46	47	48
5s	2	2	1	1	1	1	1	0	1	2
4d	1	2	4	5	6	7	8	10	10	10
5d series	La	Hf	Ta	W	Re	Os	Ir	Pt*	Au*	(Hg)
Z	57	72	73	74	75	76	77	78	79	80
6s	2	2	2	2	2	2	2	1	1	2
5d	1	2	3	4	5	6	7	9	10	10

Prop

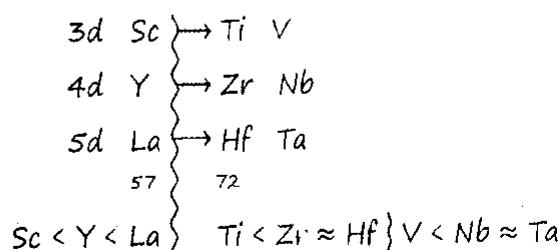
1. Size

(a) 3d series



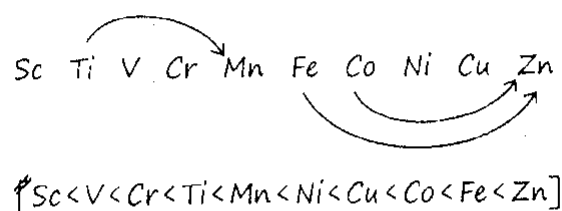
(b) Size $3d < 4d \approx 5d$

due to L.C (poor screening of 4f e⁻s)



2. Ionisation Energy (I.E)

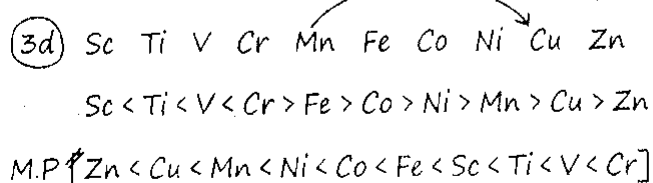
(a) 3d series



(b) IE $5d > 3d > 4d$

due to L.C

3. M.P



4. O.N

+2	+3	+4	+5	+6	+7
Ti - Zn	Sc - Ni	Ti - Ni	V - Mn	Cr - Fe	Mn
Tu zanti Hai ye	Sach nahi Hai	Tu nahi	Ve maahi	Kar fikar	Mere mann ki



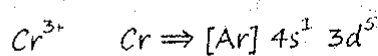
5. Magnetic nature

$$MM = \sqrt{n(n+2)}$$

n = no. of unpaired e^- s

$M.M = 0$ (Diamagnetic)

$MM \neq 0$ (Paramagnetic)



$n = 3$

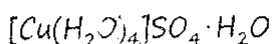
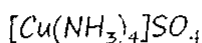
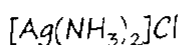


$$MM = \sqrt{3(3+2)} = \sqrt{15}$$

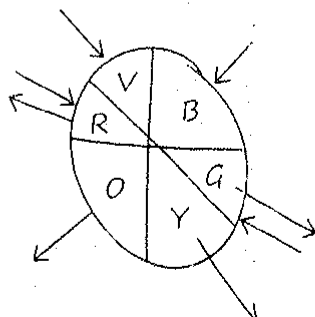
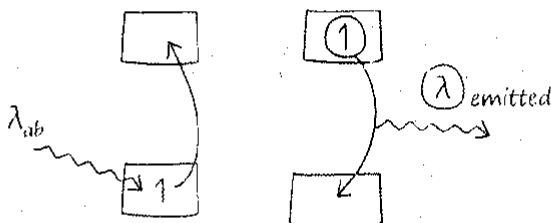
Paramagnetic

6. Complex formation tendency

d-block elements form Large no. of complexes due to small size, high ionic charge & availability of d orbitals.



7. Formation of coloured ion



8. Interstitial comp, formation

I.C are formed when small atoms like H, C, N etc are trapped in the metal crystal lattice

Ti C, Ti H_{1.7}, Fe₃ C etc

(1) Non stoichiometric

(2) Very hard

(3) High M.P than pure metal

(4) Chemically inert

(5) retain metallic conductivity

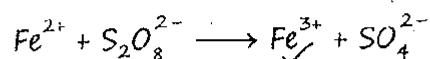
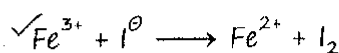
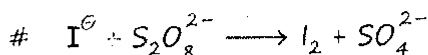
9. Catalytic prop

d-block elements are known for catalytic prop.

V₂O₅ in contact process

Fe in Haber's process

Ni in Hydrogenation



10. Alloy formation

Alloy $\Rightarrow$ Alloy is Homo solid solⁿ in which atoms are randomly distributed

Brass Cu + Zn

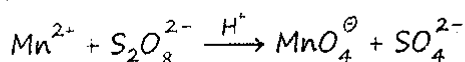
Bronze Cu + Sn

Some Imp. Compounds of d-block

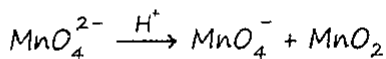
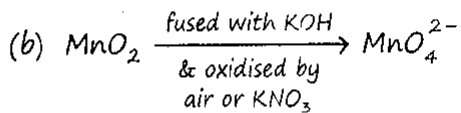
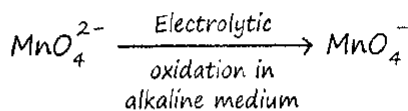
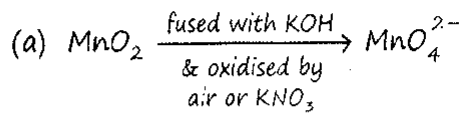
1. KMnO₄

M.O.P

(1) Lab method

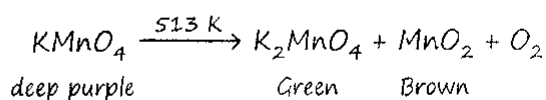


(2) Commercial Method

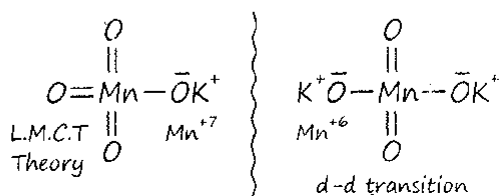


Prop

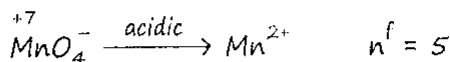
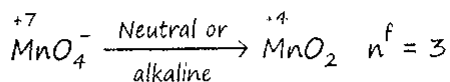
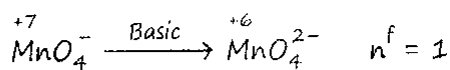
(1) Heating effect



(2) Structure

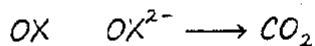
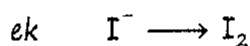


(3) Chemical Rxn

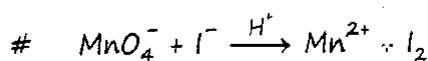
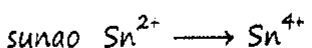
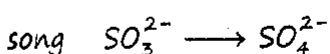
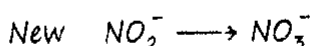
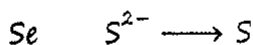
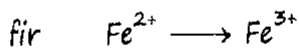


1	3	5
B	N	A

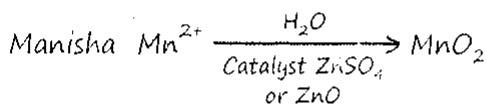
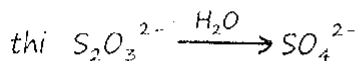
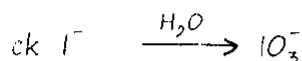
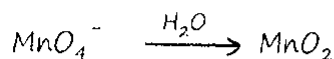
(1) In A.M



KO

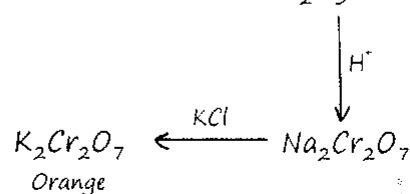
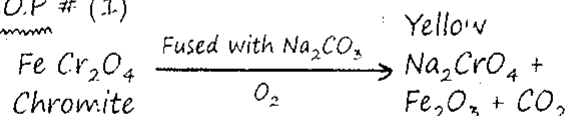


2. In Neutral or Alkaline medium



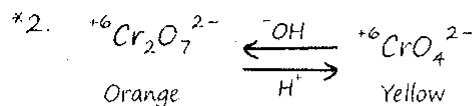
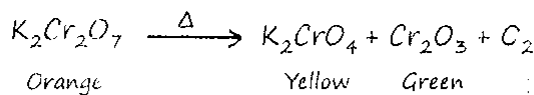
1. $\text{K}_2\text{Cr}_2\text{O}_7$

M.O.P # (1)

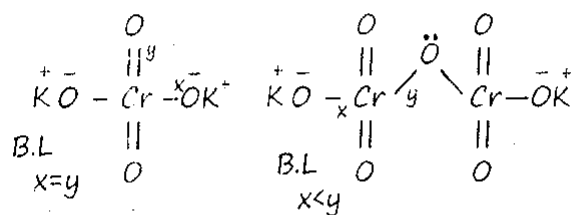


Prop

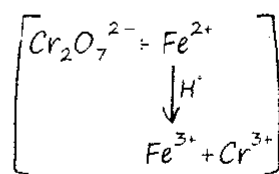
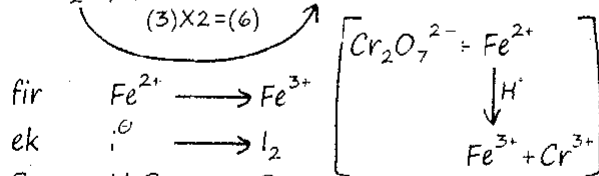
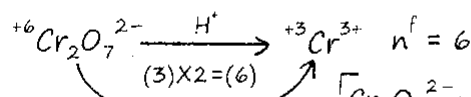
1. Heating effect



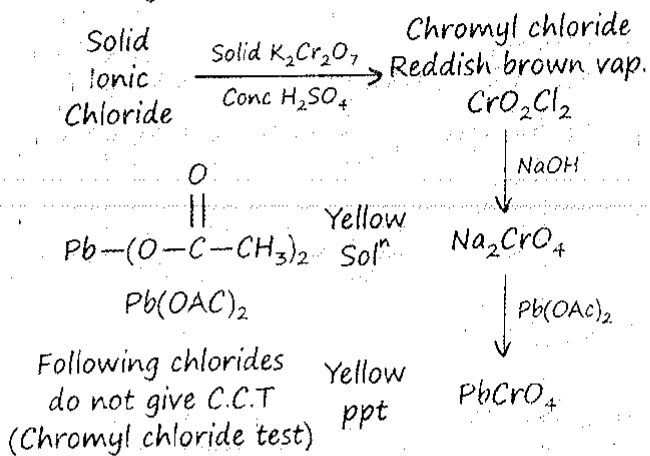
3.



4. $\text{K}_2\text{Cr}_2\text{O}_7$ acts as an O.A in acidic medium



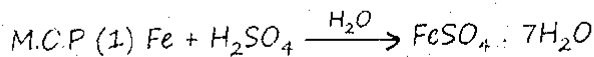
5. Chromyl chloride test



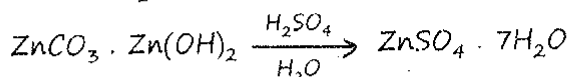
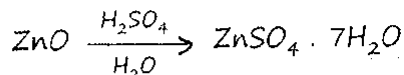
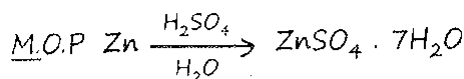
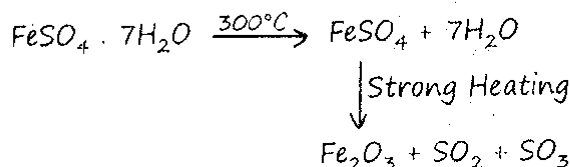
$AgCl, SnCl_4, PbCl_4, Hg_2Cl_2$ & $HgCl_2$

Ag Sn Pb Hg
 aji Suno Pyar Hogya

3. Green vitriol ($FeSO_4 \cdot 7H_2O$)

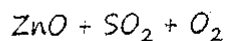
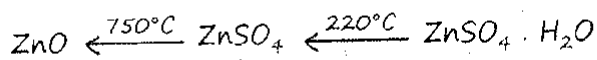
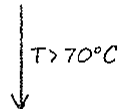
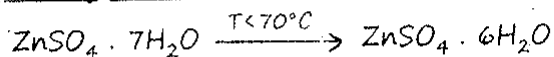


Heating effect

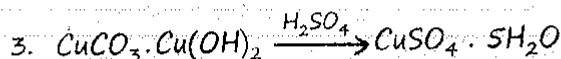
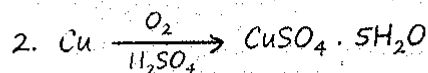
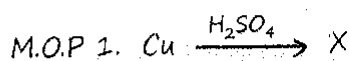


Basic Zinc Carbonate

Heating effect

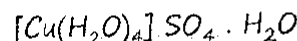


3. Blue vitriol ($CuSO_4 \cdot 5H_2O$)

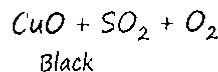
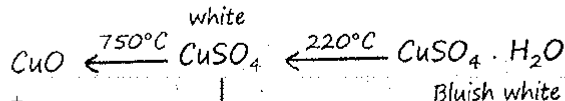
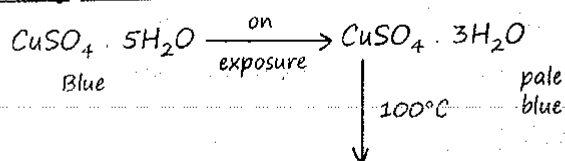


Basic Copper Carbonate

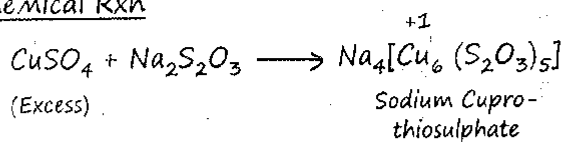
Structure # $CuSO_4 \cdot 5H_2O$



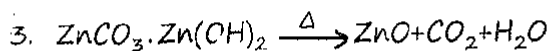
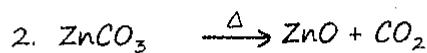
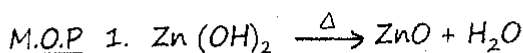
Heating effect



Chemical Rxn



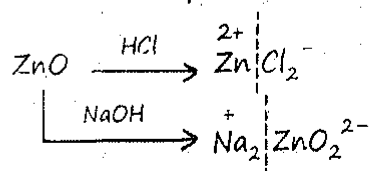
6. Philosophers wool (ZnO) or Chinese white



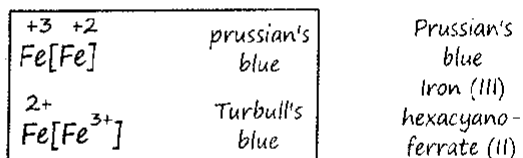
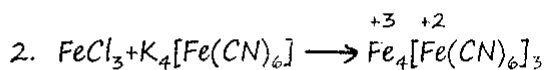
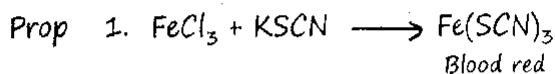
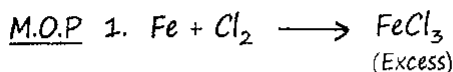
Prop

1. ZnO is white when cold & yellow when hot (Due to crystal defect).

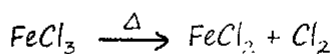
2. ZnO is an amphoteric oxide



7. FeCl₃

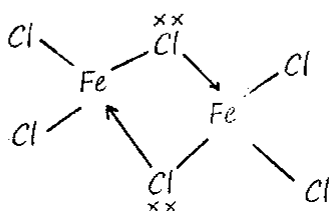


3. Heating effect

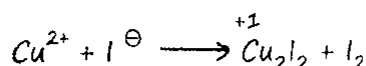


4. Struc.

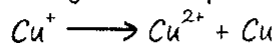
FeCl_3 in solid form exist as dimer $(\text{FeCl}_3)_2$



1. All Cu^{2+} halides are stable except iodide

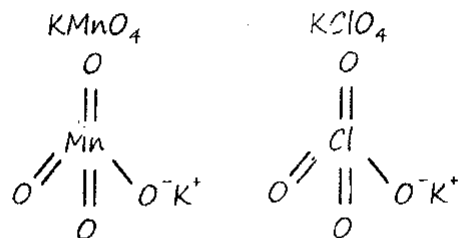


2. Many Cu^+ Comp are unstable in aq Solⁿ.



(Stability) $\text{Cu}^{2+} > \text{Cu}^+$ due to more -ve ΔH^{hyd}

which is greater than 2nd I.E.



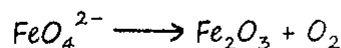
Tetrahedral

Tetrahedral

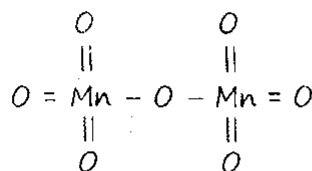
3. O.N of metal in most of the Carbonyls is zero
[Ni(CO)_4], [Mn(CO)_5] [Cr(CO)_6]

$$x + 0 = 0$$

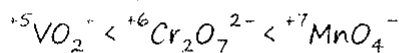
4. FeO_4^{2-} ion formed in alkaline media



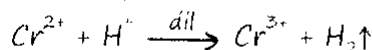
5. Highest fluoride of Mn is MnF_4 while oxide is Mn_2O_7



6. Oxidising power (O.P)

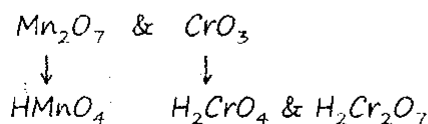


7. Mn^{3+} & Co^{3+} are strong O.A while Ti^{2+} , V^{2+} , Cr^{2+} are strong R.A.

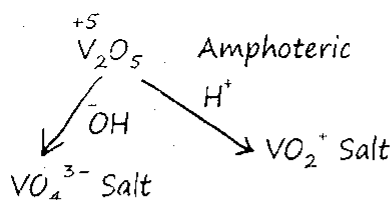
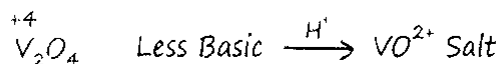


8. All metal except Sc form ionic MO.

9. Higher O.N oxides are acidic in Nature



10. V_2O_3 Basic

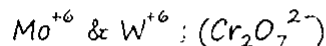


$$x - e = -3; x = +5;$$

11. Mn^{+7} halide does not exist, MnO_3F exist.

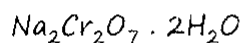
12. CrO Basic, Cr_2O_3 Amphoteric

13. Cr^{+6} is less stable than



Cr^{+6} is strong O.A while MOO_3 & WO_3 are not?

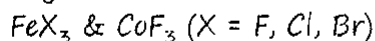
14. Sodium dichromate crystallises as



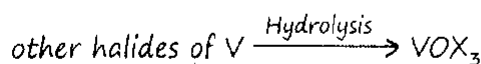
15. Stability of VF_2 is due to

Lower O.N VX_2 ($\text{X} = \text{Cl}, \text{Br}, \text{I}$)

16. Beyond Mn, no metal has trihalide except



17. Halides of V^{5+} exist as VF_5 , Cr^{+6} exist as CrF_6



Inner Transition Metal

Group (3)

Period 6 (1) Lanthanoids

$_{57}\text{La}$ | $_{58}\text{Ce}$ Pr Nd Pm Sm Eu Gd Tb Dy Ho Er Tm Yb $_{71}\text{Lu}$

Group (3)

Period 7 (2) Actinoids

$_{89}\text{Ac}$ | $_{90}\text{Th}$ Pa U Np Pu Am Cm Bk Cf Es Fm Md No $_{103}\text{Lr}$

2. Lanthanoids (1) E.C F-block $\Rightarrow$ General E.C $(n-2)f^{1-14} (n-1)d^{0-1} ns^2$
Lanthanoids $\Rightarrow 4f^{1-14} 5d^{0-1} 6s^2$

	La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
4f	0	1	3	4	5	6	7	7	9	10	11	12	13	14	14
5d	1	1	0	0	0	0	0	1	0	0	0	0	0	0	1
6s	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2

(2) O.N

+2	+3	4+	+4
$\text{Eu}^{2+}, \text{Yb}^{2+}$	Most common	Ce	Pr Nd Tb Dy
R.A	Ln^{3+}	MO_2	O.A

(3) Ionic Radii

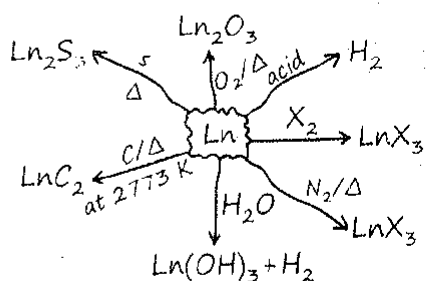
Ln^{3+} $\text{La}^{3+} > \text{Ce}^{3+} > \text{Pr}^{3+} \dots \dots \dots > \text{Lu}^{3+}$

Left to Right decreases due to increase in E.N.C owing to Lanthanoid contraction (poor screening of $4f e^-$ s)

Screening order:

$s > p > d > f$

(4) Chemical Rxns



(5) General prop.

(1) All Ln are silvery white soft metals (but Sm is steel hard) due to strong metallic bond.

(2) Except La^{3+} and Lu^{3+} all M^{3+} ions are coloured.

(3) Except f^0 type (La^{3+} & Ce^{4+} etc) type (Yb^{2+} & Lu^{3+} etc), all ions are paramagnetic

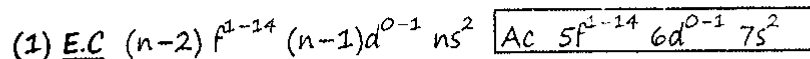
(4) Basicity of Ln decreases from L to R due to decrease in size

$\text{La(OH)}_3 \xrightarrow[\text{decreases}]{\text{Basicity}} \text{Lu(OH)}_3$

(5) Hardness $\propto$ atomic no.

(6) Alloys of Ln are called Mischmetals 95% Ln 5% Fe with some traces of S, C, Ca etc.

2. Actinoids



	Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr
5f	0	0	3	4	5	6	7	7	9	10	11	12	13	14	14
6d	1	1	0	0	0	0	0	1	0	0	0	0	0	0	1
7s	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
(2) O.N	-	3	3	3	3	3	3	3	3	3	3	3	3	3	3
		4	4	4	4	4	4	4	4						
			5	5	5	5	5								
				6	6	6	6								
					7	7									

3. Size

L to R size decreases due to increase in ENC owing to Actinoid contraction (poor screening of $5f e^-s$)

4. General Charac

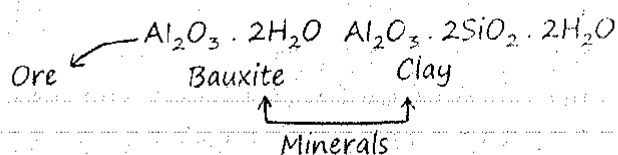
- (1) All are paramag.
- (2) All are silvery white metal.
- (3) They get tarnished on exposure to alkali.

- (4) All are radioactive (& At No 92 onward all elements are called transuranic element)
- (5) They react with boiling water to form mix. of oxides & hydrides.
- (6) All react with non metals at moderate temp.
- (7) All react with HCl but effect with HNO_3 is small due to the formation of oxide Layer.

The scientific and technological method used to extract metal from its ore is called Metallurgy

Minerals # The naturally occurring compounds of a metal (in earth's crust).

Ores # The minerals from which metal is extracted economically.



All ores are minerals but all minerals are not ores.

Metallurgy involves following steps

1. Crushing and grinding (Pulverisation) by Jaw mill or stamp mill
2. Concⁿ of the ore
3. Isolation of metal from ore
4. Purification of metal, Pulverisation
6. Concⁿ or dressing or benefaction

The removal of impurities (sand, clay etc.) from ore.

1. Gravity sep
2. Magnetic sep
3. Froth Floatation
4. Leaching

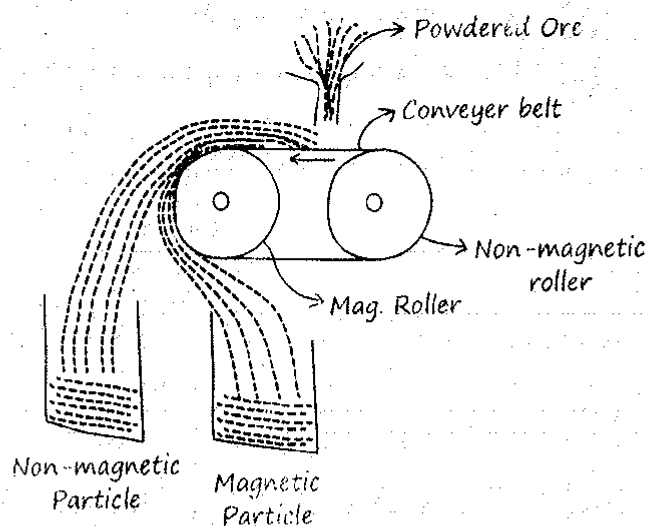
1. G.S or Hydraulic washing or Levigation.

- (a) It is Based on diff Specific gravities of ore and gangue particles.
- (b) An upward stream of running H_2O is used to wash the ore. The lighter gangue particles are washed away and heavier ore particles are left behind.
- (c) Used for the concⁿ of Fe_2O_3 (Haematite oxide ores).

2. **Magnetic Separation Method**

- (1) It is based on diff magnetic prop of ore & gangue.
- (2) Either ore or gangue particles are magnetic in nature.
- (3) Used for the concⁿ of

	SnO_2	FeWO_4
(a)	Cassiterite	Wolframite
	(non mag)	(mag)
	Ore	Impurity
(b)	Rutile	Chlorapatite
	TiO_2	$3\text{Ca}_3(\text{PO}_4)_2 \cdot \text{CaCl}_2$
	Mag ore	Non mag impurity

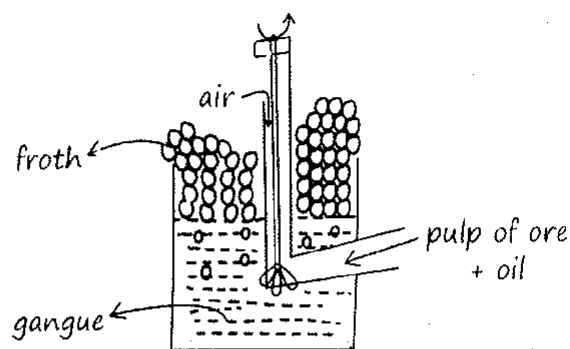


3. **Froth Floatation Method #**

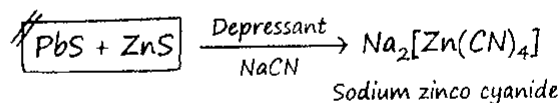
- (1) It is based on diff. wetting prop of ore and gangue particles.
- (2) Ore particles are wet by oils & gangue by water.
- (3) Collectors & Froth stabilisers are added.

Collectors $\Rightarrow$ Pine oil, Fatty acids & Xanthates

Froth stabilizers $\Rightarrow$ Cresol & aniline



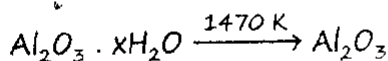
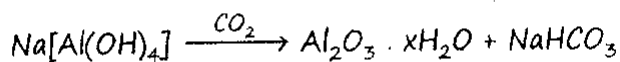
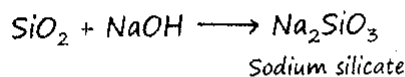
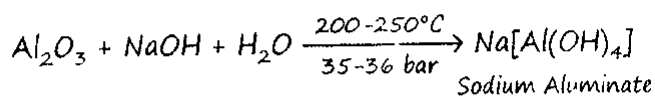
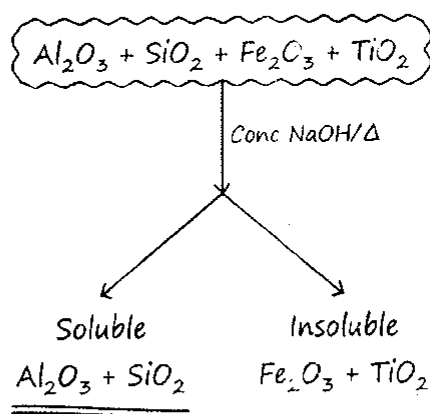
- # It is used for the conⁿ of sulphide ores
- # It is also used for the sep. of two sulphide ores



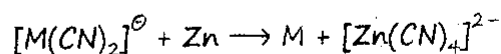
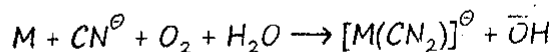
4. Leaching

- (1) Alumina from Bauxite ore

Powdered ore



- (2) Ag & Au



[M = Ag & Au]

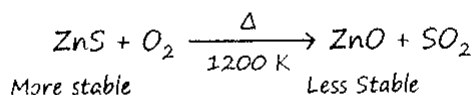
Mac Arthur-forrest cyanide process.

Extraction of Metals

1. Zn from ZnS (Zinc blende) or (Sphalerite)

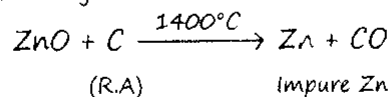
- (1) Pulverisation
- (2) F.F Method
- (3) Isolation

- (a) Roasting: Heating in presence of air at the temp below M.P of metal

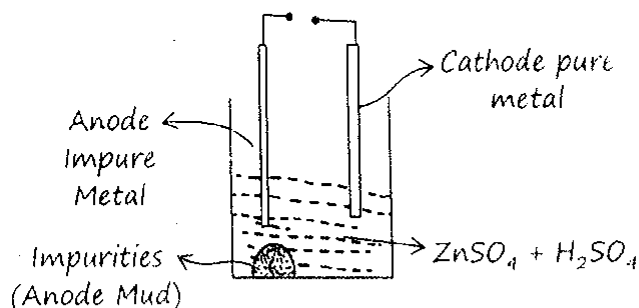


- (b) Reduction by coke

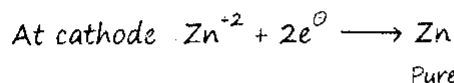
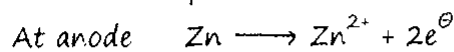
Smelting



- (4) Purification by Electrolytic Refining



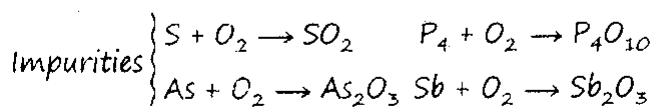
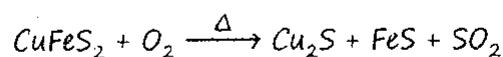
Impure

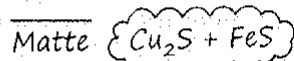
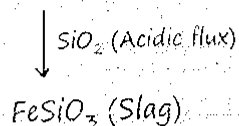
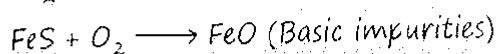
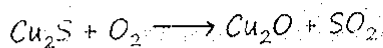


2. Cu from CuFeS₂ (Chalcopyrite) or (Copper pyrite)

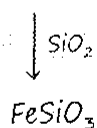
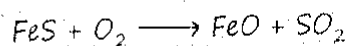
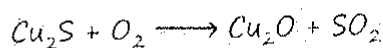
- (1) Pulverisation
- (2) F.F Method
- (3) Isolation

- (1) Roasting:

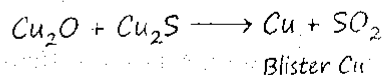




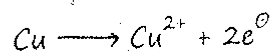
In Bessemer convertor



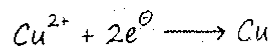
self Reduction process



(4) Purification by Electrolytic Refining



At anode:- Impure Cu



At cathode:- Pure Cu

Electrolyte:- $\text{CuSO}_4 + \text{H}_2\text{SO}_4$

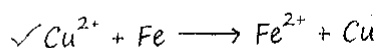
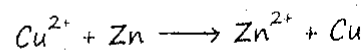
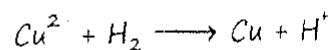
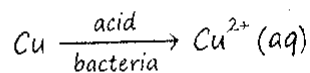
Impurities:- Sb, Se, Te

Ag, Au, Pt etc.

Anode Mud

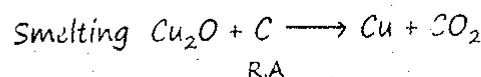
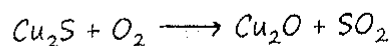
Extraction of Cu from Low grade ore

Hydrometallurgy



Scrap

Extraction of Cu from Cu_2O

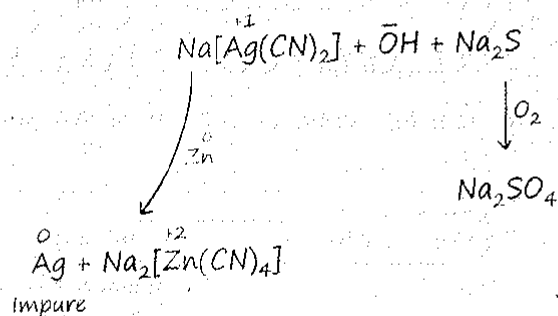
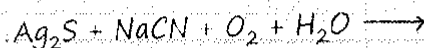


3. Ag from argentite (Ag_2S)

(1) Pulverisation

(2) F.F Method

(3) Mac-Arthur-forrest cyanide process



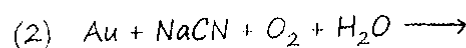
(4) Purification # At anode # Impure Ag

At cathode # Pure Ag

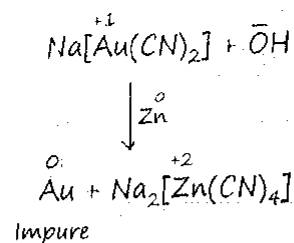
Electrolyte # $\text{AgNO}_3 + \text{HNO}_2$

4. Au

(1) Pulverisation



Leaching



Purification

At anode:- Impure Au

At cathode:- Pure Au

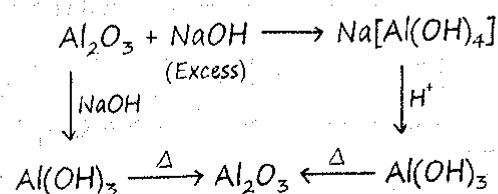
Electrolyte:- $\text{AuCl}_3 + \text{HCl}$

5. Al from Bauxite ($\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$) ($0 < x < 1$)

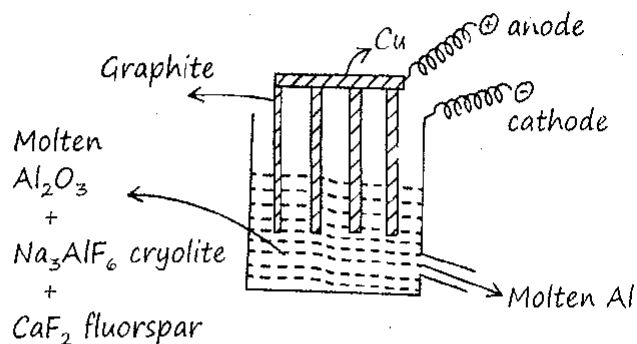
(1) Pulverisation

(2) Leaching

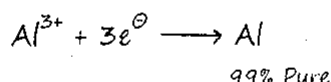
Baeyer's process



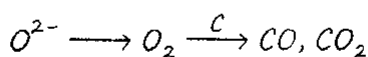
(3) Hall-Heroult's process



At Cathode



At anode

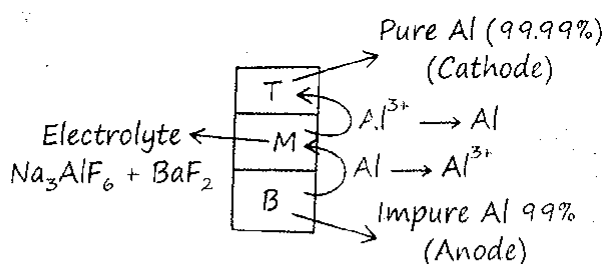


To decrease M.P

To increase conductivity

(4) Purification by Hoopes's process

it is a three Layer process



All three Layers have diff. densities

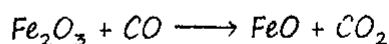
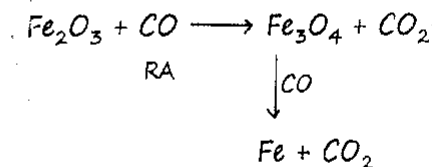
6. Fe from Haematite (Fe_2O_3)

(1) Pulverisation

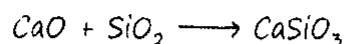
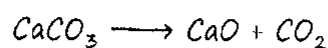
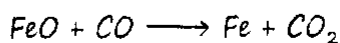
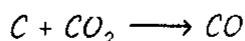
(2) Concⁿ by Gravity sep method

(3) Rxns #

(1) Low temp zone (500-800 K)

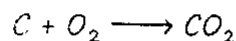


(2) Medium temp zone (900-1600 K)

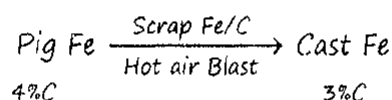


Basic impurity slag
flux acidic

(3) In High temp zone 2170 K

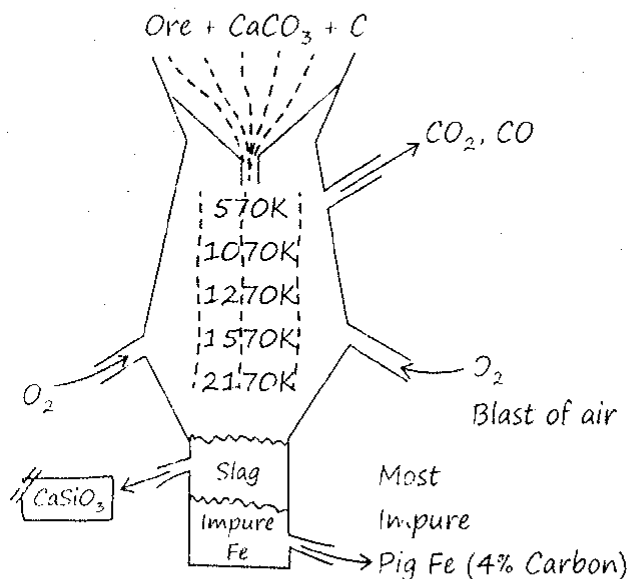
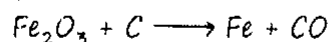


(4) Fe from Haematite (Fe_2O_3)



(P, S, Si, Mn etc)

Wrought iron is prepared by Heating of cast Fe in reverb. furnace lined with Fe_2O_3



Purification

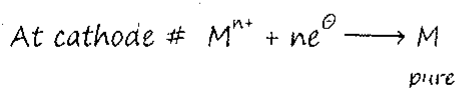
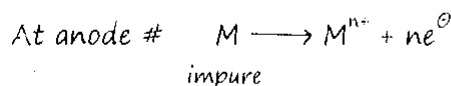
1. Distillation # (for Low Boiling metals like Zn & Hg)

Impure metal is evaporated to obtain pure metal as distillate

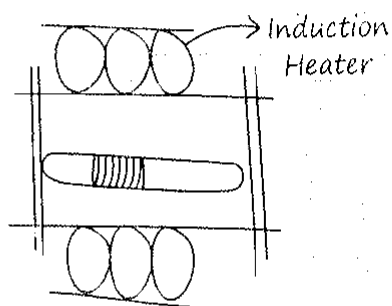
2. Liquation # (for Low melting metal like Sn)

Impure metal can be made to flow on slopping surface. In this way, metal is sep by impurities.

3. Electrolytic refining



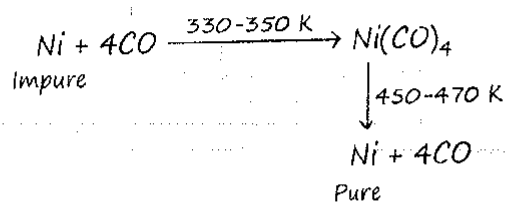
4. Zone refining (for Semi conductor & metals of High purity (like Si, Ge, B etc))



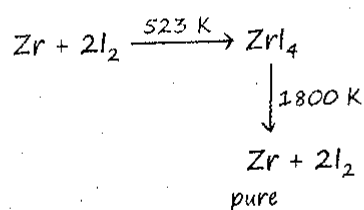
It is based on the principle that impurities are more soluble in melt than in solid state of the metal.

5. Vapour phase refining

- (a) Mond's process for Ni

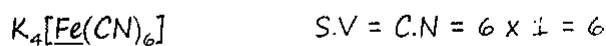


- (b) Van-Arkel process for Ti & Zr



Werner's Theory

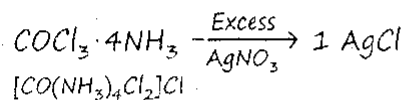
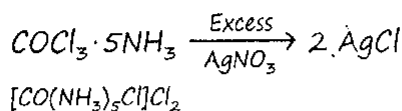
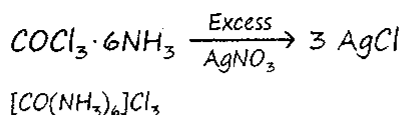
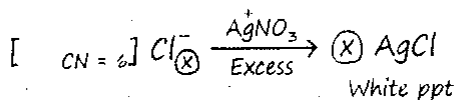
1. Werner proposed the concept of primary valency and secondary valency
2. P.V is ionisable & it is equal to O.N of central metal atom/ion
3. S.V is non ionisable & it is equal to C.N of C.A or C.M.A



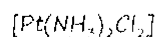
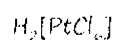
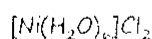
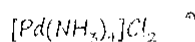
$$4 + x = 6 = 0$$

$$x = +2 \quad \boxed{PV = 2}$$

Rxn with Excess $AgNO_3$



Q. Formula Moles of AgCl ppt per mole of comp with excess $AgNO_3$



Q. Compare conductivity of following comp.

Total ions

(A) $[Co(NH_3)_6]Cl_3$ ④ 1:3 type Electrolyte

(B) $[CoCl(NH_3)_5]Cl_2$ ③ 1:2 type Electrolyte

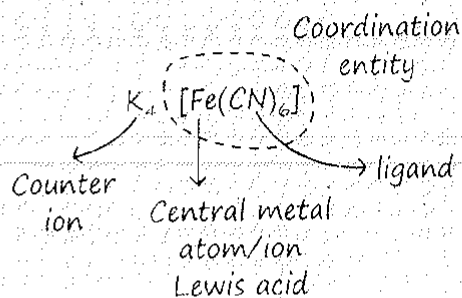
(C) $[CoCl_2(NH_3)_4]Cl$ ② 1:1 type Electrolyte

$$\boxed{A > B > C}$$

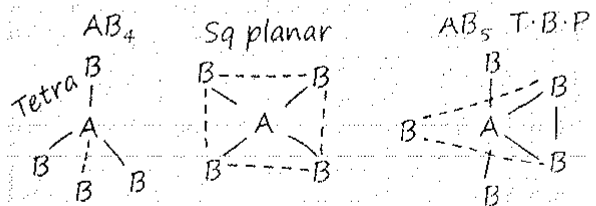
Q. Diff. b/w double salt and a complex

Double Salt	Complex
1. Retains identity in solid state but loses identity in sol ⁿ	1. Retains identity in both solid as well as in sol ⁿ
2. Completely dissociates into ions	2. Do not completely disso.
3. Give test of simple ions	3. Do not give test of simple ion
Exp: Carnallite $KCl \cdot MgCl_2 \cdot 6H_2O$	Exp: $K_4[Fe(CN)_6]$, $[Cr(H_2O)_6]Cl_3$
Mohr salt $FeSO_4(NH_4)_2SO_4 \cdot 6H_2O$	
Potash alum $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$	

Some imp. Terms of C.C



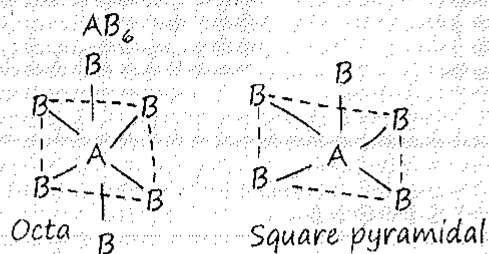
Coordination Polyhedron



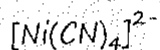
Classification

1. Based on charge

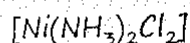
+ve	-ve	Neutral
NO_2^+ , $NH_3^+-NH_2$	X^-	NH_3
NO^+	$OH^\ominus$	H_2O
	$OX^{2-} \begin{pmatrix} COO^- \\ \\ COO^- \end{pmatrix}$	CO
	$CN^\ominus$	NO
	$NO_2^\ominus$	Py
	CO_3^{2-}	
	SU_4^{2-}	
	EDTA ⁴⁻	
	$\begin{array}{c} \text{OOCCH}_2 \\ \\ \text{N} \\ \\ \text{OOCCH}_2 \end{array} \begin{array}{c} \text{CH}_2\text{COO}^- \\ \\ \text{N} \\ \\ \text{CH}_2\text{COO}^- \end{array}$	



Homoleptic complex



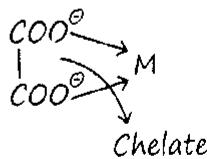
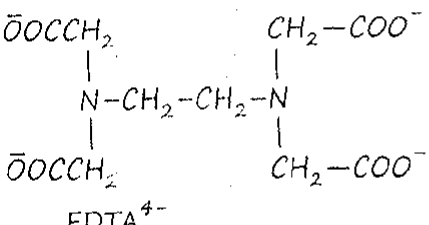
Heteroleptic complex



Ligands

- Ligands can be $1e^\ominus$ donor, $2e^\ominus$ donor (LP donor) and $\pi e^\ominus$ donor
- Ligand is Lewis Base

2. Based on donation

Monodentate (1 atom donor)	didentate 2 donor atom	Polydentate Hexadentate (6 donor atom)	Ambidentate (2 donor atoms) but only one can donate at a time
$X^{\ominus}, OH^{\ominus}, NO_2^+, NO^+$ $N^+H_3-NH_2, CO, NO, Py$ CH_3-NH_2, NH_3, H_2O etc PPh_3	$en \begin{array}{c} CH_2-CH_2 \\ \quad \\ NH_2 \quad NH_2 \end{array}$ $Ox^{2-} \begin{array}{c} COO^{\ominus} \\ \\ COO^{\ominus} \end{array}$ 	 EDTA ⁴⁻	$M \leftarrow NO_2^{\ominus}$ $M \leftarrow O^{\ominus}NO$ $M \leftarrow SCN^{\ominus}$ $M \leftarrow NCS^{\ominus}$

IUPAC Naming

RULES

1. Name of cation is written before anion.
2. Naming of ligand

(a) Anionic Ligands ending with e

$X^{\ominus}$	halido	e ↓ o
$\bar{O}H$	hydroxido	
OX^{2-}	oxalato	
$CN^{\ominus}$	cyanido etc	

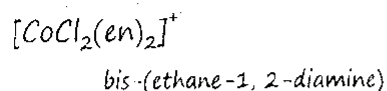
(b) Neutral

H_2O	→ aqua
NH_3	→ ammine
CH_3-NH_2	→ methyl amine
CO	→ Carbonyl
NO	→ Nitrosyl
PPh_3	→ triphenyl phosphine
en	→ ethane-1, 2-diamine
Py	→ Pyridine

(c) +vely charged Ligand (Cationic Ligand)

<u>-ium</u>	NO_2^+ nitronium
	NO^+ nitrosonium
	$^+NH_2-NH_2$ Hydrazinium

3. If prefixes mono, di, tri etc used for number of ligand has numerical prefix then bis, tris, tetrakis etc are used



- 4 Naming of C·M·A/ion

CMA	When CMA is neutral or positively charged complex	When CMA is -vely charged complex
Fe	iron	Ferrate
Co	cobalt	Cobaltate
Ni	nickel	nickelate
Pd	palladium	palladate
Pt	platinum	platinate
Cu	copper	Cuprate
Ag	silver	argentate
Au	gold	aurate
Zn	Zinc	Zincate

5. O.N of C.M.A / ion is written just after naming of C.M.A / ion in Roman in paranthesis

Q. $[\text{Cr}(\text{NH}_3)_3(\text{H}_2\text{O})_3]\text{Cl}_3$
triamminetriaquachromium (III) chloride

Q. $\text{K}_2[\text{Zn}(\text{OH})_4]$
Potassium tetrahydroxidozincate (II)

Q. $\text{Hg}[\text{Co}(\text{SCN})_4]$
Mercury (I) tetrathiocyanatocobaltate (III)

Q. $\text{K}_3[\text{Al}(\text{OX})_3]$
Potassium trioxalatoaluminate (III)

Q. $[\text{Ni}(\text{CO})_4]$
tetracarbonylnickel (0)

Q. $[\text{CoCl}_2(\text{en})_2]$
dichlorido bis-(ethane-1, 2-diamine) Cobalt (III) ion

Q. $[\text{Co}(\text{NH}_3)_5(\text{CO}_3)]\text{Cl}$
Pentamminecarbonatocobalt(III) Chloride

Q. $[\text{Co}(\text{en})_3]_2(\text{SO}_4)_3$
tris-(ethane-1, 2-diamine) cobalt (III) sulphate

Q. $[\text{Ag}(\text{NH}_3)_2]^+ [\text{Ag}(\text{CN})_2]^-$
diamminesilver (I) dicyanidoargentate (I)

Q. $[\text{Pt}(\text{NH}_3)_4 \text{Cl}(\text{NO}_2)]$
tetraamminechloridonitrito-N-platinum (II)

Q. $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3^{4-}$
iron (III) hexacyanidoferrate (II)

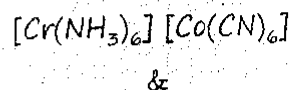
Isomerism

Structural Isomerism

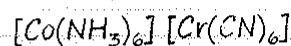
1. Coordination isomers
2. Hydrate isomers
3. Linkage isomers
4. Ionization isomers

1. Coordination iso

When cation & anion both are complex

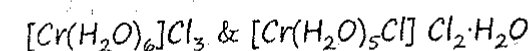


&



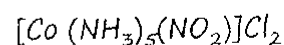
2. Hydrate iso

When H_2O acts as ligand as well as water of crystallization

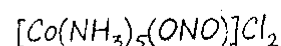


3. Linkage iso

When ambidentate Ligand is present

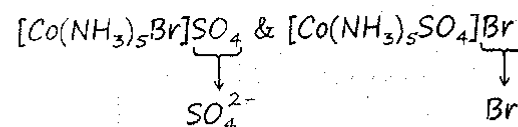


&



4. Ionisation isomers

Compounds which give diff ion in Solⁿ

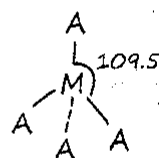


Stereo isomerism

1. Geometrical isomerism

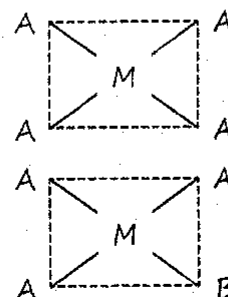
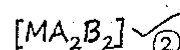
Case 1: In C.N = 4

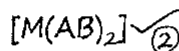
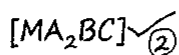
(a) Tetrahedral Geometry



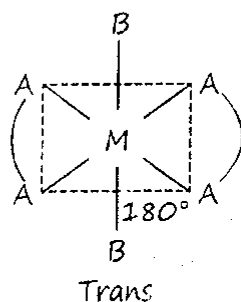
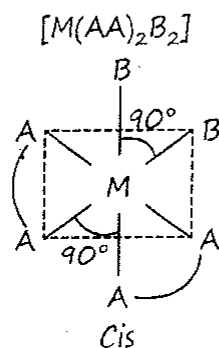
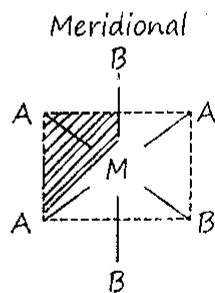
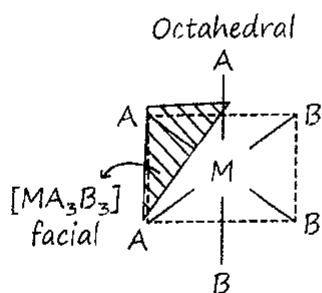
Does not Exhibit G.I

(b) In sq. planar Geo

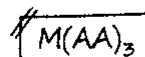




Case 2: In $C \cdot N = 6$

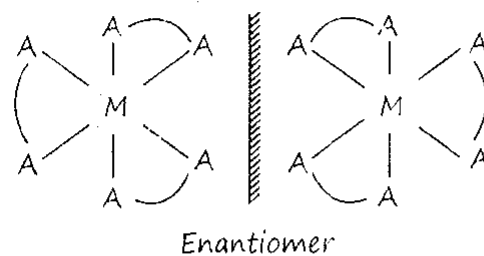


Complexes	No. of G.I.
$[MA_4B_2]$	2
$[MA_4BC]$	2
$^*[MA_3B_3]^*$	2
$[MA_3B_2C]$	3
$[MA_3BCD]$	4
$[MA_2B_2C_2]$	5
$[MA_2B_2CD]$	6
$[MA_2BCDE]$	9
$[MAB CDE F]$	15
$^*[M(AA)_2B_2]^*$	2
$[M(AA)_2BC]$	2
$[M(AB)_3]$	2



Does not Exhibit G.I.

2. Optical isomerism



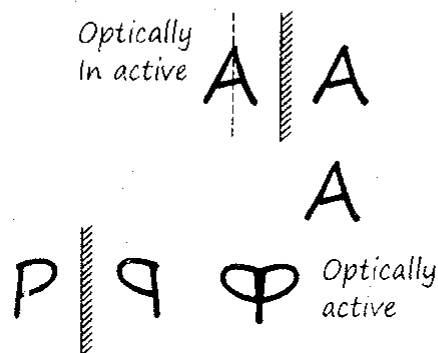
Case 1: In $C \cdot N = 4$

O.I is not exhibited in

$C \cdot N = 4$ Tetra x (unless four different ligands are present)

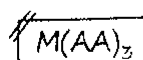
sq planar x

Enantiomer # non - superimposable mirror image



Case 2: In $C \cdot N = 6$

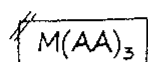
Complexes	Enantiomer pair
$[MA_4B_2]$	0
$[MA_4BC]$	0
$[MA_3B_3]$	0
$[MA_3B_2C]$	0
$[MA_3BCD]$	1
$[MA_2B_2C_2]$	1
$[MA_2B_2CD]$	2
$[MA_2BCDE]$	6
$[MAB CDE F]$	15
$[M(AA)_2B_2]$	1
$[M(AA)_2BC]$	1
$[M(AB)_3]$	2



Exhibits O.I

in C.N = 6

Stereo	G.I	Complexes	Enantiomer pair
2	2	$[MA_4B_2]$	0
2	2	$[MA_4BC]$	0
2	2	$[MA_3B_3]$	0
3	3	$[MA_3B_2C]$	0
5	4	$[MA_3BCD]$	1
6	5	$[MA_2B_2C_2]$	1
8	6	$[MA_2B_2CD]$	2
15	9	$[MA_2BCDE]$	6
30	15	$[MABCDEF]$	15
3	2	$[M(AA)_2B_2]$	1
3	2	$[M(AA)_2BC]$	1
4	2	$[M(AB)_3]$	2



Exhibits O.I

not G.I

O.I ✓

G.I X

Q. Identify which complex Exhibits G.I or O.I or both?

1. $[Pt(NH_3)_2C(NO_2)]$ (C.N = 4 sq planar)

$[MA_2BC]$ G.I ✓, O.I X

2. $[Pt(Gly)_2]$ Gly $H_2N-CH_2-COO^-$

$[M(AB)_2]$ G.I ✓, O.I X

3. $[Cu(en)_2]^{+2}$

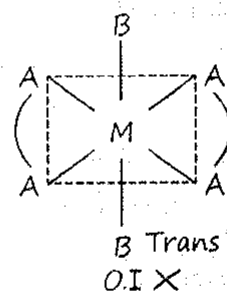
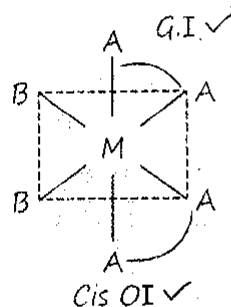
$[M(AA)_2]$ G.I X, O.I X

4. $[Pt(H_2O)(NH_3)BrCl]$

$[MABCD]$ G.I ✓, O.I X

5. $[Co(H_2O)_2(ox)_2]^-$

$[M(AA)_2B_2]$



6. $[Co(H_2O)_2(NH_3)_2(ox)]^+$

$[M(AA)B_2C_2]$ G.I ✓, O.I ✓

7. $[Zn(NH_3)_2Cl_2]$

$[MA_2B_2]$ Tetra

G.I X, O.I X

8. $[Fe(NH_3)_2(CN)_4]^-$

$[MA_4B_2]$

G.I ✓, O.I X

9. $[Co(NH_3)_5(NO_2)](NO_3)_2$

$[MA_5B]$

G.I X, O.I X

10. $[Co(en)_3]Cl_3$

$[M(AA)_3]$

G.I X, O.I ✓

Valence Bond theory

- C.M.A./ion used $(n-1)d$, ns , np or ns , np , nd for Bonding
- Ligand will donate L.P to the vacant orbital of CMA/ion

3. Pairing occurs when strong field Ligands like NH_3 , $\text{CN}^\ominus$, CO etc are present

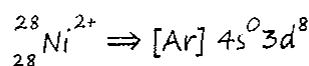
Hybri	Geo
sp^3	Tetrahedral
dsp^2	Sq. planar
sp^3d	T.B.P
dsp^3	Sq. pyramidal
sp^3d^2	Octahedral
d^2sp^3	Octahedral

Q. $[\text{Ni}(\text{CN})_4]^{2-}$

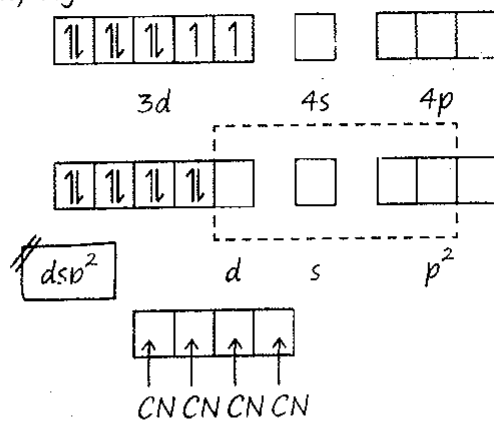
(a) $O.N = +2$

(b) $C.N = 4 \times 1 = 4$

(c) $E.C \text{ Ni} \Rightarrow [\text{Ar}] 4s^2 3d^8$



(d) Hybridisation



(e) Shape Sq planar

(f) Unpaired $e^\ominus s = 0$

$$n = 0$$

$\text{MM} = 0$ Diamagnetic

(g) $E.A.N = A.N - O.N + 2 \times C.N$

$$= 28 - 2 + 2 \times 4 = 26 + 8 = 34$$

(h) Colour Colourless

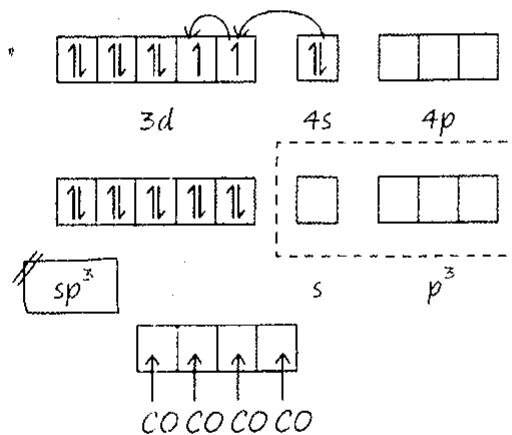
Q. $[\text{Ni}(\text{CO})_4]$

(a) $O.N = 0$

(b) $C.N = 4 \times 1 = 4$

(c) $E.C {}_{28}\text{Ni} \Rightarrow [\text{Ar}] 4s^2 3d^8$

(d) Hybridisation



(e) Shape Tetrahedral

(f) Unpaired $e^\ominus s = 0$

$$n = 0$$

$\text{MM} = 0$ Diamagnetic

(g) $E.A.N = A.N - O.N + 2 \times C.N$

$$= 28 - 0 + 2 \times 4 = 28 + 8 = 36$$

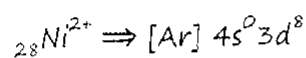
(h) Colour Colourless

Q. $[\text{Ni}(\text{Cl})_4]^{2-}$

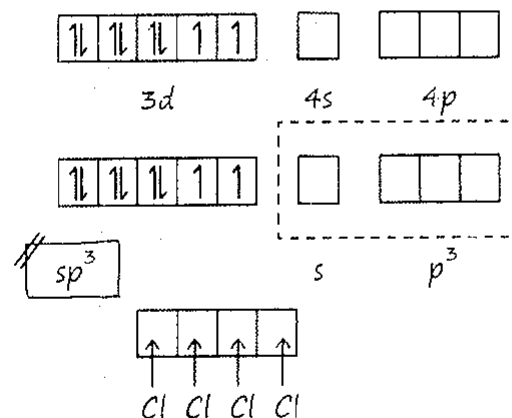
(a) $O.N = +2$

(b) $C.N = 4 \times 1 = 4$

(c) $E.C {}_{28}\text{Ni} \Rightarrow [\text{Ar}] 4s^2 3d^8$



(d) Hybridisation



(e) Shape Tetrahedral

(f) Unpaired $e^-s = 2$

$$n = 2$$

$$M.M = \sqrt{2(2+2)}$$

$$M.M = \sqrt{8} \text{ Paramagnetic}$$

(g) $E.A.N = A.N - O.N + 2 \times C.N$

$$= 28 - 2 + 2 \times 4 = 26 + 8 = 34$$

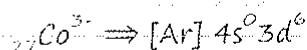
(h) Colour Coloured

Q. $[\text{Co}(\text{NH}_3)_6]^{3+}$

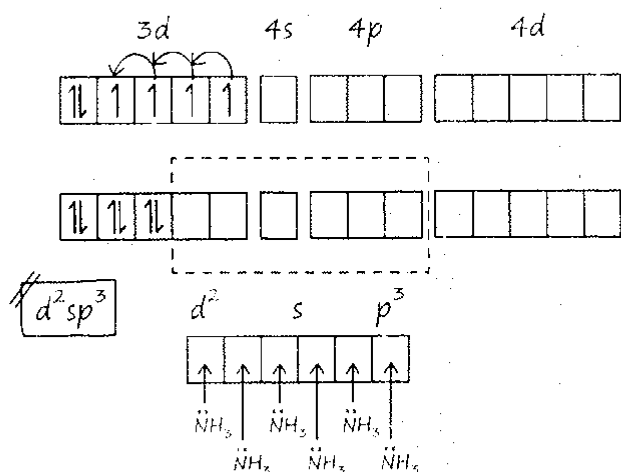
(a) $O.N = +3$

(b) $C.N = 6 \times 1 = 6$

(c) $E.C_{27}\text{Co} \Rightarrow [\text{Ar}] 4s^2 3d^7$



(d) Hybridisation



(e) Shape Octahedral

(f) Unpaired $e^-s = 0$

$$n = 0$$

$$M.M = 0 \text{ Diamagnetic}$$

(g) Colour Colourless

(h) HS/LS : Inner complex or Low spin complex or spin paired complex

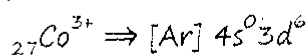
(i) $E.A.N = 27 - 3 + 2 \times 6 = 24 + 12 = 36$

Q. $[\text{CoF}_6]^{3-}$

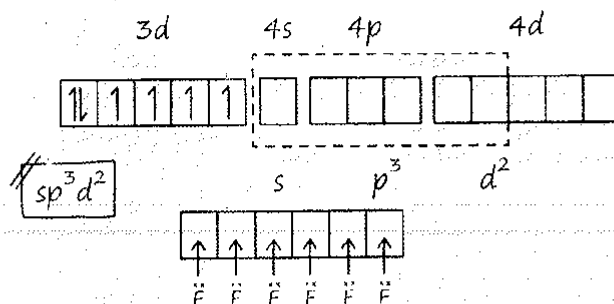
(a) $O.N = +3$

(b) $C.N = 6 \times 1 = 6$

(c) $E.C \Rightarrow {}_{27}\text{Co} \Rightarrow [\text{Ar}] 4s^2 3d^7$



(d) Hybridisation



(e) Shape Octahedral

(f) Unpaired $e^-s = 4$

$$n = 4$$

$$M.M = \sqrt{4(4+2)} = \sqrt{24} \text{ Paramag.}$$

(g) Colour Coloured

(h) HS/LS : Outer complex or High spin complex or spin free complex

(i) $E.A.N = 27 - 3 + 2 \times 6 = 24 + 12 = 36$

Q. $[\text{Ni}(\text{NH}_3)_6]^{2+}$

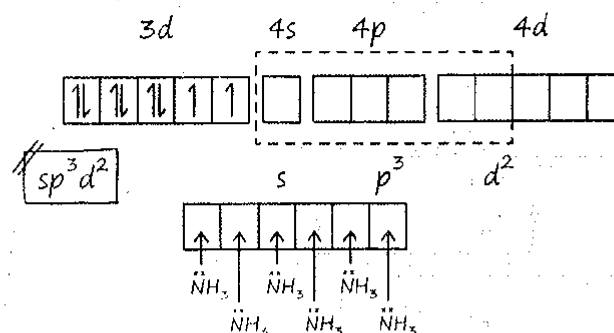
(a) $O.N = +2$

(b) $C.N = 6 \times 1 = 6$

(c) $E.C_{28}\text{Ni} \Rightarrow [\text{Ar}] 4s^2 3d^8$



(d) Hybridisation



(e) Shape Octahedral

(f) Unpaired $e^-s = 2$

$$n = 2$$

$$M.M = \sqrt{2(2+2)} = \sqrt{8} \text{ Paramag.}$$

(g) Colour Coloured

(h) HS/LS : Outer complex or High spin complex or spin free complex

(i) $E.A.N = 28 - 2 + 2 \times 6 = 26 + 12 = 38$

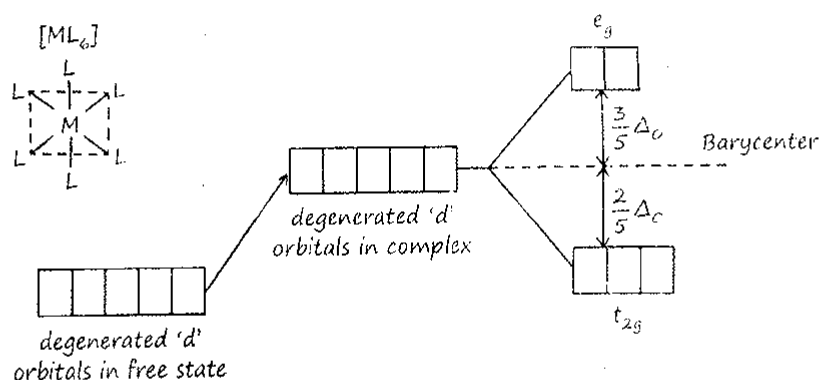
1. Pt in $C.N = 4$ always sq planar also (Pd & Au)

2. $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ is diamag d^2sp^3

C.F.T (Crystal Field Theory)

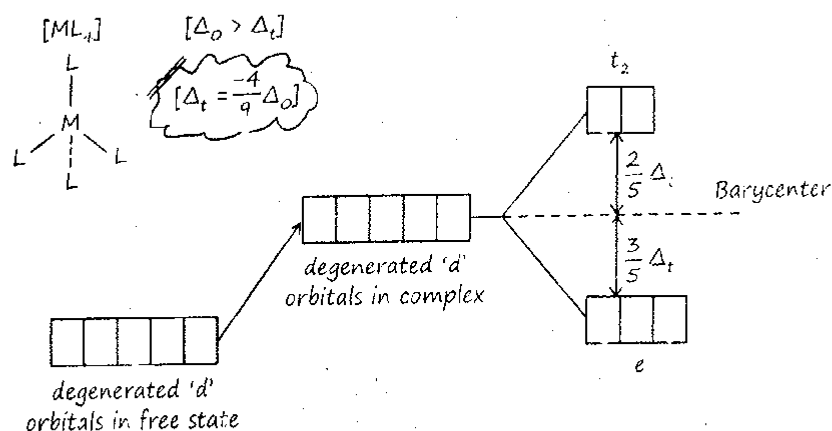
Case 1 : In octahedral field

$\Delta_o \Rightarrow$ C.F.S.E (Crystal field splitting energy)



Case 2 : In Tetrahedral field

$\Delta_t \Rightarrow$ C.F.S.E (Crystal field splitting energy)



Factors affecting C.F.S.E

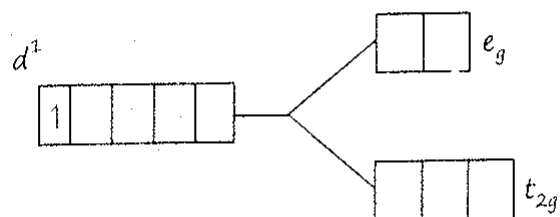
1. C.F.S.E $\propto$ C.N
2. CFSE $3d < 4d < 5d$
3. Geometry $\Delta_o > \Delta_t$ $\Delta_t = -\frac{4}{9}\Delta_o$
4. Nature of ligand
SFL CFSE $\uparrow$
WFL CFSE $\downarrow$

* Spectrochemical Series *

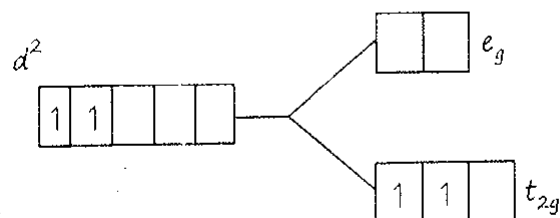
$I^- < Br^- < SCN^- < Cl^- < S^{2-} < F^- < OH^- < OX^{2-} < H_2O$
 $< NCS^- < EDTA^{4-} < NH_3 < en < NO_2^- < CN^- < CO$
 (C donor > N donor > O donor > X donor)

Calculation of C.F.S.E in octa field

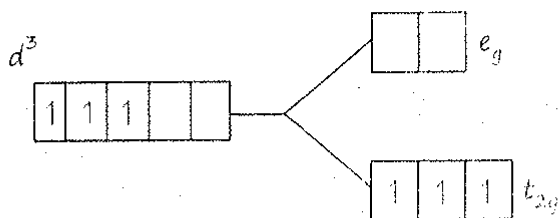
$$CFSE = x \times (-0.4 \Delta_o) + y \times (+0.6 \Delta_o) + z \times P$$



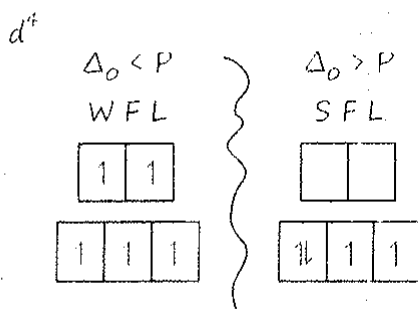
$$CFSE = 1 \times (-0.4 \Delta_o) + 0 + 0 = -0.4 \Delta_o$$



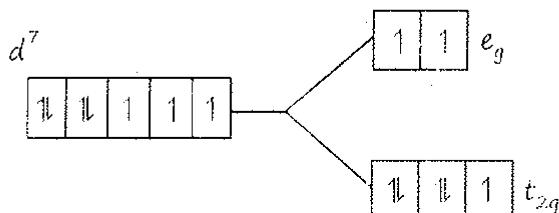
$$CFSE = 2 \times (-0.4 \Delta_o) + 0 + 0 = -0.8 \Delta_o$$



$$CFSE = 3 \times (-0.4 \Delta_o) + 0 + 0 = -1.2 \Delta_o$$



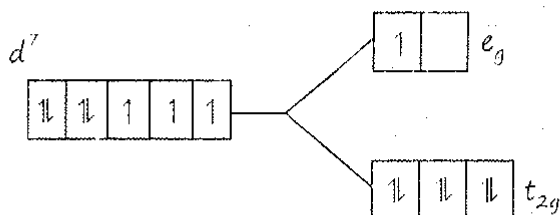
Case 1 : If $\Delta_o < P$ (WFL)



$$CFSE = 5 \times (-0.4 \Delta_o) + 2 \times (+0.6 \Delta_o) + 0$$

$$= -2 \Delta_o + 1.2 \Delta_o = -0.8 \Delta_o$$

Case 2 : If $\Delta_o > P$ (SFL)

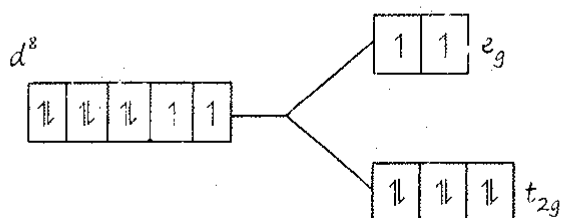


$$CFSE = 6 \times (-0.4 \Delta_o) + 1 \times (+0.6 \Delta_o) + 1 \times P$$

$$= -2.4 \Delta_o + 0.6 \Delta_o + P$$

$$= -1.8 \Delta_o + P$$

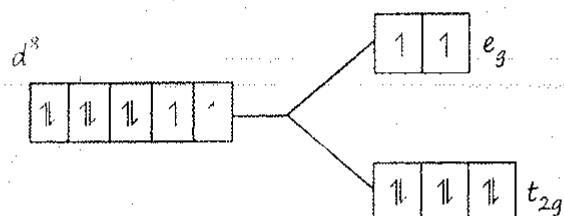
Case 1 : If $\Delta_o < P$ (WFL)



$$CFSE = 6 \times (-0.4 \Delta_o) + 2 \times (+0.6 \Delta_o) + 0$$

$$= -2.4 \Delta_o + 1.2 \Delta_o = -1.2 \Delta_o$$

Case 2 : If $\Delta_o > P$ (SFL)

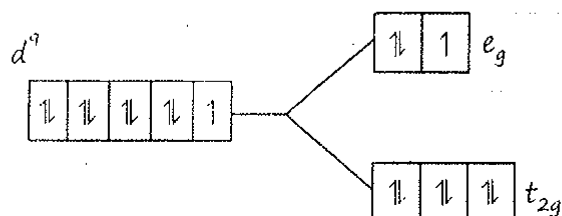


$$CFSE = 6 \times (-0.4 \Delta_o) + 2 \times (+0.6 \Delta_o) + 0 \times P$$

$$= -2.4 \Delta_o + 1.2 \Delta_o$$

$$= -1.2 \Delta_o$$

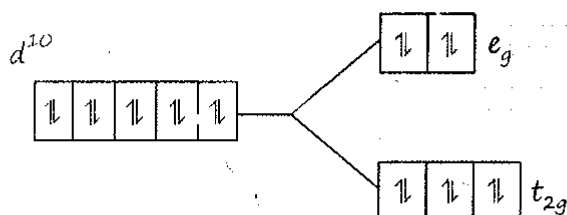
Either WFL or SFL



$$CFSE = 6 \times (-0.4 \Delta_o) + 3 \times (+0.6 \Delta_o) + 0$$

$$= -2.4 \Delta_o + 1.8 \Delta_o = -0.6 \Delta_o$$

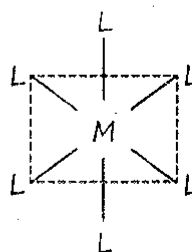
Either WFL or SFL



$$CFSE = 6 \times (-0.4 \Delta_o) + 4 \times (+0.6 \Delta_o) + 0$$

$$= -2.4 \Delta_o + 2.4 \Delta_o = 0$$

Jahn-Teller Distortion (J.T.D)



In symmetrical octahedral field, J.T.D is absent

Half filled

Fully filled

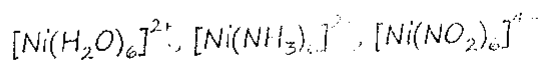
H.F + F.F

d^1		✓			
d^2	✓		d^6	✓	
d^3	✗		d^6	✗	
d^4	✓		d^7	✓	
d^5	✗		d^8	✗	
(W.F.L.)					
d^5	✓		d^9	✓	
(S.F.L.)			d^{10}	✗	

JTD is absent

~~d^3, d^5, d^6~~
WFL SFL
 ~~d^8, d^{10}~~
~~3 5 6 8 10~~

Q. Correct order of wavelength of absorption (λ_{ab}) in visible region



① ② ③

C.F.S.E ① < ② < ③

λ_{ab} ① > ② > ③

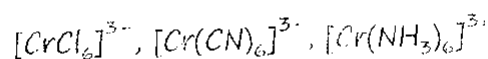
Q. J.T.E is not observed in High spin complexes of

(A) d^4 (B) d^7
(C) d^7 ~~(D) d^8~~

Q. Δ_o for $[\text{CoCl}_6]^{4-}$ is $18,000 \text{ cm}^{-1}$ find the CFSE of $[\text{CoCl}_4]^{2-}$?

$$\Delta_t = \frac{4}{9} \Delta_o = \frac{4}{9} \times 18000 = 8000 \text{ cm}^{-1}$$

Q. Arrange the order of CFSE



① ② ③

CFSE ① < ③ < ②

Organometallic comp

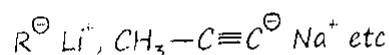
Comps in which metal is directly Bonded with Carbon.

O.M.C

Ionic
M & C $\Rightarrow$ Ionic Bond

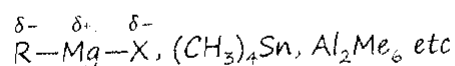
Covalent
M & C $\Rightarrow$ Covalent
a) σ Bonded
b) π Bonded
c) σ & π Both

1. Ionic O.M.C #

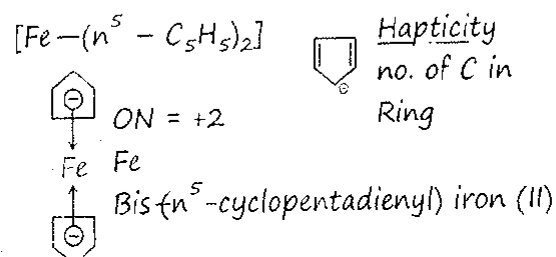


2. Covalent O.M.C

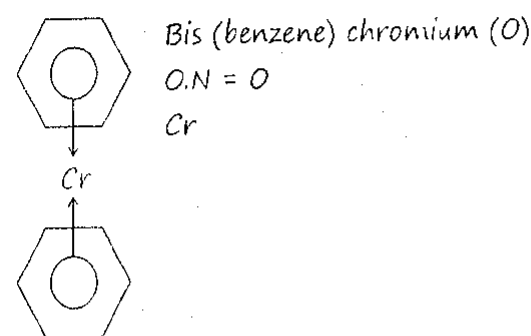
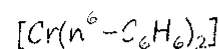
(1) σ Bonded



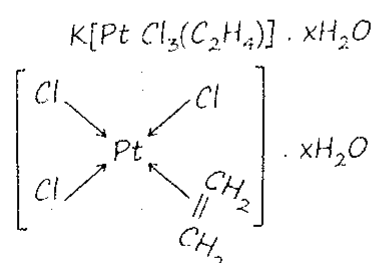
(2) Ferrocene



3. Dibenzene Chromium #

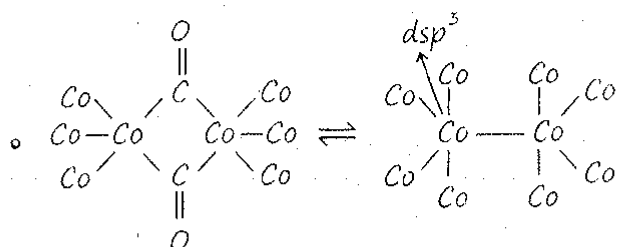
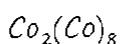
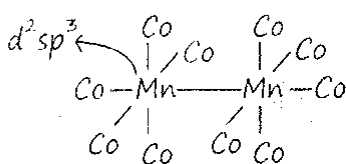
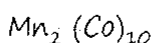
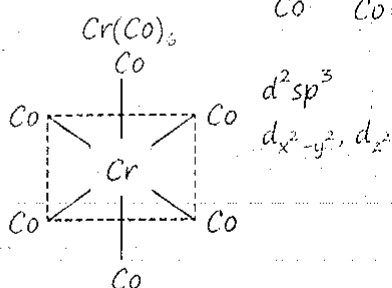
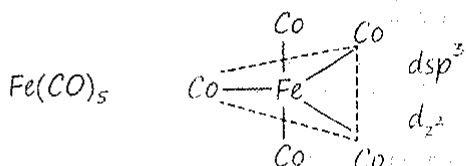
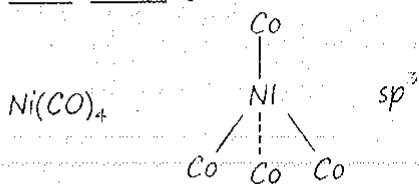


4. Zeise's salt

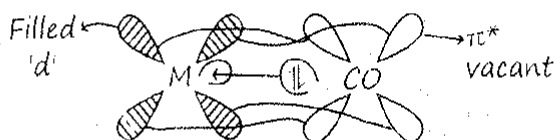


σ and π Both bonded

Metal Carbonyl #



Bonding in Metal carbonyls



1. M & C σ Bond is formed by donation of L.P of carbonyl carbon to the vacant 'd' orbital of metal.
2. M & C π Bond is formed by sidewise overlapping of M (filled 'd') & C (π^* orbitals)
3. This bonding is known as synergic bonding & effect is called synergic effect & ligands are called π -acid ligands CO, CN^- , NO, PF_3 etc.

4. due to synergic bonding

M & C B.O $\uparrow$ B.S $\uparrow$ B.L $\downarrow$

C & O B.O $\downarrow$ B.S $\downarrow$ B.L $\uparrow$

5. Charge density on metal $\uparrow$

M & C B.O $\uparrow$ B.S $\uparrow$ B.L $\downarrow$

C & O B.O $\downarrow$ B.S $\downarrow$ B.L $\uparrow$

Q. If CO Bond length in CO is 1.128 Å then what is the value of CO Bond length in Fe(CO)_5 ?

- (A) 1.15 Å (B) 1.128 Å
(C) 1.72 Å (D) 1.118 Å

Q. Compare CO Bond strength in



(1) (2) (3)

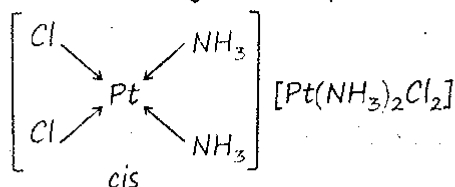
CO Bond strength

(1) > (2) > (3)

Some Imp. points

1. Chlorophyll, Haemoglobin & Vitamin B_{12} are coordination complexes of Mg, Fe & Co respectively.

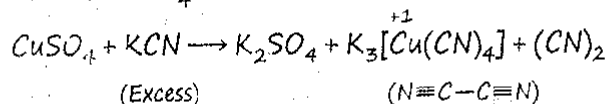
2. Anti cancer agent is cis platin



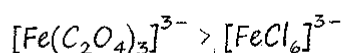
3. Wilkinson catalyst is used for hydrogenation of alkene $[\text{Rh(PPh}_3)_3\text{Cl}]$

chloridotris-(triphenyl phosphine) rhodium (I)

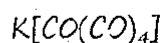
4. Rxn of CuSO_4 with Excess KCN



5. Chelate complexes are more stable than non chelate complex usually

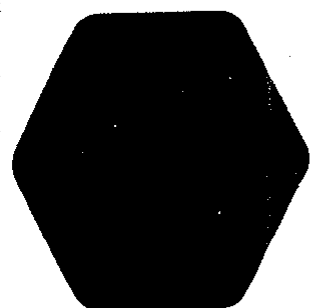


6. O.N of metal in metal carbonyls may be -ve

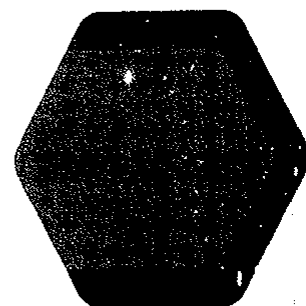


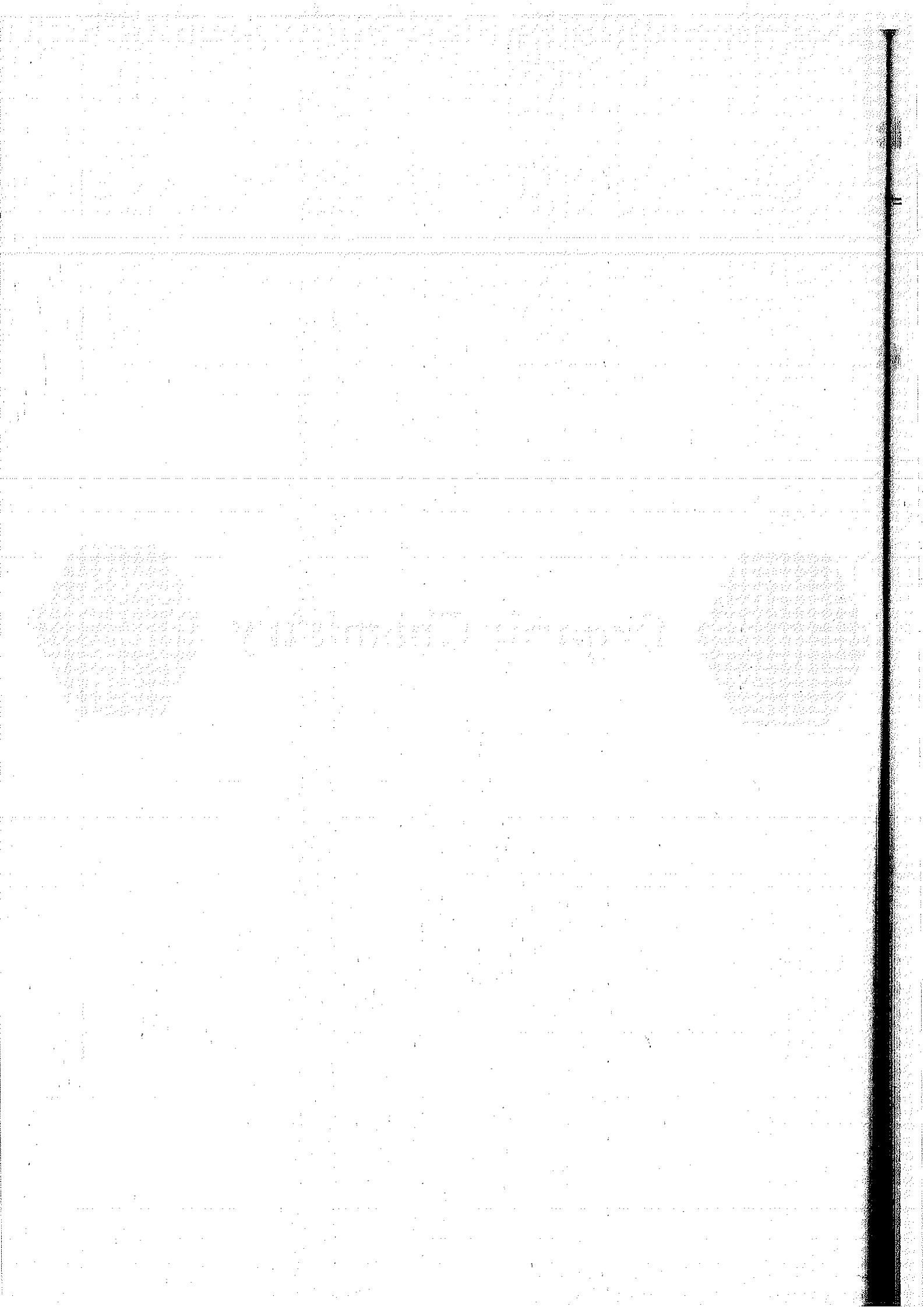
$1 + x + 0 = 0$

$x = -1$

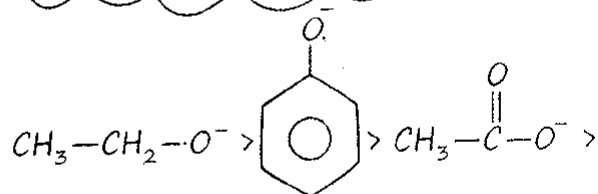
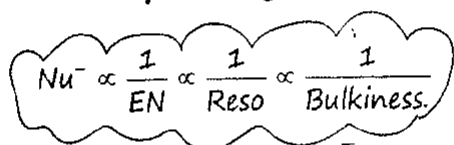


Organic Chemistry





1. Nucleophilicity

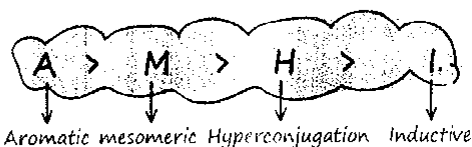


$\text{Nu}^- \propto \text{LP} \propto \text{-ve charge} \propto \text{size of C}^-$

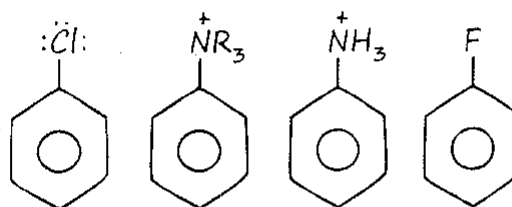
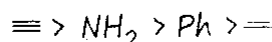
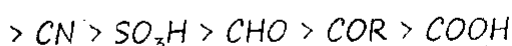
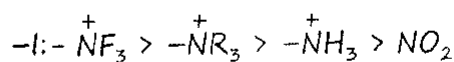
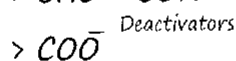
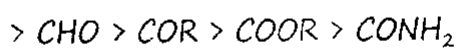
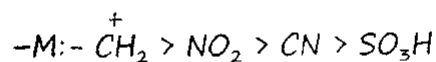
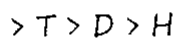
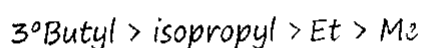
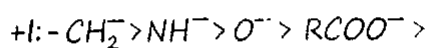
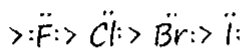
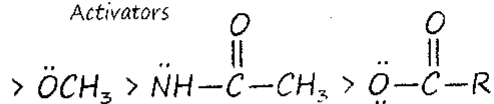
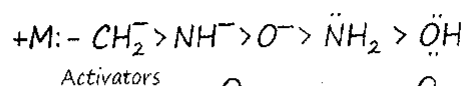
$\text{F}^- < \text{Cl}^- < \text{Br}^- < \text{I}^-$ [Polar protic solvent]

Polar Aprotic = $\text{F}^- > \text{Cl}^- > \text{Br}^- > \text{I}^-$.

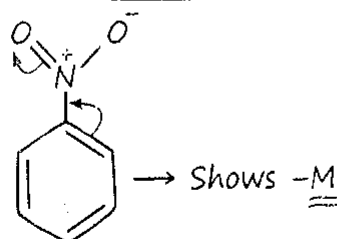
2. Priority Order



3. Orders



(No d-orbital)
-I ✓ (Not-M)



+ With respect to group already present, it's Meta position is not affected by Meso or Reso effect.

Only Inductive!

+ Cyclic > Linear > Cross conjugation.

+ More equivalent ↑ Acidic Nature ↑
resonating structure of CB

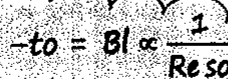
Rules for Resonating Structure

Stability:- least stable: (++) or (--) near

1. Octet complete.
2. Neutral structure > Charged ones.
3. More EN = -ve Charge
Less EN = +ve Charge.
4. +ve & -ve charge near to each other

Bredt's Rule

1. At bridge head carbon, -ve is not accepted till total no. of carbon in the ring is 8. After 8-carbon sp^2 chalcogen:



Effect of Resonance

1. Max e^- density.

2. $\text{BL} \propto \text{Reso}$

$$\text{Heat of hydrogenation} \propto \frac{1}{\text{Reso}} \propto \pi \text{ bond}$$

$$\text{Aromaticity} \propto \text{Reso Energy} \propto \text{Reso}$$

$$\alpha + M$$

Stability Order



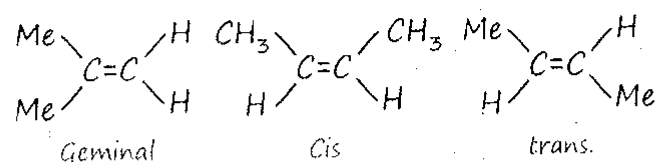
Conjugated Isolated Cumulated

HOH Order



Hyperconjugation [H > O > T]

Alkane: $\sigma - \pi^*$ overlap



Stability: Geminal > Trans > Cis.

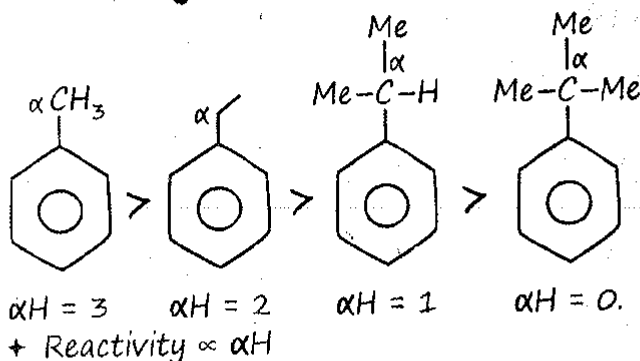
+ Heat of hydrogenation is energy required to convert double into single bond in a molecule.

Stability $\downarrow$ HOH $\uparrow$

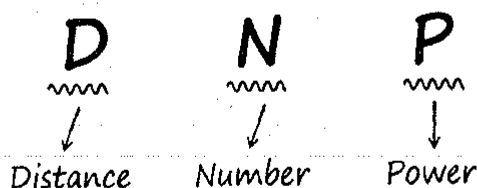
Reactivity $\propto$ Stability

$$\alpha(\alpha\text{-H}) \approx \text{H-effect} \propto \frac{1}{\text{HOH.}}$$

Reactivity of Benzene



VIMP TRICK



Only inductive

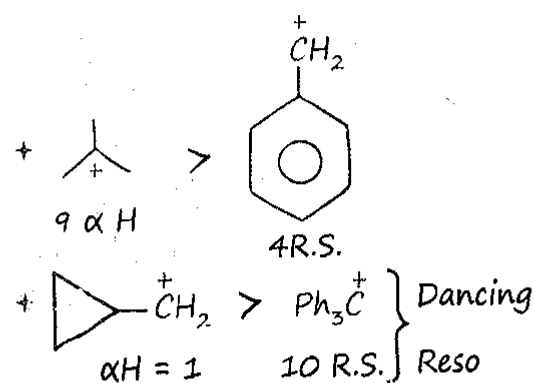
+ **(Stability)**

Hyperconjugⁿ

+ **(Stability)**

$\alpha\text{-H} = 3$ $3 \alpha \text{D}$ $3 \alpha \text{T}$

(Stability)

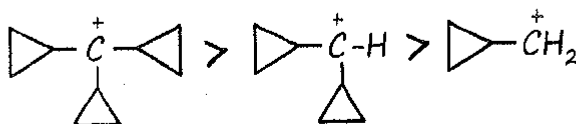


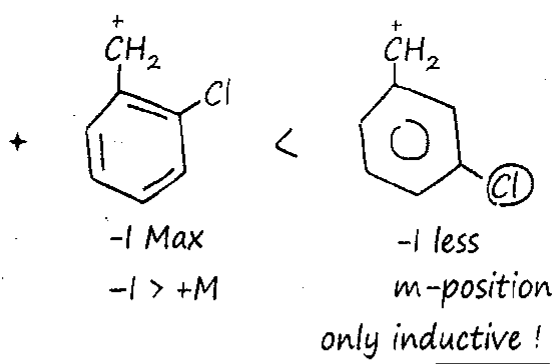
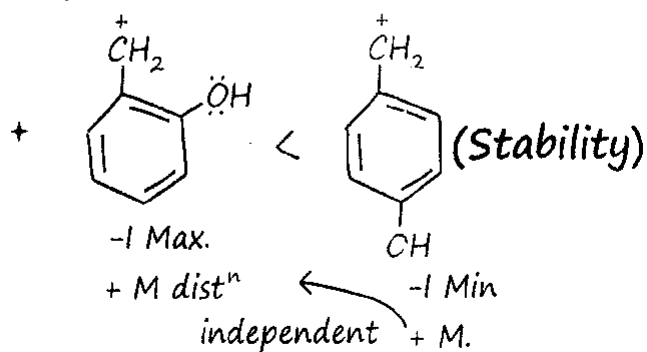
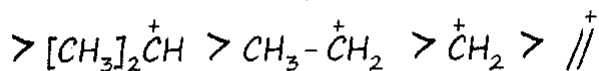
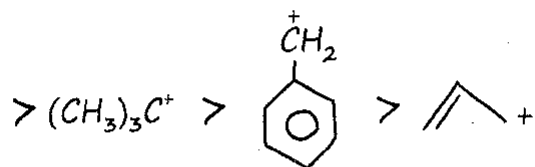
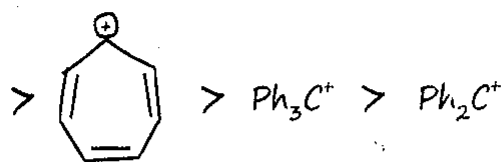
Cyclopropyl
Methyl
Carbocation.

Very Stable

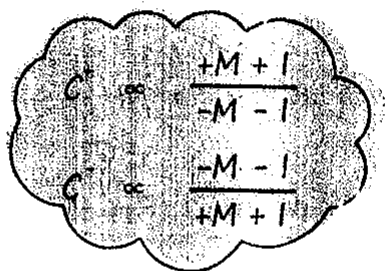
Dancing
Reso

Stability Order of Carbocation

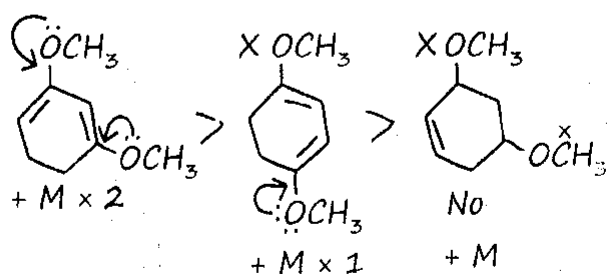




+ Stability of C^+ and C^-



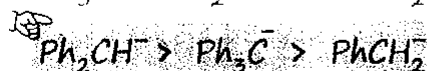
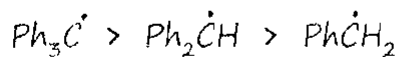
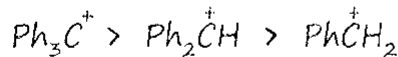
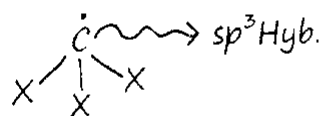
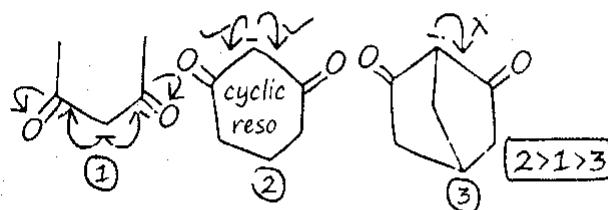
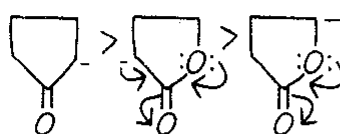
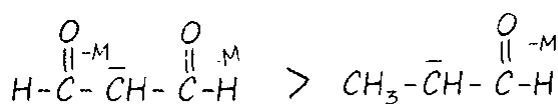
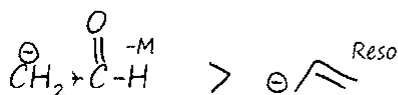
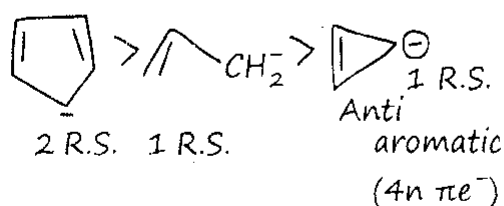
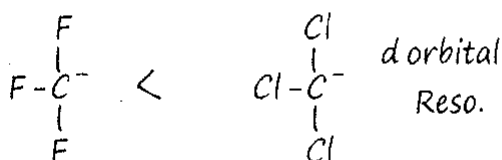
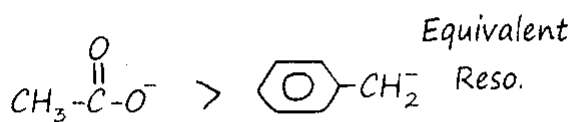
☞



+ + I order

T > D > H.

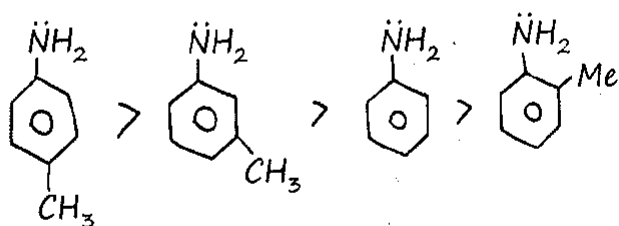
V. imp Examples



There is no repulsion b/w para group & NH_3^+ so after L.P. donation it becomes more stable.

Para Amines > Ortho Amines.
(Basic strength)

$P > m > o$

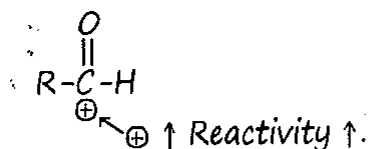


Neet

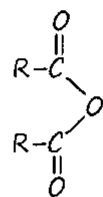
Reactivity of $\text{RX} \propto \text{C}^+$ Stability

Neet

Reactivity of carbonyl compounds



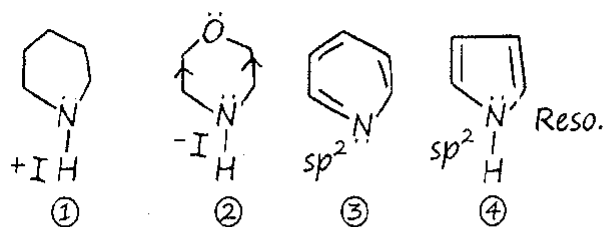
Acid anhydride :-



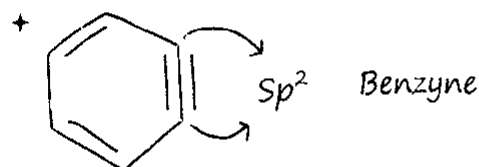
Acyl chloride:- $\text{R}-\overset{\text{O}}{\underset{\text{Cl}}{\text{C}}}-\text{Cl}$

Ester :- $\text{R}-\overset{\text{O}}{\underset{\text{OR}}{\text{C}}}-\text{OR}$

1° Amide
 $\downarrow$
 RCONH_2



$K_b :- ① > ② > ③ > ④$

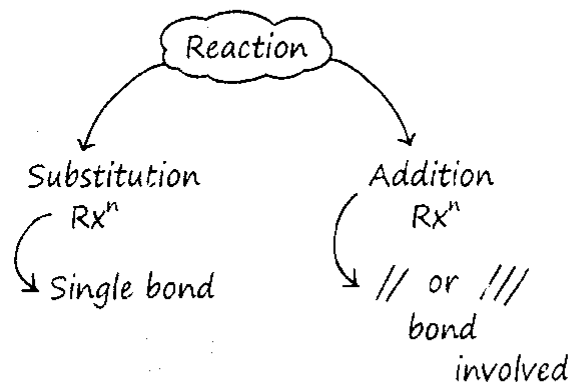


Rearrangement of carbocation

Migratory Aptitude:-

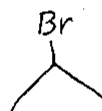
$\text{H} > \text{Ar} > \text{R}$

Benzyl.



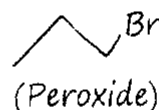
Markovnikov Rule

More part of reagent goes to multiple bonded carbon bearing less "H"



Anti Markovnikov Rule

More part goes to multiple bonded carbon bearing more "H"

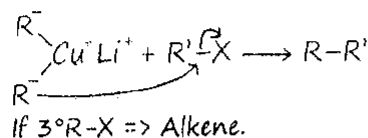


Name of reaction	Reagent	Function
1> Hydrogenation:- of alkene/alkyne	$\xrightarrow[\text{Ni/Pt/Pd}]{\text{H}_2 \text{ Raney Ni [Sabatier-Senderens Rxn]}}$	Alkane banao. 
2> Reduction of alkyl halide to alkane	$\xrightarrow[\text{(Zn + H}^+)]{\text{Zn + dil HCl}}$	X- hatao H- lagao
3> Wurtz reaction $\text{I} > \text{Br} > \text{Cl} > \text{F}$	Na/Dry Ether (New C-C bond Form)	Free Radical banao aur higher Alkane banao $\text{R}^\bullet \rightarrow \text{R-R}$
4> Decarboxylation $[\text{R-COOH} \rightarrow \text{R-H}]$	$\xrightarrow[\text{Acidic } \Delta \text{ Poison}]{\text{NaOH + CaO}}$ 3 1	COOH/COONa hatao "H" lagao. [All COOH Eliminated]
5> Kolbe's electrolysis of:- RCOONa (or RCOOK) (New C-C bond Form)	Cathode :- $\text{H}_2 \uparrow$ Anode :- Alkane (R-R) $\text{Sol}^n = \text{Basic [pH } \uparrow]$	Same like Wurtz Rx^n . Free Radical banao!
6> Halogenation:- of alkane $\text{F} > \text{Cl} > \text{Br} > \text{I}$	$\text{X}_2 / h\nu$ In case of I_2 also use HIO_3 & HNO_3	"H" - Hatao "X" - lagao.
7> Isomerisation:- of alkane	Anhyd. $\text{AlCl}_3 / \text{HCl}$	Chain isomer banao.
8> Aromatisation of:- n-hexane or higher alkane	Cr_2O_3 or V_2O_5 Mo_2O_3	Aromatic Ring banao.
9> Fittig rxn of:- aryl iodide	Na - DE. Chlorobenzene $\longrightarrow$ Ph-Ph	
10> Wurtz fittig rxn:- ($\text{Ar-X} + \text{R-X}$)	Na - DE. Chlorobenzene $\longrightarrow$ Toluene + Ph-Ph	
11> Ullmann rxn:-	Cu - DE. Iodobenzene $\longrightarrow$ Ph-Ph	

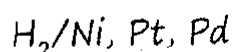
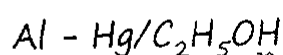
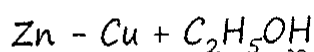
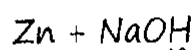
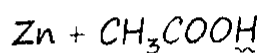
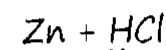
New C-C bond form

12> Corey-HOUSE
syn:-

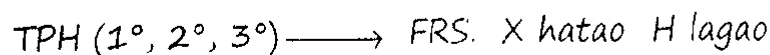
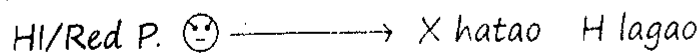
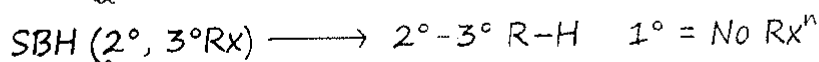
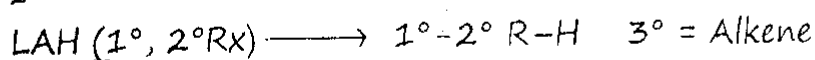
Gilman Reagent.
(R_2CuLi)



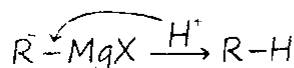
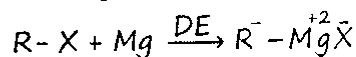
13> Reduction of R-X:-



X hatao
H lagao

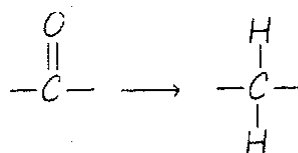
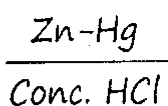


14> Grignard reagent:- $R - MgX$



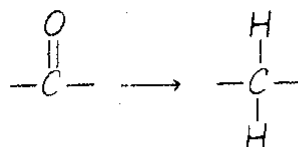
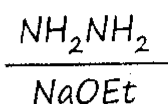
Acidic
Medium

15> Clemmenson
reduction:-

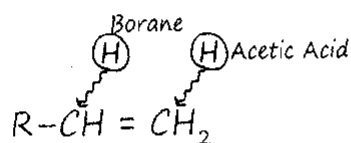
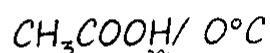
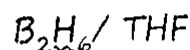


Basic
Medium

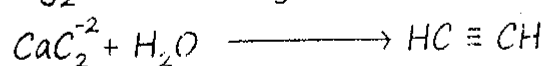
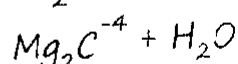
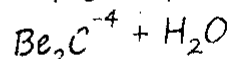
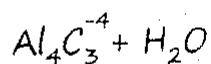
16> Wolff-kishner
reduction:-



17> HBO of Alkene:-



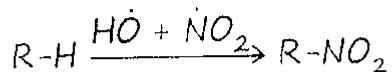
18> Hydrolysis of
carbide:-



CH₄

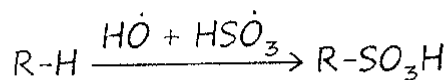
19> Nitration of alkane :-

Conc. HNO_3

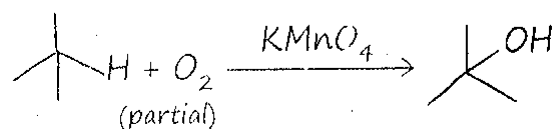
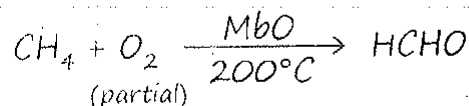
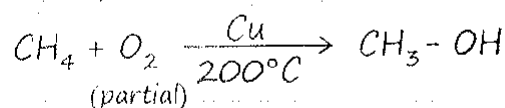
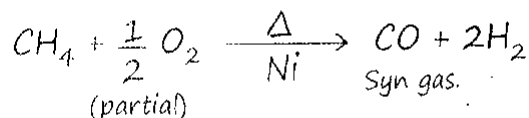
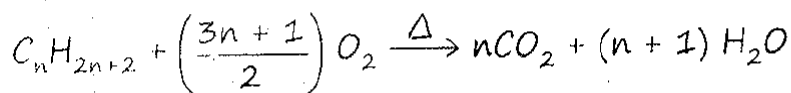


20> Sulphonation of alkane :-

Conc. H_2SO_4



21> Combustion:-



22> Dehydration of alcohols:-

Conc. H_2SO_4

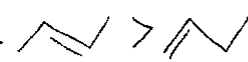
Rearrangement ✓

H_3PO_4 KHSO_4

C^+ banega

E_1 = Multi Step C^+ ✓

POCl_3 / Pyridine

Saytzeff Rule:- 

E_2 = Single Step [Anti Elimⁿ]

Jaha pr -OH laga hoga

Waha pr C^+ banakr rearrange krke product likhna!

(R-X → Alkene)

23> Dehydro-halogenation of haloalkane:-

Alc. KOH

Bad LG

(α , β -elimination)

(Anti Elimⁿ)

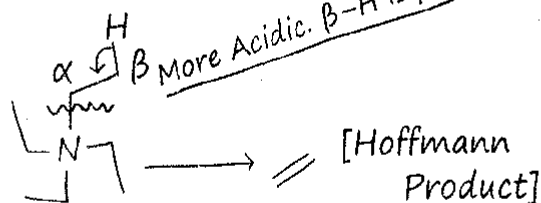
Bulky base

Hoffmann (Major)

24> Hoffmann elimination of quaternary ammonium hydroxide :-

$\text{Ag}_2\text{O}/\text{H}_2\text{O}$

(Anti Elimⁿ)



25> Copes elimination of 3° N-Oxide:-

PAA

MCPBA

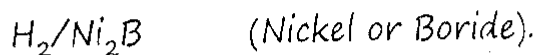
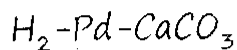
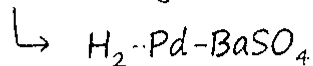
$\text{H}_2\text{O}_2/\Delta$

[Syn Elimⁿ]

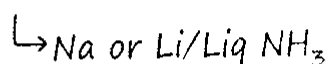
Less Crowded β -H Eliminate 1st (Hoffmann Product)

26> Partial reduction of non terminal alkynes:-

Lindlar's Catalyst



Birch Reduction



Cis Alkene

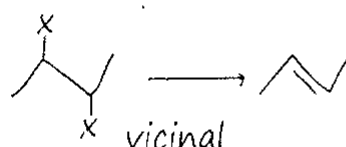
Trans Alkene

27> Dehalogenation of vicinal-dihalide :-
Free Radical banake
Product likho....

Zn dust + CH_3OH or

Nal + Acetone (ether)

Anti elimination



dihalides convert to alkene.

Tetrahalides convert to alkyne.

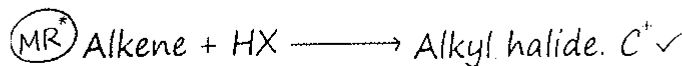
28> Addition of H_2 or alkene/Alkyne:-

Raney Ni, Pt/C, Pd/C, Rh/C, PtO_2 [Adam's Catalyst]

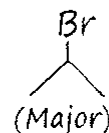
Alkene से Alkane

Alkyne से Alkane

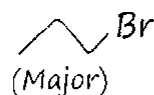
29> Addition of HX:-
[Rearrangement ✓]



(Markownikoff Obey!) Jidhar C^+ Stable hoga udhar X lagega.



(AMR) Anti Markownikoff:-
HBr/Peroxide



30> Hydration of alkene:- $H^+ \longrightarrow C^+$ Pr

C^+ ✓

(MR)

$OH^- \longrightarrow$ Stable C^+ पर lagega.

31> Addition of X_2-CCl_4 :-

Non classical C^+

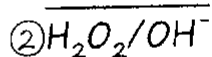
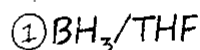
Vicinal dihalide banega.

Anti Addition

CAR, TAM, CSM, TSR.

NO^+Cl^- = Tilden Reagent

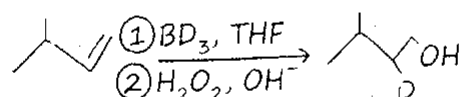
32> Hydroboration oxidation (HBO)
(Syn addition)



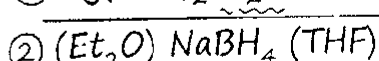
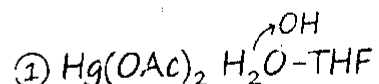
(AMR) No C^+

Less crowded pr OH^- uske Adjacent

(H) from BH_3

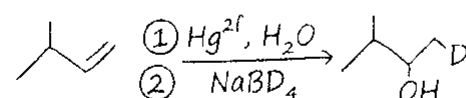


33> Oxymercuration demercuration (OMD) (Anti addition)



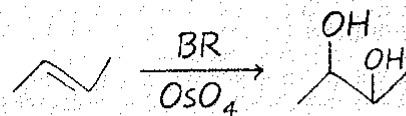
H

(MR) No C^+



More Crowded Pr OH^- (H_2O) uske adjacent H from R.A = $NaBH_4$

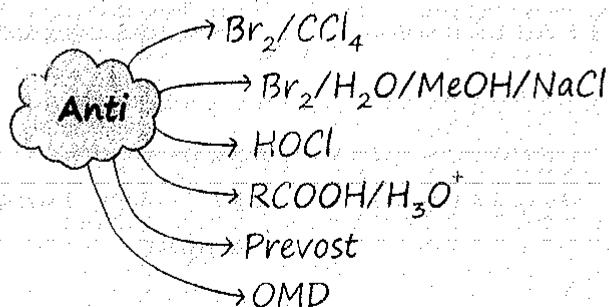
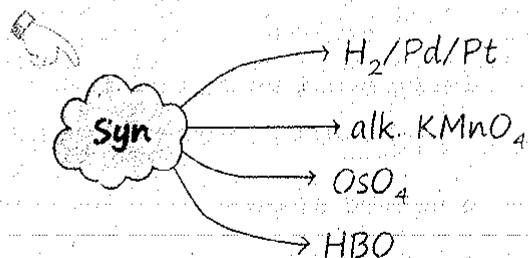
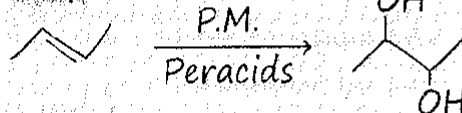
34> Bayer's reagent & + BR = Cold alkaline dil + Vicinal diol banega [1, 2-diol]
 $\text{OsO}_4/\text{pyridine}/\text{NaHSO}_3$ KMnO_4
 [syn-addition] $\rightarrow \text{OsO}_4/\text{Py}/\text{NaHSO}_3$



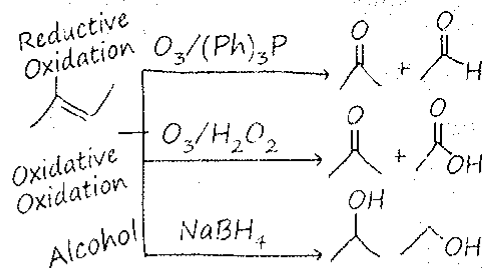
35> Peracids/prevost method
 (Anti addition) :- $\text{CH}_3-\text{C}(=\text{O})-\text{O}-\text{O}-\text{H}/\text{H}_3\text{O}^+$
 $\rightarrow \text{I}_2/\text{CH}_3-\text{COO}^-\text{Ag}^+$

+ Vicinal diol banega

+ Anti addition



36> Ozonolysis:-

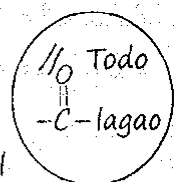


CH_3OH or $[\text{O}_3]$

CH_2Cl_2 (-78°)

Oxidative Ozo:- $-\text{COOH} + -\text{C}(=\text{O})-$
 $\rightarrow \text{H}_2\text{O}, \text{H}_2\text{O}_2, \text{Ag}_2\text{O}$

Note:-
 O_3/NaBH_4 or LAH gives Alcohol



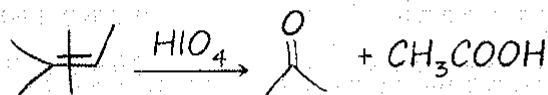
Reductive Ozo:- $-\text{CHO} + -\text{C}(=\text{O})-$
 $\rightarrow \text{Zn} + \text{H}_2\text{O}, (\text{CH}_3)_2\text{S}, (\text{Ph})_3\text{P}, \text{Zn} + \text{HCl}/\text{Acetic Acid}, \text{Thiourea}$

37> Oxidation:-

+ Hot Conc. $\text{KMnO}_4/\text{H}^+/\text{OH}^-$

Alkene & Acid Banao !

+ $\text{K}_2\text{Cr}_2\text{O}_7/\Delta + \text{NaIO}_4$
 + $\text{HIO}_4 + [\text{CH}_3\text{COO}]_4\text{Pb}$



38> Wohl-zeigler:-
 (Free-Radical)

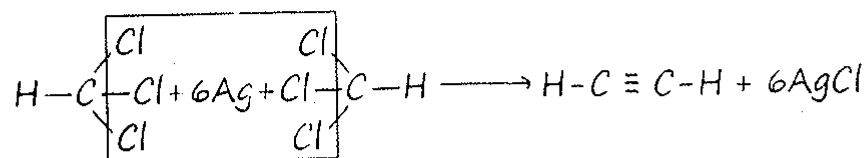
$\rightarrow \text{Br}$ lgayega.
 + $\text{NBS}/\text{CCl}_4, \text{hv}$
 + SO_2Cl_2
 + $\text{Br}_2 - 500^\circ\text{C}$
 + $\text{Cl}_2 - 500^\circ\text{C}$

Only Allylic/Benzylic Sub^s.

Jaha Kahi pr C^{*} Stable hoga waha pr -Br lagega.

39> RX^n with SeO_2 :- Allylic -OH Addⁿ.

40> 1,1,1 tri-halo compound:-

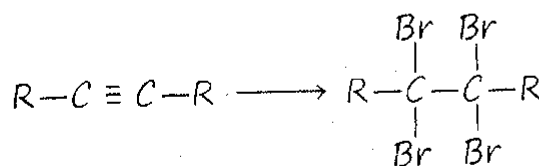


41> Dehydrohalogenation:-
 (Anti Elimination)
 Vicinal/Geminal dihalides $\xrightarrow{\quad}$ Alkynes

i. Alc. KOH/ Δ
 ii. NaNH₂ or KNH₂
 or Na/liq NH₃

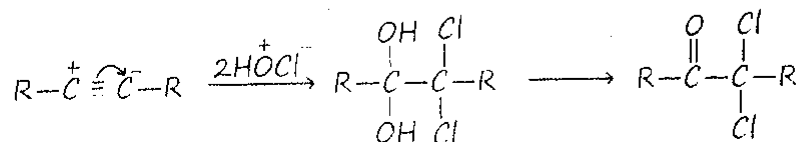
If NaNH₂ use directly
 (3) } Moles Req.
 Terminal alkyne }
 (2) }
 Intermediate vinylic halide }

42> Addition of Br₂/CCl₄ on alkyne:- Alkyne se Tetrahalide banao.



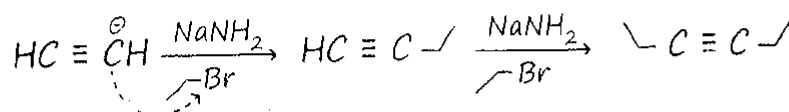
43> Addition of 2HOCl:-

Jaha Stable C⁺ waha $\text{C}=\text{O}$ uske Adjacent Geminal dihalide.

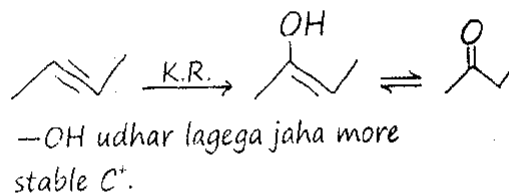
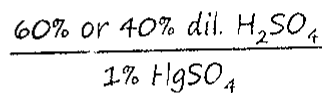


44> Addition of HBr:- Alkyne $\rightarrow$ Vicinal or Geminal dihalide.
 (Major) $\hookrightarrow \text{:Br} = +\text{M}$

45> Higher alkynes from Lower alkynes:-



46> Kucherov's rxn:-
 [Nu^o addⁿ rxn]

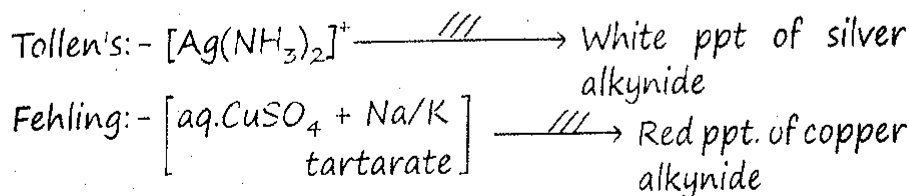


47> Polymerisation:- Red-hot Fe or Cu Tube. or Ni(CN)₂

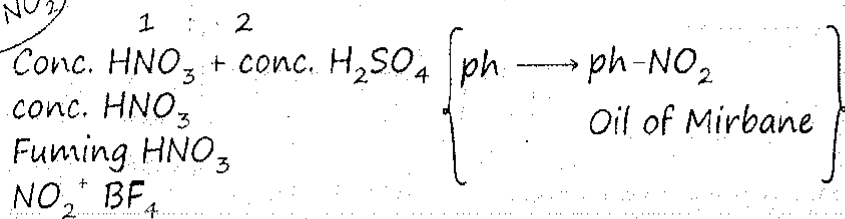
Alkyne se Aromatic Ring.

48> Tollen's or fehling Test:-

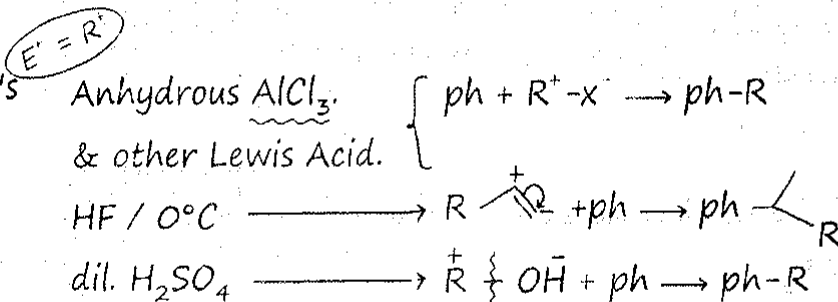
Terminal alkynes give (+)ve Test



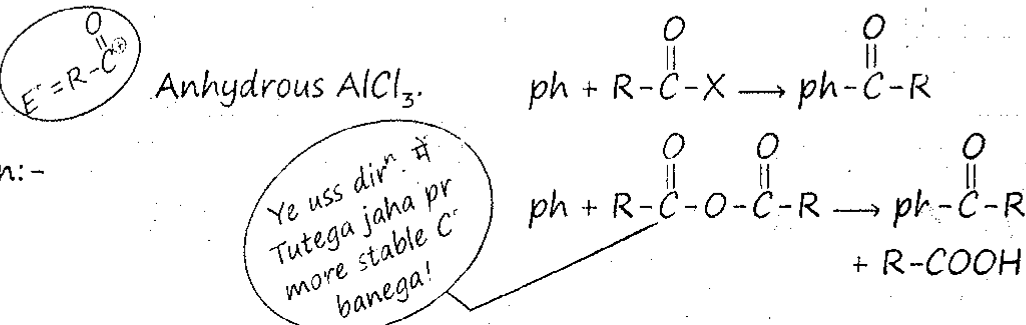
49> Nitration of benzene: -



50> Friedel craft's alkylation: -

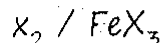


51> Friedel craft's acylation: -



52> Halogenation: -

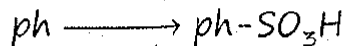
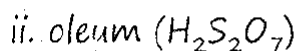
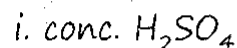
($F_2 > Cl_2 > Br_2 > I_2$)



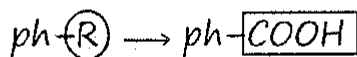
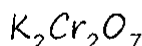
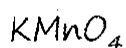
53> Sulphonation: -

(Neutral E^+)

$C_6H_6 > C_6D_6 > C_6T_6$

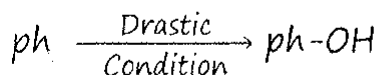


54> Oxidation of alkyl halide: -

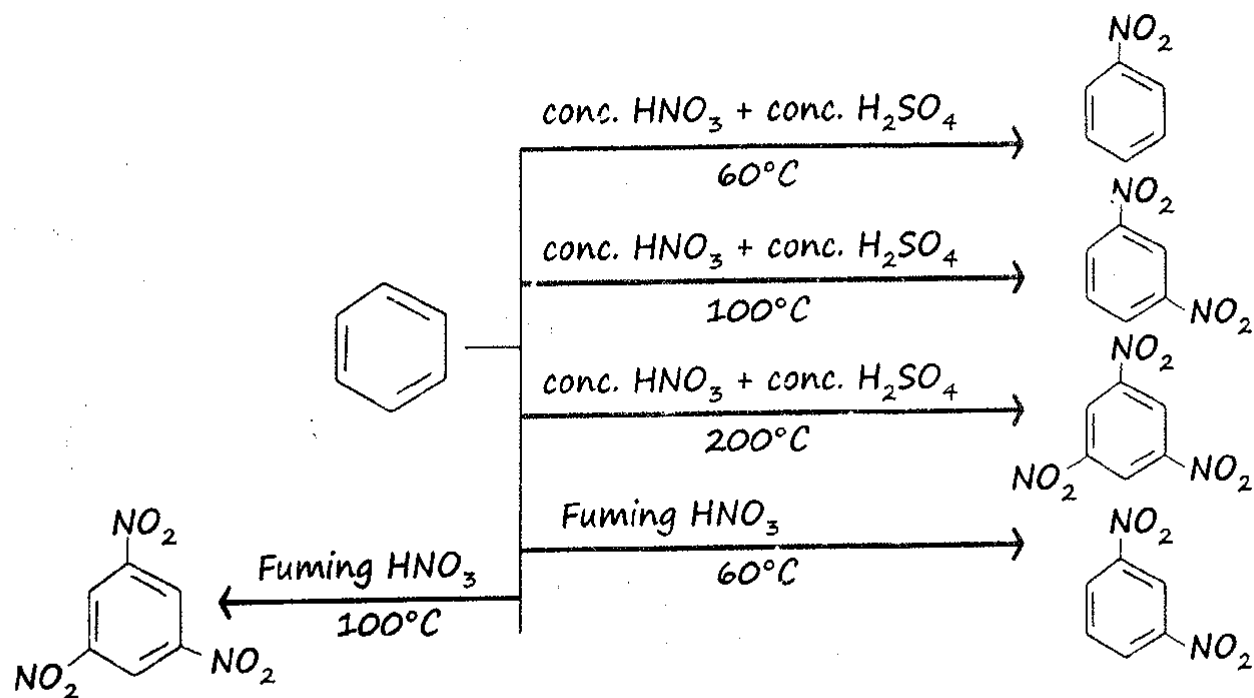


Note: - α -H must be present

55> Rxn with fused NaOH: -



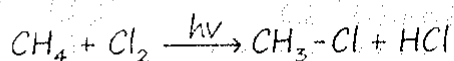
Benzene pr Good Leaving Group होना है!



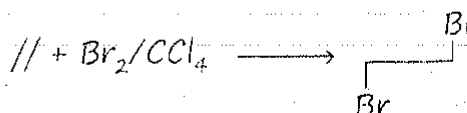
56> Gattermann Koch reaction: - (1) $\text{CO} + \text{HCl}, \text{AlCl}_3$ $\text{ph} \longrightarrow \text{ph-CHO}$

57> Gattermann aldehyde synthesis: - (1) $\text{HCN} + \text{HCl}, \text{AlCl}_3$ $\text{ph} \longrightarrow \text{ph-CHO}$
 (2) H_2O

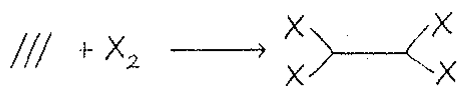
1) Halogenation of Alkane :-



2) Addition of HX to Alkenes :-

3) Addition of X_2/CCl_4 Alkene :-

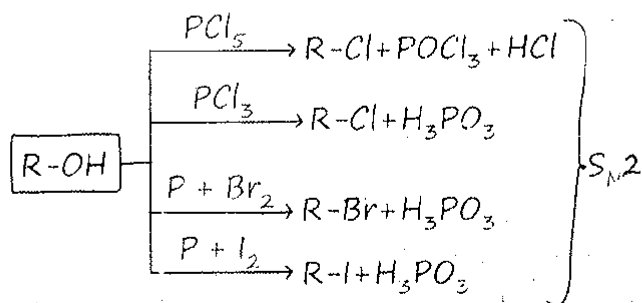
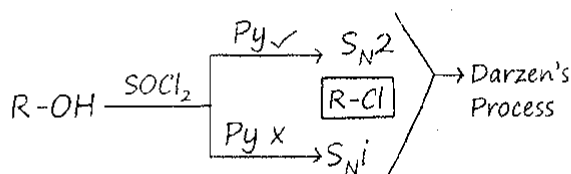
4) Addition of HX to Alkyne :-

5) Addition of X_2 to Alkyne :-

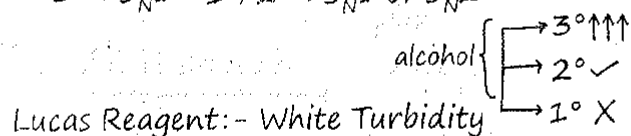
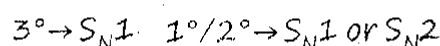
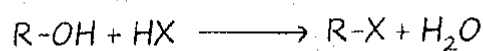
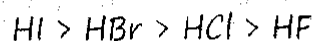
6) Reaction with NBS :-



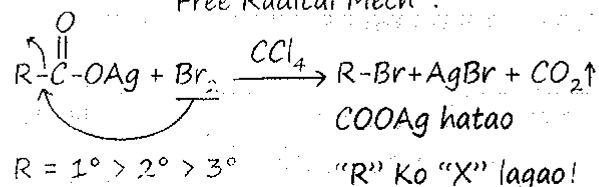
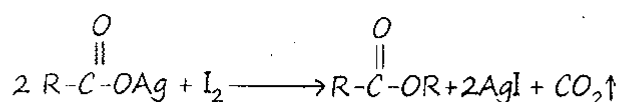
7) From Alcohols :-



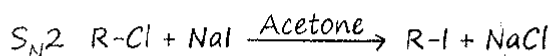
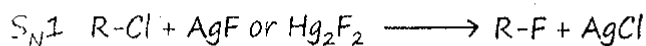
8) Reaction of Alcohol with HX :-



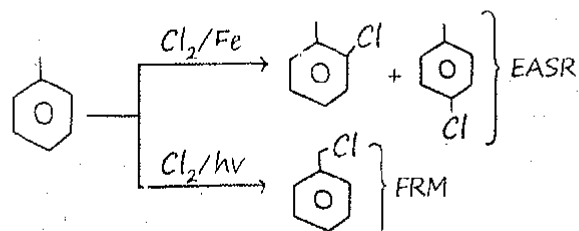
9) Hunsdiecker- Borodin Reaction :-

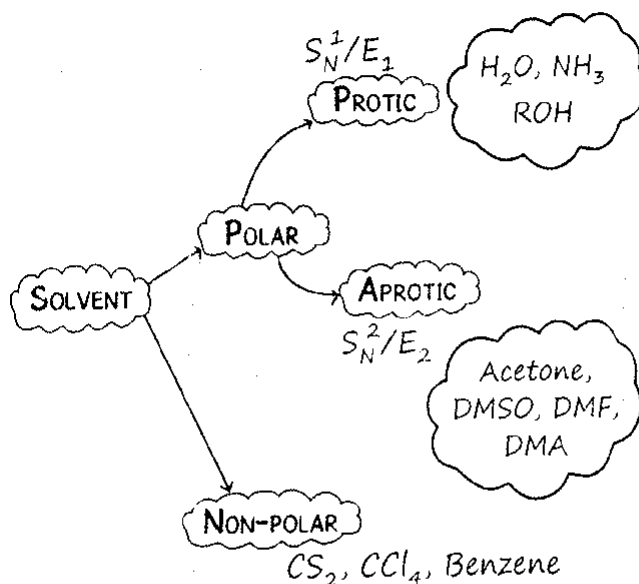
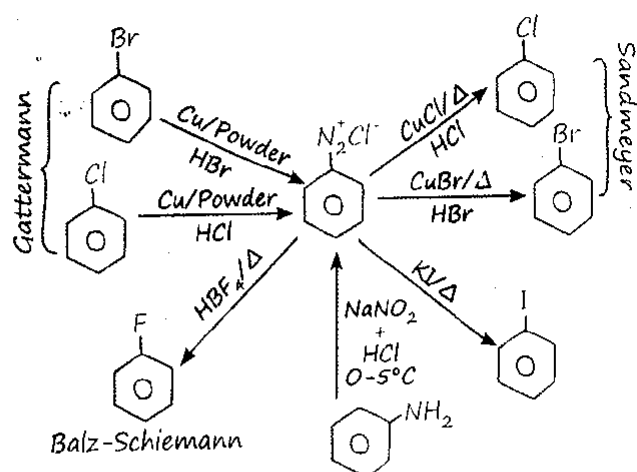
Free Radical Mech^m.Simonini RX^{II} :-

10) Halide Exchange Method :-

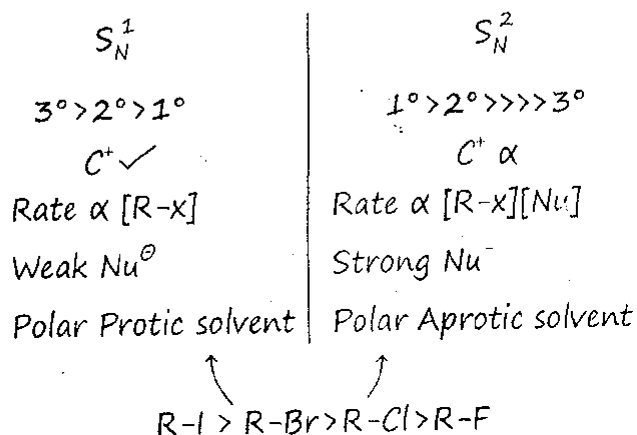
Finkelstein RXⁿ :-Swarts rxⁿ :-

11) Aryl Halides :-

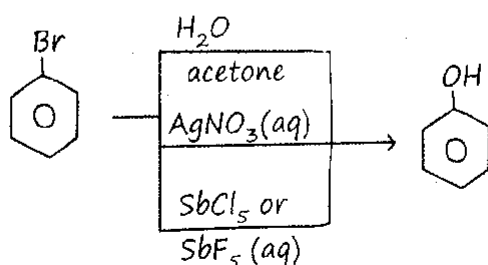
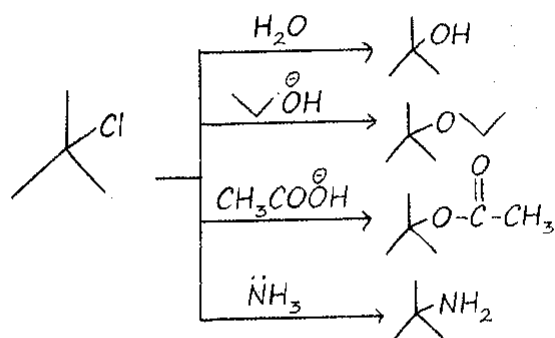




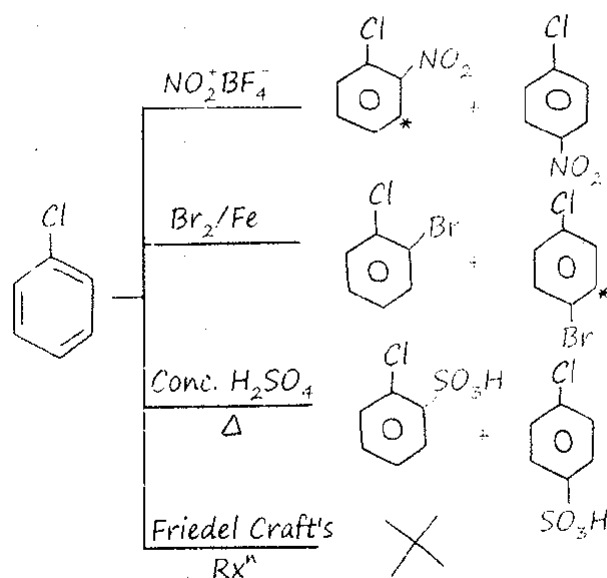
Note :-



12) Solvolysis :- S_N1



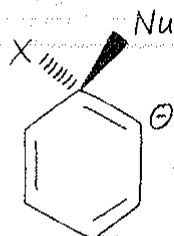
EASR :-



NASR :-

ADDITION- ELIMINATION

$F > Cl > Br > I$

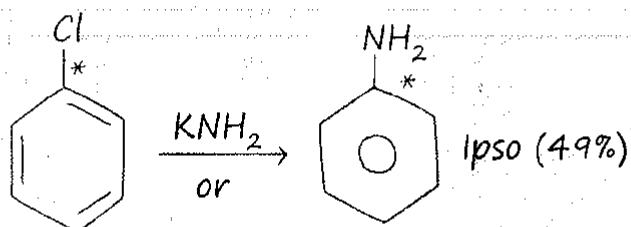


Meisenheimer Complex

- + Same as S_N2
- + Benzene must be connected to EWG.
- + "X" ko hatakr Nu^- ko lagao !

ELIMINATION- ADDITION

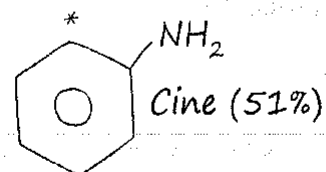
$I > Br > Cl > F$



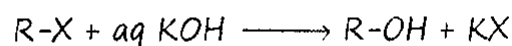
$NaNH_2$

or

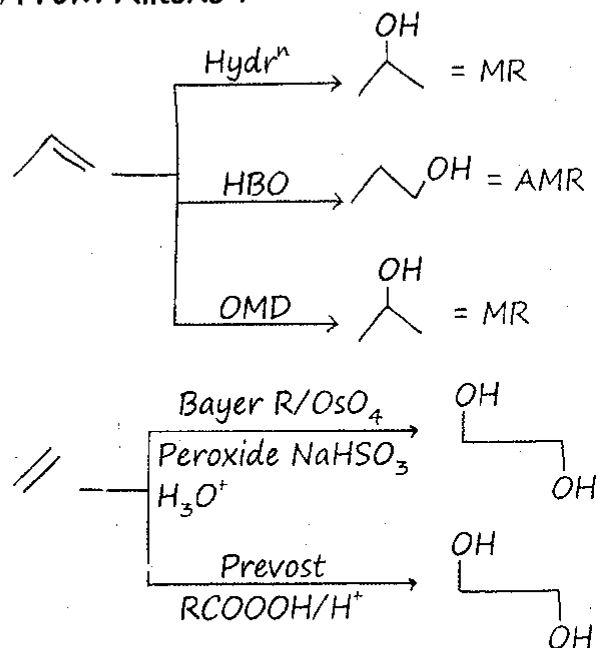
Birch
Redⁿ



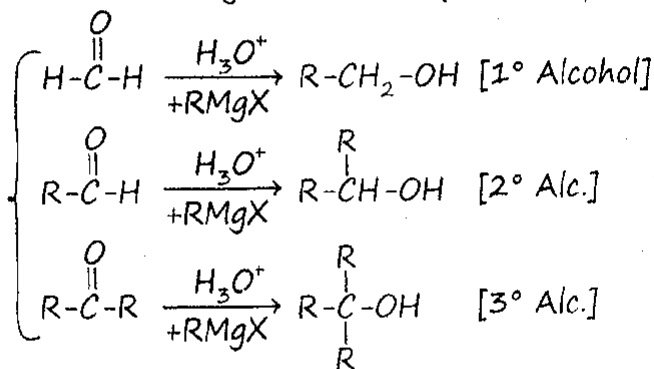
ALCOHOLS

① From Alkyl Halides :- [S_N^1 and S_N^2]

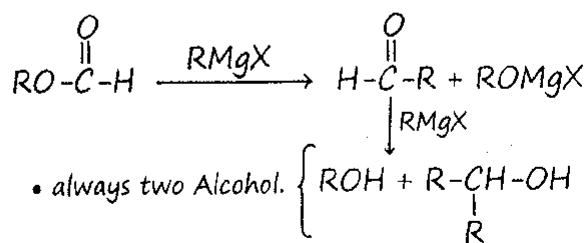
② From Alkene :-



③ Using Grignard Reagent :-



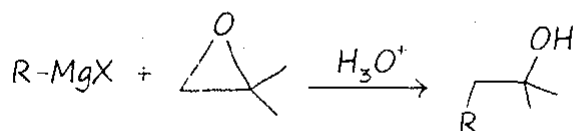
+ Ester :-



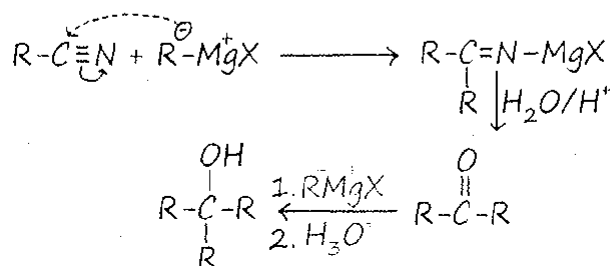
+ Epoxide :-



Cyclic Ethers are more reactive than normal ones!

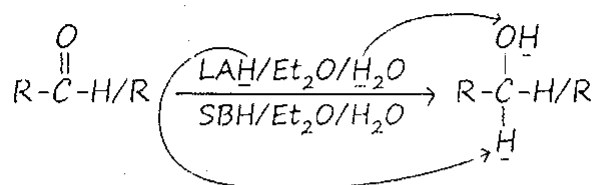


+ CYANIDE :-



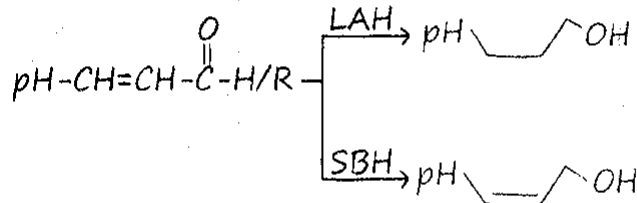
④ Reduction of Carbonyl

Compounds :- Sodium boro hydride:-
Acid halides only.



• LAH [Strong R.A.]

• Exceptionally reduce (cinnamyl condⁿ)



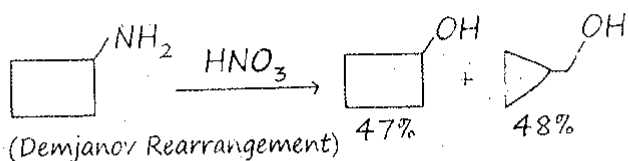
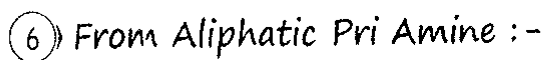
Other Reducing Agents :-

$\text{H}_2/\text{PtO}_2 \longrightarrow$ Adam's Catalyst.

$\text{Na}/\text{EtOH} \longrightarrow$ Bouevault Blanc Redⁿ.

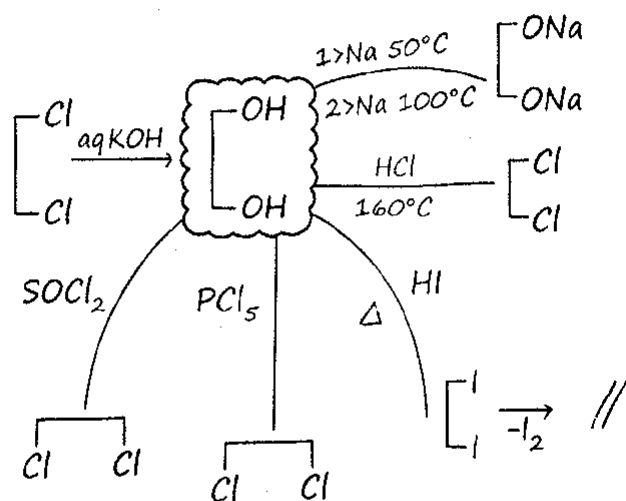
Aluminium isopropoxide $\longrightarrow$ MPV- Reduction.

(5) Hydrolysis of Esters :-

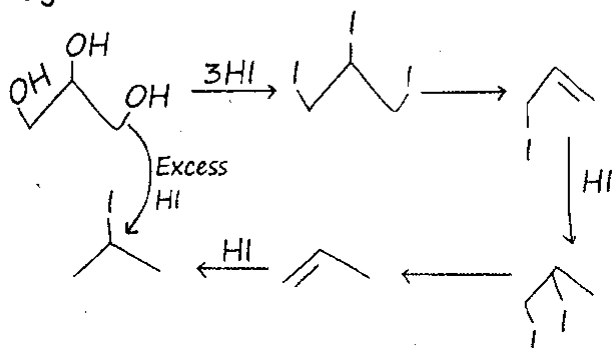


Organic Chemistry

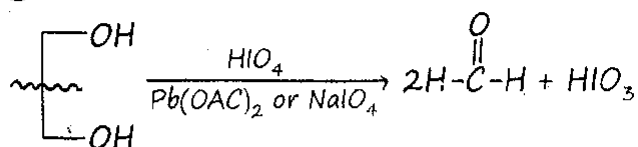
DIOLS



Glycerol :-

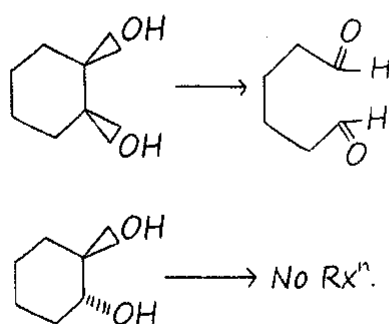


① Oxidation of DIOL :-



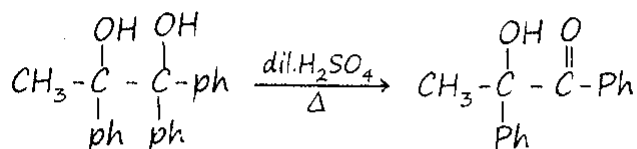
② Malaprade/Criegee RXN :-

(1,2) Vicinal diol same same hone chahiye bond break Karo dono Ke beech Ka aur $\text{O}=\text{O}$ lagao!



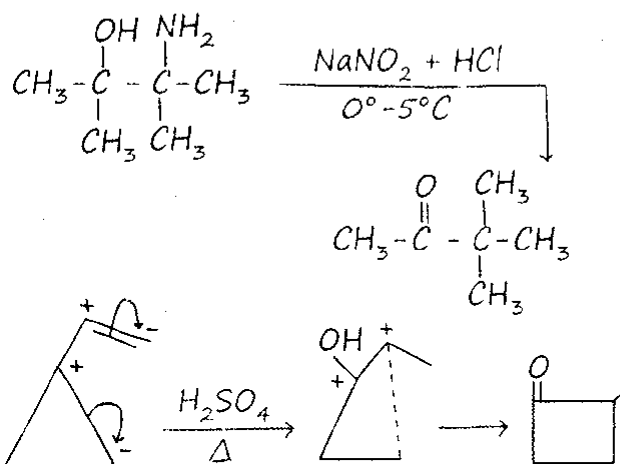
Reagent:-
 HIO_4
 NaIO_4
 Pb(OAc)_2

① Pinacol - Pinacolone Rearrangement :-



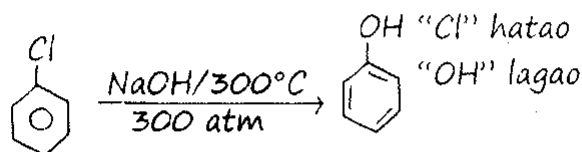
If the initially form C^+ is unstable it can undergo $\text{H} > \text{Ph} > \text{R}$ shift & C^+ will be stabilised by +M of -OH Finally pinacolone is formed!

Pinacol - pinacolone type :-

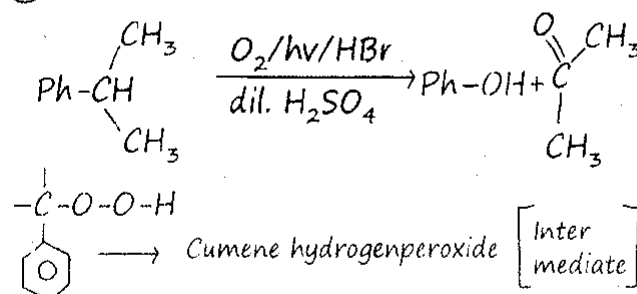


PHENOLS

① Dow's Process :-

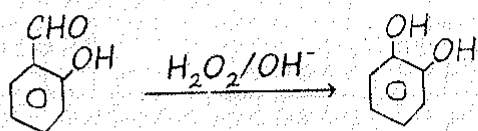


② From Cumene :-



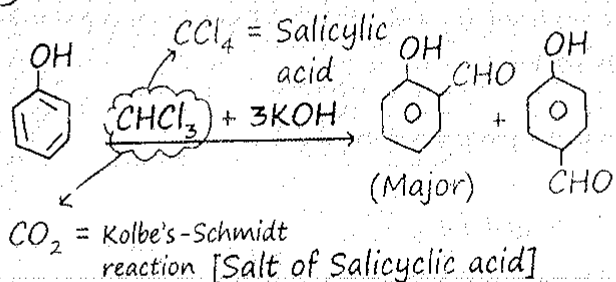
OH waha pr lagega jaha e^- density jyada hoga uske adjacent O lagado.

3) Dakin Process :-

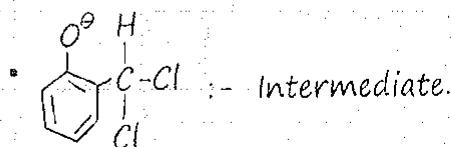


• -CHO ko -OH banado.

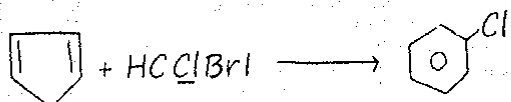
4) Reimer - Tiemann RXN :-



• $:CCl_2:-$ DCC $[E^+] \rightarrow$ (Singlet.)

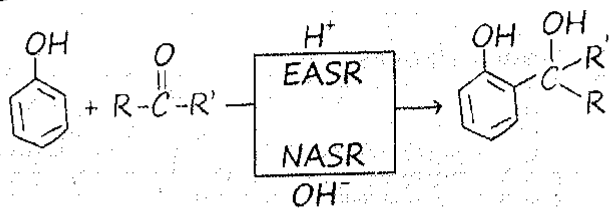


Miscellaneous-Reimann-Tiemann reaction:-

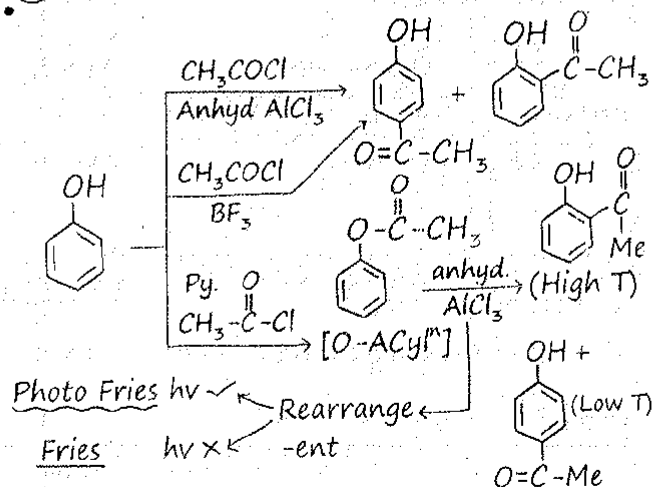


• Joh Bad L.G. hoga woh Benzene pr lagega.

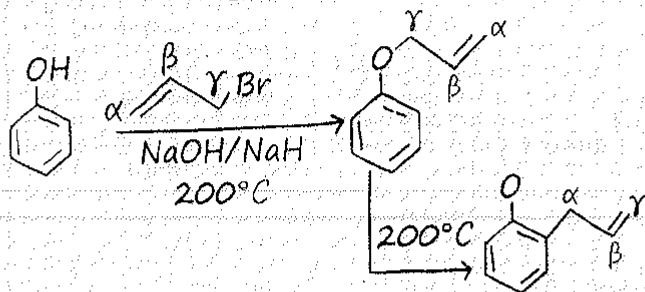
5) Lederer - Manasse Reaction :-



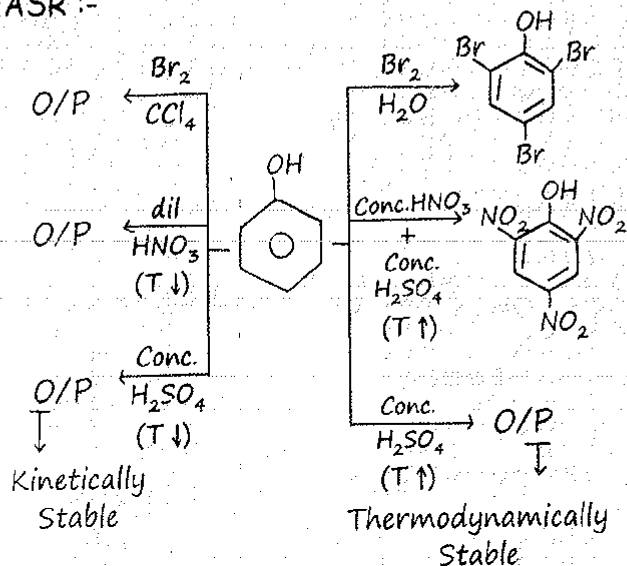
6) Friedel - Craft Acylation :-



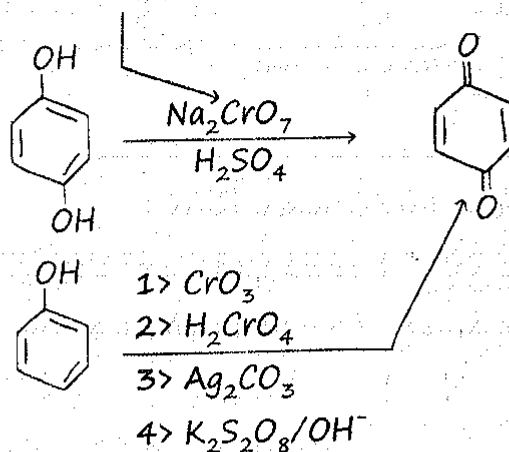
Claisen-rearrangement :-



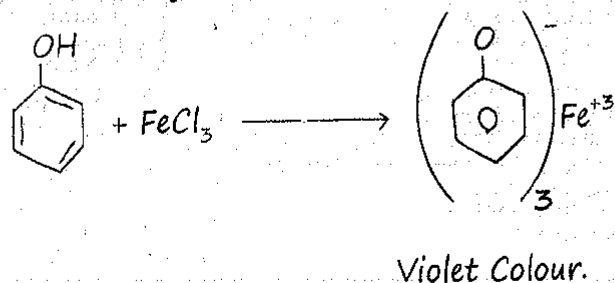
EASR :-



Elb's RXN (oxidation) :-



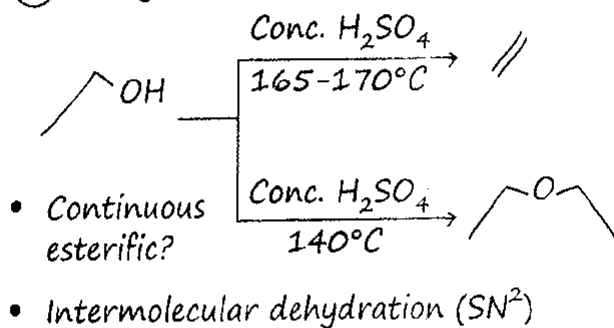
Rxn with FeCl3 :-



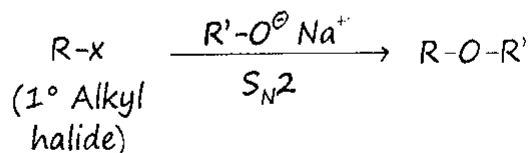
Violet Colour.

ETHERS

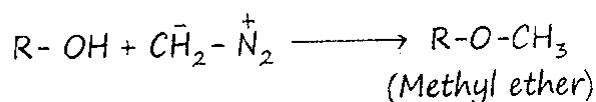
① Dehydration of Alcohols :-



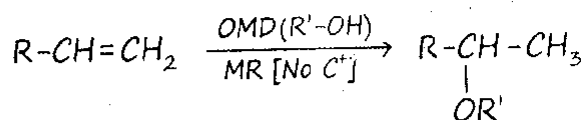
② Williamson's Ether SYN :-



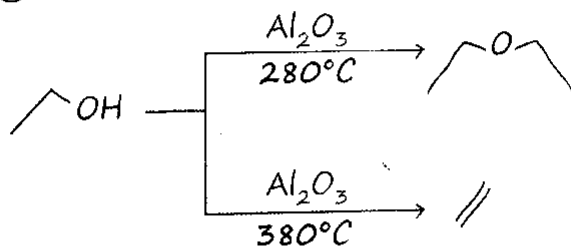
③ RXN with Diazomethane :-



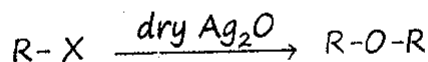
④ Alkoxymercuration :-



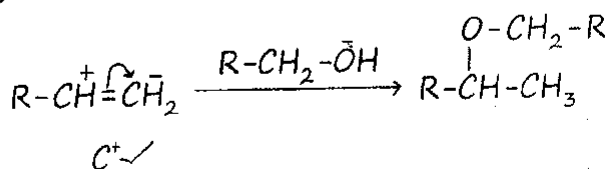
⑤ RXN with Al_2O_3 or ThO_2 :-



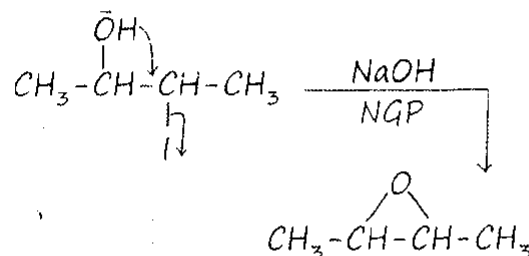
⑥ Dry Ag_2O :-



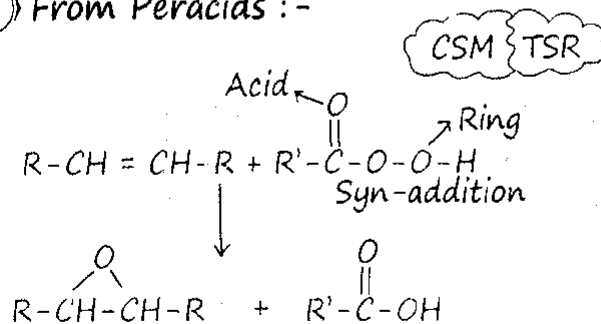
⑦ Addition of Alcohol to Alkene :-



⑧ From Hydroxyl Halo Compounds :-

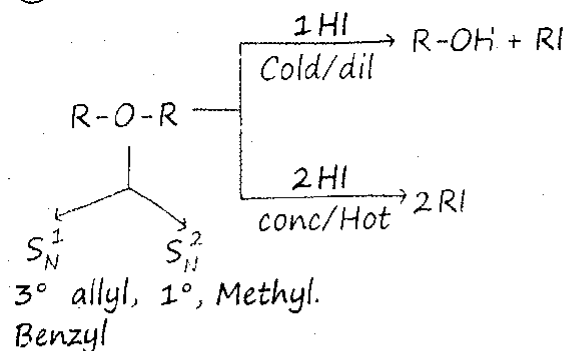


⑨ From Peracids :-



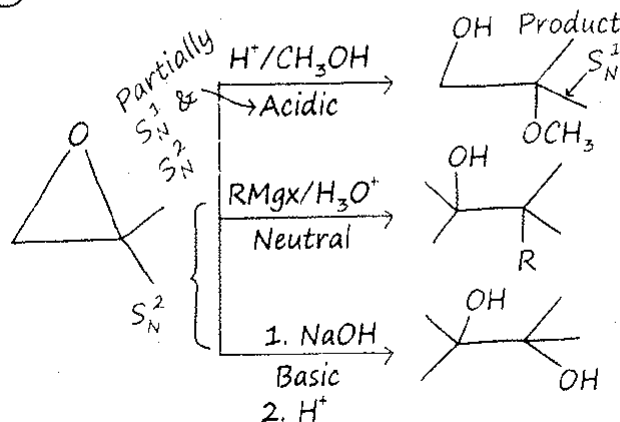
Chemical property :-

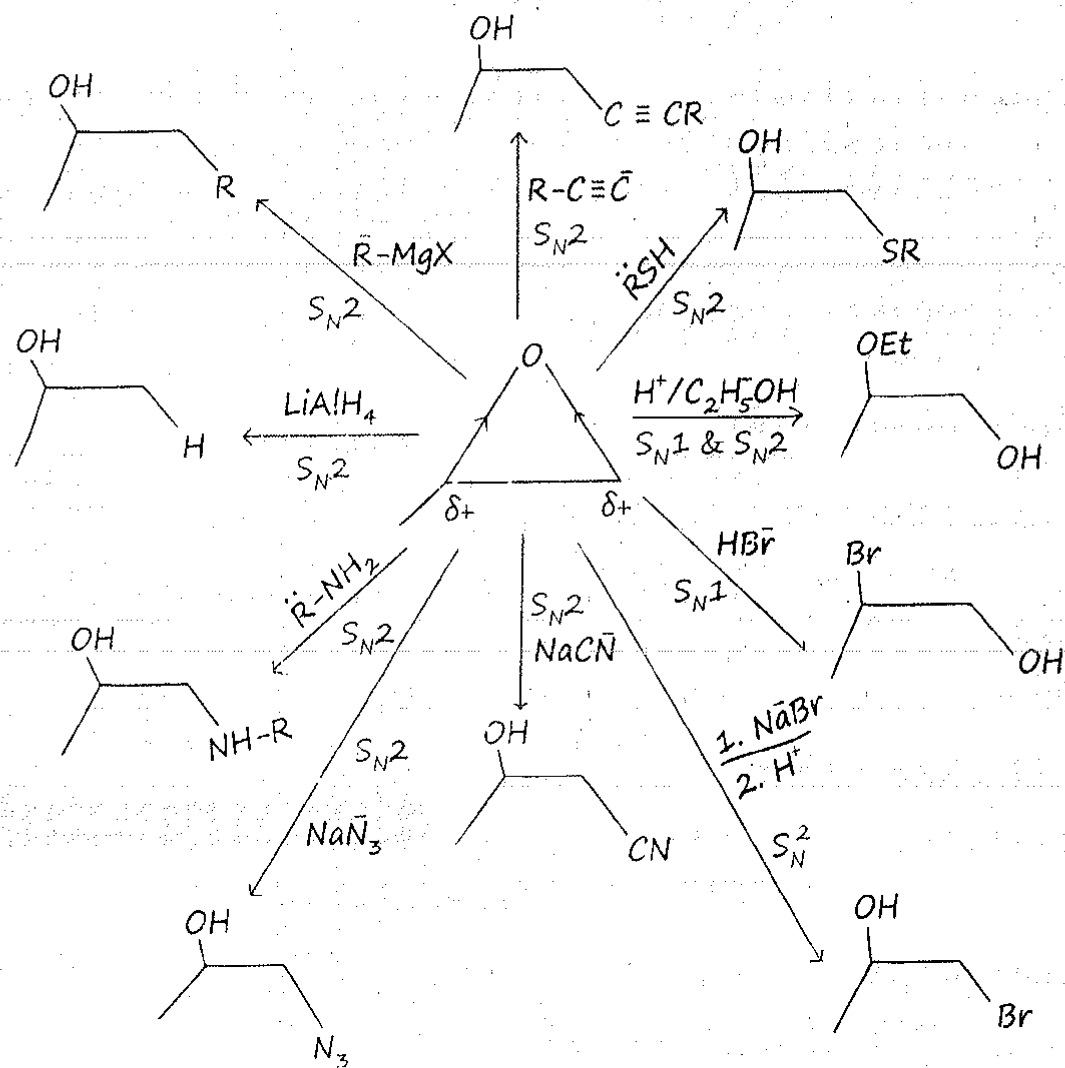
① Reaction with HI :-



S_N^1 :- I⁻ Udhar lagega jaha stable C⁺ hoga aur H⁺, O⁻ pr Ether Ke. aur S_N^2 mein just inverse.

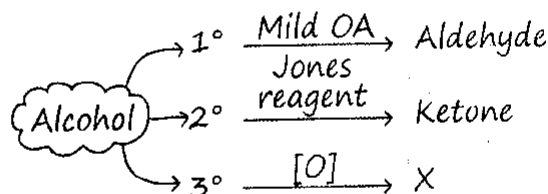
② Reaction with Oxirane :-



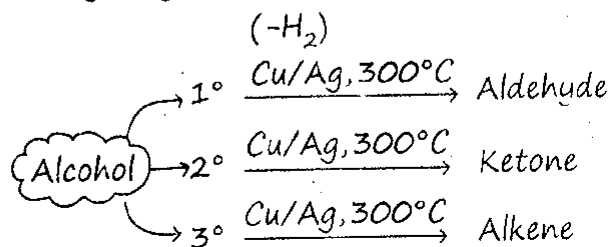


1) Oxidation of Alcohol :-

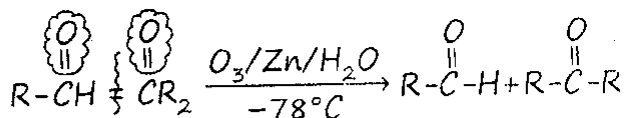
Mop



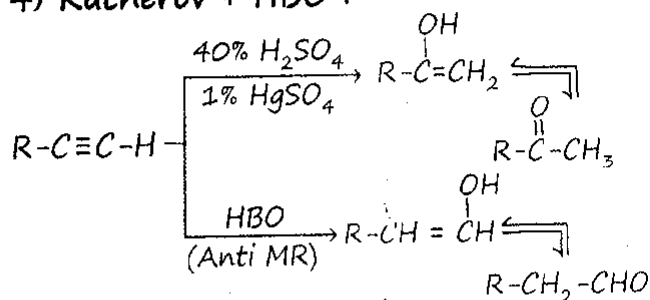
2) Dehydrogenation of Alcohol :-



3) Ozonolysis of Alkene :-

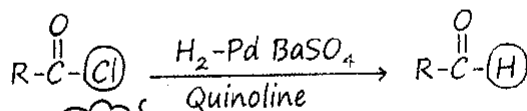


4) Kucherov + HBO :-



5) Rosenmund Reduction :-

"exclusively for aldehyde"



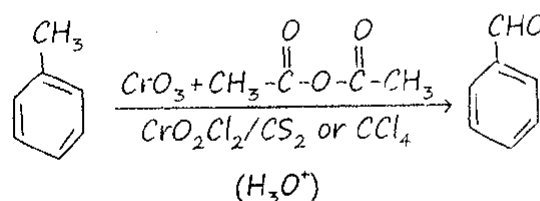
Lindlar's Catalyst

Reactivity order:-

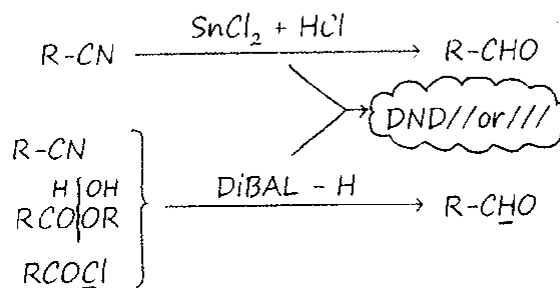
Acid halide > Alkyne > Aldehyde > Alkene > Ketone

6) Etard Reaction :-

* Aromatic Aldehydes *

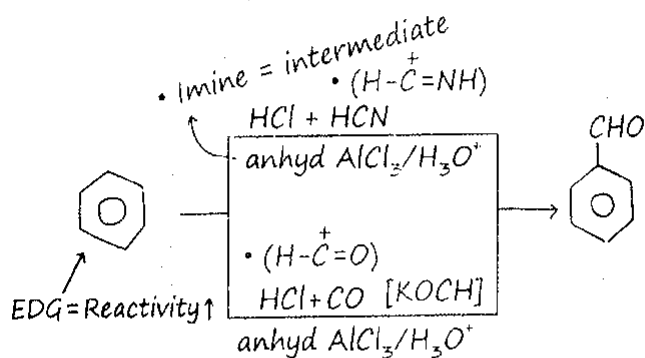
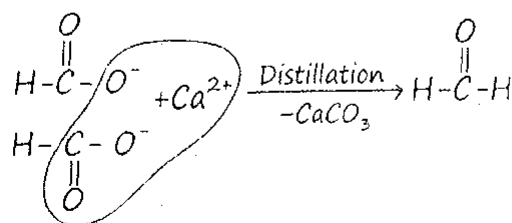


7) Stephen reduction - Aldehyde SYN:-



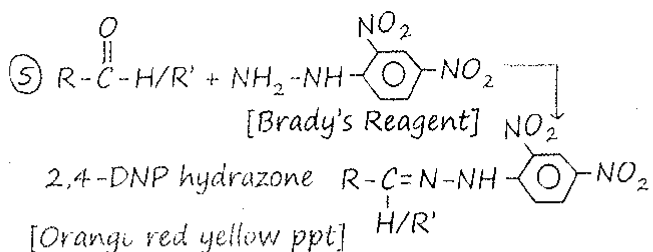
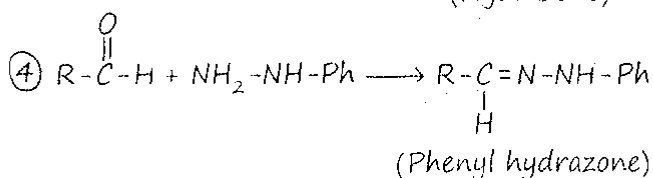
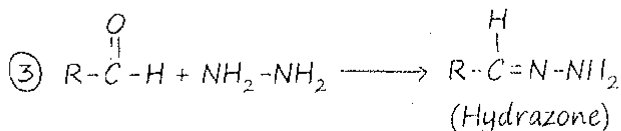
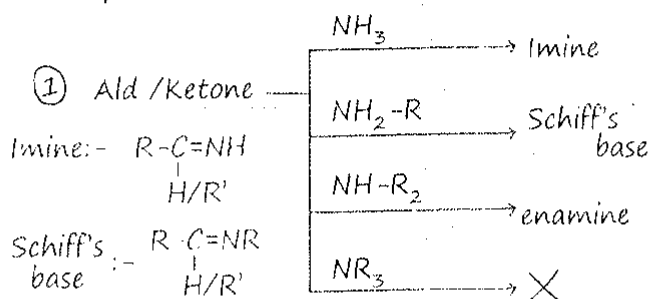
EASR

8) Gattermann :-

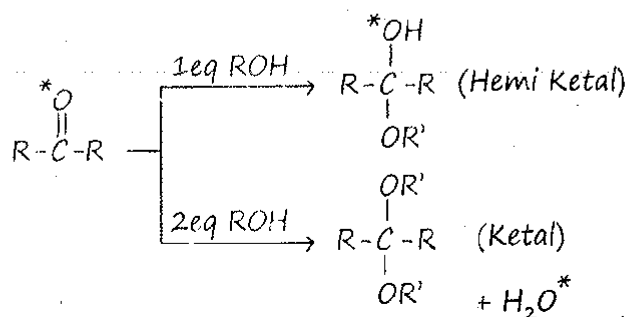
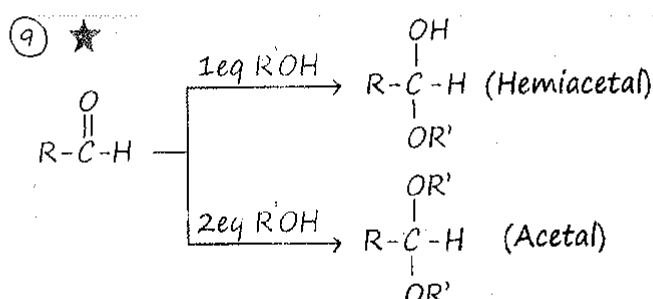
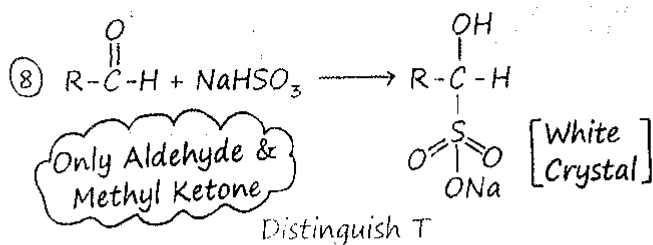
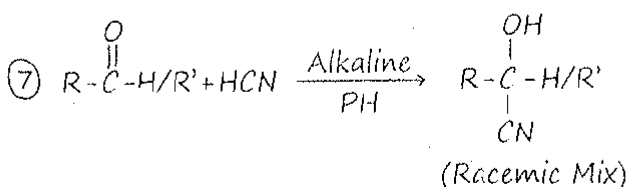
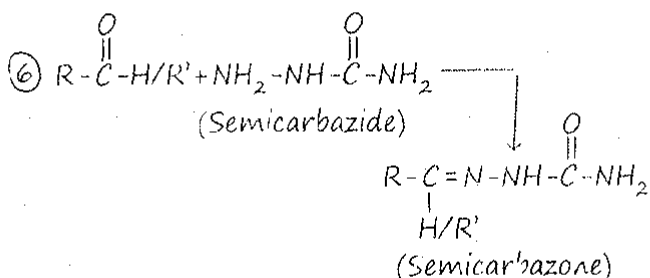
9) Distillation of Ca^{2+} & Ba^{+2} Salts of $COOH$:-

Chemical property:-

Nucleophilic Addition Reaction:-

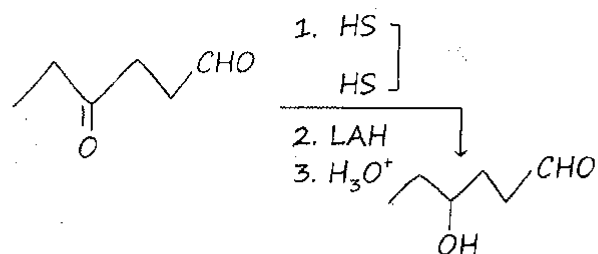


Distinguish T

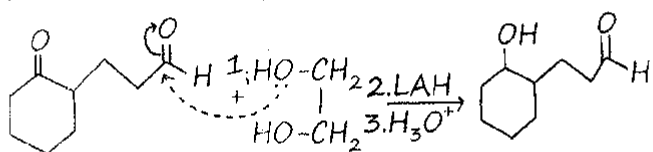


• MOZINGO Rxn

• Diols or thiols are used as protecting agents.

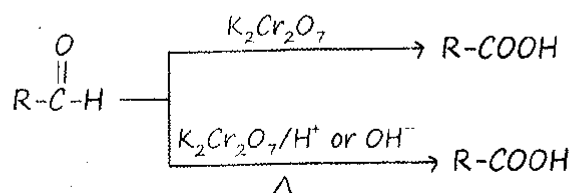


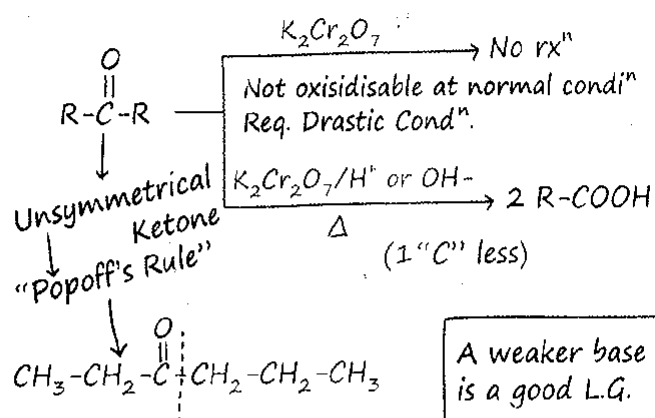
• Vicinal diols can be used as protecting groups in a compound if it has a multiple Fn groups & protect one of them.



Oxidation Rxn's :-

1> Rxn with $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+$ or $\text{OH}^-/\text{KMnO}_4$:-



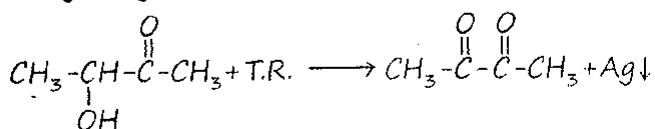


* TOLLEN'S REAGENT:- (SILVER MIRROR TEST)

Given by:- Aldehyde, Terminal Alkyne, Formic Acid, α -hydroxy Ketones.

Note:- Ketones ye test nai denge !

α -hydroxy Ketone:-

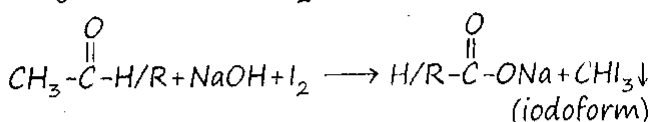


Reagent:- Ammonical Silver Nitrate (O.A.)
 $[\text{Ag}(\text{NH}_3)_2]^+ \text{NO}_3^-$

* HALOFORM TEST :-

Given by:- $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-$ $\Rightarrow$ Aldehydes & Ketones having this group.

Reagent:- $\text{NaOH} + \text{X}_2$ (excess)



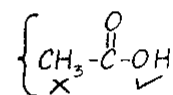
Note:-

COOH & its derivative inspite

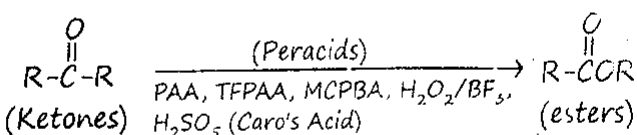
* Bright yellow crystalline ppt.*

of having $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-$ not give test!

$\rightarrow$ as we need more Acidic α -Hydrogen.



* BAYER-VILLIGER OXIDATION :-

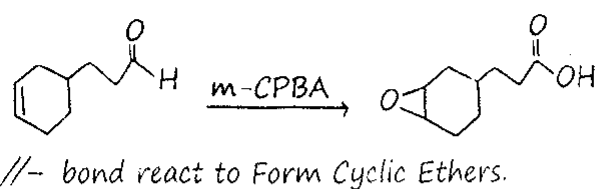


Migratory Aptitude:-

$\text{H} > 3^\circ \text{alkyl} > \text{Cyclohexyl} > 2^\circ \text{alkyl} > \text{Benzyl}$
 $> \text{Phenyl} > 1^\circ \text{alkyl} > \text{CH}_3$

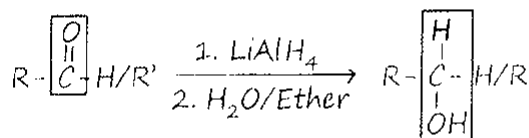
Cyclic Ketones $\Rightarrow$ Lactones.

* Jo Migrate Karega uske baju men "O".

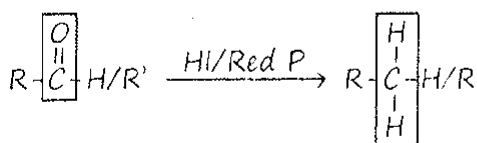


Reduction Rxn's :-

1> With LAH, SBH, H_2Pd :-

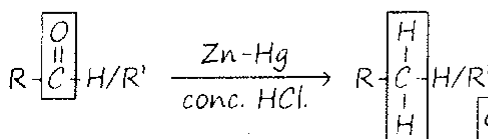


2> With HI/Red P :- 😡!



Baap of all R.A.

3> Clemmensen Redⁿ :- (Acidic Medium).

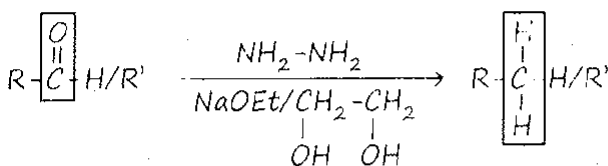


1 mole } $4e^- \leftarrow \text{Zn}$
 Aldehyde } $4\text{H}^+ \leftarrow \text{HCl}$

intermediate:- Alcohol

Grp like NO_2 , $-\text{OC}_2\text{H}_5$, $-\text{OH}$, $-\text{I}$ don't prefer C.M.R. respond to Acidic Medⁿ

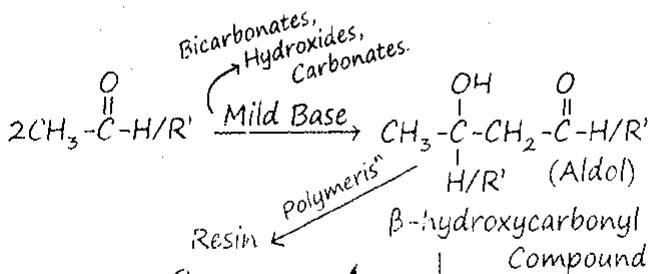
4> Wolf-Kishner Redⁿ :-



intermediate :- Hydrazone

Rxn's due to α -H :-

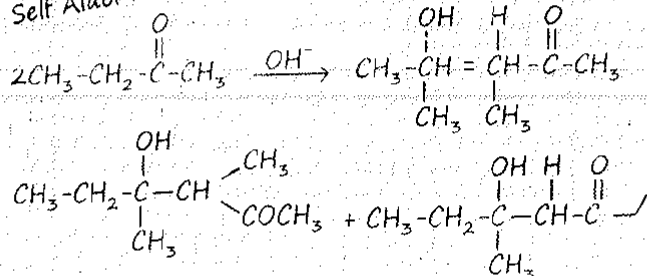
1> ALDOL Condensation :- (NAR)



Cross/Mixed Aldol Condensation :-

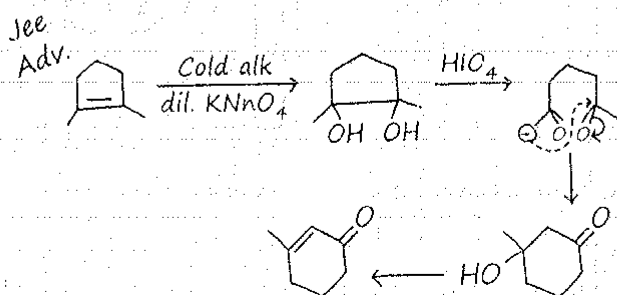
Major Product :- $\text{Nu}^\ominus = \text{Ketone}$

Self Aldol :- Reagent = Aldehyde

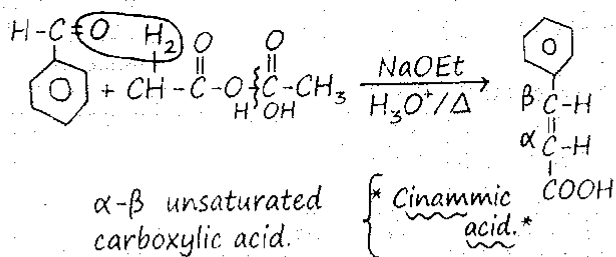


Intramolecular Aldol Condensation :-

Product :- Ring Size > Acidic "H" (Priority).

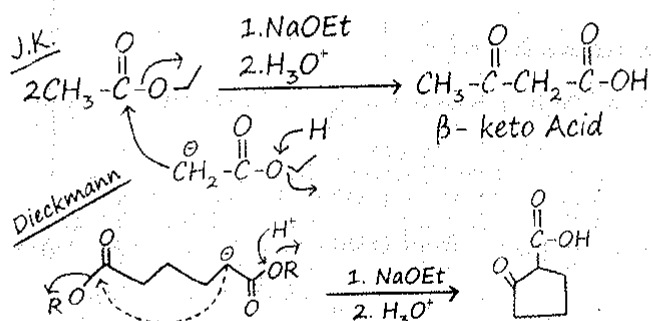


2> PERKIN RXⁿ (Condensation) :- (NAR)

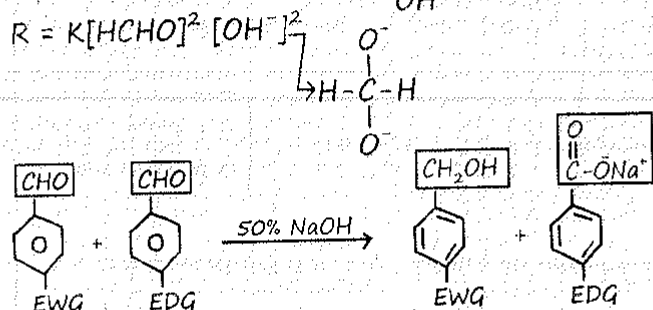
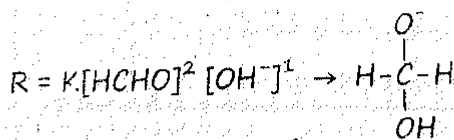
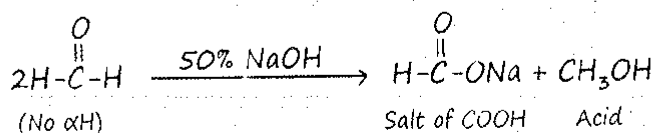


* Ar. Aldehyde with EWG. inc. Reactivity.

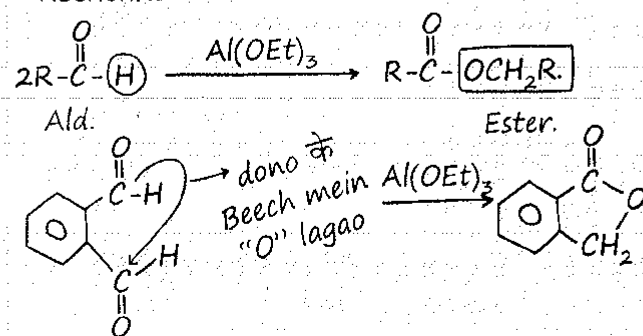
3> Claisen - Condensation :- (NSR)



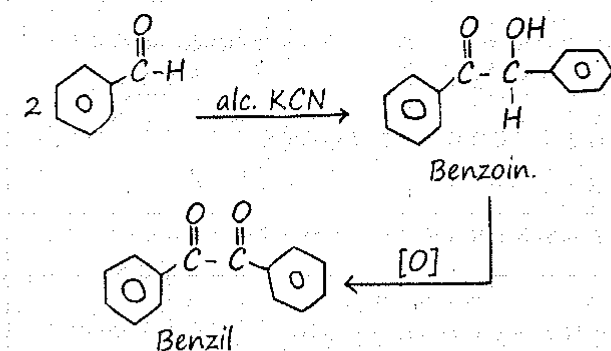
4> Cannizaro's RXⁿ :- [Disproportionation RXⁿ]



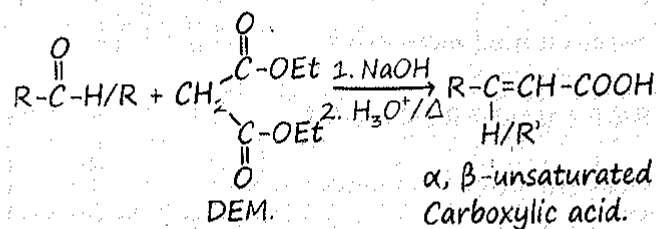
Tischenko RXⁿ :-



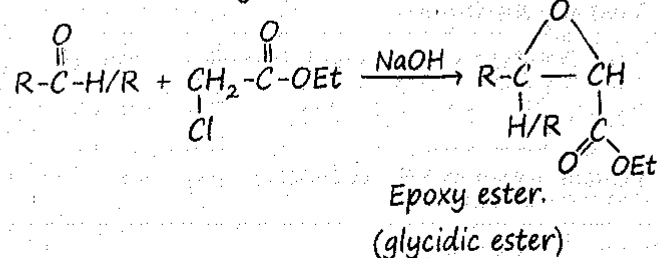
5> Benzoin Condensation rxⁿ :-



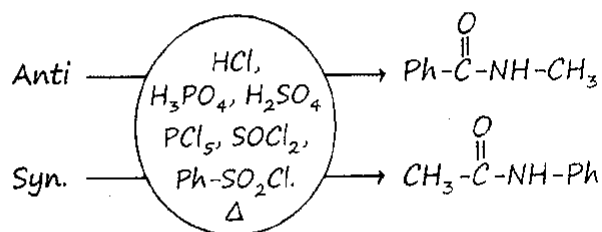
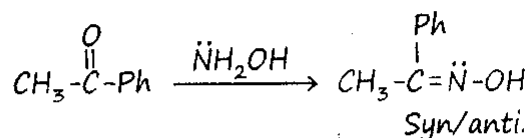
6> Knoevenagel rxⁿ. [Modified Aldol]



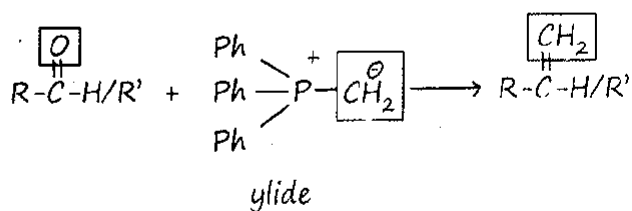
7> DARZEN Glycidic Ester condensation :-



8 > Beckmann Rearrangement :-



9 > Wittig Rxⁿ :-



DISTINGUISH TEST :-

Ketones	X	X	X	X
α -hydroxy Ketone	>	>	>	>
Terminal alkyne	>	>	>	>
HCOOH	>	>	>	>
Aromatic Acid	>	X	X	X
Aliphatic Acid	>	>	>	>
	1 > Tollen's Reagent	2 > Fehling	3 > Benedict	4 > Schiff.

1 > Tollen's Reagent :- $[\text{Ag}(\text{NH}_3)_2]^+ \rightarrow$ Silver Mirror Test.

2 > Tollen's Reagent :- (A) aq $\text{CuSO}_4 \rightarrow$ Red ppt.

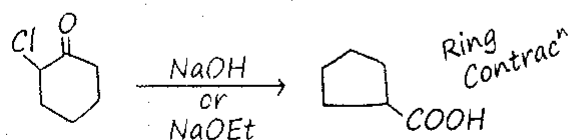
(B) Na/K Tartarate $\rightarrow$ Rochells salt

3 > Benedict Solution :- aq $\text{CuSO}_4 + \text{Na/K Citrate} + \text{RCHO}$
Magenta (Pink)

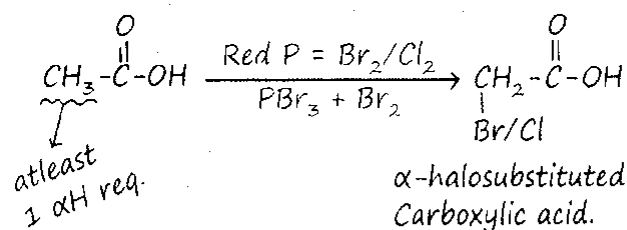
4 > Schiff's Reagent :- p-rosaniline hydrochloride

Carboxylic acids & it's derivative :-

1 > Favorskii Rearrangement :-



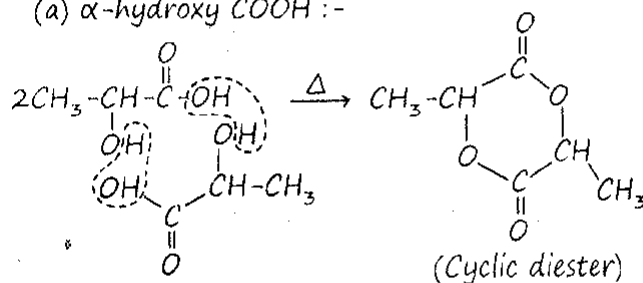
2 > Hell - Volhard - Zelinsky (HVZ) Rxⁿ :-



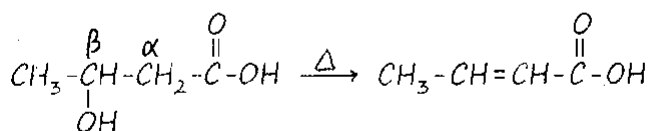
Aromatic COOH not give HVZ Rxⁿ.

3 > of Carboxylic Acid :-

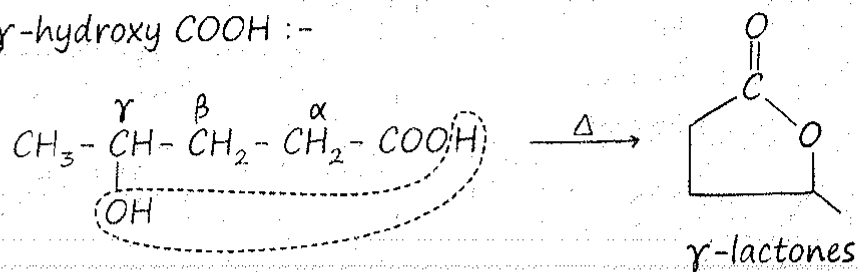
(a) α -hydroxy COOH :-



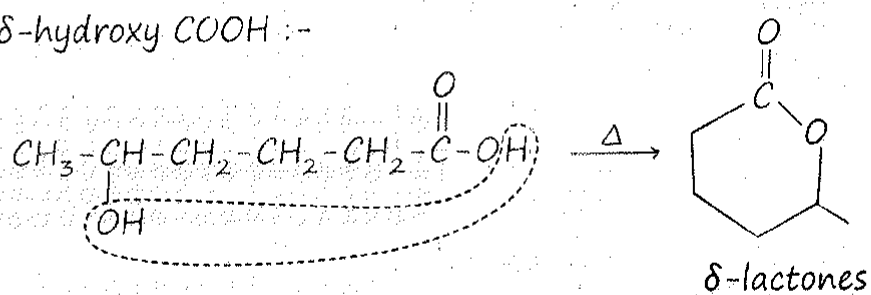
(b) β -hydroxy COOH :-



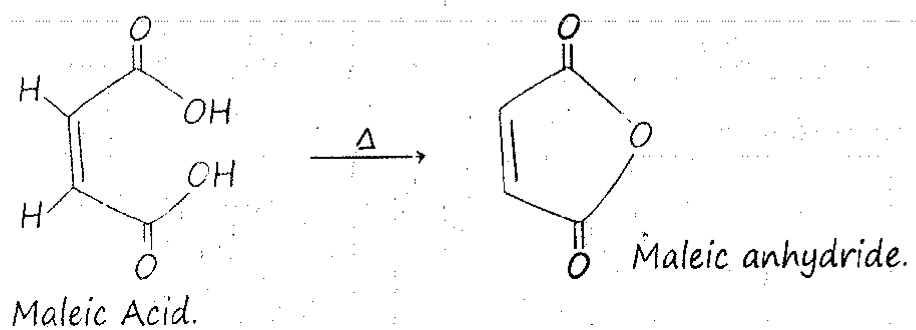
(c) γ -hydroxy COOH :-



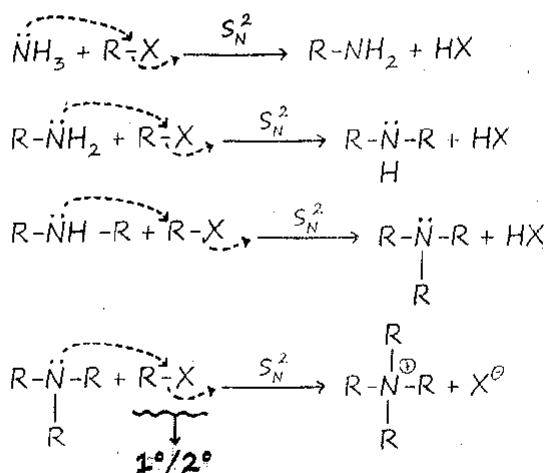
(d) δ -hydroxy COOH :-



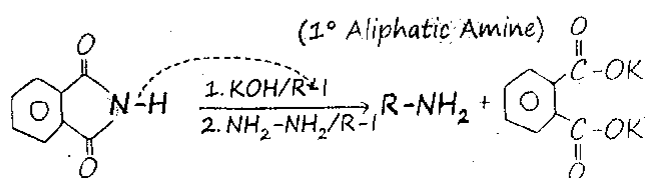
Note:-



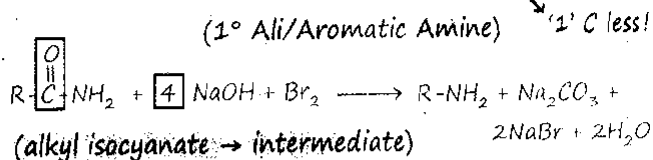
1> Hoffmann Exhaustive Ammonolysis :-



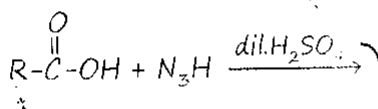
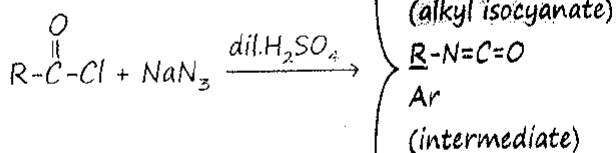
2> Gabriel - Phtalimide Synthesis :-



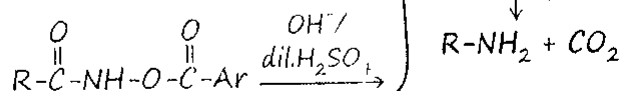
3> Hoffmann's Bromamide degradation :-



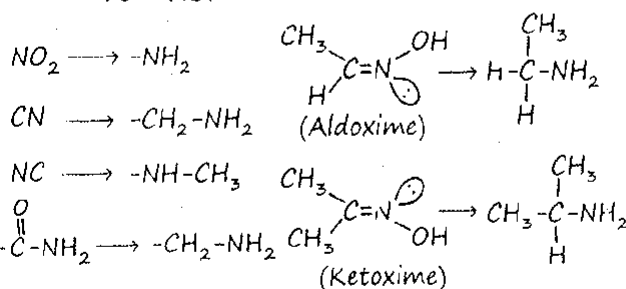
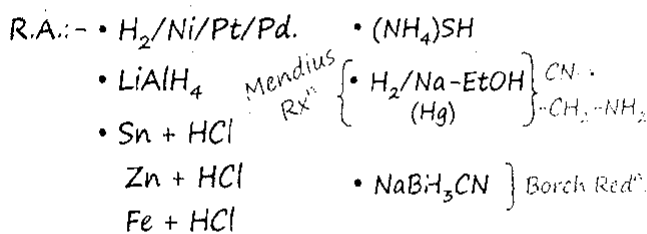
* If R = Chiral ⇒ Chirality is Retained.

4> Schmidt Rxⁿ :-5> Curtius Rxⁿ :-

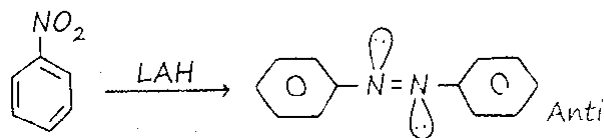
6> Lossen rxn :-



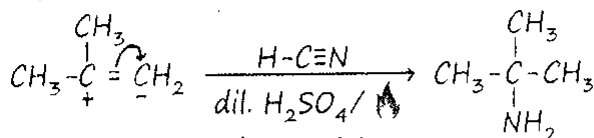
7> REDUCTION of N-Containing Compounds :-



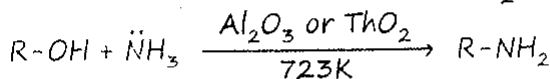
Note:-

8> Ritter Rxⁿ :-

H⁺ Med^m, Jaha C⁺ Stable banega waha NH₂ lagao.

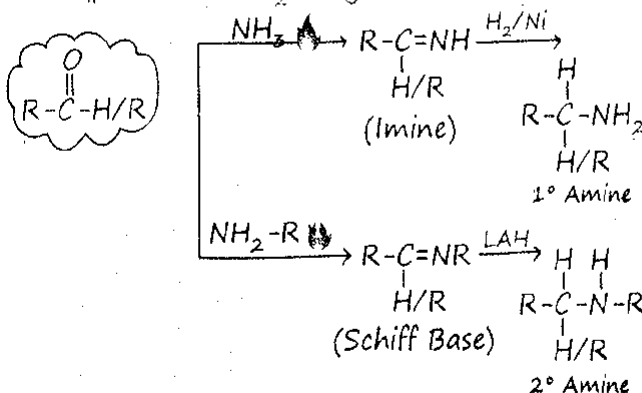


"1° amine connected to 3°C."

9> From Alcohols :- DM:- OH- hatao NH₂-lagao.

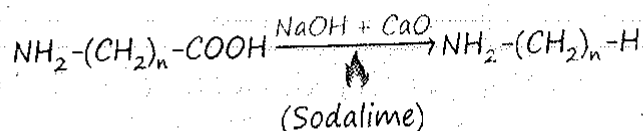
10> From Carbonyl Compound :-

"O" hatao "NH₂" lagao!



11> 🔥 of α, β, γ & δ ... amino acids:-

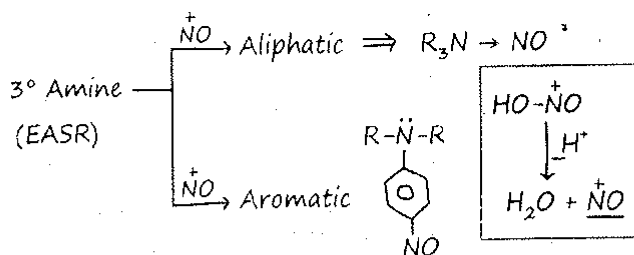
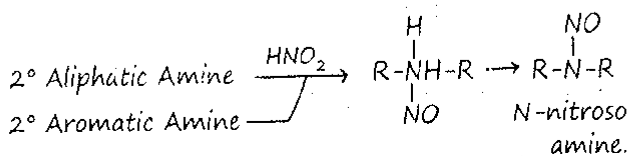
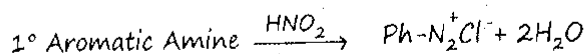
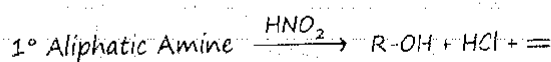
Just Remove COOH & Make it CO_2
 "COOH" hatao "H" lagao.



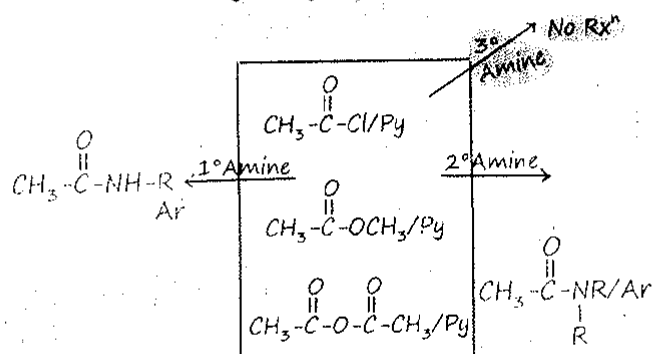
12> Rx^n with HNO_2 [$\text{NaNO}_2 + \text{HCl}$]

(Diazotisation)

Temp^r = $0-5^\circ\text{C}$.

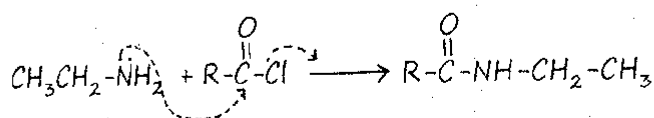


13> Rx^n with Acylating Agents :-



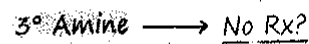
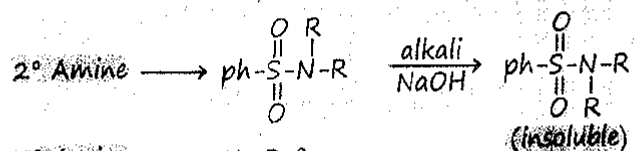
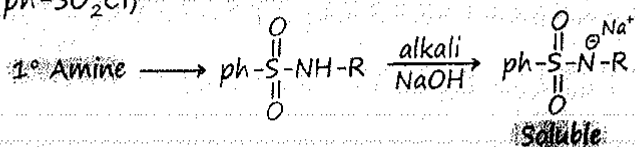
14> Schotten Baumann :-

(1° Amine with R-X)



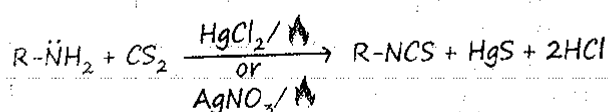
DISTINGUISH TEST :-

1. Hinsberg Test :- benzene sulphonyl chloride.
 ($\text{ph-SO}_2\text{Cl}$)



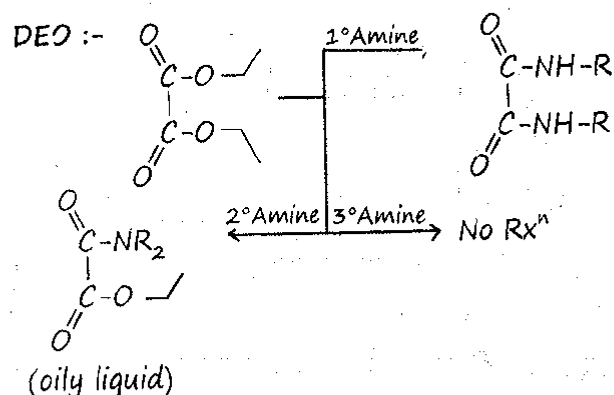
2. Hoffmann Musard Oil Test :-

(Only for 1° Amine)



intermediate :- N-alkyl isothiocyanate

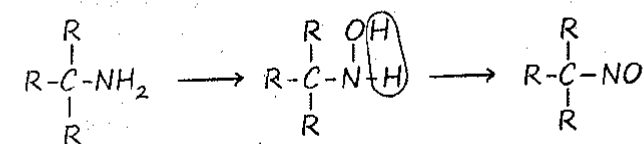
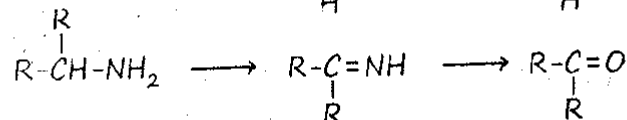
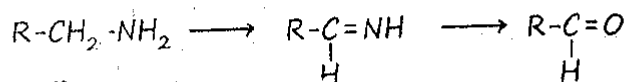
3. Hoffmann Test :- (Diethyl oxalate).



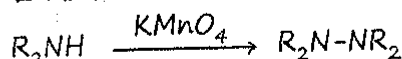
OXIDATION of AMINES :-

1. Aliphatic Amine with KMnO_4 :-

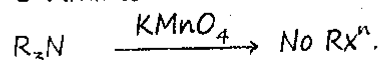
1° Amines :-



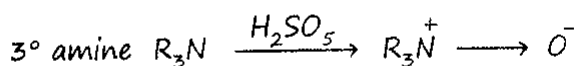
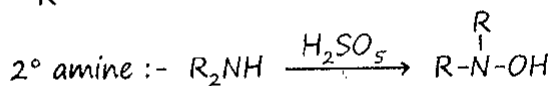
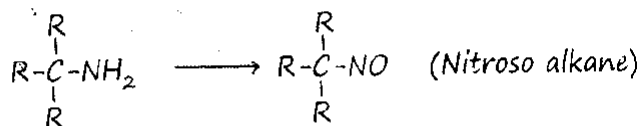
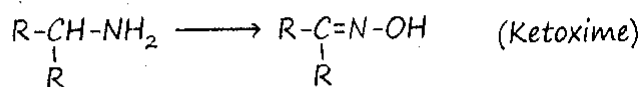
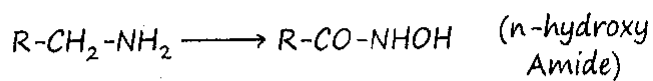
2° Amines :-



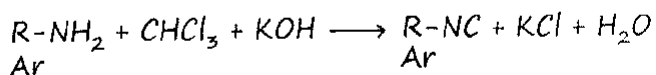
3° Amines :-



2. Oxidation with H_2SO_5 :-

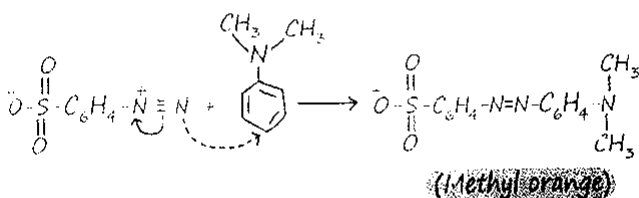
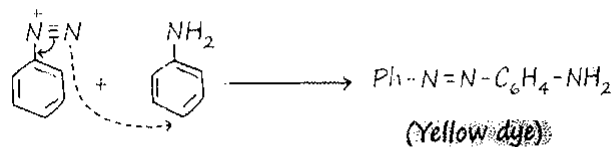
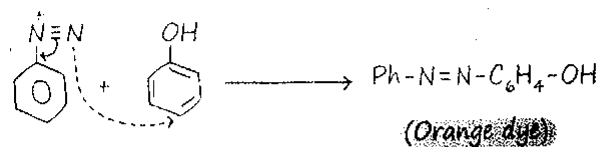
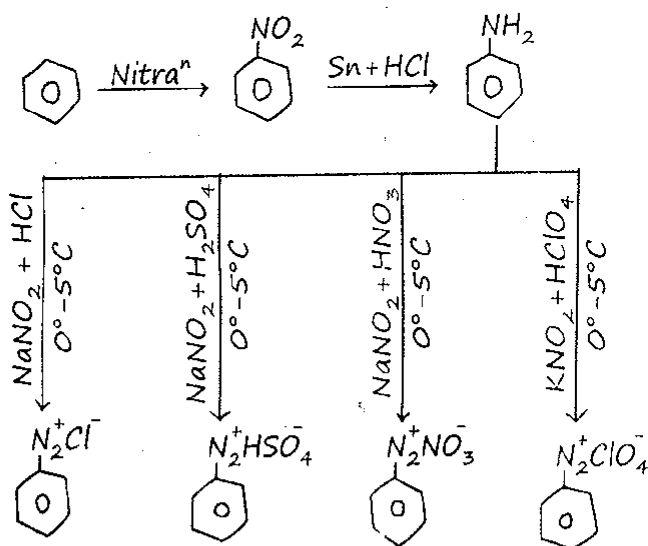


3. Carbylamine Test (Hoffmann isocyanide Test) :-



(CCl_2) = DCC. (Foul Odour)

BENZENE DIAZONIUM SALTS :-



Indicator :- Acid Base Titration.

BIOMOLECULES

Carbohydrates

Classification of Carbohydrates:

A. Monosaccharides: A carbohydrate that cannot be hydrolysed further to give simpler unit of polyhydroxy aldehyde or ketone.

Examples: Glucose, fructose, arabinose etc.

B. Oligosaccharides: These carbohydrates yield two to ten monosaccharide units on hydrolysis. They are further classified as disaccharides, trisaccharides etc, depending upon the no. of monosaccharides, they provide on hydrolysis.

C. Polysaccharides: They yield a large number of monosaccharide units on hydrolysis.

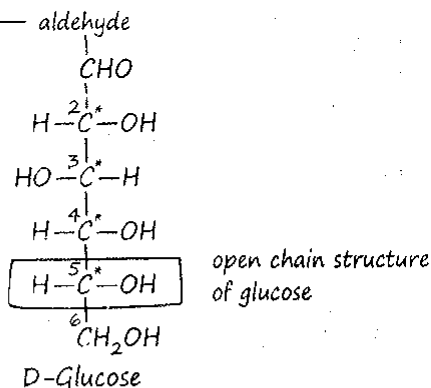
Example : Starch, Cellulose etc.

Monosaccharides :

☞ If a monosaccharide contains an **aldehyde group**, it is known as an **aldose** and if it contains a **keto group**, it is called **ketose**

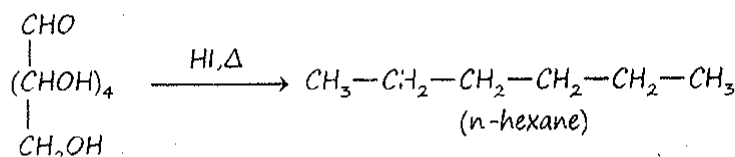
(a) **Glucose:** It is an **aldohexose**

Structure of glucose ($C_6H_{12}O_6$)

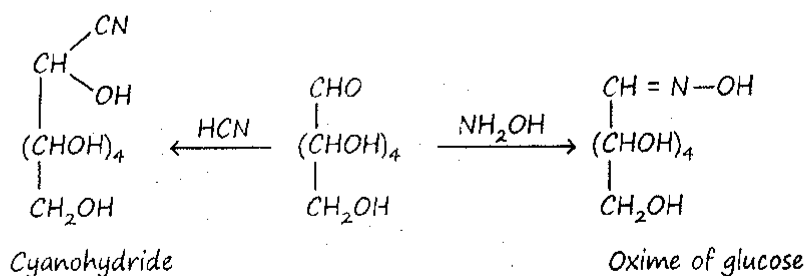


Reaction of Glucose :

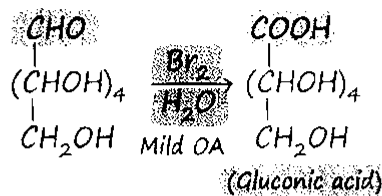
☞ On **prolonged heating with HI**, it forms **n-hexane**, suggesting carbons are linked in straight chain.



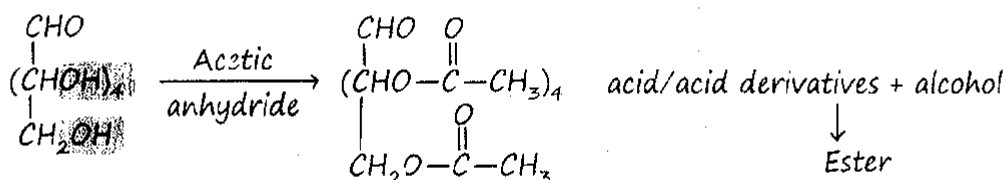
☞ Presence of **carbonyl group** ($-\text{CHO}$) is confirmed by the following reactions.



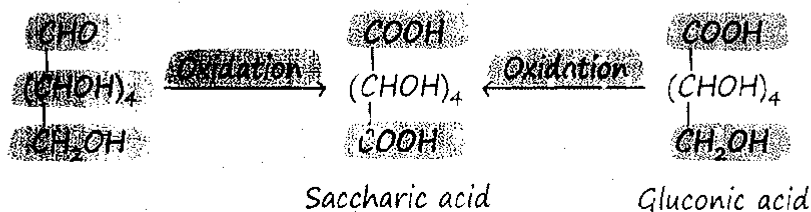
- Carbonyl group present, is an aldehyde and it is confirmed by the given reaction



- Acetylation of glucose gives glucose pentaacetate which confirms presence of five OH groups



- ~~Glucose and gluconic acid~~ is oxidised to saccharic acid by nitric acid. This indicates presence of primary -OH group



Cyclic Structure of Glucose

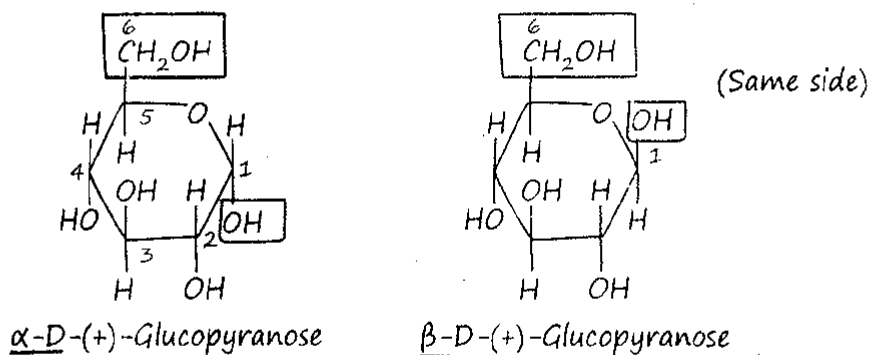
- The given observations could not explain chain structure of glucose

(i) It does not react with ~~NHHSO₃~~ or Schiff's reagent

- It does not give 2, 4-DNP test and it does not react with Grignard Reagent.

(ii) Pentaacetate of glucose does not react with NH_2OH .

(iii) Glucose exist in two different crystalline forms, i.e. α and β forms.



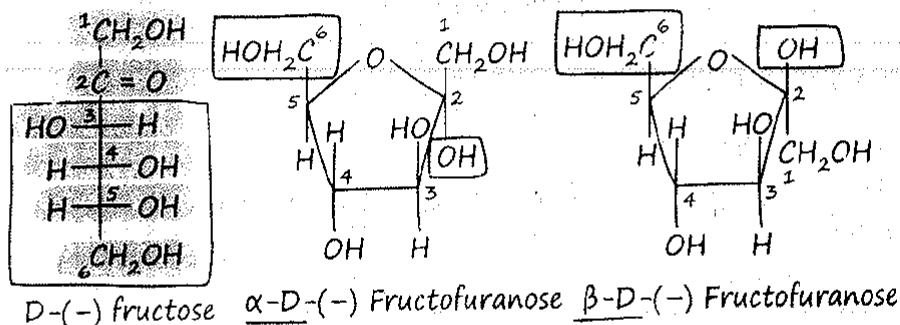
- α and β forms of glucose are called anomers as they only differ at C1 position. Six membered cyclic structure of glucose is ~~called pyranose structure~~

(b) Fructose

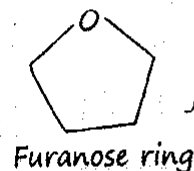
Fructose is a ketohexose



Following structure have been assigned to this molecule ($\text{C}_6\text{H}_{12}\text{O}_6$)



Fructose & glucose are functional isomers of each other

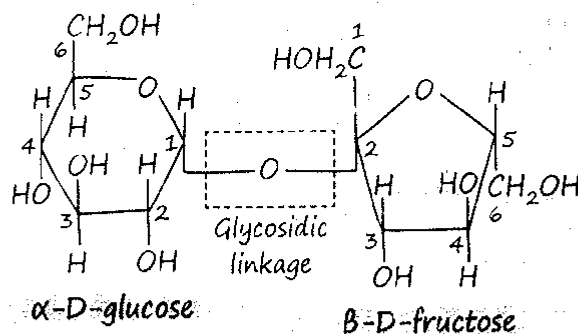


Tests of Glucose and Fructose

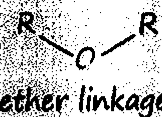
Both glucose and fructose reduce Tollen's reagent and Fehling's solution. They are also called reducing sugars.

Disaccharides

(i) Sucrose $\Rightarrow$ α -D-Glucose + β -D-Fructose



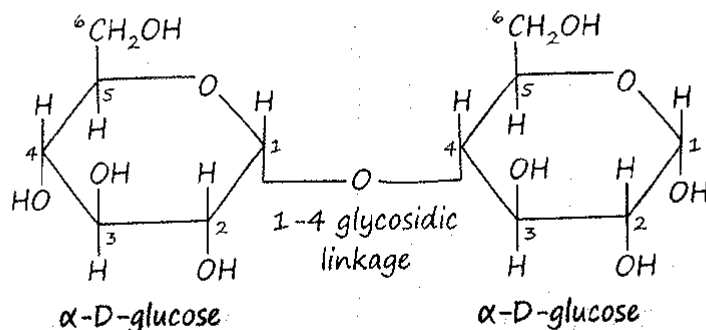
1-2 glycosidic linkage



It is dextrorotatory

It is non reducing sugar

(ii) Maltose $\Rightarrow$ α -D-Glucose + α -D-Glucose



Sucrose - α -D Glu + β -D Fru

Maltose - α -D Glu + α -D Glu

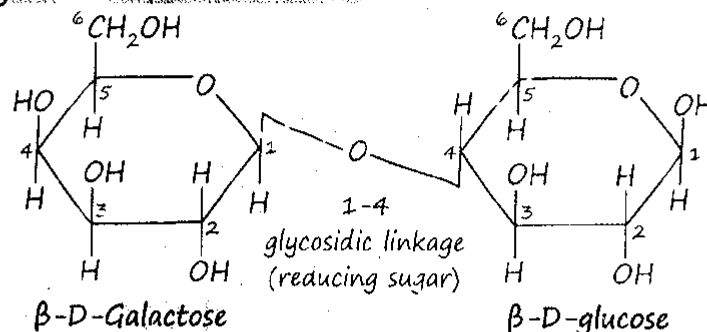
Lactose - β -D-Gal + β -D-Glu

It is dextrorotatory

It is reducing sugar

It gives positive test with Tollen's reagent and Fehling's solution

(iii) $\text{Lactose} \Rightarrow \beta\text{-D galactose} + \beta\text{-D Glucose}$



Polysaccharides

(i) **Starch** : It is Polymer of α -D Glucose. It consists of two components. Amylose and Amylopectin.

Amylose (15-20% of starch)

- Water soluble component
- Unbranched chain which is formed by C1-C4 glycosidic linkage of α -D Glucose unit

Amylopectin (80-85% of starch)

- Water insoluble component
- Branched chains which is formed by C1-C4 glycosidic linkages whereas branching occurs by C1-C6 glycosidic linkages

(ii) *Cellulose*

- It is a straight chain polysaccharide
- It is composed of β -D-glucose units

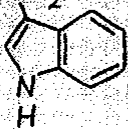
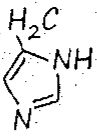
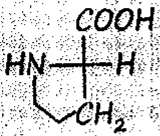
(iii) **Glycogen**

- It is called animal starch
- Its structure is similar to amylopectin and is rather more highly branched.

Proteins

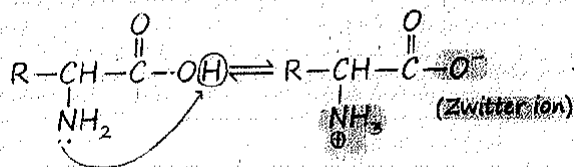
- All proteins are polymers of α -amino acid
- Structure of some commonly occurring amino acids along with their 3-letter and 1 letter symbols are given in the following table.
- The amino acids which can be synthesised in the body, are known as non-essential amino acids.

S.No.	Name of the amino acids	Characteristic feature of side chain, R	Three letter symbol	One letter code
1.	Glycine	H	Gly	G
2.	Alanine	$-CH_3$	Ala	A
3.	Valine*	$(H_3C)_2CH-$	Val	V
4.	Leucine*	$(H_3C)_2CH-CH_2-$	Leu	L
5.	Isoleucine*	$ \begin{array}{c} H_3C-CH-CH- \\ \\ CH_3 \end{array} $	Ile	I
6.	Arginine*	$ \begin{array}{c} HN=C-NH-(CH_2)_3- \\ \\ NH_2 \end{array} $	Arg	R

7.	Lysine*	$\text{H}_2\text{N}-(\text{CH}_2)_4-$	Lys	K
8.	Glutamic acid	$\text{HOOC}-(\text{CH}_2)_2-$	Glu	E
9.	Aspartic acid	$\text{HOOC}-\text{CH}_2-$	Asp	D
10.	Glutamine	$\text{H}_2\text{N}-\overset{\text{O}}{\parallel}\text{C}-\text{CH}_2-\text{CH}_2-$	Gln	Q
11.	Asparagine	$\text{H}_2\text{N}-\overset{\text{O}}{\parallel}\text{C}-\text{CH}_2-$	Asn	N
12.	Threonine*	$\text{H}_3\text{C}-\text{CHOH}-$	Thr	T
13.	Serine	$\text{HO}-\text{CH}_2-$	Ser	S
14.	Cysteine	$\text{HS}-\text{CH}_2-$	Cys	C
15.	Methionine*	$\text{H}_3\text{C}-\text{S}-\text{CH}_2-\text{CH}_2-$	Met	M
16.	Phenylalanine*	$\text{C}_6\text{H}_5-\text{CH}_2-$	Phe	F
17.	Tyrosine	$(p)\text{HO}-\text{C}_6\text{H}_4-\text{CH}_2-$	Tyr	Y
18.	Tryptophan*	$-\text{CH}_2-$ 	Trp	W
19.	Histidine*	H_2C 	His	H
20.	Proline		Pro	P

* essential amino acid

- The amino acids which cannot be synthesised in the body and must be obtained through diet are known as essential amino acids.
- In aqueous solution, the carboxyl group can lose a proton and amino group can accept a proton, giving rise to a dipolar ion known as zwitter ion. This is neutral but contains both positive and negative charges.



Structure of Proteins

(a) Fibrous proteins :

- The polypeptide chains run parallel and held together by hydrogen and disulphide bonds shows fibre like structure
- They are insoluble in water e.g. keratin (present in hair, wool, silk), myosin (present in muscles) etc.

(b) **Globulin proteins**

- ➔ In this case polypeptide chains coil around to give a spherical shape
- ➔ They are usually soluble in water e.g. insulin, globulin

Vitamins

- (i) **Fat soluble vitamins** : Vitamins are soluble in fat and oils. These are vitamins A, D, E and K.
- (ii) **Water soluble vitamins** : B group vitamins and vitamin C are water soluble vitamins.

Name of Vitamin	Deficiency Diseases
1. Vitamin A	Xerophthalmia
2. Vitamin B ₁	(Thiamine) Beri beri
3. Vitamin B ₂	(Riboflavin) Cheilosis
4. Vitamin B ₁₂	Pernicious anaemia
5. Vitamin C	(Ascorbic acid) Scurvy
6. Vitamin D	Rickets (in children) osteomalacia (in adults)
7. Vitamin K	Increased blood clotting time

POLYMER

Classification of Polymers

➤ Classification based on source

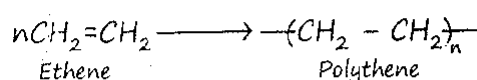
- (a) **Natural polymers**: These polymers are found in plants and animals. Examples are proteins, starch, rubber etc.
- (b) **Semi-synthetic polymers**: Cellulose derivatives as cellulose acetate and cellulose nitrate are semi-synthetic polymers.
- (c) **Synthetic polymers**: These are man made polymers. Examples are Buna-S, nylon 6, 6, polythene etc.

Type of Polymerisation Reaction

(A) Addition polymerisation or chain growth^o polymerisation

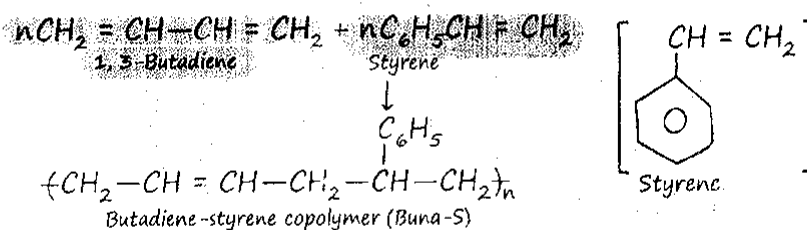
- The molecules of the same monomer or different monomers add together on a large scale to form a polymer.
- Monomers used are unsaturated compounds, e.g. alkenes, alkadienes and their derivatives.
- Chain growth can take place through the formation of either free radicals or ionic species.
- The addition polymers formed by the polymerization of a single monomeric species are known as homopolymers.

Example : Polythene



- The polymers made by addition polymerisation from two different monomers are termed as copolymers.

Example : Buna-S



➤ Preparation and uses of important addition polymers

- (i) Polythene : There are two types of polythene

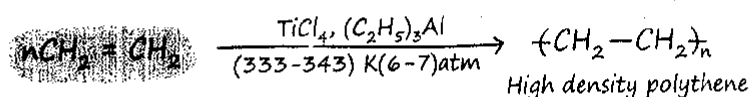
Low density polythene :

It is obtained by the polymerisation of ethene under high pressure of 1000 to 2000 atmospheres

at a temperature of 350 K to 570 K in presence of a peroxide initiator.

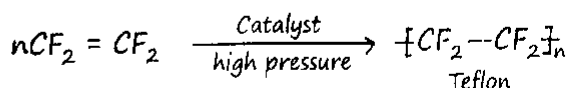
- It is obtained through free radical addition and H-atom abstraction.
- It has highly branched structure.
- They are chemically inert, tough but flexible and poor conductor of electricity.

High density polythene : HDPE - Low pressure 6-7 atm



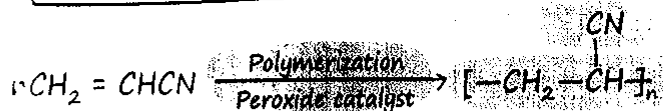
- Catalyst $\text{TiCl}_4, (\text{C}_2\text{H}_5)_3\text{Al}$ is called **Ziegler-Natta catalyst**.
- The polymer obtained has high density due to close packing.
- These are also called linear polymers.
- It is also chemically inert, more tough and hard.

(ii) **Polytetrafluoroethene (Teflon)**



Uses : It is used in making oil seals, gaskets and non-stick surface coated utensils.

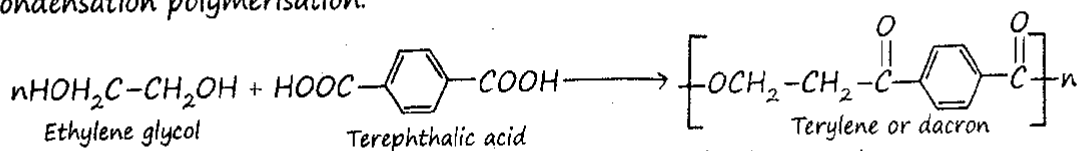
(iii) **Polyacrylonitrile :**



Uses : It is used as a substitute for wool in making commercial fibers as orlon or acrilan.

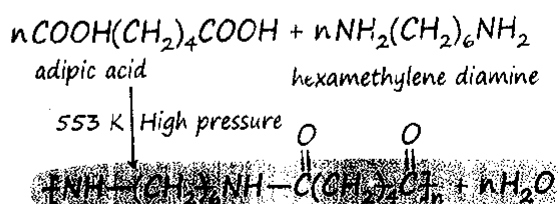
(B) **Condensation polymerisation or step growth polymerisation**

- It generally involves a repetitive condensation reaction between two bi-functional or tri-functional monomeric units.
- This is also called step growth polymerisation. Preparation and uses of some polymers obtained by condensation polymerisation.



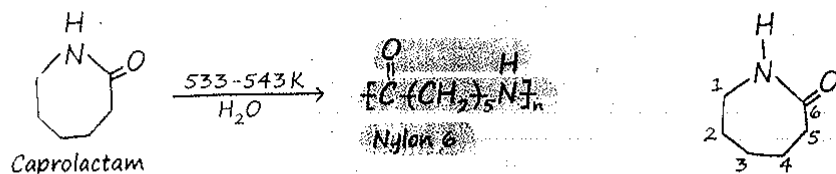
(i) **Polyamides :** These polymers possess amide linkages (-C(=O)-NH-) and are called nylons.

➤ **Nylon 6,6 :**



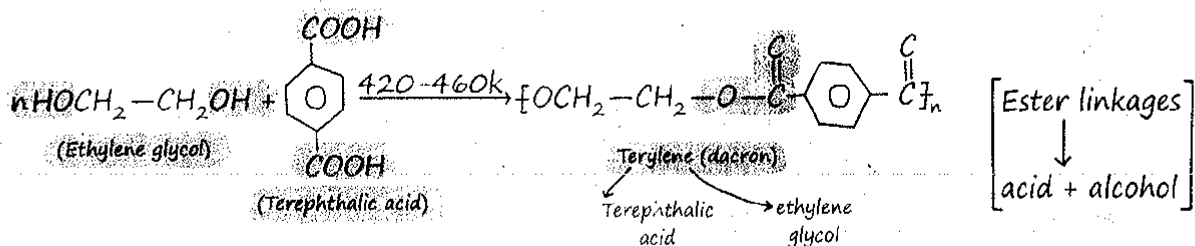
Uses : It is used in making sheets, bristles for brushes and in textile industry.

➤ Nylon 6 :



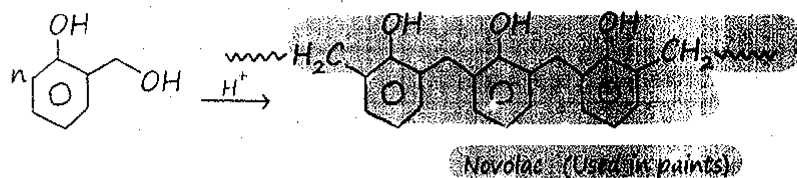
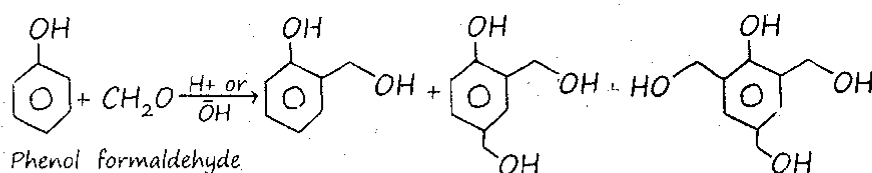
Uses : It is used in manufacture of tyre cords, fabrics and ropes.

(ii) **Polyesters** : These are condensation products of dicarboxylic acids and diols.



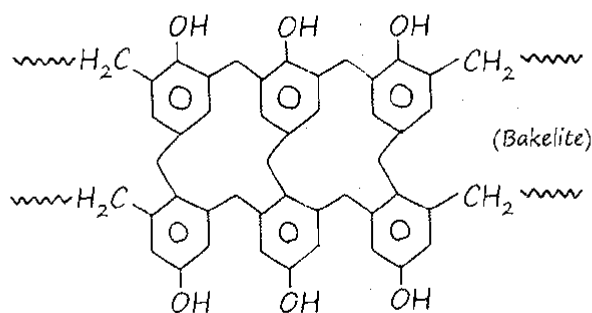
Uses : It is used in blending with cotton and wool fibres.

(iii) **Phenol formaldehyde polymer** (Bakelite)



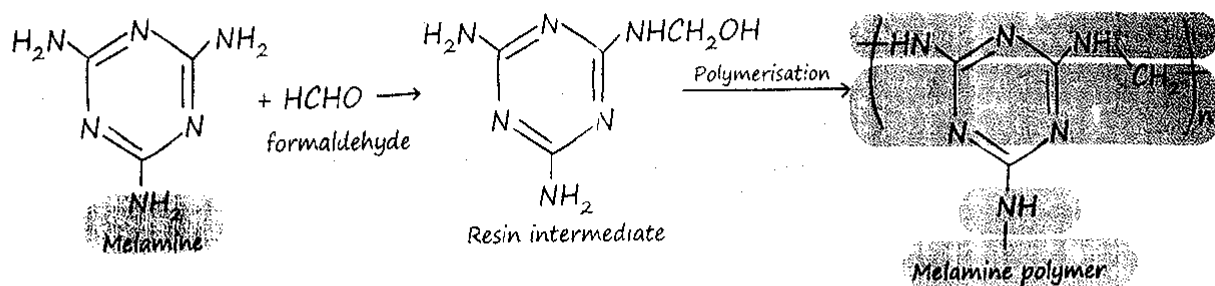
➤ Novolac on heating with formaldehyde undergoes cross linking to form bakelite

➤ Bakelite is thermosetting polymer which can not be reused or remoulded



Uses : It is used for making combs, electrical switches and handles of various utensils.

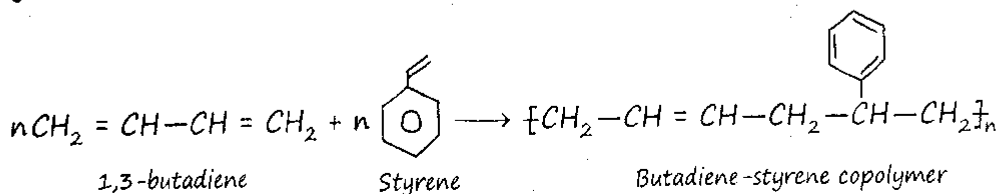
(iv) **Melamine-formaldehyde polymer**



Uses : It is used in the manufacture of unbreakable crockery.

Copolymerisation

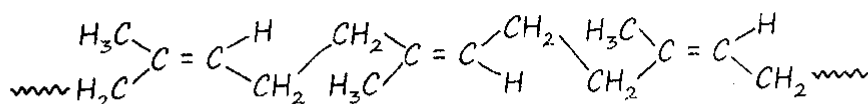
- Here a mixture of more than one monomeric species are allowed to polymerise and form a copolymer
- The copolymer can be made both by chain growth (addition) and by step growth (Condensation) polymerisation.



Uses : It is used in manufacture of autotyres, floortiles, cable insulation etc.

Rubber (Elastomer)

- (i) **Natural rubber**: It is a linear polymer of isoprene (2-methyl-1, 3-butadiene) and is also called as **cis-1, 4-polyisoprene**.

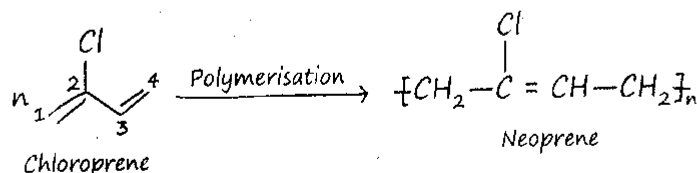


Vulcanisation of rubber: This process consists of heating a mixture of raw rubber with sulphur and appropriate additive 273 K to 415 K.

- Sulphur forms cross links at the reactive sites of double bond.
- Due to vulcanisation rubber gets stiffened.
- 5% sulphurised rubber is used in tyre manufacturing.

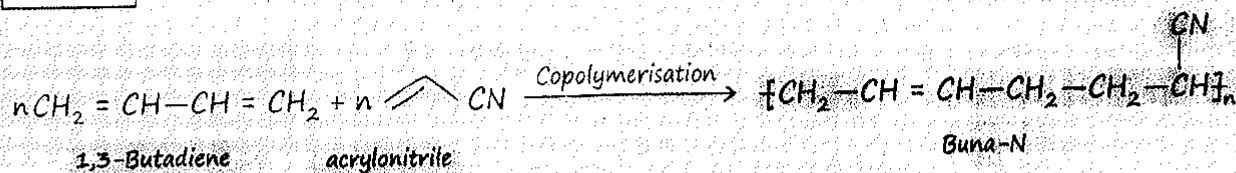
(ii) **Synthetic rubbers :**

(a) **Neoprene**



Uses : It is used for the manufacture of conveyor belts, gaskets and hoses.

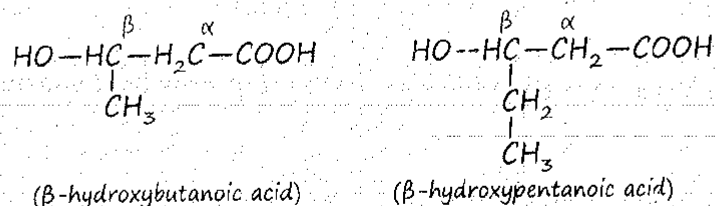
(b) **Buna-N**



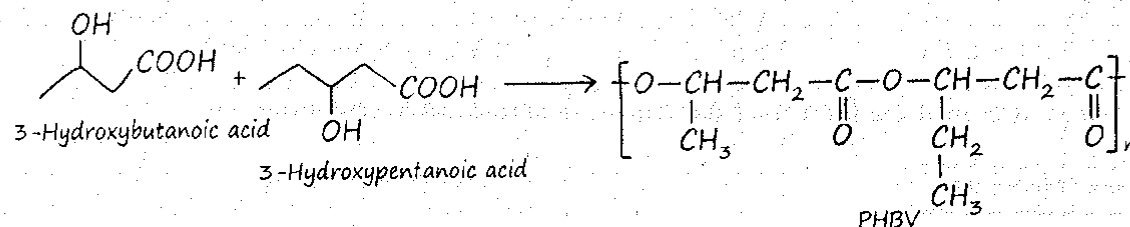
Uses : It is used in making oil seals, tank lining etc.

BIODEGRADABLE POLYMERS

Classification of Polymers

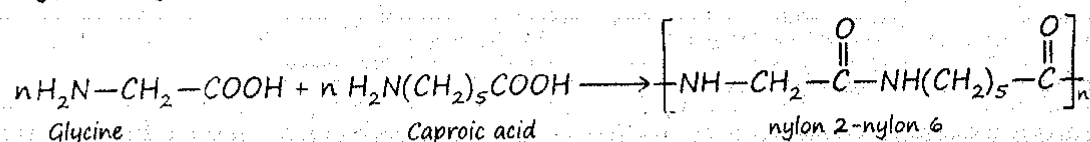


➤ **Poly β-hydroxybutyrate -co- β-hydroxy valerate (PHBV)**



Uses : It is used in orthopaedic devices and in controlled release of drugs.

➤ **Nylon 2-nylon 6**



POLYMERS OF COMMERCIAL IMPORTANCE

Some commercially important polymers along with their structures and uses are given below in table.

Name of polymer	Monomer	Structure	Uses
Polypropene	Propene $\text{CH}_3 - \text{CH} = \text{CH}_2$	$\left(\text{CH}_2 - \underset{\text{CH}_3}{\text{CH}} \right)_n$	Manufacture of ropes, toys, pipes, fibres, etc.
Polystyrene	Styrene $\text{CH} = \text{CH}_2$ 	$\left(\text{CH}_2 - \underset{\text{C}_6\text{H}_5}{\text{CH}} \right)_n$	As insulator, wrapping material, manufacture of toys, radio and television cabinets.

Polymers Polymers	Vinyl chloride $\begin{array}{c} \text{Cl} \\ \\ \text{CH}_2 = \text{CH} \end{array}$	$\text{-(CH}_2\text{--}\overset{\text{Cl}}{\underset{ }{\text{CH}}}\text{)}_n$	Manufacture of rain coats, hand bags, vinyl flooring, water pipes.
Urea-formaldehyde Resin	(a) Urea (b) Formaldehyde	$\text{-(NH--CO--NH--CH}_2\text{)}_n$	For making unbreakable cups and laminated sheets.
<u>Glyptal</u> glycol	(a) <u>Ethylene glycol</u> (b) <u>Phthalic acid</u>	$\text{-(OCH}_2\text{--CH}_2\text{OOC} \begin{array}{c} \diagup \quad \diagdown \\ \text{C}_6\text{H}_4 \end{array} \text{CO)}_n$	Manufacture of paints and lacqures.
Bakelite	(a) Phenol (b) Formaldehyde	$\text{-(} \begin{array}{c} \text{O--H} \\ \\ \text{C}_6\text{H}_4 \end{array} \text{--CH}_2\text{--} \begin{array}{c} \text{O--H} \\ \\ \text{C}_6\text{H}_4 \end{array} \text{--CH}_2\text{)}_n$	For making combs, electrical switches, handles of utensils and computer discs.

(1) Some Important Terms

- 💡 **Drugs** : Drugs are chemical of low molecular masses ($\sim 100-500u$). These interact with macromolecular targets and produce a biological response. When the biological response is therapeutic and useful, these are called medicines.
- 💡 **Enzyme** : Proteins which perform the role of biological catalysts in the body are called enzymes.
- 💡 **Receptors** : Proteins which are crucial to communication system in the body are called receptors.
- 💡 **Enzyme inhibitors** : Drugs can block the binding site of the enzyme and prevent the binding of substrate, or can inhibit the catalytic activity of enzyme. Such drugs are called enzyme inhibitors.
- 💡 **Competitive inhibitors** : Drugs compete with natural substrate for their attachment on the active site of enzymes. Such drugs are called competitive inhibitors.
- 💡 **Allosteric site** : Some drugs do not bind to the enzyme's active site. These bind to a different site of enzyme which is called allosteric site.
- 💡 **Chemical messengers** : In human body, message between two neurons and that between neurons to muscles is communicated through certain chemicals. These are called chemical messengers.
- 💡 **Antagonists** : Drugs that bind to the receptor site and inhibit its natural function are called antagonists.
- 💡 **Agonists** : Drugs that mimic the natural messenger by switching on the receptor are called agonists.

(2) Therapeutic Action of Different Classes of Drugs

Examples

(I) Antacids:

- Over production of acid in the stomach causes irritation and pain
- Histamine stimulates the secretion of pepsin and HCl in the stomach
- Antacids prevent the interaction of histamine with receptors present in the stomach wall. This results in the release of lesser amount of acid.

Examples of Antacids

Cimetidine (Tegamet), ranitidine (Zantac)

(II) Antihistamines :

- Histamine is a potent vasodilator
- Histamine is also responsible for the nasal congestion associated with common cold and allergic response to pollen
- Synthetic drugs, brompheniramine (Dimetapp) and terfenadine (Seldane), act as antihistamines

(III) Neurologically active drugs :

- Tranquilizers and analgesics are neurologically active drugs. These affect the message transfer mechanism from nerve to receptor.

(a) Tranquilizers :

- These drugs are used for the treatment of stress, and mild or even severe mental diseases.
- These relieve anxiety, stress, irritability or excitement by inducing a sense of well being.
- They form an essential component of sleeping pills.

Examples are : Iproniazid, phenelzine (Nardil), Meprobamate, Chlordiazepoxide, Equanil

- Derivatives of barbituric acids also constitute important class of tranquilizers.

Examples: Veronal, amytal, nembutal, luminal and seconal

(b) Analgesics : These drugs reduce or abolish pain. These are of two types :

- (i) Non-narcotic analgesics : These drugs inhibit synthesis of prostaglandins which stimulate inflammation in the tissue and cause pain.

Examples: Aspirin and paracetamol

- (ii) Narcotic analgesics : Morphine when administered in medicinal dose, relieve pain and produce sleep.

(IV) Antimicrobials :

- An antimicrobial destroy/prevent development or inhibit the pathogenic action of microbes such as bacteria, fungi, virus, parasites selectively
- Antibiotics, antiseptics and disinfectants are antimicrobial drugs

(A) Antibiotics : Antibiotics are used as drugs to treat infections because of their low toxicity for humans and animals.

- Arsphenamine also known as salvarsan was used to treat syphilis caused by bacteria spirochete.

o Antibiotics are of two types :

(a) **Bactericidal:** These antibiotics have killing effect on microbes

Examples: Penicillin, Aminoglycosides, Ofloxacin

(b) **Bacteriostatic:** These antibiotics have inhibitory effect on microbes.

Examples: Erythromycin, Tetracycline, chloramphenicol

Antibiotics which kill or inhibit a wide range of Gram-positive and Gram-negative bacteria are said to be broad spectrum antibiotics.

Examples: Ampicillin, Amoxycillin, Chloramphenicol, vancomycin, ofloxacin

Antibiotics effective mainly against Gram-positive or Gram-negative bacteria are narrow spectrum antibiotics.

Example: Penicillin G

(B) **Antiseptics and disinfectants:**

(a) **Antiseptics:** These are applied to the living tissues such as wounds, cuts, ulcers and diseased skin surfaces.

Examples: Furacine, soframidine

o Dettol is a mixture of chloroxylenol and terpineol.

o Bithional is added to soaps to impart antiseptic properties

o 2-3 per cent solution in alcohol-water mixture is known tincture of iodine. It is applied on wounds.

o Boric acid in dilute aqueous solution is weak antiseptic for eyes.

o 0.2 per cent solution of phenol is an antiseptic.

(b) **Disinfectants :**

o These are applied to inanimate objects such as floors, drainage system, instruments etc.

o 1% solution of phenol is used as disinfectant

o Chlorine in the concentration of 0.2 to 0.4 ppm in aqueous solution and SO_2 in very low concentrations, are disinfectants.

(V) **Antifertility drugs :**

o Antifertility drugs are used to control birth in human beings

o Birth control pills essentially contain a mixture of synthetic estrogen and progesterone derivatives.

o Norethindrone and novestrol are used as antifertility drugs.

(3) Chemicals in Food

(I) Artificial Sweetening Agents :

Natural sweeteners, e.g., sucrose add to calorie intake and therefore many people prefer to use artificial sweeteners.

Artificial sweetener

Sweetness value in
comparison to cane sugar

(a) Aspartame	100
(b) Saccharin	550
(c) Sucralose	600
(d) Alitame	2000

As
Saccharin
Scores
all time



Aspartame is limited to cold foods and soft drinks because it is unstable at cooking temperature.

Funnel-shaped opening provided with many finger-like Fimbriae. It helps to collect the ovum.

Sucralose is trichloro derivative of sucrose. It is stable at cooking temperature.

(II) Food Preservatives :

- Food preservatives prevent spoilage of food due to microbial growth.

Examples : Table salt, sugar, vegetable oils, C_6H_5COONa etc.

Class - I
preservatives

Class - 2
preservatives

(III) Antioxidants in Food :

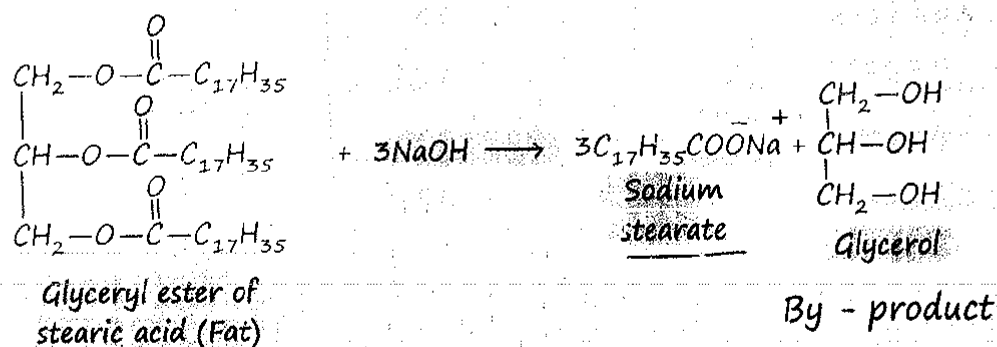
- These help in food preservation by retarding the action of oxygen on food.
- These are more reactive towards oxygen than food materials which they are protecting.
- Butylated hydroxytoluene (BHT) and butylated hydroxy anisole (BHA) are most familiar antioxidants.
- SO_2 and sulphite are useful antioxidants for wine and beer, sugar syrups and cut, peeled or dried fruits and vegetables.

(4) Cleansing Agents

Two types of detergents are used as cleansing agents.

(i) Soaps :

- Soaps are sodium or potassium salts of long chain fatty acids.
- Soaps containing sodium salts are formed by heating fat with aqueous NaOH. This reaction is called saponification

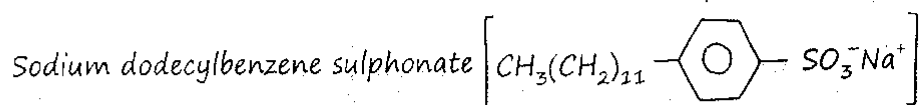


(ii) Synthetic Detergents : These are cleansing agents which have all the properties of soaps, but can be used both in soft and hard water.

These are of Three Types

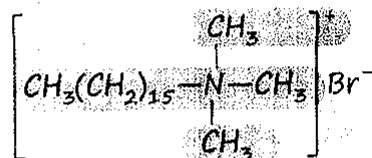
(a) Anionic detergents : Anionic detergents are sodium salts of sulphonated long chain alcohols or hydrocarbons.

Examples : Sodium laurylsulphate ($\text{CH}_3(\text{CH}_2)_{10}\text{CH}_2\text{OSO}_3^- \text{Na}^+$),



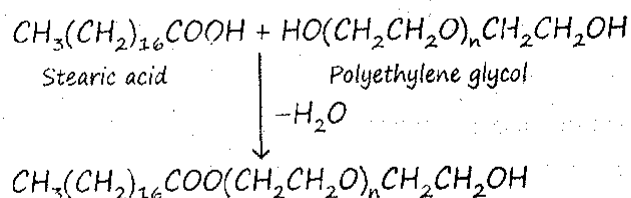
(b) Cationic detergents : Cationic detergents are quaternary ammonium salts of amines with acetates, chlorides or bromides as anions.

Examples : Cetyltrimethyl ammonium bromide



(c) Non-ionic detergents : Non-ionic detergents do not contain any ion in their constitution.

Examples : Detergent formed when stearic acid reacts with polyethylene glycol.



- Liquid dishwashing detergents are non-ionic type
- Detergents and soap remove grease and oil by micelle formation